



**GREEN
CLIMATE
FUND**

Meeting of the Board

16 – 19 March 2021

Virtual meeting

Provisional agenda item 14

GCF/B.28/02/Add.10

23 February 2021

Consideration of funding proposals - Addendum X

Funding proposal package for FP163

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Sustainable Renewables Risk Mitigation Initiative (SRMI) Facility";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Programme title:	Sustainable Renewables Risk Mitigation Initiative (SRMI) Facility
Multiple Countries:	Botswana, Central African Republic (CAR), Democratic Republic of Congo (DRC), Kenya, Mali, Namibia and Uzbekistan
Accredited Entity:	World Bank
Date of first submission:	2020/05/19
Date of current submission	2021/02/15
Version number	V.12



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A. PROGRAMME SUMMARY					
A.1. Project or programme	Programme	A.2. Public or private sector	Public		
A.3. Request for Proposals (RFP)	Not applicable				
A.4. Result area(s)	<i>Check the applicable <u>GCF result area(s)</u> that the <u>overall</u> proposed project/programme targets. For each checked result area(s), indicate the estimated percentage of <u>GCF budget</u> devoted to it. The total of the percentages when summed should be 100%.</i> <table border="1"> <tr> <td> <u>Mitigation:</u> Reduced emissions from: ☒ Energy access and power generation: ☐ Low-emission transport: ☐ Buildings, cities, industries and appliances: ☐ Forestry and land use: <u>Adaptation:</u> Increased resilience of: ☐ Most vulnerable people, communities and regions: ☐ Health and well-being, and food and water security: ☐ Infrastructure and built environment: ☐ Ecosystem and ecosystem services: </td> <td> <u>GCF contribution:</u> 100% </td> </tr> </table>			<u>Mitigation:</u> Reduced emissions from: ☒ Energy access and power generation: ☐ Low-emission transport: ☐ Buildings, cities, industries and appliances: ☐ Forestry and land use: <u>Adaptation:</u> Increased resilience of: ☐ Most vulnerable people, communities and regions: ☐ Health and well-being, and food and water security: ☐ Infrastructure and built environment: ☐ Ecosystem and ecosystem services:	<u>GCF contribution:</u> 100%
<u>Mitigation:</u> Reduced emissions from: ☒ Energy access and power generation: ☐ Low-emission transport: ☐ Buildings, cities, industries and appliances: ☐ Forestry and land use: <u>Adaptation:</u> Increased resilience of: ☐ Most vulnerable people, communities and regions: ☐ Health and well-being, and food and water security: ☐ Infrastructure and built environment: ☐ Ecosystem and ecosystem services:	<u>GCF contribution:</u> 100%				
A.5. Expected mitigation impact	88,980,000 CO ₂ eq tons	A.6. Expected adaptation impact			
A.7. Total financing (GCF + co-finance)	USD 1,563,500,000	A.9. Project size	Large (Over USD 250 million)		
A.8. Total GCF funding requested	USD 280,000,000				
A.10. Financial instruments requested for the GCF funding	☒ Grant USD 54,000,000 ☒ Reimbursable Grant USD 30,000,000 ☒ Loan USD 176,000,000 ☐ Results-based payment n/a ☒ Guarantee USD 20,000,000				
A.11. Implementation period	12 years	A.12. Total lifespan	25 years for solar and wind projects / 50 years + for grid infrastructure		
A.13. Expected date of AE internal approval	February 2, 2021	A.14. ESS category	A		
A.15. Has this FP been submitted as a CN before?	Yes ☐ No ☒	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒		
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐		

A.19. Complementarity and coherence	Yes ☒ No ☐
A.20. Executing Entity information¹	Republic of Botswana represented by the Ministry of Finance and acting through the Ministry of Energy and the Botswana Power Corporation (BPC – the State Utility) Central African Republic (CAR) represented by the Ministry of Economy, Planning and Cooperation and acting through the Ministry of Energy Democratic Republic of Congo (DRC) represented by the Ministry of Finance and acting through the Ministry of Energy Republic of Kenya represented by the National Treasury and acting through the Ministry of Energy Republic of Mali represented by the Ministry of Finance and acting through the Ministry of Energy, the Société de Gestion de l’Energie de Manantali (SOGEM), and <i>Electricité du Mali</i> (EDM – the State Utility) Republic of Namibia represented by the Ministry of Finance and acting through the Ministry of Energy and NamPower (the State Utility) Republic of Uzbekistan represented by the Ministry of Finance and acting through the Ministry of Energy
A.21. Executive summary (max. 750 words, approximately 1.5 pages)	
While the proportion of renewable energy (RE) generation is rising every year, it remains a small percentage of the total world-wide electricity production and far from the targets needed to meet the Paris Agreement objectives. The proportion of the world electricity generated by wind, solar – together Variable Renewable Energy (VRE), biomass and waste-to-energy, geothermal, marine and small hydro rose from 6 percent in 2010 to 12.9 percent in 2018. Large solar and wind energy deployment is needed for enhancing energy access and energy security and to stay below the Climate Change Paris Agreement 2C° scenario. According to the Sustainable Development Scenario of the International Energy Agency (World Energy Outlook 2018), 950 GW of solar PV and 580 GW of wind need to be installed by 2025 in developing countries. It represents another 690 GW solar and 330 GW wind installation from 2019’s level. The limited installed capacity of cost-competitive solar and wind in developing countries reveals important regulatory, structural, and technical constraints. The main constraints identified are (i) limited generation and transmission planning capacity, (ii) inadequate regulatory framework that constrains the mobilization of sustainable private investments, (iii) limited procurement capacity to select the Independent Power Producers (IPPs), (iv) limited financial viability of off-takers and (v) VRE grid integration challenges due to weak grids especially in Sub-Saharan Africa. The World Bank through its Energy Sector Management Assistance Program (ESMAP), developed the Sustainable Renewables Risk Mitigation Initiative (SRMI) to address these challenges and propose a comprehensive financial and technical support to countries. Launched in 2018 for the COP	

¹ For the purposes of the loans and grants, the Executing Entities (EEs) would be the Borrowers/Recipients of the loans/grants. For the purposes of the guarantees contemplated under Component 3.2, the EEs would be the private sector RE project companies.

24 under the leadership of the World Bank in partnership with the *Agence Française de Développement* (AFD), the International Renewable Energy Agency (IRENA) and the International Solar Alliance (ISA), SRMI – previously known as the Solar Risk Mitigation Initiative – aims to support countries in developing and implementing sustainable renewable energy programs that will be more conducive to private investments to reduce reliance on public sector financing while maximizing socio-economic benefits for the countries (including from women empowerment's perspective).

Under the context of the Covid-19 crisis, SRMI's unique, integrated and replicable approach can support countries unlock a green economic stimulus through targeted public investments in order to leverage private sector investments at scale. SRMI aims to offer development and climate financing for (i) technical assistance to help countries develop evidence-based VRE targets, implement a sustainable renewable energy program, and maintain robust procurement processes with transaction advisors; (ii) critical public investments to enable integration of VRE, finance solar/wind park infrastructure, and increase access to electricity; and (iii) risk mitigation instruments to cover residual risks perceived by private investors. This financial support for renewable energy development is needed more than ever in the current situation alongside government economic support measures (such as short-term job allowances, direct cash transfers to citizens or targeted liquidity support to small and medium size enterprises).

The SRMI Facility to be funded by the GCF aims to provide technical and financial support to seven initial countries (Phase 1), namely Botswana, Central African Republic (CAR), Democratic Republic of Congo (DRC), Kenya, Mali, Namibia and Uzbekistan. If in 2017 those countries represented 1.2 percent of the total global greenhouse gas (GHG) emissions, under a BAU scenario they would on average triple their emissions between 2017 and 2030. With international support, the seven countries are committed to contribute to GHG emissions reductions especially focusing on their energy sector which is the main contributing sector. However, the countries have now prioritized immediate emergency response to manage the current Covid-19 crisis. The question of how the economic recovery is designed remains crucial in shaping the long-term pathways for emissions and determining whether the NDC targets can be achieved. If governments do not roll out low carbon development strategies and policies in response to the coming economic crisis, emissions could rebound and even overshoot previously projected levels by 2030, despite lower economic growth in the period to 2030.

It is therefore critical in this context to support policymakers to develop green stimulus interventions with catalytic concessional climate financing and support a medium-term strategy. The SRMI Facility, through the seven projects, is expected to contribute to the shift to low-emission sustainable development pathways as follows:

- **Directly support the development of an estimated 2.5 GW of RE,** (i) 1.8 GW of solar PV, (ii) 400 MW of concentrated solar plants (CSP), and (iii) 300 MW of onshore wind. Of this 2.1 GW are expected to be built during the lifetime of the projects and 400 MW expected to be built after the end of the projects. To unlock the RE power plants expected to be built during the lifetime of the projects, around USD 1 billion in public investment of which USD 250 million from GCF would need to have been disbursed. These public investment projects will also enable the integration of another 1.9 GW of RE. The integration of other projects is enabled due to the VRE integration analysis and grid investments.
- **Directly reduce GHG emissions by an estimated 97.93 m metrics tons of CO₂,** more precisely 88.98m reduction for the 2.1 GW RE built under the projects over the 25 years of their operation, and 8.95m metric tons of CO₂ for 0.4 GW of RE power plants over the 25 years of their operation that are expected to be prepared under the SRMI Facility projects but not built during the lifetime of the projects.

- **Unlock an estimated USD 3.3 billion of private investments**, including USD 3 billion for the 2.1 GW built during the lifetime of the projects and USD 300 million for the 0.4 GW built after the end of the projects.
- **Provide access to clean low-emission electricity to an estimated 4.2 million people through the electrification investments.**
- **Strengthen institutional and regulatory systems for planning and development.**
- **Reduce locking-in long-lived, climate-vulnerable infrastructure by increasing resilience of grid infrastructure.**

The SRMI Facility aims to be scaled up to a total USD 400 million in GCF funding amount by including additional countries, namely Indonesia and Vietnam, with whom significant and extensive engagements have been had to date. These efforts are expected to result into NOLs from these two countries post submission of this funding proposal. Vietnam and Indonesia are expected to have projects that meet the agreed eligibility criteria as defined in the present funding proposal. The Phase 2 of the Facility that is to be complementary to the current Phase 1 aims to –based on current project information - (i) avoid a total of 286.32 m metric tons of CO₂, including 60.6 m metric tons of CO₂ during project lifetime and 225.7 m metrics tons built after the end of the project, (ii) unlock a total of 6.1 GW, including in the short-term 2 GW and in the medium-term 4.1 GW of RE, and (iii) leverage a total of USD 10.2 bl, including USD 1.2 bl during the project lifetime and USD 9 bl built after the end of the project. Once and if approved by GCF Board, Phase 2 will be complementary to the present Funding Proposal.

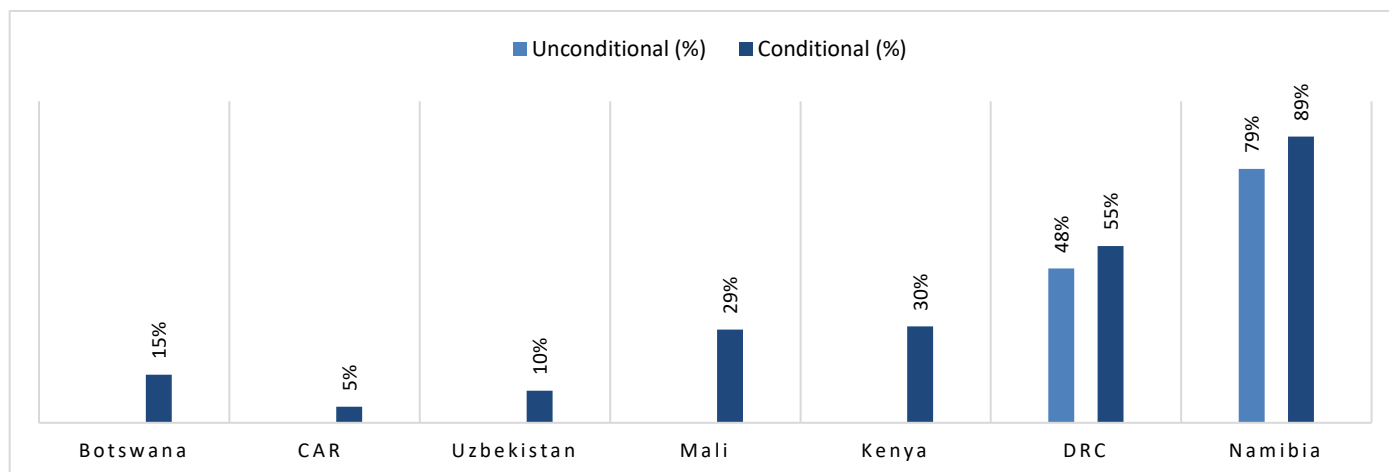
B. PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

The SRMI Facility aims at deploying the SRMI approach to unlock RE deployment by leveraging sustainable private investments starting with seven countries, namely Botswana, CAR, DRC, Kenya, Mali, Namibia and Uzbekistan. These countries were selected based on the willingness and interest of their respective governments to embrace renewable energy deployment and combat the challenges they face in the implementation of their NDCs and low-emissions development pathways. In addition, those countries represent a large sample of developing countries – with around 200 million inhabitants – facing different constraints, from very small to very large grids, low to high electrification rates, availability or not of domestic fossil fuels, and fragile and post conflict countries to medium income countries. The lessons learnt from the implementation of SRMI in those countries will enable to replicate the approach at scale in a sustainable and affordable manner, contributing to fast-track the Sustainable Development Goals (in particular SDG 13 on Climate Action and SDG 7 on Affordable and Clean Energy).

Under the context of the Covid-19 crisis, the SRMI Facility's projects aim to support countries in their energy transition and contribute to the overall mitigation agenda at a time where the economic recovery design will determine whether the Paris Agreement targets can be achieved. In that regard, macro-level economic stimulus packages need to be combined with catalytic climate finance for specific programs such as this one to ensure that this is achieved. If in 2017 those countries represented 1.2 percent of the total global GHG emissions, under a BAU scenario they would on average triple their emissions between 2017 and 2030. With international support, the seven countries are committed to contribute to GHG emissions reductions. As per their NDCs presented in *Figure 1*, Botswana, CAR, DRC, Kenya, Mali, Namibia and Uzbekistan committed to reduce, conditionally to external financing with the exception of Botswana, compared to their BAU scenario their emissions by 15 percent, 5 percent, 55 percent, 30 percent, 30 percent, 89 percent and 10 percent, respectively. However, the priority of countries is the immediate emergency response to the current Covid-19 crisis. The question of how the economic recovery is designed remains crucial in shaping the long-term pathways for emissions and determining whether the NDC targets can be achieved. If government do not roll out low carbon development strategies and policies in response to the coming economic crisis, emissions could rebound and even overshoot previously project levels by 2030, despite lower economic growth in the period to 2030.

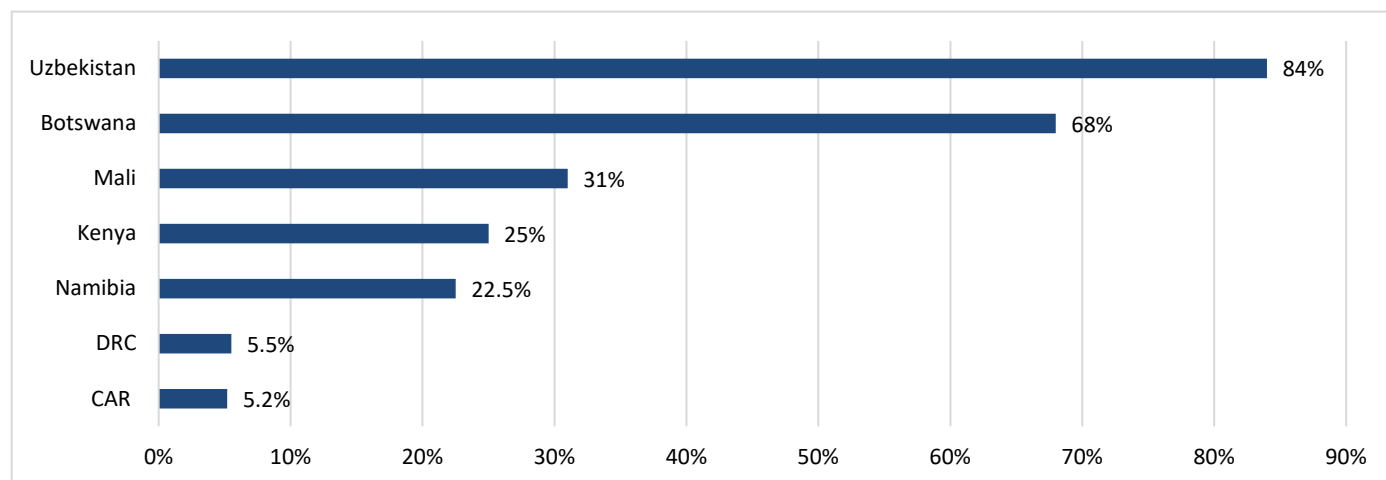
Figure 1: NDC target (conditional and unconditional – when applicable)



Sources: countries' NDC UNFCCC submissions

To reach their NDC targets, the seven countries will need to transform their energy sector. The energy sector represents in most of these countries the main contributor of emissions. The energy sector represents 84 percent of GHG emissions in Uzbekistan, 68 percent in Botswana and Mali 31 percent.

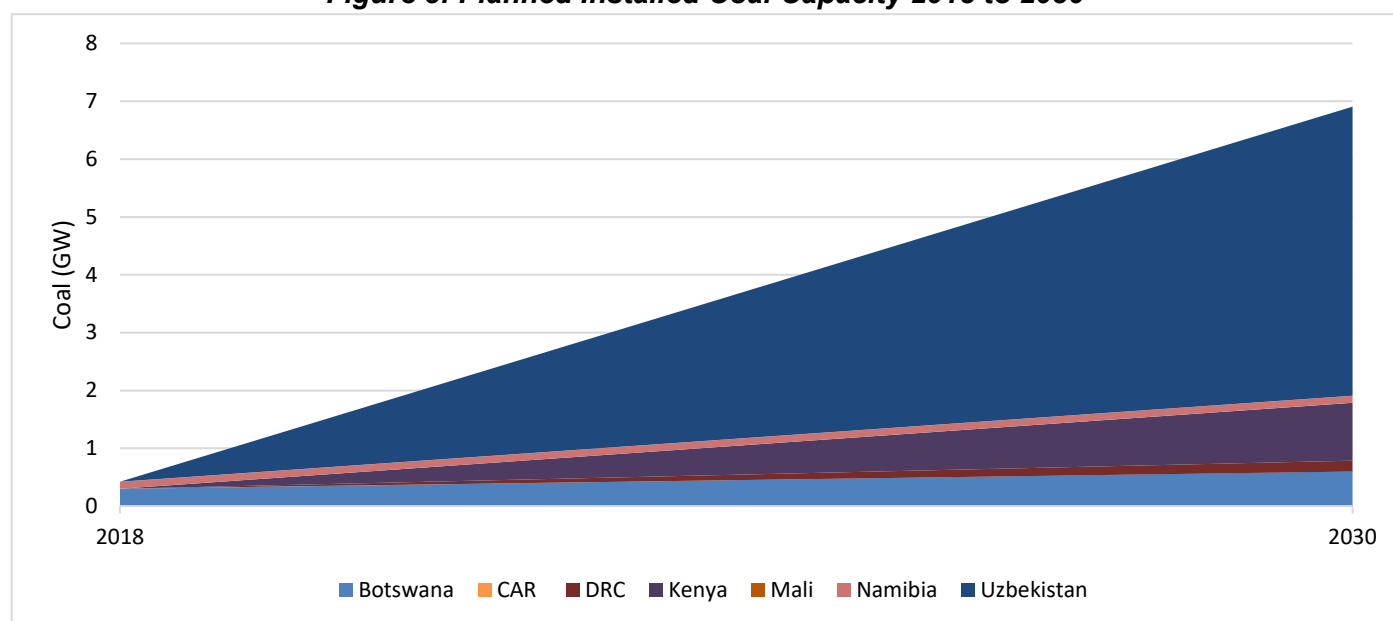
Figure 2: Energy Sector GHG Contribution per Country



Sources: countries' NDC submissions and various online sources

However, if the seven countries have ambitious GHG emissions reduction objectives as part of their NDCs, their current energy policies in place, the planned deployment of coal and the Covid-19 crisis may inhibit the NDC achievements. In the seven countries, total installed coal capacity is expected to increase from 0.5 GW to 7 GW from today's levels to 2030. As per their current plans by 2030, 8.5 GW of solar and wind generation will be built out of a total of 60 GW installed capacity – representing 14 percent. If representing these shares in GWh energy production, the solar and wind share would drop to 6-7 percent.

Figure 3: Planned Installed Coal Capacity 2018 to 2030



Sources: countries existing generation plans

It will be critical in this context to support policymakers to develop green stimulus interventions with targeted public investments in order to unlock large private investments for economic recovery.

According to the IEA and estimations by the World Bank, around 20 GW and 45 GW of RE (including large hydropower) by 2025 and 2030, respectively, will need to be deployed in those seven countries as part of their electricity mix to reach the 2C° targets and universal access. This represents around USD 15 billion in investments needed just for new solar and wind generation by 2025. To reach this objective, large amounts of private funding will have to be unlocked to complement the limited public sector financing available. Yet most developing countries still lack a pipeline of bankable solar and wind projects for consideration by the private sector. To develop one, countries must take a series of key steps to tackle critical risks perceived by the private sector such as grid constraints and procurement while also minimizing risks for the public sector.

The seven countries have been receiving support from other multilateral development banks (MDB) and bilateral banks to deploy RE in their countries outside of SRMI and of the scope of the present Funding Proposal – with a coordination at World Bank sector strategy level to ensure complementarity of programs, avoid overlaps and enhance synergies. More precisely:

- a. **Botswana:** The new World Bank-supported solar program is aligned with current support provided by Power Africa and the AfDB under a memorandum of understanding (MOU) on solar deployment.
- b. **CAR:** MDBs and donors are focusing their support in CAR – which is all grant – on access and generation: (i) the World Bank is financing an ongoing project consisting of a 25 MWp solar plant with 25 MWh battery storage – the PURACEL project, (ii) AfDB is currently financing the extension of Boali 2 hydropower system from 10 MW to 20 MW, plus the upgrade of the (only) transmission line between Boali and Bangui to 110 kV, (iii) AFD is working on refurbishing the existing 10 MW of Boali 2, (iv) the Chinese Government is supporting another solar project in Bangui (15 MWp), and eventually the construction of Boali 3 hydropower plant (5 MW) and (v) a Saudi fund is supporting investment in thermal generation for Bangui (10 MW). The new World Bank grant-funded project aims to transform the current 100 percent publicly owned energy sector by supporting the first steps towards mobilization of private investments and such, while supporting further deployment of disruptive technologies such as battery storage – CAR having the largest battery storage project being built in the African continent under the PURACEL project.
- c. **DRC:** The World Bank has been financing investments and technical assistance in DRC focusing on grid-connection, private investment mobilization for electrification, technical and financial support to the utility and interconnection to the neighboring countries. The World Bank program is aligned with the current portfolio of other MDBs such as AfDB, AFD, United Nations Development Program (UNDP), Japan International Cooperation Agency (JICA), the UK Department for International Development (DFID), USAID and the KfW. As the electrification agenda is critical to DRC while there are limited domestic financial resources, support for PPP RE mini-grids have been promoted under a DFID technical assistance financed by AfDB and GCF. The new USD 500 million DRC World Bank-supported project aims to leverage the lessons learnt from this PPP program and develop at a larger-scale fully privately financed mini-grids in some of the biggest regional capital cities. To unlock private financing for mini-grids, the project will propose tailor-made guarantees to the selected private investors to reduce the risk perceived.
- d. **Kenya:** The World Bank has been working on last mile electrification, generation (mostly geothermal) and utility strengthening/capacity building in Kenya as part of a large program funded by various donors. Since 2016, the World Bank Group, AFD and EIB are supporting solar and wind development with AFD financing a strategy on solar auction. The solar auction program has not yet been implemented due to limited appetite of the utility. However, the government has requested the World

Bank to support the modification of the solar auction program under its new RE program which will include least-cost generation plan, VRE integration and IPP procurement support. EIB financed IPPs in renewables and accessed GCF funding for its Global Energy Efficiency and RE Fund, and World Bank Group, through MIGA, is assessing providing support to RE IPP projects.

- e. **Mali:** The World Bank energy portfolio in Mali is focusing on access, sector strengthening and new generation. As part of its support to the West Africa Power Pool (WAPP), the World Bank is supporting the deployment of two regional solar with battery plants. These are being developed in coordination with other MDBs that are also supporting the utility and the energy sector, namely AfDB, the Islamic Development Bank (IsDB), the European Union, the West African Development Bank (*la Banque Ouest Africaine de Développement*, BOAD), AFD and KfW. These partners are preparing specific projects in the electricity sector in Mali and working in partnership on supporting the deployment of solar generation – with the World Bank leading the support to the VRE grid upgrades investment and the first deployment of competitive selection of IPPs. The ongoing OMVS 225 kV transmission line from Bamako to Manantali, Kayes to Senegal, and Kayes to Mauritania is funded by the AFD, IDA, and China Exim Bank. The IsDB is funding the mini-grids with productive uses in rural remote areas; the AFD is preparing a rural electrification program for some 60 localities; and the AfDB, AFD, and BOAD are co-financing the Guinea-Mali interconnection with the World Bank. GCF financing was accessed by BOAD for its climate finance facility to scale-up solar energy investments in francophone West Africa and its Mali solar rural electrification project. Clean Technology Fund (CTF) financing is being sought for the new investment project to support Mali in this transformative program with some of the largest solar and battery plants in the world (subject to CTF approvals).
- f. **Namibia:** The new World Bank-supported solar program is aligned with current support provided by Power Africa and the AfDB under a MOU on solar deployment. KfW and GIZ have also been supporting NamPower – the utility – in developing their least-cost generation plans and grid analysis as well as a battery storage pilot. Proparco provided debt financing for the 37 MWp PV. The Deutsche Bank accessed GCF funding for its universal green energy access programme. Together with the other MDBs and Development Finance Institutions (DFIs), the World Bank is coordinating its RE support to ensure complementarity of the different work streams.
- g. **Uzbekistan:** The World Bank Group has been very active in solar deployment and such with the support of ADB and EBRD. ADB financed a solar roadmap that was implemented with the support of the World Bank Group Scaling Solar Program with 1 GW of solar PV targeted. Asian Development Bank (ADB) is further supporting the deployment of solar and EBRD is supporting wind deployment in the country. Strong cooperation is ongoing to ensure that the tenders have similar bankable risk allocation. The new World Bank-supported VRE program in Uzbekistan will enable the integration of the VRE being deployed under the three MDB supported programs.

Outside of the scope of the present Funding Proposal, to facilitate the development of the Initiative's core objectives, a collaborative multi-stakeholder group has been established. The multi-stakeholder group comprises of AfDB, KfW, and EIB in addition to AFD, ISA and IRENA. The multi-stakeholder group is an informal high-level platform where stakeholders are expected to discuss SRMI's core principles during workshops sharing each stakeholder's vision to support the development of renewables in developing countries. A high-level collaboration with the Initiative stakeholders is ongoing (outside of the present Funding Proposal), including with AfDB's team working on their initiative Desert to Power and their new SEFA initiative on developing RE aligned with SRMI's principles, and AFD's work program on RE

deployment. Discussions are also on-going with the Energy Transition Coalition and in particular the Climate Investment Platform (CIP) to enhance cooperation and maximize synergies.

B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

The SRMI Facility aims at tackling the lack of sustainable and bankable pipeline of RE projects in developing countries to achieve low-emission development pathways. The SRMI theory of change is for the entire Programme and draws on the lessons learned from the successes and failures of national electricity policies and IPP selection processes in developing countries. Under the current context, it also aims to support the development of renewable energy in parallel with green stimulus packages for these countries to provide core support in the economic recovery post Covid-19. The SRMI Facility aims at supporting multiple countries in their energy transition starting with seven pilot countries and quickly after adding other countries via a complementary funding proposal presenting information on the new countries, such as Indonesia and Vietnam, once NOLs are received, for projects following the Facility's eligibility criteria presented in the present Funding Proposal.

The main barriers identified that need to be addressed to leverage private investments can be grouped into two broad categories: (i) risks occurring during the development phase, that is, prior to construction and operation; and (ii) those that arise once the project begins to operate. Both types of risk are integrated into the IPPs and lenders' cost of capital. These risks have been exacerbated under the current crisis, and comprehensive risk support will need to be provided to leverage private investments.

a. **Development risks** encompass:

- grid risk, including connection risks;
- land risk, including availability, permitting, and environmental and social (E&S) aspects;
- legal risk, including the applicable regulatory, dispute resolution, and judicial frameworks;
- procurement risk; and
- integrity risk.

b. **Operational risks** encompass:

- off-taker credit risk (including the off-taker's record of performance and timely payment) and risk of contract termination;
- the country's power sector risk (including sector financial sustainability risk, reform risk, regulatory risk, and delay in the government's construction work;
- market risk (including currency risk and interest rate risk;
- country and macroeconomic risks; and
- political risk (including risks of breach of contract; expropriation; transfer restriction and currency inconvertibility; and war and civil disturbance).

To tackle those risks, the SRMI Facility developed activities around four components – that are adapted to the needs of each country as part of the Facility, namely:

- a. **Component 1: Technical Assistance** – concentrates on: (i) comprehensive and integrated planning support to develop least-cost generation plans combined with VRE integration analysis and dispatch and planning capacity building, (ii) legal, financial and procurement aspects capacity building and transaction advisory support to implement competitive and transparent tendering of privately-financed RE project, and (iii) technical and environmental and social (E&S) studies to prepare solar and wind parks for future tenders;

- b. **Component 2: Public Investments** – concentrates on public investments for: (i) solar and/or wind park common/shared infrastructure, (ii) VRE integration grid upgrades (including publicly-owned battery storage) and enhancing grid resilience to climate change, and (iii) electrification to increase population's resilience;
- c. **Component 3: Risk Mitigation Instruments** – concentrates on providing risk mitigation instruments to private investors for grid-connected and off-grid renewable energy/storage projects; and
- d. **Component 4: Project Management** – concentrates on the management of the projects and coordination and implementation of their activities, in compliance with the Executing Entities (EEs)' and Program Implementing Entities (PIEs)' contractual obligations included or referred to in the legal agreements entered into with the Accredited Entities (AE).

The technical assistance component focusses on the country/macro level risks looking at (i) developing sustainable RE targets to reduce planning risks and provide evidence-based targets for low-emission pathways, (iii) reducing procurement risks through a robust and transparent selection of IPPs; and (iii) increasing government's internal capacity, and reducing developing risks for IPPs by providing land and permits through solar/wind park schemes.

The public investment component focusses on the grid and infrastructure level risks looking at (i) ensuring minimal curtailment and risk to the grid stability due to high penetration of VRE, (ii) increasing grid resilience to climate change impacts, finance the public infrastructure needed for the IPPs and (iii) providing access to electricity to vulnerable populations.

The risk mitigation instrument component focusses on the residual risks for private investors looking at ensuring a sustainable leveraging of private investments in grid-connected and off-grid RE projects.

Table 1: Components Financing per Project (indicative financing in million USD)²

Components	Total Financing Instrument	Botswana	CAR	DRC	Kenya	Mali	Namibia	Uzbekistan
Component 1: Technical Assistance								
Sub-Component 1.1 Generation plans with VRE integration and Capacity Building (technical)	Grants	0.5		1	1	1	1	0.5
Sub-Component 1.2 Transaction Advisory and Capacity Building (procurement, legal)		1.5	1	1.5	1	1	1	1.5
Sub-Component 1.3 RE Park Infrastructure studies		1.5	0.5	1	1	2.5	1.5	1.5
Component 1 indicative budget per country (all)	23	3.5	1.5	3.5	3	4.5	3.5	3.5
Component 1 indicative budget (GCF)	21.5	3.5		3.5	3	4.5	3.5	3.5
Component 2: Public Investments								
Sub-Component 2.1 Public investments in solar and wind park infrastructure	Loans (with the exception of CAR which is grant)	30	20			25	21	50
Sub-Component 2.2 Public investments for VRE integration grid upgrades and grid resilience		68	10		75	46	40	250
Sub-Component 2.3 Public investments for electrification (PPP mini-grids, grid extension)		49	60	500	125	62	35	
Component 2 indicative budget per country (all)	1466	147	90	500	200	133	96	300
Component 2 indicative budget (GCF)	206	30	30		45	38	20	43
Component 3: Risk Mitigation Instruments								
Sub-Component 3.1 Risk mitigation instrument for grid-connected RE	Reimbursable Grants			30				
Sub-Component 3.2 Risk mitigation instrument for mini-grids	Guarantees	10					10	
Component 3 indicative budget per country (all)	50	10		30			10	
Component 3 indicative budget (GCF)	50	10		30			10	
Component 4: Project Management (Grants)								
Component 4 indicative budget per country (all)	24.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
Component 4 indicative budget (GCF)	2.5	0.5	0	0.5	0.5	0.5	0	0.5
Total indicative budget per country	1563.5	164	95	537	206.5	141	113	307

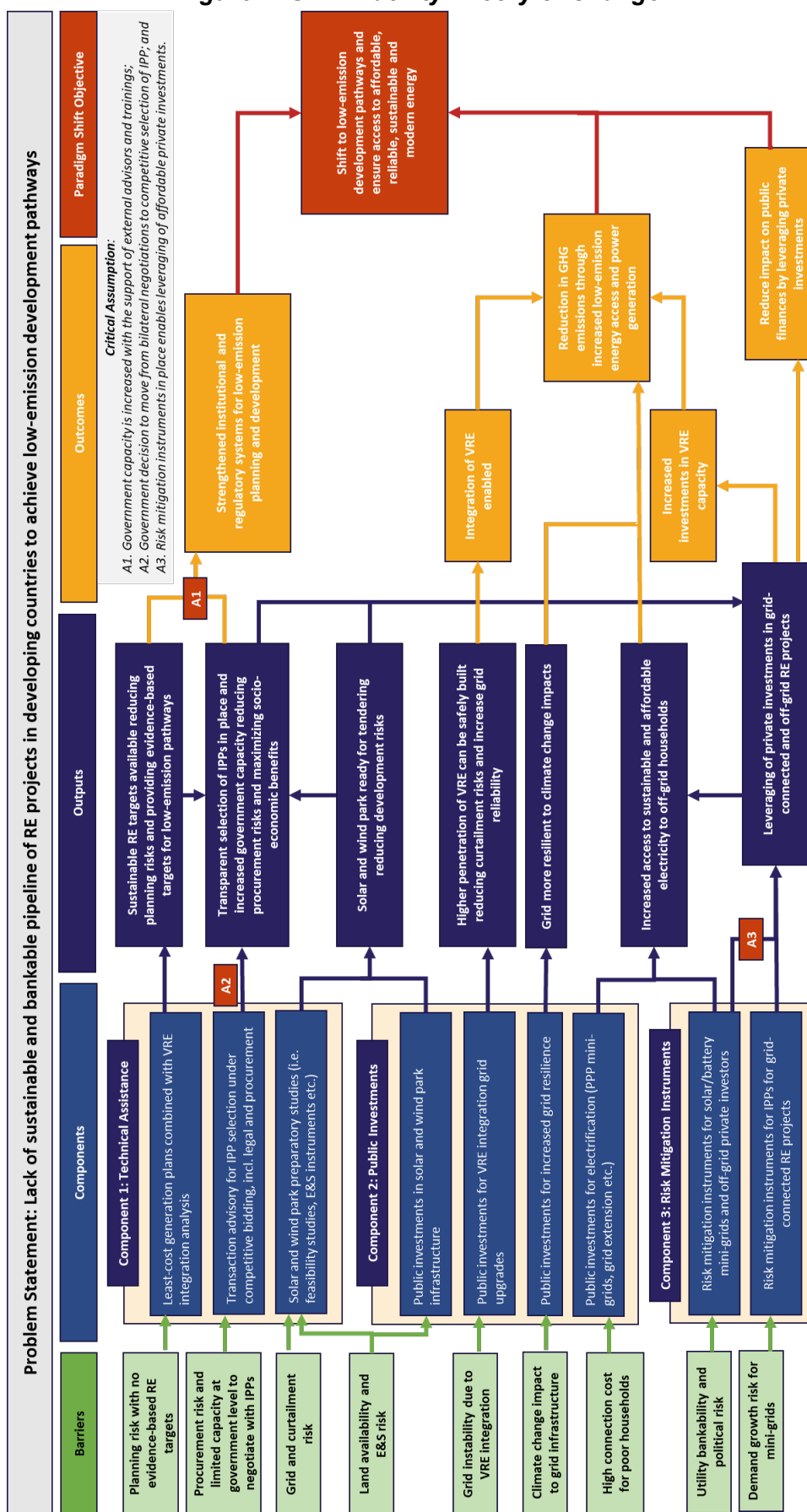
The main expected outcomes under the SRMI Facility are:

- a reduction in GHG emissions thanks to an increase in investments in VRE capacity through private investments in grid-connected and off-grid projects, and enabled VRE integration,
- the leveraging of private investments reducing the burden on public sector finances,
- improvement of grid resilience as well as resilience of local communities around the projects, and
- strengthening of institutional and regulatory capacity.

The SRMI Facility's objective is to support the eligible countries to shift to low-emission sustainable development pathways and increase access to affordable, reliable, sustainable and modern energy to its populations.

² All the financing presented in Table 1 are estimates based on current scope of the projects. These are indicative and subject to approval by the World Bank Board and agreements with the EE.

Figure 4: SRMI Facility Theory of Change



B.3. Programme description (max. 2000 words, approximately 4 pages)

■ Programme Overall Background

Solar and wind energy generation, as part of a green stimulus package post covid-19, could become game changers as they are already least-cost source of generation in most countries and are modular with shorter construction timeline than other power plants. Solar and wind energy generation can play a critical role from different perspectives: (i) climate change, (ii) energy security, (iii) energy access as it is a modular source of energy, (iv) responding to fast growing demand as they have short construction timeline, and (v) local development with industrial integration and community development (including from women empowerment's perspective). Moreover, solar and wind can help countries deal with their potential power shortages while ensuring they are not dependent on imported fuels.

However, large penetration of VRE is usually rapidly constrained due to grid technical limitations and lack of flexibility in the grids. The main challenges related to the integration of VRE into the grid are its variable nature, uncertain availability and that it can only produce power during the day for solar and when the wind is blowing. Three issues must be considered when deploying VRE into a grid: (i) the capacity for the generation mix to meet demand at any hour of the year, considering VRE variability; (ii) the economic optimum in terms of power system operating costs, after considering both cost reductions thanks to VRE and the investment costs required for deploying and integrating VRE; and (iii) the limitations of solar PV/wind capacity to maintain grid stability, due to its variability and its limited capacity to contribute to the balancing of demand and generation. The flexibility needed in the grid to enable a sustainable penetration of VRE can be constrained due to (i) technical limitations, i.e. can the grid have the dispatch and transmission capacity to integrate such variability, and (ii) commercial arrangements, i.e. even if there is in a given grid very flexible generators such as gas plants, they may not be able to provide flexibility to the grid as ancillary services may not be priced in their power purchase agreement (PPA) or they would have "take or pay" arrangements pushing the utility to dispatch them prior to considering new renewables.

Investment for grid modernization and upgrade will be required to ensure large penetration of VRE in the seven countries. Taking advantage of variable renewable generation requires significant expansion and modernization of electrical grids. Specific technologies and processes may be used to support the gradual transition of power systems into "VRE-friendly" grids that will significantly reduce integration costs in the long term. The penetration of VRE requires power system planning and grid management to adapt to the particular characteristics of VRE. It also requires better forecasting methods and stringent grid code requirements. In addition, this may provide the opportunity for the country to also look at its grid from a climate change resilience perspective and adapt it to future shocks.

If the technical aspects are critical, the deployment schemes selected to mobilize and select IPPs are also crucial. Since 2015, most countries that moved into competitive selection of IPPs directly benefited from the reduction in solar construction and equipment, and in cost of capital, through lower PPA prices compared to countries with FIT policies or bilaterally negotiated deals. As of the date of this Funding Proposal, Vietnam, Kenya and Mongolia have FIT prices of USD 9.35 cents, USD 12 cents and USD 15 cents per kWh, respectively. In comparison, in 2019, under competitive selection, Tunisia, Ethiopia, Zambia and Philippines announced PPA prices of USD 2.44, USD 2.56, USD 3.9 and USD 4.5 cents per kWh, respectively. The main difference between FIT and competitive bidding is that under FIT, Governments are fixing the price whereas under a competitive bidding process the private investors are.

Under competitive bidding, governments can access lower prices as IPPs will reflect in their proposed PPA price: (i) the capital expenditure (CAPEX) cost at the time of the bid – since 2010 CAPEX in wind and solar decreased rapidly, in particular solar module cost and it is difficult to assess its cost a few years ahead, (ii) high competition between solar IPPs leading to reduced equity returns expectations, (iii) bankable contractual arrangements with fair risk allocation between the private and public stakeholders that enable IPPs to attract long-term non-/limited recourse project finance that lower the cost of capital, and (iv) the scale of the market in the country and the region. Therefore, a bankable well-organized competitive selection can attract cost-competitive IPPs and ensure low cost of capital.

Based on the identified development challenges and in particular grid curtailment risks for VRE, two deployment schemes, namely substation-based competitive bidding and solar/wind park competitive bidding, are recommended. Both schemes aim to reduce the development risks perceived by IPPs, thus reducing the risk premium in the cost of capital and reduce the utility's cost of integrating solar generation into the grid. The main outcome expected is a reduced PPA tariff proposed by developers. More precisely:

- **Under the substation-based competitive bidding scheme**, the government sets a target of solar PV capacity to be procured at a certain time. It identifies substations/feeders with available MW capacity, and a certain MW capacity at each substation/feeder is opened for bidding to achieve the target. IPPs compete through a competitive bidding process to obtain the right to connect their projects to the identified substations/feeders based on a set of pre-determined selection criteria comprising technical, financial and safeguard aspects. IPPs are responsible for selecting the land, obtaining the investment license and have to follow the E&S requirements as required in the bidding documents. This scheme support countries better manage the volume of new solar PV/wind installation connected to the grid and develop RE in selected areas. It will enable the utility to optimize the use of existing transmission capacity, reducing the potential curtailment and/or additional costs to integrate the solar PV and wind projects. It can also proactively drive grid investments needed for new VRE generation.
- **The solar/wind park concept**, with pre-arranged comprehensive de-risking measures (including land, power evacuation and permits secured), can contribute to reach fully cost-effective electricity tariff. The government identifies project site(s), conducts land clearance, and constructs shared infrastructure for the solar/wind park that can range from the evacuation line to basic elements, such as the fence, roads, street lighting. The government conducts the E&S assessments as well as the geotechnical studies. Once the project is ready for competitive bidding, the bidding procedure begins and the winning IPP is responsible for the financing, construction, and operation of the solar/wind energy project. This scheme is critical when IPPs are identifying risk of curtailment and land ownership issues as paramount to the deployment of solar and wind in the country.

RE deployment can also be an opportunity for countries to maximize socio-economic benefits which is particularly important in the Covid-19 context. The lack of visibility on the local opportunities, the tight timeframe of the tenders, the existence of suppliers already tested, and the costs to assess the potential of the local industries do not encourage the bidders to explore what could be done competitively in a given country. Assessing the positioning of the local players on the solar/wind value chain and identifying the measures needed to improve their involvement in a competitive manner on such a value chain will increase their chances to benefit from the opportunities offered by the new RE projects. Factoring these measures in an RE roadmap and in the bidding processes launched can increase the participation of local players in a competitive manner, fostering the transfer of knowledge and the skills

development and allowing them to build a track record to the extent possible. In addition, expanding the RE roadmap to assess the needs of the local communities around the RE projects implemented and to design adequate programs to increase their resilience and enhance their livelihoods will ensure that they effectively benefit from the implementation of the projects. Sites with good solar/wind resources and large plots of lands are usually located in areas where the level of living is below the national average. The local communities there face numerous challenges hindering their development. Including tailor-made development programs in an integrated manner in the RE roadmap of a given country and operationalizing them through the RE projects deployed will ensure local communities benefit positively from them.

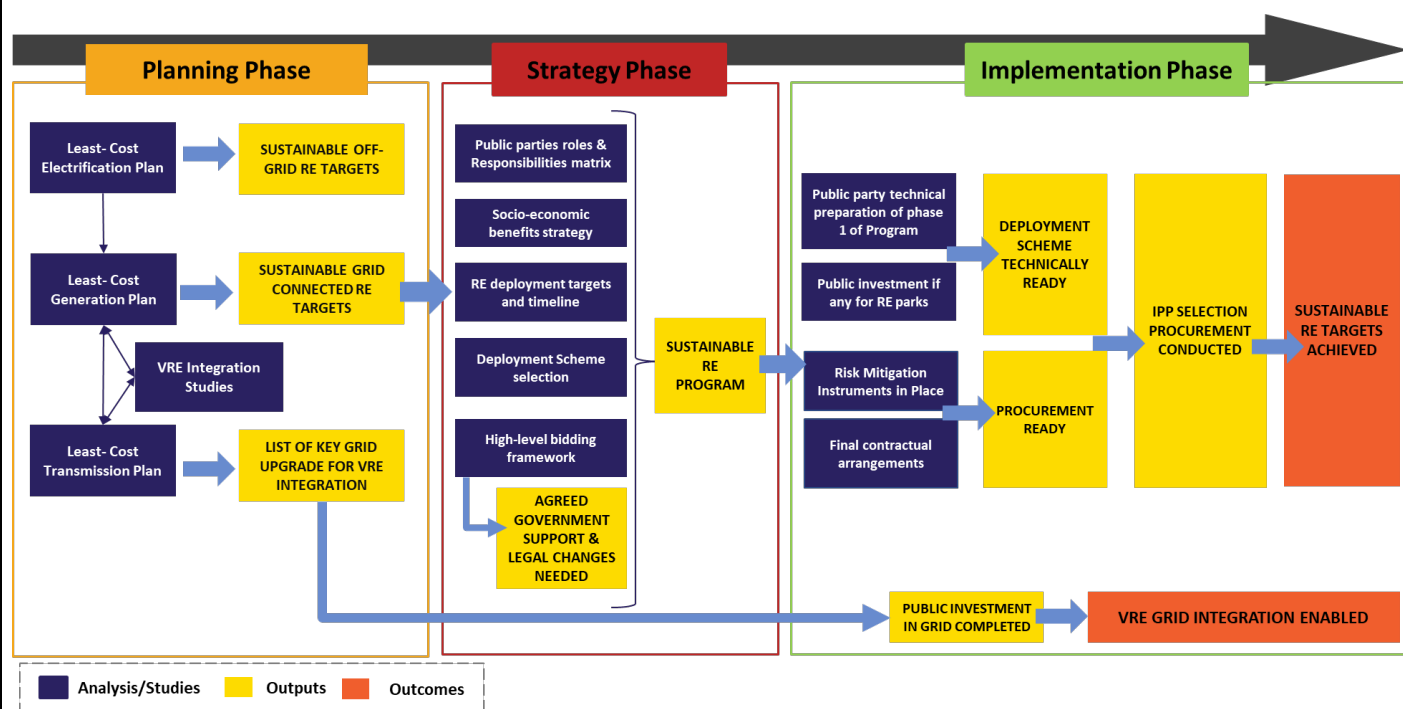
These programs aim to maximize the socio-economic benefits of the RE projects from the local industry as well as from the local community perspectives will pay a specific attention to women empowerment. Employment opportunities created by large-scale solar and wind infrastructure projects often benefit men more than women, resulting in a significant employment gap in the RE industry. Also, in the remote areas where large projects are developed, women, if employed, are more exposed to lower-paid, lower-quality employment in the informal sector. As a result, without a proactive policy to promote female employment throughout the RE project-cycle, men may benefit disproportionately from new employment opportunities across the RE value chain.

The World Bank SRMI aims at addressing in a comprehensive manner the different challenges that are limiting the uptake of sustainable privately financed RE and the associated socio-economic benefits. Launched in 2018 for the COP 24 under the leadership of the World Bank in partnership with the AFD, IRENA and ISA, SRMI aims to support countries in developing sustainable solar and wind programs designed to attract private investments and so reduce reliance on public sector financing, while maximizing the socio-economic benefits triggered by the projects deployed. Its unique and integrated approach offers development and climate financing for (i) technical assistance to help countries develop evidence-based RE targets, implement a sustainable renewable energy program, and set up and maintain transparent and competitive procurement processes with transaction advisors; (ii) critical public investments to enable integration of VRE, finance solar/wind park infrastructures, and increase access to electricity; and (iii) risk mitigation instruments to cover residual risks perceived by private investors. Its framework is supporting the creation of jobs, the development of skills as well as increasing the resilience and enhancing the livelihoods of the local communities around the projects.

SRMI aims to support countries to:

- develop sustainable RE programs to leverage private investment at scale for grid-connected and off-grid RE projects,
- implement the country's RE programs through transparent and robust competitive tenders,
- improve their grid capacity to integrate VRE and resilience to climate change,
- leverage private investments for RE grid-connected and off-grid projects,
- build RE capacity,
- avoid GHG emissions, and
- maximize the socio-economic benefits triggered by the RE projects developed.

Figure 5: SRMI Methodology



Country Context

The SRMI approach aims to support countries in a comprehensive multi-dimensional risk mitigation approach, yet in a flexible manner adapting to their current situation and constraints. Therefore, it is critical to understand the different constraints and starting points of each of the countries currently identified to benefit from the Facility:

- **Botswana's** power sector currently relies on high-carbon electricity, largely produced from coal at the Morupule power station using the country's large coal reserves (around 200 billion tons). To meet domestic demand, 30 percent of electricity is imported from South Africa (92 percent coal) and the SAPP market (74.5 percent coal) contributing to coal-fired emissions and unnecessary fiscal outflow as tariffs do not cover peak power prices. Only sixty percent of national households in Botswana have access to electricity, and the government wants to reach universal access by 2030. Under a BAU scenario, Botswana could see increased output of coal-fired power generation, and annual emissions of greenhouse gases will be 2.5 times their 2012 levels by 2030. Yet, Botswana enjoys one of the highest irradiation densities in the world, with over 3,200 hours of sunshine per year and an average global irradiation throughout the country. This resource, while largely underexploited today could place Botswana as a leader for developing solar. Botswana also benefits from a good wind resource, complementary with the solar resource, with a potential currently estimated between 300 MW and 500 MW. No solar/wind projects are built to date in the country.
- **CAR's** electricity sector is characterized by inadequate infrastructure, a weak policy and regulatory framework and a utility that is struggling to recover costs and thus maintain and expand services. The overall rate of access is 8 percent in the country. Total installed capacity is 28 MW, of which 23 MW is available, whereas the current load is 45 MW, and the projected unmet demand is estimated at around 60 MW. The majority of the generation is hydropower and diesel plants. Therefore, to improve reliability in the power system, support stability, and promote

peacebuilding in a context of fragility, the country is in dire need of sizeable and urgent increases in its electricity generation capacity. Solar with battery storage is being sought as a solution with a World Bank financed utility-owned solar plant under construction and a deployment of solar/battery mini-grid, and off-grid solutions. The country does not have yet any privately-owned generation but has a provision for IPPs in its new Electricity Law and started working on the decrees for the implementation of the law.

- **DRC** has an access rate of around 9 percent. The well-distributed potential of both hydro and solar resources remains mainly untapped, as power sector development has focused on a few large hydropower plants over the last decades, leaving most of the country's population unelectrified. SNEL, the national state-owned utility, has a critical role to play in access, in particular in Kinshasa, but is caught in a vicious cycle of mounting commercial losses, deteriorating assets and mounting debt that hinder its ability to drive electrification outside of Kinshasa. Therefore, it will be critical for the future of the electrification in DRC to develop large solar/battery mini-grids in off-grid zones by leveraging private investments. To enable private investments, the government must improve its institutional and regulatory framework.
- After decades of shortages, **Kenya** now has enough generation capacity to meet its demand, and supply is predominantly from renewable energy sources. The current installed generation capacity is 2,819 MW with a peak demand of 1,912 MW, resulting in a peak reserve margin of over 30 percent. Renewable energy (geothermal and hydro) account for 65 percent of total installed capacity and almost 90 percent of total electricity generation. Geothermal energy contributes more than 45 percent of total energy generated, making Kenya one of the global leaders in use of this firm, renewable resource. Kenya has a FIT since 2008, and revised in 2012, for wind, biomass, small-hydro, geothermal, biogas and solar with a standardized PPA. However, little solar and wind generation were built under the FIT, with only 65 MW and 335 MW, respectively.
- Despite significant progress over the last decade, **Mali's** access to modern energy services remains low and overall unreliable. Mali relies heavily on expensive thermal power generation (heavy fuel oil (HFO) and diesel) and have an access rate of around 39 percent. The development of interconnections with its neighboring countries is enabling the country to reduce its cost of electricity, while improving its reliability. However, to ensure energy security, the country have been bilaterally negotiating with IPPs for new solar projects. No solar/wind projects are built to date as prices negotiated are well above the regional standards (with prices well above USD 0.10 per kWh). In addition, Mali's grid is under development with key backbones under construction and therefore, will face issues with integrating large amounts of VRE without any critical investments put in place first.
- **Namibia** imports 73 percent of its electricity needs and is particularly reliant on South Africa's imports. High imports also lead to high foreign exchange, high electricity costs which place a burden on the balance of payments and budget, and vulnerable electricity supply. The government has taken an important policy decision to increase domestic supply and reduce electricity import-dependency through the procurement of public and private investments in electricity generation and transmission. Namibia aims to achieve 80 percent self-sufficiency through domestic electricity production by 2022. Namibia is one of the best countries in the world for CSP, PV and wind production. Namibia has an experience in FIT for very small VRE projects

and a limited experience in competitive bidding. Namibia has an installed 120 MW of PV capacity and 5 MW of wind.

- **Uzbekistan's** energy system is characterized by high losses and low reliability of supply due in part to the rapidly aging supply infrastructure that was mostly constructed during the Soviet era. The energy sector is highly dependent on natural gas, which is a major source of commodity exports but getting depleted and making the system vulnerable, thereby prompting the country to seek sustainable energy transition pathways. Natural gas accounts for 86 percent of the total primary energy supply. Diversification of Uzbekistan's electricity generation is also limited, as more than 80 percent of the total installed capacity of 12.5 GW in Uzbekistan is natural gas based, followed by hydropower (16 percent) and coal (11 percent). The least-cost power generation expansion plan for Uzbekistan provides a path to solar and wind transition with solar and wind accounting for around 30 percent of the total capacity mix (7.7 GW). Uzbekistan has already a very successful tendering system in place with the support of the World Bank Group Scaling Solar that was implemented and reached prices of US\$ 3.9 cents per kWh. However, to be able to reach their ambitious VRE targets, Uzbekistan needs to tackle the limitation on its grid availability/flexibility which is expected once a higher penetration of VRE will be reached.

■ Programme's Objective & Components

The SRMI Facility's objective is to support the eligible countries to shift to low-emission sustainable development pathways and increase access to affordable, reliable, sustainable and modern energy to its populations. The SRMI Facility is developed around four components that are complementary. Depending on the country specificities such as access rate, available plans and grid flexibility, sub-components may not all be required for each project. These components are based on the identified risk mitigant that will enable to reduce the cost of RE deployment, contributing to low-emission development pathways. At individual project level, the Facility development objectives will be adjusted to reflect what is specifically done under each project and which objective and impacts the activities in each project are particularly designed to focus on.

The Facility and its Components have been developed with enough flexibility to be able to add countries to the Facility once (and if) approved by GCF Board under subsequent phases. The design of the Facility, its components as well as its instruments are applicable to all countries that aim to apply the SRMI approach to sustainably move forward with their energy transition.

Component 1: Technical Assistance

Grant amounts will be allocated by the EE (Host Country) to the financing of the following types of activities to be carried out by their Ministry or state utility responsible for energy, on an as country-needed basis:

Sub-Component 1.1: Generation plans with VRE integration and Capacity Building (technical)

- Preparation of a new national (or adaptation of an existing) least-cost power generation plan, which includes reiterative VRE integration analyses based on technical and commercial constraints, an evaluation of the dispatch capabilities and the limits of grid integration and determines a cost-optimized electricity mix that can meet demand at any time;
- Preparation of a new national (or adaptation of an existing) least-cost transmission/distribution plan, which includes capacity expansion models simulating generation and transmission capacity

investment, given assumptions about future electricity demand, fuel prices, technology cost and performance, policy and regulations, and providing informed solar targets and indication of where best to locate future projects as well as a list of key investments needed to improve grid VRE integration capacity; and

- Provision of institutional capacity building support to government institutions and/or utilities/dispatching entities (as identified and designated by the EE) for energy planning and dispatch (including using grid-connected battery storage).

Sub-Component 1.2: Transaction Advisory and Capacity Building (procurement, legal)

- Provision of technical assistance (including legal, procurement and transaction advisory services) and implementation of capacity building activities to support: (i) the assessment of the existing legal framework and the development of a legal framework for IPPs and a competitive bidding scheme that is conducive to private sector investment; (ii) the design of energy projects (including key terms of the PPA and the government support agreements) with fair risk allocation among stakeholders and contractual arrangements acceptable to potential investors; (iii) the competitive selection of IPPs and mini-grid developers; and (iv) the negotiations of the final contractual arrangement with the selected investors and closing.. If the relevant Project includes Component 3, the development, structuring and preparatory work for Sub-component 3.1 and Sub-component 3.2 will be expected to be financed and carried out under this Sub-component 1.2. Under such technical assistance, a specific attention will be brought to inform clients on how to leverage an effective and efficient competitive bidding process (including in terms of evaluation formula to select the IPPs) to maximize the socio-economic benefits, building on the results of the associated studies, lessons learnt and best practices.

Sub-Component 1.3: Solar and wind park preparatory studies

- Preparation of preparatory and feasibility studies and environmental and social risk mitigation instruments, for the development of solar PV, solar PV with battery storage, CSP, onshore wind and offshore wind parks in support of investment de-risking and investor (IPPs and mini-grids developers) selection process. Preparatory and feasibility studies may include, as needed, geospatial analyses of the land around the substation, topography and geotechnical analyses, environmental and social analyses, site-specific grid interconnection studies, solar irradiation/wind measurement analyses and other studies as needed.

Component 2: Public Investment

Loan amounts (or grant for grant only countries) will be allocated by the EE (Host Country) to the financing of the following types of activities to be carried out by their Ministry or state utility responsible for energy, on an as country-needed basis:

Sub-Component 2.1: Public investments in solar and wind park infrastructure.

- Carrying out infrastructure investments in support of the development of solar and wind parks, such as acquisition of land, construction of transmission lines and fencing of such infrastructure for solar and wind parks.

Sub-Component 2.2: Public investments for VRE integration grid upgrades and grid resilience

- Investment for grid modernization and upgrade to improve grid's flexibility to integrate larger amounts of VRE, limit curtailment or grid instability and adapt to climate change related shocks,

including power system planning, grid management, strengthening of forecasting methods and grid code requirements. Grid reinforcements that support VRE integration may include (i) addition or replacement of lines and transformers for grid extension and capacity enhancements (both for answering growing demand and for integrating VRE power), (ii) equipment for smoothing the voltage and frequency issues, such as capacitor banks, battery storage and other reactive power compensators, together with flexible alternating current transmission systems, (iii) equipment for faster and more efficient grid operation, such as monitoring systems, demand and production forecasting systems, and automats for controlling generation units and grid operations through automatic generation control; and (iv) other innovative technologies whose commercial viability may not have been established yet such as load shifting battery storage.

Sub-Component 2.3: Public investments for electrification

- Investments as needed to support increased electrification, in partnership with the private sector when feasible, including through solar home systems, mini-grids and grid connection, as best appropriate to the country context, with a particular focus on women.

Component 3: Risk Mitigation Instruments

Provision of financial instruments (in the form of guarantees or reimbursable grants whose reimbursement is conditional and modular based on verified performance, or other instruments derived from their financial engineering) to reduce the following risks of investors or their commercial financiers and enhance financial sustainability, in the following areas:

Sub-Component 3.1: Risk mitigation instruments for solar/battery mini-grids and/or off-grid private investors

- Provision of reimbursable grant to mitigate the critical risks related to private capital investment in the off-grid space (mini grids and/or SHS – for which Multi-Tier Framework (MTF) data will be used when available), such as for instance demand off-take risk affecting the investors for the first few years of operation and consequently financial risks affecting pricing and tenor of financing available to the investors.

Sub-Component 3.2: Risk mitigation instruments for grid-connected RE projects

- Provision of guarantees to mitigate the critical risks related to private capital investment and provision of credit enhancement to private sector investors (and financiers) and electricity off-takers as needed to leverage private sector investments in grid-connected RE projects. If GCF so approves, these risk mitigation instruments may be provided without member country counter-guarantee, meaning that any amounts paid under the risk mitigation instrument following the occurrence of a covered risk would not be reimbursed to the AE or to GCF.

Component 4: Project Management

Management of the projects and coordination and implementation of their activities, in compliance with the EEs' and PIEs' contractual obligations included or referred to in their subsidiary agreements entered into with the AE.

▪ **Programme's Project Eligibility Criteria**

Each of the Projects which will be included in the Program – under the present Funding Proposal and any subsequent phases - will have to meet the following eligibility criteria that are aligned with and organized along the GCF Investment Criteria, prior to being approved by the Accredited Entity's approval body and being able to access the Facility financing:

- **Impact potential:**
- The project will directly support: (i) the financing and the integration of solar and wind energy representing a minimum of 15 percent of future total installed capacity by 2025 in the given country (or grid), or (ii) the financing and the integration of a minimum of 60 percent of future solar and wind installed capacity by 2025 in the given country (or grid), or (iii) the financing of a minimum of 1 GW of solar and wind in a given Host Country (*excluding* projects that are not requesting financing for Sub-component 2.1 and 2.2).
- The project will support the country in its effort to leverage private investments in a competitive and efficient manner. This criterion's applicability is dependent on the current security and macro-fiscal constraints that the country may have. However, even in Fragile, Conflict-affected Situations (FCS) countries such as CAR, DRC and Mali, it is critical to develop a path to private sector leveraging to reduce the impact of large-scale infrastructure financing onto the government's budget.
- The project will not directly finance nor support non-RE sources and will instead support VRE deployment by financing (i) investment for grid modernization and upgrade to improve grid's flexibility to integrate larger amounts of VRE (using technology such as new SCADA systems, STATCOM, capacitor banks, substation upgrades with transformer, battery storage, transmission line upgrades/increased voltage, and new transmission lines), limit curtailment or grid instability (using technology such as battery storage and other climate friendly technologies - and other technical solutions which may be developed after the date of this FP, whose reliability and efficiency, based on the technical data available at the time, have been considered sufficient by the AE at the time of its internal approval of the project) and adapt to climate change related shocks (technology such as lines and substation following climate change resilience guidelines) (ii) and/or solar/wind park infrastructure (such as land, substations, new transmission line connecting the park to the grid, fencing, security system and all infrastructure needed to complete the park as the case may be), and/or (iii) renewable energy-based electrification (including solar home systems and mini-grids or any climate friendly technologies).
- If the project includes activities under Sub-component 3.1 or Sub-component 3.2, it will cover risks related to long-term mobilization of private capital selected in a competitive manner, and in RE investments in utility-scale or mini-grids.
- The estimated installed capacity (MW) as a result of infrastructure financed under the Facility shall not deviate more than 15 percent less than the initially planned installed capacity for the whole Facility as indicated in Table 2 below.
- The estimated mitigation impact (tCO₂eq) as a result of infrastructure financed under the Facility shall not deviate more than 15 percent less than the initially planned mitigation impact for the whole Facility as indicated in Annex 3 of the Funding Proposal.
- The project will support solar and/or wind (with or without storage) technologies.
- **Paradigm shift potential:**

- The project will focus on increasing RE deployment in the country (grid-connecting and/or off-grid) in a transformative manner – from different angles such as legal, procurement, technical, economic and social development etc., depending on the starting point of the country.
- The project will include a capacity building plan to be financed under the project to support the long-term impact of the change promoted under the project.
- **Sustainable development potential:**
 - The project will, to the extent possible, support the development of an industrial integration plan to support local industries, skills development and long-term jobs' creation.
 - The project will, to the extent possible, support the development of a local development action plan to increase the resilience and enhance the livelihoods of the local communities.
- **Country ownership:**
 - The relevant Host Country will demonstrate a strong willingness to develop a comprehensive and streamlined strategy to deploy and integrate VRE and leverage in a sustainable manner private investment.
 - The VRE targets and revised policy supported by the project will be aligned with the NDC/low-emission development plans as reflected in the no-objection letter issued by the Host Country.
- **Recipients' needs:**
 - The Host Country will be vulnerable to climate change and in need to fundamentally adapt its energy systems by harnessing its domestic renewable resources to mitigate climate change.
- **Efficiency and effectiveness:**
 - The project will aim at substantially reducing GHG emissions in the given country compared to BAU in the energy sector of the Host Country(ies) and support the achievement of the Host Country(ies)' NDC.
 - The project's expected ratio of total USD per carbon dioxide invested³ is maximum 45, and per GCF USD invested – the GCF Proceeds – is maximum 10 for each project *excluding projects that are not requesting financing for Sub-component 2.1 and 2.2, and full grant projects*
 - The SRMI Facility budget (GCF Proceeds) allocated per Host Country shall not deviate more than 15% percent from the initial budget allocated per Host Country as indicated in Table 2 below.
 - The project's financing will be structured with a minimum co-financing ratio GCF to co-financing of 3 (one to two co-financing) in each country.
 - The project will have been prepared in compliance with the Accredited Entity's applicable policies and procedures and the relevant provisions of the AMA and the FAA.
 - The project will be approved by the World Bank within 3 years after the Facility's GCF Board approval.
- **Gender:**

³ Defined as cost per ton CO₂ equivalent, calculated as total investment cost divided by expected lifetime emissions reductions.

- The project will contribute to closing gender gaps in the energy subsector by (i) identifying gaps relevant to the four pillars of the World Bank Group (WBG) Gender Strategy⁴, and (ii) aiming to address these gaps through specific actions supported by the project.

The projects presented in this Facility are aligned with the eligibility criteria proposed, as illustrated in the feasibility study annexed to this Funding Proposal. The World Bank will work with each Executing Entity covered under this Facility to finalize the scope and co-financing of these projects, taking into account (i) the needs of the clients as they may evolve, (ii) the results of the studies at project's level and (iii) the national dialog involving all relevant stakeholders. A multi-disciplinary World Bank team will closely coordinate with the operational teams to ensure that the preparatory work carried-out at the project's level is adequate for the projects to meet all the eligibility criteria as agreed under the Facility with the GCF.

▪ Identified Project's Scope and Financing

Subject to IBRD/IDA/ESMAP corporate approval as appropriate in due course, the projects currently considered for inclusion in the Programme are expected to be financed through GCF financing and a co-financing from IBRD/IDA in the form of loans and IDA and ESMAP in the form of non-reimbursable grants. Co-financing does not include any other sources of financing for separate (even if coordinated/complementary) projects/activities that fall outside the scope of this Funding Proposal nor private investments that are only leveraged. CTF financing may be sought for Mali – however, this financing is not yet approved. Given the limited CTF resources expected to be available, the contemplated CTF financing is expected to substitute a very limited part of the World Bank resources if the CTF resources are eventually approved. The CTF portion is currently included in the WB financing described in the present Funding Proposal and therefore there is no financing gap in what is presented in the Funding Proposal. For the avoidance of doubt, in the event that the CTF resources are approved, CTF resources will replace IBRD/IDA/ESMAP resources. In the event that the CTF resources are not approved, IBRD/IDA/ESMAP resources will continue to be used so there will not be any financing gap. The World Bank targets to submit a request for CTF financing for the Mali project upon opening of the new CTF window supporting VRE integration. However, as this window is expected to be operational by late Q3 2021 (as the operationalization of the associated modalities is still on-going), it is likely that the Mali project will be submitted to the CTF Trust Fund Committee for approval post the GCF Board targeted for the SRMI Facility (B28). Meanwhile, these projects are included in the CTF pipeline and the World Bank will be monitoring every opportunity to advance CTF financing to these projects.

Table 2: Indicative Projects Scope and Financing Needs (in million USD)⁵

Country	Scope of project	Comp. 1	Comp. 2	Comp. 3	Comp. 4	Total cost
Botswana	a. park shared infrastructure for 200 MW CSP and 100 MW wind,	3.5 (GCF: 3.5)	147 (GCF:30)	10 (GCF: 10)	3.5 (GCF: 0.5)	164 (GCF 44)

⁴ The four pillars: (i) improving human endowments, (ii) removing constraints for more and better jobs, (iii) finding solutions that help increase women's ownership of and control over assets, and (iv) enhancing women's voice and agency and engaging men and boys.

⁵ All the financing presented in Table 2 are estimates based on current scope of the projects. These are indicative and subject to approval by the World Bank appropriate bodies and agreements with the EEs. Component 1 financing instrument is grants, Component 2 financing instrument is loans except for CAR which will be full grant, and Component 3 is a mix between reimbursable grants for DRC mini-grids and the others will be GCF guarantees. For further information on the split between sub-components please see Table 1.

	b. VRE integration and grid upgrades including 50 MWh of storage, and c. electrification for estimated 600,000 people; and d. risk mitigation instrument covering liquidity/termination for private investors competitively selected for CSP and/or wind projects developed under SRMI.					
CAR	a. solar PV (15 MWp) with battery storage (15 MWh) and VRE integration upgrades, b. mini-grid (20 MW PV with 20 MWh storage) and off-grid RE solutions for provincial cities and rural areas to electrify estimated 250,000 people, and c. support energy sector and utility's financial performance to attract private sector participation in renewable energy generation.	1.5	90 (IDA grant) (GCF: 30)	-	3.5	95 (GCF: 30)
DRC	a. Public-private-partnership (PPP) 100 MW PV and 140 MWh storage mini-grids to electrify estimated 1.5m people, b. grid infrastructure and resilience, c. risk mitigation instrument to cover demand risks for private investors for mini-grids.	3.5 (GCF: 3.5)	500	30 (GCF: 30)	3.5 (GCF: 0.5)	536.5 (GCF: 34)
Kenya	a. park shared infrastructure for 300 MW grid-connected solar, b. VRE integration including 150 MWh of storage, enabling integration of 250 MW solar and wind, and c. electrification for estimated 420,000 people.	3 (GCF: 3)	200 (GCF: 45)	-	3.5 (GCF: 0.5)	206.5 (GCF: 48.5)
Mali	a. park shared infrastructure for 300 MW grid-connected solar with 300 MWh storage, b. VRE integration incl. 50 MWh storage enabling integration of 150 MW PV, and c. electrification for estimated 400,000 people.	4.5 (GCF: 4.5)	133 (GCF: 38)	-	3.5 (GCF: 0.5)	141 (GCF: 43)
Namibia	a. park shared infrastructure for 200 MW CSP and 200 MW wind, b. VRE integration incl. 200 MWh storage, c. electrification for estimated 1m people, and	3.5 (GCF: 3.5)	96 (GCF: 20)	10 (GCF: 10)	3.5 (GCF: 0)	113 (GCF: 33.5)

	d. risk mitigation instrument covering liquidity/termination for private investors competitively selected for CSP and/or wind projects developed under SRMI.					
Uzbekistan	a. park shared infrastructure for 1 GW solar PV, b. VRE integration including 50 MWh enabling integration of 1.5 GW VRE, c. grid upgrade and deployment.	3.5 (GCF: 3.5)	300 (GCF: 43)	-	3.5 (GCF: 0.5)	307 (GCF: 47)
		23 (GCF:21.5)	1,466 (GCF: 206)	50 (GCF: 50)	24.5 (GCF: 2.5)	1,563.5 (GCF:280)

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

The following section describes the overall Facility management, the management process to ensure the most transformational projects are selected, and to deploy the legal arrangements between GCF, World Bank and the countries.

1. SRMI Facility as a GCF Program

The SRMI Facility is a program to support a series of projects, that currently proposes to support seven projects subject to the results of their appraisal but possibly more. All projects which will ultimately be supported by the SRMI Facility will have the common purpose to scale-up RE in a systematic and integrated manner. This Programme aims to enhance coordination and strategic alignment of the projects in a manner that can increase the prospects of transformational change, provide operational efficiencies and accelerate implementation of the SRMI methodology across countries and regions. The Facility is being developed as a Programme to ensure enough flexibility to include countries under subsequent phases, such as Indonesia and Vietnam when the NOLs are received and upon confirmation of them meeting the eligibility criteria.

This Funding Proposal includes indicative information for each of the currently proposed seven projects. Under the programmatic approach, the GCF would rely on the AE's due diligence on the proposed projects included in the Program. Subject to the outcome of such due diligence, the World Bank will approve funding for such projects that meet the eligibility criteria as presented in Section B3.

The EE are responsible for ensuring the implementation of the projects and the fulfilment of their development objective. The AE will provide implementation support to the EE to the extent permitted under its policies.

2. Implementation Arrangements

As part of its role as GCF AE and co-financier of the projects to be supported under the Facility, the World Bank will assess and confirm the eligibility of each such Project to be included in the Facility and benefit from the Bank's and GCF Financing. In order to do so, for each project proposed to be included in the Facility, the World Bank will comprise energy and climate finance experts to manage and be responsible for the Facility's overall objectives and ensure that the projects selected meet all the eligibility criteria (and their limitations) as presented in Section B3.

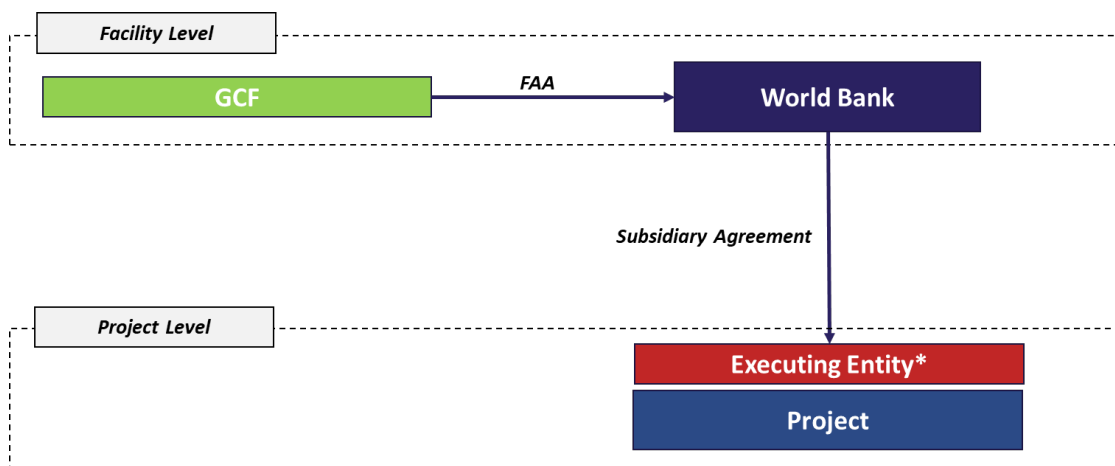
Following the GCF Board approval, the World Bank and the GCF will, based on the Accreditation Master Agreement (AMA), enter into one Funded Activity Agreement – FAA for the provision of the funds under all GCF instruments supporting the Program. The World Bank will be solely responsible for the management and administration of GCF resources and will carry out such management and administration in accordance with its policies, procedures and practices, and the relevant provisions of the FAA and AMA. The World Bank will apply its own fiduciary principles and standards relating to any integrity checks, anti-corruption, countering of financing of terrorism, fraud, financial sanctions, embargoes and anti-money laundering.

The World Bank will inform the GCF Secretariat about projects selected in accordance with the Eligibility Criteria as part of the periodic reporting outlined in Section E7. During the implementation of the Facility, the World Bank will be responsible for providing the necessary governance, oversight and quality assurance in accordance with its policies, procedures and any specific requirements in the AMA and FAA.

The World Bank will use its operation systems and established processes to review individual project proposals that would meet the eligibility criteria as presented in the present Funding Proposal and approve allocations. It would also facilitate the reporting in an integrated way to the GCF on the SRMI Facility's projects, as well as the learning across projects. Following approval of Projects, the World Bank will sign Subsidiary Agreements with the EEs and other agreements as applicable with any loan/grant PIEs and guarantee obligors and beneficiaries. These transactions will make available World Bank finance as well as GCF finance for investments in line with the provisions of the FAA.

There is no capacity assessment of the countries presently in the Facility, as those will be conducted during the appraisal stage of each project. When the projects go through the appraisal stage, some of the capacity assessment information will be summarized in the projects' appraisal documents and will be publicly disclosed. As part of the World Bank's due diligence in accordance with its own policies and procedures, WB ensures that adequate fiduciary (procurement and financial management [FM]), safeguards, legal and M&E frameworks are in place: (i) the different experts within the World Bank are assessing the client's capacity, (ii) they provide recommendations in the event where the capacity is not sufficient and (iii) an effective project management unit (PMU) with the core experts is usually part of the legal covenants for EEs to meet.

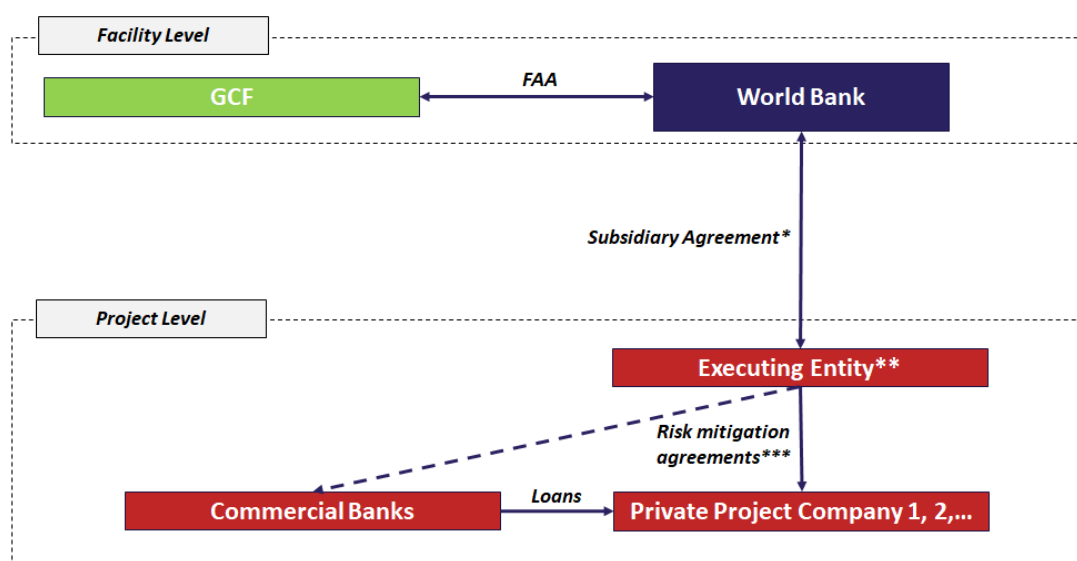
Figure 7: Indicative Implementation Arrangements for Component 1 and 2



*If a project has a Project Implementing Entity (PIE), there may be additional arrangements including a project agreement between the World Bank and such PIE to provide on-lending/on-granting.

With regards to risk mitigation instruments, depending on the final arrangements that will be developed as part of the preparation and appraisal of each project, figure 8 presents the links between the country, commercial banks and private developers for Component 3.1 and figure 9 for Component 3.2.

Figure 8: Indicative Implementation Arrangements for Component 3.1



* Including commitment to reimburse the grant remaining at the end of the implementation period.

**If a project has a Project Implementing Entity (PIE), there may be additional arrangements including a project agreement between the World Bank and such PIE to provide on-granting.

*** Payments structured to mitigate key risks, such as early demand risk.

Overview of the indicative mechanism contemplated for the reimbursable grants, subject to future due diligence, market sounding, and World Bank management approval:

- a. **Eligible sub-projects:** RE private sector projects selected by the Executing Entity, with the technical assistance of the World Bank to ensure that they meet both the eligibility criteria as well as the Bank's mandatory requirements (including safeguards and fiduciary) for sub-projects, as stated in the GCF financing agreement between the Accredited Entity and the Executing Entity
- b. **Executing Entity(ies):** a Host Country, expected to be the Democratic Republic of Congo, or a state-owned company mandated by the Host Country
- c. **Beneficiaries:** private sector developers preparing a public private partnership for the development of mini-grid sub-projects with an agreed business plan
- d. **Covered Risk:** mitigation of the critical risks related to private capital investment in the off-grid space, mainly demand off-take risk affecting the investors for the first few years of operation and consequently financial risks affecting pricing and tenor of financing available to the investors (to support viability during the initial years of the sub-projects)
- e. **Financial terms:**
 - Disbursement to the project company established by the private developers for the implementation of its sub-project, during a pre agreed number of years at the beginning of the operation period during which demand/revenues are ramping up and may not be sufficient to reach viability of the mini grid projected in the pre-agreed business plan
 - Reimbursement during a pre agreed number of years starting not later than the date when the demand/revenues in the pre-agreed business plan are sufficient to achieve viability as reflected in the pre-agreed business plan
 - Triggering events to draw down expected to be set by reference to the verified level of the project company revenues below the agreed viability level in the audited business plan during initial (and riskier) stages of the operation of the sub-project
 - Triggering event for reimbursement expected to be set by reference to the verified level of the project company revenues above the pre-agreed viability threshold

Figure 9. Indicative structure for the contemplated reimbursable grants

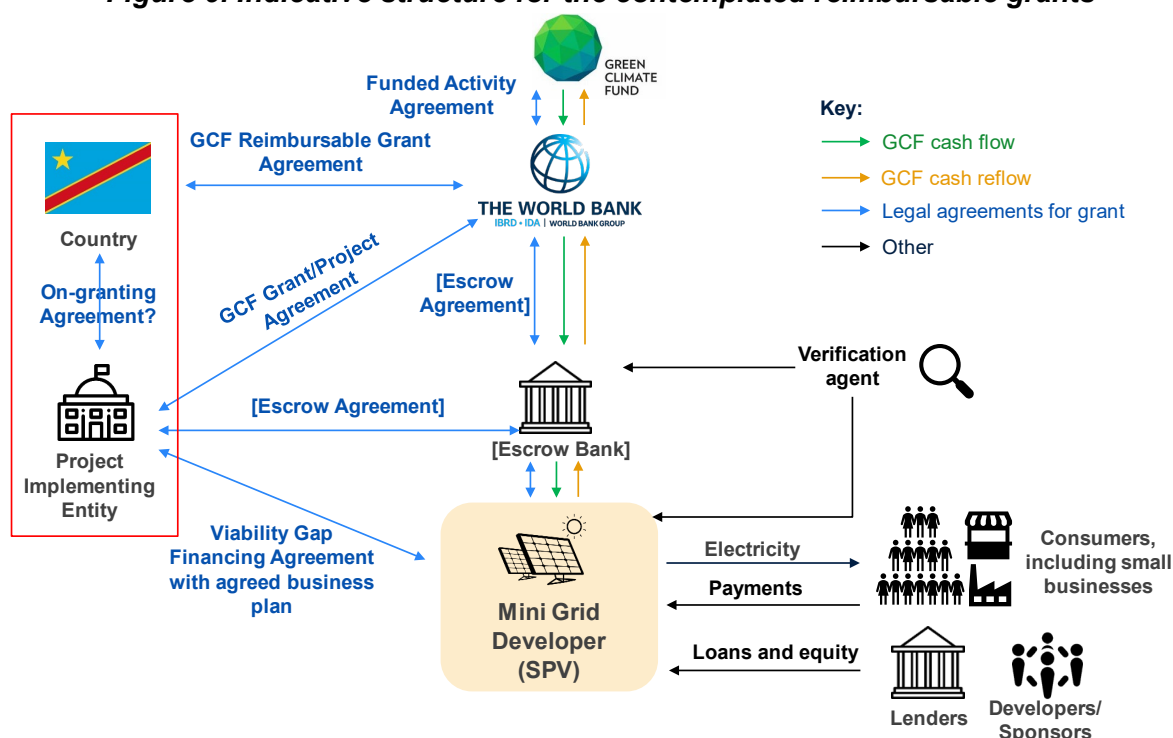
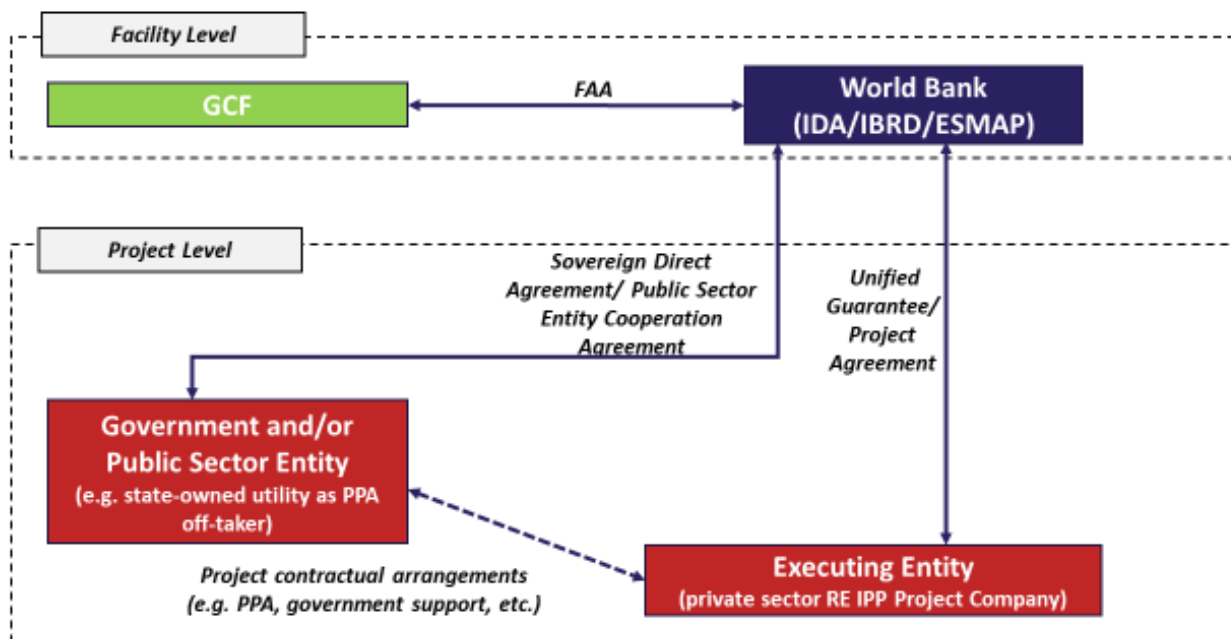
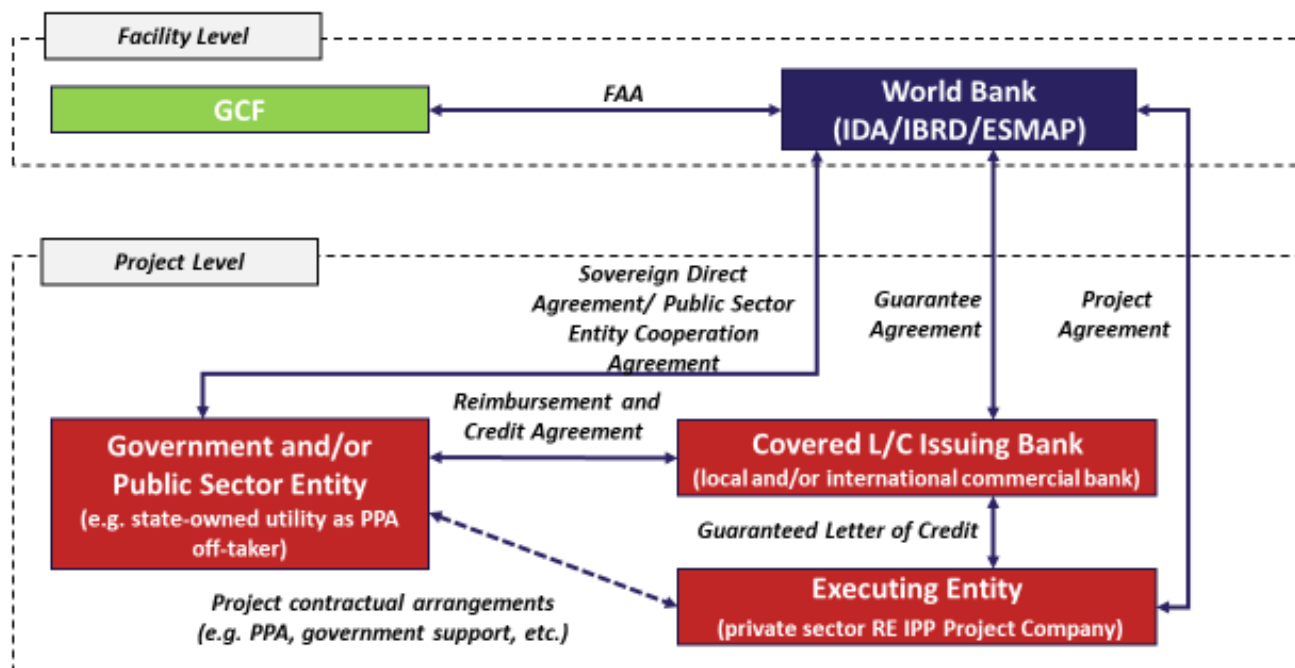


Figure 10: Indicative Implementation Arrangements for Component 3.2

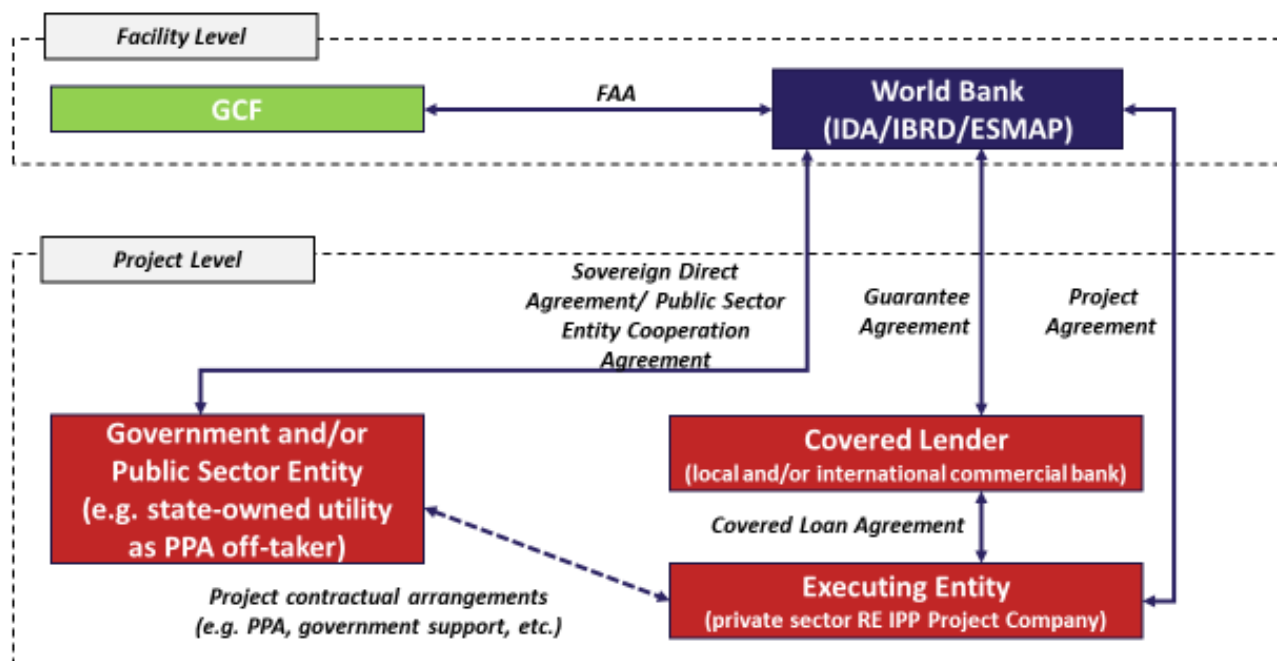
Direct payment guarantee for a private sector project (generic structure)



Payment guarantee (with a letter of credit) for a private sector project (generic structure)



Loan guarantee to a private sector project (generic structure)



Overview of the indicative mechanism contemplated for the guarantees:

- Eligible projects:** RE private sector projects competitively selected by the Host Country or a state-owned company mandated by the Host Country, with the technical assistance of the World Bank
- Executing Entity(ies):** private sector RE project company
- Beneficiaries:** (i) private sector RE project companies (in the case of the direct payment guarantee) or (ii) commercial international or local letter of credit issuing banks providing payment security to such private sector RE project companies (in the case of a letter-of-credit payment guarantee) or (iii) in the case of a loan guarantee, commercial lenders to the private sector RE project company.
- Covered Risk:** Two of the three structures (the letter-of-credit payment guarantee and direct payment guarantee) would support payments under a power purchase agreement (PPA) to an Executing Entity from a public-sector off-taker (i.e. liquidity risk), and the nature of the structure provided would depend on the project itself as it takes shape. The third structure, a loan guarantee, could provide appropriate support to commercial lenders (against public-sector risks related to long-term mobilization of private capital) to private investors (that is, Executing Entity RE project companies) for a RE project.
- Financial terms:** The conditions of guarantee would reflect World Bank's policy and procedural requirements but given the nature of guarantee projects, would be specifically negotiated with the participants in each project. In accordance with World Bank's policies and procedures, guarantees would only be provided to the extent necessary to mobilize private financing for the projects and/or to mitigate payment risks of the projects, taking into account country, project and market circumstances.

3. Projects Arrangements and Reporting Requirements

World Bank-financed projects follow a robust framework to design, prepare, implement, and supervise projects, as assessed by GCF prior to entering into its AMA with the WB. All the potential Facility-supported

projects will follow the WB project cycle, as well as the Bank's operational policies and procedures. For each Project included in the Programme, including those for which a GCF financing would be requested, the World Bank will carry out its due diligence and submit an appraisal-stage Project Appraisal Document (PAD) for quality control covering the IBRD/IDA financing and GCF financing, following standard WB procedures. Once all the relevant internal World Bank's requirements are met, the project moves to negotiations and World Bank Board approval stage, which cover the GCF financing and all WB or WB administered co-financing. The implementation of the Facility-supported projects (project implementation through completion and evaluation) follows the World Bank's applicable policies and procedures. For each project, a project management unit (PMU) will be established with a team expected to include at least a team lead, technical expert, procurement specialist, financial management specialist, environmental specialist and social/gender specialist, and a lawyer. Each project will have its own monitoring and evaluation (M&E) and reporting process, and the EE will be accountable for its implementation.

Figure 11: Key Personnel (Staff or Consultants) Typically Included in a Project Management Unit



B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

GCF funding is critical to address the current RE market failures identified and to ensure that countries continue their low carbon development pathway. Today, the deployment of solar and wind is constrained due to (i) lack of existing government strategy to adapt the current energy sector to the new disruptive technologies, (ii) weak grids that require upfront investments by public parties, (iii) government financing constraints, (iv) conflicting views on short-term vs medium-term economic growth needs, and (v) domestic production/import of fossil fuel. VRE deployment is also expensive when accounting the integration cost and when there is no organized competitive selection of IPPs, VRE deployment is not always competitive with coal generation. This may lead to a tough politico-economic decision when a government has to choose between promoting least-cost coal production versus slightly more expensive and more complex to integrate VRE as the case may be.

This situation is aggravated by the Covid-19 crisis which is impacting the RE deployment today. It is more difficult for the governments to spare public financing for critical public investments needed to unlock private investments as public financing is prioritized to tackle critical challenges hindering health and jobs. In addition, the utilities themselves are facing unprecedented challenges with payments issues and lower demand affecting their revenues, and therefore, private investors are perceiving them as riskier, increasing the premium on their cost of capital which in turn would increase the electricity price.

To face these challenges, the SRMI Facility is providing an integrated risk mitigation framework to (i) reduce the cost of privately-owned generation (lowering the risk premium embedded in the tariff) making

the electricity price more affordable, (ii) provide concessional funding leveraging climate finance to unlock critical public investments needed to enable private investments making them more affordable for the countries, and (iii) maximizing socio-economic benefits fostering jobs creation, skills development and women empowerment, leveraging the bidding processes.

Catalytic GCF concessional financing will be critical as (i) some countries among the poorest in the world are undergoing dramatic tensions on their budgets due to Covid-19; (ii) all will turn towards the cheapest source of energy with the lowest CAPEX under the current budgetary constraints excepts if it is not immediately impacting their budgets; and (iii) with the investment environment improving, concessionality is the last mile to raising private investors' interest. Under the current Covid-19 crisis and the risk to have coal and gas stranded assets supported by fossil fuel subsidies, climate financing and grant technical assistance to support developing countries in their energy transition are more critical than ever. Access to concessional public climate financing and DFI financing would enable solar and wind generation to be cheaper than the operating costs of most fossil fuel plants. In a context where the governments grapple with the critical task of dealing with the very real impacts of Covid-19 on the health and welfare of their population, countries may fall into a fast recovery strategy that would promote fossil fuel generation unless there is strong parallel support for renewable energy development. Therefore, countries need to be supported financially and technically to revise their current plans and implement the revised ones to avoid building new coal and gas plants. These are critical times to support new solar and wind projects in competing with the billions of USD spent on fossil fuel subsidies. In addition, by using concessional financing to reduce perceived risks by private sector investors, countries would be able to unlock at scale privately financed RE projects especially at a time where the utilities are weakened due to the financial situation of the countries. The GCF, thanks to its mandate and scale, will enable a scale-up of the SRMI approach that will enable real impact. In addition, supporting the scale-up of RE through a systematic, integrated and open approach such as SRMI will ensure the replicability in a timely manner of low-emission development pathways.

The use of climate concessional financing under the Initiative is also essential to support innovative technologies with a significant mitigation potential that are not yet commercially viable such as battery storage. GCF's risk appetite will be critical to provide innovative support to these countries. The industry supporting these paradigm- shifting disruptive technologies is at a nascent stage and will also suffer due to lack of market uptake unless there is continued demand from countries. Without additional investments in dispatch and system operation – including battery storage, VRE generation is likely to increase the risks of load shedding and systems' defaults. However, given the high cost of technologies and the risk arising from lack of experience in deploying these technologies at scale in the context of developing countries, private sector investors may not be willing to invest in these innovative technologies without development support and strong risk coverage.

Private investments in the off-grid RE space requires adapted risk mitigation instruments that can be developed with climate financing that is more flexible than DFI guarantees. A risk mitigation mechanism, as developed under Component 3.1, that covers critical risks such as demand off-take risk will be an important catalyst in helping attract more private sector investment and scale up mini grids and SHS. This demand risk is not covered by traditional DFI risk mitigation mechanisms like MIGA insurance or WB guarantees. GCF's support will be crucial to ensure that the right risk mitigation instrument is being provided.

With the exception of countries which receive full grants from the World Bank for instance due to their level of GDP per capita or post-conflict status (such as CAR in the current list of projects proposed to be supported by the SRMI Facility), the allocation between grants and loans is done between technical assistance/capacity building and infrastructure. IDA is providing grants to CAR

because it is an FCS country and is recovering from the recent major domestic political and security crisis in 2013. The ensuing insecurity led to massive forced displacement, the collapse of state institutions, and devastation of the country's economy. The country is in dire needs of IDA grant to recover its infrastructure and its economy. CAR remains among the poorest countries in the world and ranks second to last out of 188 countries on the United Nations Human Development Index. The other seven countries are split between three IDA and three IBRD. As per the SRMI methodology, the activities promoted under the technical assistance component are critical requirements for the development of a well-organized, sustainable and affordable generation mix as well as developing an enabling environment to private sector mobilization. These plans and studies are too expensive or complex in general for the countries to do without concessional support. Therefore, without targeted grant technical assistance, those studies are usually not being completed which leads to far from optimal decision with regards to new generation or transmission investments and they are usually not compatible with the current climate urgency. GCF's financing will be needed to ensure that the most fragile countries are able to reduce their debt weight and the cost of the interest.

Public investments are critical to enable the sustainable unlocking of private investments. However, governments are often stretched in their budget due to overwhelming social sector priorities. In this context, catalytic climate financing will help support more investments in RE. The approach promoted under SRMI targets private financing for the generation of renewables with the public investment being limited to critical public investment required for a higher penetration of renewables such as VRE integration investment, and solar parks infrastructures. However, governments need to continuously look at the budget allocation between different sectors, especially driven by pressing needs in the social sectors which is being exacerbated by the Covid-19 crisis. The possibility to access climate financing for those critical infrastructure investments, means that those countries would not need to reduce their IDA/IBRD envelopes for education or health sectors. The level of GCF participation in the financing of the currently proposed projects has been determined based on discussions with countries and their possibility to increase their financing for their access and RE generation agendas.

Another critical aspect of the SRMI Facility that will support an increased public investment allocation towards critical investments is the concessionality of GCF compared to IBRD and to a certain extent IDA as well. Under the SRMI Facility, GCF high concessional terms for vulnerable countries will be sought to be able to not increase the cost for the IDA countries while being attractive enough for the IBRD countries to make the investment towards low-carbon infrastructure fostering low-emission development pathways.

The GCF high concessional terms will enable the investments to be viable for both IDA and IBRD countries – and in turn the concessionality will fully benefit the electricity households of these countries. The GCF high concessional loan terms with a maturity period of 40 years and 10 years grace period with an all-in fee of 0.25 percent, i.e., 0.00 percent interest rate, 0.25 percent service fee (and waiver of 0.50 percent commitment fee) will result in an increase of 2 percent in project internal rate of return (IRR).

Given the abovementioned rationale, the GCF financing will be critical from the additionality perspective to unlock the sustainable programs presented by the seven countries included in this Funding Proposal from different perspectives, including (i) making the critical public investments needed more affordable, (ii) encouraging de-risking measures meeting the needs of the private sector through the availability of this concessional financing (e.g. mitigating the development risk for the private sector by financing solar/wind park infrastructures), (iii) adapting to the specific needs of some countries who need innovative risk mitigation approaches that are complex to provide by DFIs but that are within the mandate of the GCF and (iv) unlocking sustainable and bankable projects under SRMI at scale.

With the support of the GCF financing, the integrated risk mitigation framework promoted under the SRMI Facility is expected to play a key role on (i) a short-term basis by contributing to a green stimulus package for recovery and (ii) on a medium-term basis by ensuring a clean development path, mitigating significantly the GHG emissions trajectory.

B.6. Exit strategy and sustainability (max. 500 words, approximately 1 page)

Sustainability is at the heart of SRMI, mainstreamed in a cross-cutting integrated approach. Supporting countries to design and implement a sustainable low-emission development pathway, from a multi-dimensional perspective, fostering a transformational change with projects replicable at the level of the countries themselves:

- a. From a technical perspective, ensuring through comprehensive planning that sound RE targets are defined based on least cost planning, factoring in integration of VRE and grid resilience, and promoting a sector-coupling approach to the extent possible.
- b. From a financial perspective, reducing the reliance on public sector financing by supporting the design and implementation of bankable projects that could be financed by the private sector.
- c. From a strategy perspective, ensuring upfront that the RE program designed is aligned with the countries' priorities and strategic considerations (including in terms of risk allocation and roles and responsibilities of public parties).
- d. From an environmental perspective, financing countries' NDC in an economically viable manner while ensuring their implementation.
- e. From institutional and regulatory perspective, strengthening the countries' framework and developing the skills needed (including by financing the transaction advisors for the first round of projects under the RE program defined allowing 'a learning by doing' with hands-on experience), to give to the seven or more countries the ability to replicate the projects, and fully integrate the changes they are facing in our energy systems into their long-term strategy.
- f. From a socio-economic perspective, maximizing the positive impacts on the local industries, (positioning the local players on the solar/wind value chain in a competitive manner to the extent possible) and supporting local community development (through actions increasing their resilience and enhancing their livelihoods).
- g. From a gender perspective, ensuring that men and women benefit from the RE projects launched with a proactive policy to promote female employment throughout the RE project-cycle as otherwise men may benefit disproportionately from new employment opportunities across the RE value chain.
- h. From a financial perspective, SRMI's core investments in infrastructure for the grid and with regards to electrification are long-term investments that have long-term potential impact and will be critical to the economic development and transition to a new energy paradigm. The grid upgrades for VRE integration is expected to enable a shift on average from 7 percent penetration of VRE to over 35 percent by 2025 and electrification investments are core to the development of a population.

Therefore, the approach used by the SRMI Facility is supporting countries in developing a multi-dimensional sustainable RE program and operationalizing the first round of bankable projects. By laying strong foundations and implementing an adequate capacity building program, key public stakeholders

will be able to capitalize on the work done, the track-record built, and the skills developed for their next round of RE projects.

The SRMI Facility is expected to act as a catalyst to unlock these sustainable RE programs in the countries and pave the way to operationalize them through bankable projects privately financed. With a strong framework in place, a successful track-record and a sustainable ecosystem implemented, countries would be in a position to replicate these projects and to attract private investment in optimized conditions.

C. FINANCING INFORMATION							
C.1. Total financing							
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount			Currency		
		280			million USD (\$)		
GCF financial instrument		Amount	Tenor	Grace period	Pricing		
(i)	Senior loans	176	40 years	10 years	0.25%		
(ii)	Subordinated loans	n/a	n/a years	n/a years	n/a %		
(iii)	Equity	n/a			n/a %		
(iv)	Guarantees	20	25 years				
(v)	Reimbursable grants ⁶	30	15 years				
(vi)	Grants	54					
(vii)	Results-based payments	n/a					
(b) Co-financing information		Total amount			Currency		
		1,283.5 ⁷			million USD (\$)		
Name of institution		Financial instrument	Amount	Currency	Tenor & (grace) ⁸	Pricing	Seniority
World Bank (ESMAP)		Grant	1.5	million USD (\$)			
World Bank (IBRD)		Senior Loans	459.5	million USD (\$)	20 years	3.4%	
World Bank (IDA)		Senior Loans	750	million USD (\$)	30-40 years (5-10 years)	0.75% ⁹	
World Bank (IDA)		Grant	72.5	Million USD (\$)			
(c) Total financing (c) = (a)+(b)		Amount			Currency		
		1,563.5			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)		CTF co-financing is considered for Mali to complement efficiently the GCF financing and benefit from the synergies between the CTF and the GCF. However, the CTF financing is not yet allocated for either projects. Therefore, it is not included in the co-financing section.					
C.2. Financing by component							

⁶ The reimbursable grant will be used as a risk mitigation instrument

⁷ The World Bank (IBRD/IDA/ESMAP) co-financing amounts are indicative and subject to approval by the World Bank Board and agreement with the EEs.

⁸ The pricing information for IBRD/IDA is indicative as the Bank Board periodically approves updated IBRD/IDA financing terms and they may change for a given country depending on when the project is submitted for Board approval. The prices presented in the table are for general IBRD and IDA projects.

⁹ The IDA pricing presented in the table refers to the standard service charge on IDA credits to SDR credits. As mentioned above, the pricing is indicative and additional interest charges and service charges may be imposed on borrowers depending on their categorization under the World Bank Directive on IBRD/IDA financing terms.

Component	Sub-Component	Activity	Indicative cost million USD (\$)	GCF financing		Co-financing		
				Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Component 1: Technical Assistance	Generation plans with VRE integration and Capacity Building (technical)	Generation plans with VRE integration and Capacity Building (technical)	5	5	Grants	0	Grants	Click here to enter text.
	Transaction Advisory and Capacity Building (procurement, legal)	Transaction Advisory and Capacity Building (procurement, legal)	8.5	7.5	Grants	1	Grants	World Bank (ESMAP)
	Solar and wind park preparatory studies	Solar and wind park preparatory studies	9.5	9	Grants	0.5	Grants	World Bank (ESMAP)
Component 2: Public Investments	Public investments in solar and wind park infrastructure	Public investments in solar and wind park infrastructure	126	28	Senior loans	98	Senior loans	World Bank (IDA/IBRD)
	Public investments for VRE integration grid upgrades and grid resilience	Public investments for VRE integration grid upgrades and grid resilience	479	128	Senior loans	351	Senior loans	World Bank (IDA/IBRD)
	Public investments for electrification	Public investments for electrification	771	20	Senior loans	751	Senior loans	World Bank (IDA/IBRD)
	[All sub-component 2 activities for FCS countries (CAR)]	[All sub-component 2 activities for FCS countries (CAR)]	90	30	Grants	60	Grants	World Bank (IDA)
Component 3: Risk Mitigation Instruments	Risk mitigation instrument for solar/battery mini-grids and off-grid private investors	Risk mitigation instrument for solar/battery mini-grids and off-grid private investors	30	30	Reimbursable grants	tbd	Guarantees	World Bank (IDA/IBRD)

	Risk mitigation instrument for grid-connected RE projects	Risk mitigation instrument for grid-connected RE projects	20	20	Guarantees	tbd	Guarantees	World Bank (IDA/IBRD)
Project Management	Project management	Project management	24.5	2.5	Grant	22	Grant/Loans	World Bank
Indicative total cost (USD)			1,563,500,000	280,000,000	1,283,500,000			

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

Promoting a sustainable environment for private investments as part of the energy sector is critical to the SRMI approach. Capacity building is key from that perspective to ensure that local stakeholders are effectively empowered in relation to the solar/wind value chain to the extent possible. A multi-dimensional capacity building program is considered under SRMI for that purpose, covering from the design of the RE strategy to their implementation.

To increase the effectiveness of capacity building, the approach promotes a strategic analysis upfront to identify the targeted roles and responsibilities of the key stakeholders and assess the local industries' potential as to their positioning on the value chain. The objective being to define a concrete action plan to develop the targeted skills of the key stakeholders and improve their chances to benefit from their own RE projects. For example, by defining a cross-cutting country program to position the local players on the operation and maintenance of the solar/wind plants, allowing them to benefit from the transfer of knowledge needed as well as from the hands-on experience thanks to adequate arrangements reflected in the tender documents. This plan will benefit from the support of transaction advisors financed under SRMI to accompany the countries in this pathway throughout a 'learning by doing' approach for the pre-identified stakeholders.

In addition, as integration of VRE is a key aspect, SRMI is also promoting a transfer of knowledge in favor of key local stakeholders in relation to storage and more particularly with respect to battery storage. SRMI is leveraging the Energy Storage Partnership (ESP) launched by the World Bank in collaboration with partners to support this technology, not yet mature but presenting a significant potential to enable higher penetration of VRE in the grid. The SRMI Facility is directly contributing to the development of the battery storage technology by enabling the countries to implement it, building a track-record as well as crucial knowledge.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

The seven proposed projects are expected to contribute to the shift to low-emission sustainable development pathways with the following:

- **Directly support the development of 2.5 GW of RE**, (i) 1.8 GW of solar PV, (ii) 400 MW of CSP, and (iii) 300 MW of onshore wind. Of this 2.1 GW are expected to be built during the lifetime of the projects and 0.4 GW expected to be built after the end of the projects. The direct support is enabled through Sub-Component 1.3 financing the solar and wind park analysis and Sub-Component 2.1 financing the public infrastructure of solar and wind park infrastructure and Sub-Component 1.2 for the hiring of transaction advisors to select IPPs. The SRMI Facility will also enable the integration of another 1.9 GW of RE. The integration of other projects is enabled thanks to the VRE integration analysis conducted under Sub-Component 1.1 and Sub-Component 2.2 under which core grid infrastructure are being financed.
- **Directly reduce GHG emissions by 97.93m metrics tons of CO₂**, more precisely 88.98m reduction for the 2.1 GW RE built under the projects over the 25 years of their operation, and 8.95m metric tons of CO₂ for 0.4 GW of RE projects that were prepared under the projects but not built during the lifetime of the projects¹⁰. The GHG emission reduction is based on RE projects being built.
- **Unlock USD 3.3 billion of private investments**, including USD 3 billion for the 2.1 GW built during the lifetime of the projects and USD 300 million for the 10.4 GW built after the end of the projects. The private investment is leveraged thanks to Component 1.2 transaction advisory and the general environment and bankable support provided to them through the general SRMI support of technical assistance to governments under Component 1, critical infrastructure under Component 2 and guarantees and risk mitigation instruments under Component 3.
- **Provide access to clean low-emission electricity to 4.2 million people thanks to the electrification investments**. Component 2.3 is critical in enabling public investments in clean energy mini-grids and connection to the grid that would have been made greener thanks to Sub-Components 2.1 and 2.2, and Sub-Component 3.1 that will enable unlocking private investments for green mini-grids.
- **Strengthen institutional and regulatory systems for planning and development.**
- **Reduce locking-in long-lived, climate-vulnerable infrastructure by increasing resilience of grid infrastructure.**

Table 3: SRMI Facility Proposed Projects Expected Outcomes

Country	VRE installed capacity (MW)	VRE projects prepared (MW)	VRE integration (enabled MW)	Battery Storage (MWh)	Private sector directly leveraged (mUSD)	Private sector enabled (mUSD)	Access (people)	Total GHG Emissions Reduced (Mtons/CO ₂ eq) during/after project
Botswana	300	0	0	50	820	0	600,000	52.3/0

¹⁰ GHG emissions calculations were made following the World Bank standards using the official combined margin grid emission factor per country was used.

CAR	35	0	0	35	10	0	250,000	0.4/0
DRC	100	0	0	140	200	0	1,500,000	0.06/0
Kenya	300	0	250	150	225	0	450,000	5.8/0
Mali	300	0	150	350	350	0	400,000	8.5/0
Namibia	400	0	0	200	940	0	1,000,000	8.4/0
Uzbekistan	600	400	1,500	50	450	300	0	13.5/9
Results	2035	400	1,900	975	2,995	300	4,200,000	89/9

The SRMI Facility aims to be scaled up to a total USD 400 million by adding countries, namely Indonesia and Vietnam, that would have projects following the agreed upon eligibility criteria as per the term sheet and the present funding proposal. The Phase 2 of the Facility – that is to be complementary to the current Phase – aims to – and such based on current project information - (i) avoid a total of 286.32 m metrics tons of CO₂, including 60.6 m metrics tons of CO₂ during project lifetime and 225.7 m metrics tons built after the end of the project, (ii) unlock a total of 6.1 GW, including in the short-term 2 GW and in the medium-term 4.1 GW of RE, and (iii) leverage a total of USD 10.2 bl, including USD 1.2 bl during the project lifetime and USD 9 bl built after the end of the project.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

By tackling in a coordinated manner planning, procurement, development and curtailment risks to attract affordable private investments, SRMI aims at supporting countries in their shift to low-emission development pathways. Most countries have reached a point where most of their “easy” RE projects have been implemented. To move from a situation with a low VRE penetration to one where the majority of the electricity is produced from RE sources, governments will need to put in place sustainable generation plans, upgrade their grids and develop the right legal and procurement framework to leverage private investments at scale. This represents large investments from the public sector to enable such framework and will require strong governments buy-in and ownership. The SRMI Facility aims at accompanying governments in their path to low-emission development while ensuring it does not become an economic burden.

Building on a sound methodology¹¹, SRMI aims at supporting countries transform their strategy to low-emission development allowing them to harness their domestic resources at scale.

- The first step to a sound deployment is the setting of sustainable long-term targets for RE based on concrete analysis for demand, domestic resources and grid flexibility and stability. These targets will be the basis for a realistic but also ambitious plan to deploy RE in the given country. Through the SRMI Facility, the World Bank is supporting the seven countries to revise their generation plans and to identify the critical investments needed to integrate VRE into their grids while improving its reliability and resilience. These targets will then need to be integrated into their NDC revisions ensuring they are implementable.
- After the targets are set, the SRMI Facility is supporting the seven countries to develop a long-term strategy to implement those targets. The strategy looks at who and how to finance the deployment of RE and the associated infrastructure. It also identifies all the legal changes that need to be implemented to attract private investments and how the domestic lending sector can be leveraged to finance the debt part of the privately-owned power plants to reduce the risk of foreign exchange to

¹¹ “A Sure Path to Sustainable Solar : Solar Deployment Guidelines” was released during the 2019 United Nations Climate Summit in New York and are accessible online through the following link: <http://documents.worldbank.org/curated/en/244251575642432241/A-Sure-Path-to-Sustainable-Solar-Solar-Deployment-Guidelines>

the utility. In addition, socio-economic benefits can be integrated into the long-term strategy. It can support countries in leveraging the deployment of solar and wind to enable the population to gain opportunities for jobs and development.

- The first main pillar to the paradigm shift is capacity building. As part of SRMI, it is critical to increase capacity within the government on all fronts so that afterwards they will be able to conduct analysis without the support of consultants and to negotiate with the private sector on fair grounds.

After supporting the development of the country's long-term targets, its strategy to implement them and increase capacity, the SRMI Facility is financing the implementation its strategy. It is critical to countries to have a support which encompasses all from the target setting to its financing. Without such combined approach, the countries may not be able to finance the targets or have the right target in the first place.

As a consequence, countries will be able to operationalize their NDCs in a sustainable and timely manner in spite of the effect of the Covid-19 crisis, reducing significantly their GHG emissions trajectory by shifting to a lasting low-emission development pathway. Built on strong foundations through this comprehensive and integrated approach with a multi-dimensional sustainability perspective at its heart. This low-emission development pathway will be replicable, fast-tracking the road towards the Sustainable Development Goals for these countries.

The seven countries are representative of different types of markets/energy sector, with low to high electrification rates, low to high cost of generation, small to large grids etc. The success and lessons learnt from those seven countries will enable to replicate the SRMI approach across countries and enable to unlock sustainable RE markets for private investments. SRMI is currently being implemented/considered for another 30 countries in Sub-Saharan Africa and Asia which will benefit from the results and lessons learnt of this first phase.

Tailored support for each country will be provided under the SRMI approach. The status of RE deployment is very different from one country to another and will be further assessed at Project's level to leverage the existing related activities. The technical assistance deployed under the SRMI Facility will be tailored for each country needs and will, to the extent possible, build on the current RE activities deployed, in coordination with all stakeholders involved (including other DFIs as applicable) to enhance synergies. This will include for example adjusting existing least cost plans in place if needed or including the lessons learnt from existing RE projects in terms of risk allocation, government support, support to jobs' creation etc.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Sustainability is at the heart of the multi-dimensional approach of SRMI. The SRMI approach aims at anchoring RE projects of the targeted countries in a sustainable ecosystem ensuring a lasting and replicable low-emission development pathway, fast-tracking in particular the SDG 13 Climate Action and the SDG 7 Affordable and Clean Energy.

Sound planning associated with adequate grid investments to enable a higher penetration of renewables while enhancing resilience will lead to the development of sustainable RE targets. It will avoid the lock-in of the countries in long-term high-emission development pathways. Harnessing domestic resources such as solar or wind for generation will also reduce the need for fuel imports and the potential economic impact of fuel cost fluctuations. It will increase the energy security of the countries, diversifying

their mix et improving their energy independence while reducing the price volatility (as electricity tariffs are set for a long-term period, around 25 years).

The operationalization of these RE targets through bankable projects with a fair risk allocation, building on a strengthened institutional and regulatory framework with adequate risk mitigation arrangements, aims to ensure the timely implementation of these projects. It aims to leverage private investments at scale, ensuring the availability of the commercial financing and the appetite of the private sector to take these projects forward. Hence, it will spare limited public sector finance for the countries which could be directed to other sectors in need such as healthcare, in particular in the context of COVID-19.

The strategic analysis upfront to define an action plan to maximize the socio-economic benefits of the RE program (embedded in the SRMI methodology), will be mainstreamed during the operationalization of the projects. It will enhance the involvement of the local players on the solar/wind value chain in a competitive manner to the extent possible, maximizing jobs creation, skills development and transfer of knowledge. It will also support local community development in the areas of implementation of the projects through actions increasing their resilience and enhancing their livelihoods. A specific attention will be brought to women empowerment as otherwise men may benefit disproportionately from new employment opportunities across the RE value chain.

Capacity building will be mainstreamed throughout the RE projects cycle, strengthening the countries' framework and developing the skills needed (including through 'a learning by doing' approach with the support of transaction advisors), to give to the seven countries the ability to replicate efficiently the projects. Building on the strong foundations of the program, key public stakeholders will be able to capitalize on the work done, the track-record built, and the skills developed for their next round of RE projects.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

The seven countries have different economic development level and the Covid-19 crisis will impact their economic growth projections. GDP per capita is below USD 1000 per capita for CAR, DRC and Mali, between USD 1000 to 4,000 for Kenya and Uzbekistan and above USD 5000 for Botswana and Namibia. Their GDP growth is on average quite strong with the exception of Namibia that was hit with droughts in the last few years. The GDP growth forecast for 2020 will be reviewed with the current Covid-19 pandemic crisis. The countries also have quite different country investment ratings, with only Botswana considered as investment grade.

Table 4: SRMI Facility Proposed Countries Macroeconomic Overview

Country	Population (million)	GDP / capita 2018 (USD)	Inflation 2018 (%)	GDP growth 2017 (%)	GDP growth 2018 (%)	GDP growth forecast 2019 (%)	Country Investment Rating
Botswana	2.25	8,258	0.9	2.9	4.6	3.9	BBB+
CAR	4.7	475	1.6	4.5	3.8	3.0	n/a
DRC	84	562	29	3.7	5.8	4.4	CCC+
Kenya	51.4	1,710	4.7	4.9	6.3	5.6	B+
Mali	19	899	1.7	5.0	5.2	5.1	n/a
Namibia	2.5	5,931	4.3	-0.1	0.3	-1.4	BB+

Uzbekistan	32.9	1,532	17.5	4.5	5.4	5.6	BB-
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Source: World Bank, IMF, S&P

The seven countries have also quite different social development levels. The results of the Human Development Index (HDI) are also a good comparing instruments between the seven countries, with Botswana and Uzbekistan that have good HDI, Namibia and Kenya that have average HDI, and CAR, DRC and Mali with very low HDI.

Table 5: SRMI Facility Proposed Countries Key Social Development Index Overview

Country	Corruption Index (/180)	Human Development Index	Gini coefficient
Botswana	34	0.728	53.3
CAR	153	0.381	56.2
DRC	168	0.459	42.1
Kenya	137	0.579	40.8
Mali	130	0.427	33
Namibia	56	0.645	59.1
Uzbekistan	153	0.71	35.3

Sources: World Bank, Transparency International, United Nations

The seven proposed countries have clear future climate change impacts and a need to fundamentally adapt their energy systems by embracing disruptive technologies and transition to the grid of the future increasing resilience and sustainability while adopting low-development emission pathways. Further noting that Namibia and Botswana are not eligible for climate finance provided by the Climate Investment Funds (CIF).

SRMI aims at limiting the needs for public resources. Climate financing will be critical in providing sufficient financial incentives for those countries to change their business as usual way to tomorrow's energy system. This investment will be critical for countries to implement a clear strategy without impeding economic development or putting at risk their grid and therefore, their industrial development.

The countries considered will need strong support to strengthen their institutions and implementation capacity to transition to a disruptive technology grid. SRMI aims to provide upstream support to work with the countries on their technical and regulatory constraints and as a second step managing the implementation of their new strategy, such as with regards to managing battery storage into their grid, procuring IPPs and adapting their grid to electric mobility development.

D.5. Country ownership (max. 500 words, approximately 1 page)

While ensuring alignment with NDCs, relevant national plans, and/or enabling policy and institutional frameworks, Component 1 will be critical in providing inputs to the NDC/low-emissions development plans' objectives and regulatory frameworks and to enable competitive selection of IPPs with fair risk allocation. In each country, the World Bank has extensive engagement with relevant stakeholders, including national designated authorities. The proposed countries all have NDCs with the following commitments:

- Botswana:** The Government of Botswana has been spending a significant portion of its national resources to adapt to the impacts of climate variability over the years. The country intends to achieve

an overall emissions reduction of 15 percent by 2030, taking 2010 as the base year. The emission reduction target was estimated based on the baseline inventory for the three GHGs being CO₂, methane (CH₄) and nitrous oxide (N₂O). The reductions will be realized from the energy sources which is categorized as the stationary and mobile sources. The country is still working on developing initiatives for emission reductions in terms of long-term low carbon strategy.

- b. **CAR:** The annual emissions of the Central African Republic, estimated at 116,285.49 kt eq-CO₂ in 2010, or 26 tons eq-CO₂/person, represent less than 0.002 percent of global emissions. Despite its low rate of GHG emissions, the Central African Republic reaffirms its adherence to the principle of collective but differentiated responsibility and, in accordance with Decision 1CP/19, is taking the present ambitious measures to respond to this challenge without impeding its economic, social and environmental development. The Central African Republic aspires to reduce its emissions by 5% and 25 percent, respectively, in the 2030 and 2050 horizons in comparison to its reference BAU emissions and to increase its sequestration potential. With international support, it will emit around 33,076.1 kt eq-CO₂ less in 2050 than the annual reference emissions. The Central African Republic also aspires to reduce emissions of short-lived climate pollutants (SLCP), which science has shown have a significant short-term climate-warming potential and harmful effects on health, agriculture and ecosystems. Greenhouse gas emissions total 116,285.49 kt eqCO₂ with sectoral contributions of 89.46 percent for land use change and forestry; 5.26 percent for agriculture; 5.19 percent for energy (of which 4.91 percent is wood energy); 0.09 percent for waste and marginal amounts for industrial processes and use of solvents.
- c. **DRC:** As one of the most fragile states and poorest countries in the world, DRC faces a number of challenges which it seeks to address through its climate targets and focus on the SDGs. The contribution of the DRC to global GHG emissions is low (around 0.5 percent). Moreover, GHG intensity relative to GDP is also low. DRC has committed to reduce its GHG emissions by 17 percent by 2030, focusing specifically on three sectors – energy, agriculture and land use, land use change, and forestry (LULUCF) - which collectively produce nearly 98 percent of its GHG emissions.
- d. **Kenya:** Kenya's planning process on mitigation and adaptation hinges on the National Climate Change Action Plan (NCCAP) and the National Adaptation Plan (NAP). The two shall be reviewed every five years to inform the Kenya's Vision 2030 Medium Term Plan (MTP). The adaptation actions are further amplified in the NAP. Mitigation and adaptation actions are implemented across the various sectors at both the national and county government levels. The Ministry of Environment and Forestry coordinates the country's climate change affairs through the National Climate Change Secretariat (NCCS). The planning process follows the Climate Change Act (2016) that introduced several institutional reforms to enhance coordination of climate change adaptation and mitigation. These include the establishment of:
 - A high-level National Climate Change Council chaired by the President. The Council shall provide an overarching national climate change coordination mechanism and, among other roles, ensure the mainstreaming of climate change functions by the national and county governments, and approve and oversee the implementation of the NCCAP.
 - A Climate Change Directorate as the lead agency of the government on national climate change plans and actions to deliver operational coordination; and provide technical assistance on climate change actions and responses to County governments.

- Climate Fund to be a financing mechanism for priority climate change actions and interventions approved by the Council. In addressing climate change issues, public entities are required to undertake public awareness and consultations, and ensure a focus on gender, in line with the Constitution and the Climate Change Act (2016).
- e. **Mali:** The Ministry of Environment is responsible for adoption of NDC targets. There is no stakeholder involvement or multisector commission for the development of NDC targets. Currently, the country has no climate policies in place to mitigate climate change. However, Mali has requested technology transfer for means of implementation.
- f. **Namibia:** Namibia aims at a reduction of about 89 percent of its GHG emissions at the 2030-time horizon compared to the BAU scenario. The projected GHG emissions to be avoided in 2030 is of the order of 20000 Gg CO₂-eq. The Cabinet of Namibia is the Government entity responsible for approving policies. The Parliamentary Standing Committee on Economics, Natural Resources and Public Administration which usually advises Cabinet on relevant policy matters caters for the NDC. The Ministry of Environment and Tourism, which is responsible for all environmental issues in the country, is also the National Focal Point to the UNFCCC. It is the coordinating body for all climate change activities through its Climate Change Unit (CCU) of the Directorate of Environmental Affairs.
- g. **Uzbekistan:** In the long-term perspective up to 2030, Uzbekistan intends to strengthen measures and actions to struggle against climate change, in order to achieve the mitigation objectives: (i) to decrease specific emissions of greenhouse gases per unit of GDP by 10 percent by 2030 from level of 2010. Achievement of the INDC long-term objective envisages support from the international organizations and financial institutions, ensuring access to the advanced energy saving and environmentally sound technologies, resources for climate financing. There are currently some legally binding programs and acts aimed at ensuring by 2020 the implementation of measures for decrease in energy consumption, introduction of energy saving technologies in various sectors of the economy, social sphere and development of renewable energy: Decree of the President of the Republic of Uzbekistan, "On Measures for Further Development of Alternative Energy Sources"; Decree of the President of the Republic of Uzbekistan, "On Program of Measures for Ensuring Structural Reformation, Modernization and Diversification of Production for 2015-2019"; Resolution of the President of the Republic of Uzbekistan "On Program of Measures for Reduction in Energy Consumption, Introduction of Energy Saving Technologies in Economy Sectors and Social Sphere"; Program of Actions for Environmental Protection of the Republic of Uzbekistan; Resolution of the Cabinet of Ministers of the Republic of Uzbekistan, "On Integrated Program of Actions for Mitigation of the Aral Sea Disaster Impacts, Rehabilitation and Socio-Economic Development of Priaralie Region for 2015-2018"; Program for Further Development of Agricultural Production; Program for Further Irrigated Lands Improvement and Rational Use of Water Resources.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

Since this is a Program level facility, economic and financial viability will vary from project to project. The Program efficiency and effectiveness is further presented in terms of cost-effectiveness, co-financing and leveraging, financial viability, and application of best practices.

- **Cost-effectiveness and efficiency**

The GCF will have a catalytical role to enable VRE at scale through the comprehensive and integrated risk mitigation framework promoted under SRMI by: (i) financing the critical public investments needed in an affordable manner and such by reducing the burden for the countries in particular in the context of the Covid-19 crisis where limited public financing is available; (ii) encouraging de-risking measures meeting the needs of the private sector through the availability of this concessional financing (e.g. solar/wind park infrastructure to mitigate the development risk); (iii) adapting to the specific needs of countries that need innovative risk mitigation approaches that are complex to provide by DFIs but are within the mandate of the GCF (e.g., to unlock large-scale mini-grids business models); and (iv) unlocking sustainable and bankable projects under SRMI at scale.

As described in B5, the concessionality of GCF financing will be critical to ensure that the countries are to implement those projects and avoid a path involving deployment of more coal and gas with limited RE. The cost of the public loans is a crucial parameter in the political decision to implement or not a more sustainable path. The GCF additionality is therefore paramount to a successful result. To change the course of a given political path, these countries are requesting financial support through their NDCs and as part of our ongoing dialogue. Therefore, the SRMI Facility is requesting the highest level of concessionality for the seven countries that are part of the Program.

The financial structure is a mix between grants for technical assistance, concessional loans for critical public infrastructure – with the exception of CAR which will get full grants due to its FCS situation – and risk mitigation instruments for private investors. Each financial instrument is critical in order to achieve the Program's objective to support countries in their low-emission pathways: (i) grants for key studies and capacity building will be critical to ensure that the right targets for RE are set and capacity is built within the government, (ii) concessional financing for public investments will support countries in increasing their penetration of VRE and improve grid reliability and resilience, and (iii) risk mitigation instruments are enabling the mobilization at scale of private investments. These are critical especially in countries where the counterfactual is cheap coal production such as in Botswana and Uzbekistan.

The Facility aims to unlock over 2.1 GW of RE in the short-term and 0.4 GW in the medium-term. The division between short-term and medium-term is mostly due to the planning for demand increase and preparation of parks. The expected GHG emission reduction is 89 million tons of CO₂eq which represents an average GCF dollar per GHG emission reduced of USD 3.15 and USD 17.57 if compared to the total public investments.

▪ **Co-financing, leveraging and leveraged long-term investments**

The USD 280 million GCF will be complemented by an expected USD 1.3 billion in World Bank financing which in turn is expected to leverage around USD 3 billion in private investments. The indicative leveraging between public investments (World Bank and GCF financing) and private investments is 1 to 3. The private investments are a result of the projects' public investments and therefore, are not considered as part of the co-financing and the Funded Activity.

▪ **Financial viability**

The GCF concessional financing will be critical to improve the economic and financial rate of returns of the projects. Keeping in mind that fuel taxes and subsidies are included in the economic analysis when data is available, the GCF financing increases the financial IRR on average by around 2 percent but

more importantly makes possible those projects that would not be accepted by the countries without the highly concessional climate financing support. GCF financing is a game changer for the governments as because it is concessional countries which would otherwise select polluting energy would be incentivized to join the program.

The financial analysis conducted for Botswana, Kenya, Mali, Namibia and Uzbekistan for Component 2.1 and 2.2 activities resulted in financial IRR of 13 percent, 18 percent, 27 percent, 12 percent, and 14 percent respectively. The level of concessionality proposed for each of these projects enables the financial IRRs to be financially viable for the countries.

▪ **Application of best practices**

To ensure that the best available technologies are considered, the government will hire transaction advisors that will develop the request for proposals to IPPs and ensure that the IPPs only select Tier 1 technologies. The best practice with regards to selection of IPPs will also be implemented to ensure financial sustainability of all the contracts and develop a track-record within the countries in selection of IPPs.

E. LOGICAL FRAMEWORK

This section refers to the programme's logical framework in accordance with the GCF's [Performance Measurement Frameworks](#) under the [Results Management Framework](#) to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Paradigm shift objectives

- ☒ Shift to low-emission sustainable development pathways
- ☐ Increased climate resilient sustainable development

E.2. Core indicator targets

Provide specific numerical values for the GCF core indicators to be achieved by the project/programme. Methodologies for the calculations should be provided. This should be consistent with the information provided in section A.

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (mitigation and cross-cutting only)	Annual	3,559,000 t CO ₂ eq ¹²
	Lifetime	88,978,000 t CO ₂ eq
E.2.2. Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total project financing	USD 1,563,500,000
	(b) Requested GCF amount	USD 280,000,000
	(c) Expected lifetime emission reductions	88,978,000 t CO ₂ eq
	(d) Estimated cost per t CO₂eq (d = a / c)	17.57 USD / t CO ₂ eq
	(e) Estimated GCF cost per t CO₂eq removed (e = b / c)	3.15 USD / t CO ₂ eq
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	(f) Total finance leveraged	4,278,500,000 USD
	(g) Public source co-financed	1,283,500,000 USD
	(h) Private source finance leveraged	2,995,000,000 USD
	(i) Total Leverage ratio (i = f / b)	15.3
	(j) Public source co-financing ratio (j = g / b)	4.6
	(k) Private source leverage ratio (k = h / b)	10.7
E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	n/a
	Indirect	n/a
	For a multi-country proposal, indicate the aggregate amount here and provide the data per country in annex 17.	
E.2.5. Number of beneficiaries relative to total population (disaggregated by sex)	Direct	Click here to enter text. (Expressed as %) of country(ies)
	Indirect	Click here to enter text. (Expressed as %) of country(ies)
	For a multi-country proposal, leave blank and provide the data per country in annex 17.	

¹² The disaggregation of targets per country is provided in Annex 17.

E.3. Fund-level impacts

Select the appropriate impact(s) to be reported for the project/programme. Select key result areas and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected impact result. The result areas indicated in this section should match those selected in section A.4 above. Add rows as needed.

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
	Tonnes of carbon dioxide equivalent (t CO ₂ eq) reduced as a result of Fund-funded projects/programmes	ex-ante and ex-post analyses, electricity generation data for individual power plants	0	7.12 ml t CO ₂ eq	28.47 ml t CO ₂	- Methodology: detailed in Annex 3 to the FP - Project lifetime: 25 years (which corresponds to the duration of the operation of the RE power plants by the private IPPs selected as expected to be reflected in the power purchase agreement to be signed) - Annual emission reductions: 3,559,000 t CO₂ eq - Lifetime emission reductions: 88,978,000 t CO₂ eq - Project implementation period: 12 years - Given the country investment project development is expected to take 1-3 years, the tender and IPP selection 1.5 years, and the construction is expected to 1.5 years, the average commercial operation date of the power plants is expected to start 5 years as from the
	Cost per tonne of CO ₂ -equivalent reduced	Third party report and ex-ante and ex-post analyses	0	17.57 USD	17.57 USD	
	Volume of public and private funds catalyzed by the GCF	Third party report	0	4,278,000,000 USD	4,278,000,000 USD	
M1.0 Reduced emissions through increased low-emission energy access and power generation	M1.1 Tonnes of carbon dioxide equivalent (t CO ₂ eq) reduced or avoided - gender-sensitive energy access power generation	ex-ante and ex-post analyses, electricity generation data for individual power plants	0	7.12 ml t CO ₂ eq	28.47 ml t CO ₂	

						start of the project implementation period (should all the projects start at the same time). Lifetime costs of the programme: USD 1,563,000 (capacity building costs for USD 22.5 ml are included as they directly lead to GHG emission reductions) - Total financing is expected to be confirmed at project's stage (under the assumption that all the projects start at the beginning of the implementation period of the facility) and to be fully leveraged by the mid-term. At the stage of the Facility, it is expected as follows: ▪ Expected co-financing from the public source: USD 1,283,000,000 Expected private financing to be leveraged by the private sector IPPs selected during the tender process: USD 2,995,000,000 (expected to become available at the financial close of each project)
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E.4. Fund-level outcomes

Select the appropriate outcome(s) to be reported for the project/programme. Select key expected outcomes and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected outcome. Add rows as needed.

Expected Outcomes	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
	<i>Number of technologies and innovative solutions transferred or licensed to support low-emission development as a result of GCF support</i>	ex-ante and ex-post analyses, project installation reports or any other source of information available	0 technologies	1 technology: BESS 487.5 MWh of BESS installed with dedicated capacity building	1 technology: BESS 975 MWh of BESS installed with dedicated capacity building	Technical studies confirming the needs of BESS and associated sizing Specific measures will be deployed under the Facility to support integration of variable renewable energy in the grid, among which the deployment of 1 technology the battery energy storage systems (BESS). 1 associated measure is expected: a dedicated capacity building covering VRE integration training in particular BESS
M5.0 Strengthened institutional and regulatory systems	<i>M5.1 Institutional and regulatory systems that improve incentives for low-emission planning and development and their effective implementation</i>	planning and strategic instruments developed in each country	0 systems	Adoption or adjustment (to the extent needed) of 6 integrated roadmaps to deploy RE projects through a competitive procurement scheme	6 RE roadmaps are operationalized through sustainable and bankable RE programs leading to a significant increase in the supply of RE by IPPS selected through the competitive procurement scheme put in place	The countries will have developed integrated roadmaps to harness their RE potential, updating their institutional and regulatory framework to the extent needed (in particular to allow for competitive RE tenders) and operationalizing these roadmaps into sustainable and bankable projects expected to be implemented by private IPPs selected through a competitive procurement scheme (except in some cases detailed in the FP such as in the case of CAR where a public procurement is contemplated given the FCV status of the country)

M6.0 Increased number of small, medium and large low-emission power suppliers	<i>M6.3 MWs of low-emission energy capacity installed, generated and/or rehabilitated as a result of GCF support</i>	ex-ante and ex-post analyses, project installation reports or any other source of information available	0	1,027.5 MW	2,035 MW	The studies under Component 1.3 are carried-out on time The implementation of the projects is completed as planned (no delay) The targets expressed in MW refer to the VRE capacity constructed (new grid-connected solar and wind power generation systems installed)
	<i>M6.2 Number of households and individuals (males and females) with improved access to low-emission energy sources</i>	ex-ante and ex-post analyses, household data, surveys, or any other source of information available	0 households 0 individuals, including 0 males 0 females	420,000 households 2,100,000 individuals, including 1,050,000 males 1,050,000 females	840,000 households 4,200,000 individuals, including 2,100,000 males 2,100,000 females	The pre-requisite studies are carried-out on time and the associated electrification plan is adopted on time The implementation of the projects is completed as planned (no delay) A total of 840,000 household representing 4,200,000 people out of which 2,100,000 women (precise numbers to be confirmed at projects' stage) is expected to benefit from access to electricity through different techniques of electrification (PPP mini-grids, grid extension etc.)

E.5. Project/programme performance indicators

The performance indicators for progress reporting during implementation should seek to measure pre-existing conditions, progress and results at the most relevant level for ease of GCF monitoring and AE reporting. Add rows as needed.

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Sustainable RE targets available and presented in a government-owned least-cost generation plan	Evidence-based targets for low-emission pathways available in government-owned least-cost generation plan	Ex-ante and ex-post analyses, national plans and strategy documents, or any other source of information available	0	Sustainable RE targets in place in 3 countries under the Facility (3 plans with RE targets available)	Sustainable RE targets in place in 6 countries under the Facility (plans with RE targets available): Botswana, CAR, Kenya, Mali, Namibia and Uzbekistan	Least cost plans demonstrating the RE capacity targeted are in place and owned by the countries
Transparent and competitive selection of IPPs is in place	Successful competitive selection of IPPs	Ex-ante and ex-post analyses, procurement documents, or any other source of information available	Limited to no number of IPPs procured competitively	1 competitive process launched and successfully completed per country in 2 countries under the Facility (2 competitive biddings launched)	1 competitive process launched and successfully completed per country in 6 countries under the Facility (competitive biddings launched): Botswana, CAR, Kenya, Mali, Namibia and Uzbekistan including risk mitigation instruments proposed in Namibia and Botswana	The sub-components 1.2 and 1.3 are carried-out on time The design and bidding of the projects are completed as planned (no delay) Institutional and regulatory framework for competitive procurement of IPPs in place in the country Solar/wind projects procured competitively to select private IPPs in charge of the design, financing, construction, operation and maintenance of the plants with a bankable risk allocation Where needed the risk mitigation instruments requested by IPPs are provided, and they enable a reduction in the cost of PPA

Solar and wind parks ready for tendering	Solar and wind park critical studies are completed and the parks are ready for tendering	Ex-ante and ex-post analyses, construction completion reports, or any other source of information available	0	2 solar/wind parks are ready for tendering (2 sets of associated studies completed)	5 solar/wind parks are ready for tendering (5 sets of associated studies completed)	The sub-component 1.3 is carried-out on time RE projects tendered under a solar/wind park scheme (post completion of the critical solar and wind park studies)
Higher penetration of VRE	Enable an increase penetration of VRE in the grid	Ex-ante and ex-post analyses, energy statistics, country energy sector reports, studies or any other source of information available	% of VRE in the grid that can be integrated safely before implementation of the Facility projects (baseline values per country will be provided as part of the baseline studies)	Public investments in the grid enable a higher penetration of VRE to be safely integrated in the grid of 2 countries (2 countries with assessments showing a potential increase in VRE penetration from baseline)	Public investments in the grid enable a higher penetration of VRE to be safely integrated in the grid of all countries except DRC (countries with assessments showing a potential increase in VRE penetration from baseline)	No material changes to the least cost plans adopted RE projects deployed as planned Public investments for VRE integration grid upgrades implemented as planned Higher penetration of VRE is 15 percent and above of installed generation
Increased capacity by the Government and/or utility/dispatching entity with regards to VRE deployment and dispatch	Increased capacity of key staff designated by the EE	Ex-ante and ex-post analyses, surveys, or any other source of information available	Limited capacity in VRE deployment and dispatch by the Government and/or utility/dispatching entity	One training conducted in each country under the Facility	Two trainings conducted in each country under the Facility	Staff and experts have been designated by the EE for training and have adequate background and technical expertise to benefit from this capacity building and be able to apply the skills from the training in their operations Relevant staffs and experts designated by the EE have participated diligently in the training proposed
Private capital is leveraged for mini-grids in	Volume of private capital leveraged as a result of	Ex-ante and ex-post analyses, financial reports, or	0	0	USD 500 million	The studies under Component 1.3 are carried-out on time

DRC, and utility-scale RE in Botswana and Namibia	risk mitigation instruments	any other source of information available				The implementation of the projects is completed as planned (no delay) and the competitive selection of IPPs and mini-grid managers was successfully implemented
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E.6. Activities

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in the implementation timetable. Add rows as needed.

Activities	Description	Sub-activities	Deliverables ¹³
Component 1: Technical Assistance			
1.1.1 Generation plans with VRE integration analysis and capacity building (technical)	Least-cost generation and transmission plans are done jointly with VRE integration analysis and capacity building of technical planning teams and dispatch teams by each country EE	1.1.1.1 Preparation of a new (or adaptation of an existing) least-cost generation plan, which includes reiterative VRE integration analyses based on technical and commercial constraints, an evaluation of the dispatch capabilities and the limits of grid integration and determines a cost-optimized electricity mix that can meet demand at any time; 1.1.1.2 Preparation of a new (or adaptation of an existing) least-cost transmission/distribution plan, which includes capacity expansion models simulating generation and transmission capacity investment, given assumptions about future electricity demand, fuel prices, technology cost and performance, policy and regulations, and providing informed solar targets and indication of where best to locate future projects as well as a list of key investments needed to improve grid VRE integration capacity; and 1.1.1.3	A least-cost generation plan for 6 countries in the Facility (including grid investment needed to enable the targeted VRE) and at least two trainings per 6 countries, namely Botswana, DRC, Kenya, Mali, Namibia and Uzbekistan

¹³ Deliverables are indicative.

		Provision of institutional capacity building support for energy planning and dispatch (including using grid-connected battery storage).	
1.2.1 Transaction advisory and Capacity Building (procurement, legal)	Transaction advisory for IPP selection under competitive bidding, including legal and procurement capacity building by each country EE	1.2.1.1 Provision of technical assistance (including legal, procurement and transaction advisory services) and implementation of capacity building activities to support: (i) the assessment of the existing legal framework and the development of a legal framework for IPPs and competitive bidding conducive to private sector investment; (ii) the design of energy projects with fair risk allocation among stakeholders and contractual arrangements acceptable to potential investors; (iii) the competitive selection of investors; and (iv) the negotiations of the final contractual arrangement with the selected investors and closing.	Set of bidding documents setting the capacity of the solar and wind power plants to be installed by the private sector (Request for Proposals package of documents informed by the legal and structuring analysis done) available per country in the Facility, namely Botswana, CAR, DRC, Kenya, Mali, Namibia and Uzbekistan
1.3.1 Solar and wind park preparatory studies	Technical preparatory studies for solar and wind park infrastructure (i.e. feasibility studies, E&S instruments) by Botswana, CAR, Mali, Namibia, and/or Uzbekistan' EE	1.3.1.1 Preparation of preparatory and feasibility studies and environmental and social risk mitigation instruments, for the development of solar PV, solar PV with battery storage, CSP, onshore wind and offshore wind parks in support of investment de-risking and investor selection process. Preparatory and feasibility studies may include, as needed, geospatial analyses of the land around the substation, topography and geotechnical analyses, environmental and social analyses, site-specific	5 solar and wind parks ready for auction (technical studies completed) in the following countries: Botswana, CAR, Mali, Namibia, and/or Uzbekistan

		grid interconnection studies, solar irradiation/wind measurement analyses and other studies as needed.	
Component 2: Public Investments			
2.1.1 Public investments in solar and wind park infrastructure	Public investments in fencing, connection line, road, water access etc. for solar/wind parks by Botswana, CAR, Mali, Namibia, and/or Uzbekistan's EE	2.1.1.1 Carrying out infrastructure investments in support of the development of solar and wind parks, such as acquisition of land, construction of transmission lines and fencing; and	5 solar and wind park public investments completed in the following countries: Botswana, CAR, Mali, Namibia, and/or Uzbekistan
2.2.1 Public investments for VRE integration grid upgrades and grid resilience	Public investments for VRE integration grid upgrades (FACTS, transformer, doubling line etc.) and public investments for climate change grid resilience (vegetation control, doubling line, tsunami resilience etc.) by Botswana, CAR, Kenya, Mali, Namibia, and/or Uzbekistan's EE	2.2.1.1 Investment for grid modernization and upgrade to improve grid's flexibility to integrate larger amounts of VRE, limit curtailment or grid instability and adapt to climate change related shocks, including power system planning, grid management, strengthening of forecasting methods and grid code requirements. Grid reinforcements that support VRE integration may include (i) addition or replacement of lines and transformers for grid extension and capacity enhancements (both for answering growing demand and for integrating VRE power), (ii) equipment for smoothing the voltage and frequency issues, such as capacitor banks, battery storage and other reactive power compensators, together with flexible alternating current transmission systems, (iii) equipment for faster and more efficient grid operation, such as monitoring systems, demand and production forecasting systems, and automats for controlling generation units and grid	Reinforced grids enabling further safe VRE integration and grid more resilient to climate change in the following countries: Botswana, CAR, Kenya, Mali, Namibia, and/or Uzbekistan

		operations through automatic generation control; and (iv) other innovative technologies whose commercial viability may not have been established yet such as load shifting battery storage.	
2.3.1 Public investments for electrification	Public investments for electrification (PPP mini-grids, grid extension etc.) by Botswana, CAR, DRC, Kenya, Mali and Namibia's EE	2.3.1.1 Investments in solar home systems (SHS), mini-grids and grid connection, as best appropriate to the country context, with a particular focus on women.	Supply of electricity through grid expansion, mini-grids and/or SHS to 840,000 households across 5 countries: Botswana, CAR, DRC, Kenya, Mali and Namibia

Component 3: Risk Mitigation instruments

3.1.1 Risk mitigation instruments for solar/battery mini-grids and off-grid private investors	Risk mitigation instruments for RE mini-grids and off-grid private investors by DRC's EE	3.1.1.1 Technical assessment (informed to the extent possible by a market sounding) to ensure that the coverage proposed meets the needs of the private sector investors, allowing long-term mobilization of private capital matching the needs of the projects	Identification of the terms and conditions of the reimbursable grants available for inclusion in the sets of bidding documents for RE mini-grid selections in DRC
3.2.1 Risk mitigation instruments for grid connected RE projects	Risk mitigation instruments for RE grid connected projects by Namibia and Botswana's EE	3.2.1.1 Technical assessment (informed to the extent possible by a market sounding) to ensure that the coverage proposed will meet the needs of the private sector investors, allowing long-term mobilization of private capital matching the needs of the projects.	Identification of the terms and conditions of the guarantees available for inclusion in the sets of bidding documents for RE competitive bidding in Namibia and Botswana

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Besides the arrangements laid out in the **AMA between the World Bank and the GCF**, the **implementation of the SRMI Facility involves** (i) monitoring of program performance indicators (as per Section E5); (ii) periodic progress reports; and (iii) midterm review of implementation progress. It is noted that the end targets of the Programme are best estimates as it is not known upfront exactly the results of each project that will be financed, and the targets may be revised during AE's due diligence, project preparation and the mid-term review as necessary. The AE will report to GCF the implementation status at both Facility and country level by aggregating specific country-level information and data.

▪ **Monitoring**

Specifically, the implementation of each project under the SRMI Facility will be managed and monitored at project and Facility level by World Bank's in-house staff with inputs from, as applicable, Project Implementation Units (which may include staff and consultants). The World Bank has dedicated staff in its energy team, operation departments, Environment and Social departments and country offices that will conduct due diligence, monitor compliance and performance risks, implementation of the Environmental and Social Commitment Plans and the stakeholder engagement plans.

The World Bank will follow its standard monitoring and reporting policies and procedures, which will include semi-annual reports on the implementation status and performance of each project. By the end of each project supported by the SRMI Facility, the Bank will conduct jointly with each respective Executing Entity an implementation completion assessment and prepare an implementation completion report (ICR) which will review the performance of the project, assess effectiveness and efficiency of project implementation, the achievement of the project development objective, and provide relevant lessons learned. Financial reporting is further discussed in Section G3.

▪ **Reporting**

Reporting of EEs to World Bank. As specified in the subsidiary agreements between the EEs and the World Bank, the EEs are obliged to report on the implementation of the Project and the use of proceeds of the Facility in accordance with the terms and conditions of the subsidiary agreements, including on the environmental and social performance of the project and the achievement of the indicators agreed between the AE and the EE to the World Bank on an annual basis. For each project supported by the SRMI Facility, the respective EE will be responsible for overall M&E of the project, including collection of project performance information and reporting on project impacts and results.

Reporting of World Bank to GCF. The World Bank will provide to the GCF (i) annual activity performance reports (APRs) on the status of the Facility throughout the relevant reporting period, (ii) mid-term evaluation reports at the midpoint of the implementation period of the Facility and (iii) final evaluation reports at the end of the implementation period of the Facility as per the FAA provisions. The APRs will provide information specific to projects supported under the Facility, as relevant.

▪ **Evaluation**

The evaluation methods for mid-term and final evaluations will include:

- a. Key informant interviews with relevant beneficiary staff and clients, as well as other key stakeholders, based on stakeholder mapping; and
- b. Desktop review of relevant Facility documentation, including Project Implementation Unit reports, based on an agreed list of evidence to be provided by the client.

Please refer to Annex 11 for further details.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

The main risks identified to the successful implementation of the Facility are linked to Governance (political and governance, macroeconomic and sector strategies and policies), Technical and Operational (technical design of projects, institutional capacity for implementation and sustainability, fiduciary and stakeholders), uncertain depth and duration of Covid-19 pandemic (with direct impact on macro-economic environment and possibly project preparation and implementation context in various countries) as well as Security in specific project countries.

Selected Risk Factor 1

Category	Probability	Impact
Governance	Medium	Medium

Description

Political and Governance. As DRC, CAR and Mali are facing complex political situations, it may impact their political commitment to the projects and the governance aspects of the implementation of the components:

- DRC is still recovering from a series of conflicts that broke out in the 1990s, creating a protracted economic and social slump. Joseph Kabila served as head of state between 2001 and 2019. Presidential and parliamentary elections, initially slated for November 2016, were postponed until December 30, 2018 to facilitate an overhaul of the electoral register. Felix Antoine Tshisekedi—the son of Etienne Tshisekedi, the country's longstanding opposition leader—secured 38.5 percent of the vote and was declared the winner by the national independent electoral commission. This victory was contested, including by the National Episcopal Conference of Congo (CENCO), which had deployed 40,000 election observers.
- CAR's elections in 2016 brought an end to three years of political transition and turmoil. Since 2016, CAR has been governed by President Faustin-Archange Touadéra and Prime Minister Simplicie Sarandji. The next presidential elections are scheduled to be held between December 2020 and March 2021.
- Mali has been experiencing instability and conflict since the military coup of 2012 and the occupation of the northern regions by armed groups. The operations of the United Nations Multidimensional Integrated Stabilization Mission in Mali (MINUSMA) date back to July 2014.

Macroeconomic. This risk category includes external and domestic macroeconomic risks that may derail the achievement of operations development results during either preparation or implementation. Overall, the seven countries are quite stable from a macroeconomic perspective. GDP growth in 2017 and 2018 has been negative in Namibia, between 2 and 4 percent in Kenya and Botswana and above 5 percent in DRC, Mali and Uzbekistan. Their inflation levels are aligned with their GDP growth levels with rates in 2018 between 0 to 5 percent. However, it is important to note that it is currently unknown how the covid-19 crisis will impact their economies. The emerging view in the case of Kenya is that the Covid-19 crisis is having a large negative impact on the economy, there has been a significant widening of the fiscal and external financing gaps and consequently, GDP growth has been downgraded.

Sector Strategies and Policies. It is critical to have in place the right regulatory framework: (i) Kenya are working on new legislation to implement competitive selection of IPPs, (ii) DRC does not have the full policy in

place to enable privately-owned mini-grid operations, and (iii) Botswana, Namibia and Mali are missing clear generation plans with implementation strategies linked to it.

Mitigation Measure(s)

The World Bank team is working closely with the different Governments to ensure the sustainability and relevance of the projects to the given country. In general, good governance of the projects will be ensured by the adoption of transparency, accountability and participation principles and attention to fraud, corruption and other unethical practices resulting from government failures. These will be enabled thanks to the World Bank implementation and M&E systems.

The projects are expected to strengthen the country's macro-fiscal situations as through the deployment of least-cost renewable energy it will reduce the cost of imported fossil fuel while reducing energy security risks and improving national capacity.

In addition, the World Bank is working with the seven countries to ensure that a comprehensive regulatory framework is in place when the new investments come in play aiming to ensure the sustainability of the new generation projects.

Selected Risk Factor 2

Category	Probability	Impact
Technical and operational	Medium	Medium

Description

Technical Design of projects. As solar and wind generation are well known technologies and there are clear investments related to VRE integration, the technical design risk is minimal. In addition, with a clear and fair risk allocation between the private and the public stakeholders, RE projects are least-cost compared to other generation projects.

Institutional Capacity for Implementation and Sustainability. The institutional capacity of the implementing agencies, implementation arrangements, and M&E arrangements is quite different from one country to the next. Indeed, Namibia, Botswana, Kenya and Uzbekistan have strong implementing agencies whereas CAR, DRC and Mali have weaker institutional capacity.

Risk of RE Projects Not Materializing. Most RE power plants under the Facility will be prepared thanks to public investments. However, the financing for the power plants themselves will be financed by private investors. The private investment is not covered under the Facility.

Mitigation Measure(s)

To ensure that this risk is mitigated, the World Bank is supporting the proposed seven countries in their selection of the private investors. For the countries with limited capacity, larger capacity building components will be provided with additional external support for project implementation. A transparent and well-organized selection of private investors combined with the reduction in risks from curtailment, land and E&S perspective

thanks to the park scheme will ensure that private investment is indeed leveraged as a result to the Facility's projects.

Each country has a different fiduciary risk and it is being mitigated by previous experience with each client, and World Bank procurement and financial management requirements.

Selected Risk Factor 3

Category	Probability	Impact
Other	Medium	Medium

Description

Security. Security risks are key in countries such as Mali, CAR and DRC:

- Mali's security, which is critical for economic recovery and poverty reduction, remains fragile in the face of continued attacks by armed groups on UN peacekeepers, the Malian army, and civilians, mainly in the north and central regions. In July 2017, Germany, the African Development Bank, the World Bank, France, the European Union, and the United Nations Development Programme launched the Sahel Alliance in order to provide a coordinated and tailored response to the challenges faced by the G5 Sahel member countries (Burkina Faso, Chad, Mali, Mauritania, and Niger). Since that time, Denmark, Italy, Luxembourg, the Netherlands, Spain, and the United Kingdom have joined the Alliance.
- On February 6, 2019, the Government of CAR signed an African Union-mediated peace agreement with 14 armed groups. This agreement has been endorsed by the international community. The African Union and the Economic Community of Central African States (ECCAS) serve as guarantors of the agreement, while the UN peacekeeping force (MINUSCA) plays a critical support role in the background.
- DRC is fraught with political instability, armed clashes, and human rights violations. Conflict erupted in 2016 in the Kasai region, which includes five provinces in the center of the country. It is another instance of fighting between the military and splintered ethnic militias. Nationally, 2.1 million people were newly displaced in 2017 and 2018, making the DRC the African country with the highest number of internally displaced people — 4.5 million.

Mitigation Measure(s)

The security risk will be monitored, and the projects will rapidly adapt to any change in security levels in the three countries.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

The World Bank has a new Environmental & Social Framework (ESF) that will be used by all projects.¹⁴ The ESF is divided in ten E&S standards (ESS) that are each assessed based on their relevance depending on the project's concept. The main aspects considered at this stage from an E&S perspective are summarized below. However, each project will have its own assessment completed during project preparation. As a few of the projects aim to finance 500kV lines to unlock GWs of VRE, these projects as part of the Facility will be Category A following the GCF classification and may be classified as high or substantial risk as per the World Bank ESF classification.

▪ **ESS1: Assessment and Management of E&S Risks and Impacts**

The projects are expected to bring substantial direct environmental benefits by increasing the deployment of renewable energy. Transitioning to a higher percentage of renewable energy in the electricity mix has benefits for countries with regards to GHG emissions, air and water pollution, and use of water resources. The projects are likely to result in significant employment generation, including construction jobs, followed by a smaller number of long-term jobs associated with operation and maintenance (O&M) activities. There are substantial opportunities to promote female employment (direct and indirect) through training and incentives.

If not properly managed, the downstream development of the infrastructure for the solar/wind parks, as well as the construction and operation of the RE power plants may result in potential land conversion (and potentially degradation), habitat loss, excessive water use, discharge of hazardous wastes in manufacturing of PV solar cells and wind turbines and disposal of solar cells, and risks to worker and community health and safety. However, most of these potential impacts are low to moderate in intensity/significance and are reversible and localized in nature, and therefore can be easily prevented, e.g. by aligning site selection with ESF requirements, and mitigated, e.g. by sound operational practice during operation.

Under ESS1, an E&S Commitment Plan (ESCP) will be required, and it will specify all other E&S plans and instruments that would be required for each project. The ESCP is a legally binding document that is referred to in the legal agreements between the Bank and the EEs/implementing agencies.

▪ **ESS2: Labor and Working Conditions**

Project workers are expected to be mobilized under all the projects. The EEs/implementing agencies will have to comply with the ESS2 requirements as presented in Labor Management Procedures (LMP) to be prepared for each project.

▪ **ESS3: Resource Efficiency and Pollution Prevention and Management**

The downstream risks and impacts related to potential water scarcity (also considering seasonal water availability and any cumulative effects by multiple demands), the release of pollutants, waste generation, the management of disposal materials and hazardous wastes, impacts on communities, and resource use

¹⁴ Depending on the final design of the projects to be financed out of Component 3, the World Bank Group Performance Standards may apply to some of the financed activities under these projects, in accordance with the World Bank's relevant policies and procedures.

efficiency will be assessed for each project. However, most of these potential impacts are low to moderate in intensity/significance and are reversible and localized in nature, and therefore can be easily mitigated.

During the construction phase of the solar and wind parks, there may be some potential negative environmental and social impacts including, air quality deterioration due to dust and exhaust emissions during construction activities. These impacts require appropriate assessments and management measures to contain them.

▪ **ESS4: Community Health and Safety**

ESS4 requirements will be taken into account in the implementation of project's activities. While the construction and operation of solar/wind power plants has risks accompanying both normal operations and potential accidents, there have to date not been any accidents of the severity that have occurred in other forms of energy production (e.g. in hydropower). There is a possibility of noise emissions from the running of construction machinery and related transportation activities, community nuisance etc. These impacts require appropriate assessment and management measures to contain them.

▪ **ESS5: Land Acquisition, Restrictions on Land Use and Involuntary Resettlement**

Land needs for solar and wind parks are substantial. Any potential land acquisition or land use restriction will be assessed and managed in a manner consistent with ESS5 requirements, in order to avoid, minimize, reduce/mitigate and compensate at replacement cost land acquisition based on due diligence and plans prepared in accordance with ESS5.

▪ **ESS6: Biodiversity Conservation and Sustainable Management of Living Natural Resources**

The proposed solar/wind projects, if not properly designed and sited, may result in loss, degradation or fragmentation of habitats and other adverse impacts on biodiversity and living natural resources. Care will be taken to ensure that the impacts of such projects on natural habitats, terrestrial and aquatic ecosystems and species are well assessed and managed.

▪ **ESS7: Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities**

The presence of indigenous people in the project areas or their collective attachment to such areas will be screened and confirmed by the World Bank in consultation with the national authorities. Once it is confirmed, consistency with ESS7 objectives and requirements will be integrated into the design of the project and E&S documents, including the development of an indigenous peoples' development plan or other types of plans depending on the country and project context, and Free, Prior and Informed Consent in the circumstances specified in ESS7.

▪ **ESS8: Cultural Heritage**

Each project will screen the project area and assess potential impacts on cultural heritage, both tangible and intangible. The site selection for each project will be mindful of cultural heritage and the application of ESS8 will be further assessed under each project.

▪ **ESS9: Financial Intermediaries**

The projects are not expected to be financial intermediary projects.

▪ **ESS10: Stakeholder Engagement and Information Disclosure**

Project stakeholders can include communities or individuals affected by the project and their formal and informal representatives, national or local government authorities, politicians, religious or community organizations and civil society groups with special interest, academic communities, and businesses. The identification of project-affected parties (individuals or groups) will also include those who, because of their circumstances, may be disadvantaged or vulnerable; i.e. those who may be more likely to be adversely affected by the project impacts and/or more limited than others in their ability to take advantage of the project benefits.

A Stakeholder Engagement Plan (SEP) will be developed for each project, to ensure transparency and meaningful consultation with the project-affected and interested parties. Stakeholder engagement and consultations will be conducted throughout the project cycle. This will include discussions of project design and impacts as well as multi-stakeholder discussions on these issues during the preparation phase. The SEP, along with other E&S instruments, will be subject to public consultation and disclosure per requirements of ESS10 and will be treated as a living document to be regularly updated as needed during project implementation.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

The Facility aims to enhance female economic participation in the energy sector and create income-generating opportunities for women in project countries. Proposed interventions under the Gender Action Plan are both aligned with the World Bank Group Gender Strategy (2016-2021): (i) advancing equality between men and women through climate change mitigation and adaptation actions, and (ii) minimizing gender-related risks in climate change actions. Employment opportunities created by large-scale solar and wind infrastructure projects often benefit men more than women, resulting in a significant employment gap in the RE industry. Female access to electricity can at times also be limited in target countries, and specific attention to women's connectivity will be key in ensuring men and women equally gain from energy services. Under the SRMI approach, issues related to female employment and access to electricity will be effectively embedded across the RE project-cycle, from the procurement of the RE project to its implementation, going beyond whenever possible to also enhance livelihood in affected areas.

These issues will be tackled in a sustainable manner, raising awareness, improving gender policies, building capacities and generating knowledge on the program's differentiated impact on men and women. Preliminary gender assessments for each of the seven countries under the Facility highlighted the following gender gaps and their potential mitigation actions:

- a. **Female employment and skill gaps in the energy sector:** the main barriers to women's participation in the energy sector include social norms that confine women to unpaid household and care work, occupational gender segregation, limited capacities and skills, as well as discriminatory work environments. Worldwide, the RE industry employs only about 32 percent of women. Therefore, without a pro-active focus on institutional reform to promote female employment gender-policy, men may benefit disproportionately from new employment opportunities across the RE value chain,

particularly in construction, operation and maintenance. This gap will be addressed through skills building and, outreach efforts, as well as the promotion of equal opportunities practices at the corporate level (including through programs to afford women the ability to gain hands-on professional experience).

- b. **Female employment in the areas of implementation of RE projects:** RE projects are frequently located in remote areas with good solar/wind resources and large plots of lands, but where living standards are below the country average. Women, if employed, are more exposed to lower-paid, lower-quality employment in the informal sector. Under the SRMI approach, the socio-economic needs of the local communities will be assessed, and actions empowering women prioritized. Those may include investments in education and skill-building, for instance in traditional home-based activities such as handicrafts, or in agricultural activities. Special attention will be paid to the promotion of local women-run businesses along the project development cycle.
- c. **Gender-based violence:** women routinely experience sexual harassment and intimate partner violence in target countries, and the risk of gender-based violence may increase during the realization of large-scale RE projects. This risk will be mitigated through implementing mitigation and response approaches and the adoption of internal prevention and response mechanisms in compliance with the World Bank Group GBV practices. These interventions are expected to efficiently complement actions empowering local women and men and aim to create a safe environment at the project site and in communities.
- d. **Women access to services as energy consumers:** among project countries with lower electrification rates and under-developed grids, women suffer from time-poverty and increased health-issues, as they bear the responsibility of food preparation, as well as fuel and water collection. Limited access to electricity among poor households, including female-headed households (FHH) can also be explained through affordability constraints regarding electricity connection or use of energy services. Improving access to electricity services (off-grid and on-grid) may reduce the time and labor burden of women, generate health and education benefits, increase female labor participation and incomes. For these reasons, the EEs will be required to (i) pay attention to women and men's differentiated needs in the design of electrification programs, (ii) conduct inclusive stakeholders' consultations to ensure women's voice is heard and included in decision-making processes, (iii) undertake outreach and capacity-building initiatives to increase women's knowledge about new forms of energy, and (iv) prioritize FHH in the design of actions to assist poorer households with connection charges.

The Facility will follow a systematic approach and provide cross-support to projects interventions with a focus on data collection, policy engagement, project level actions and capacity-building activities. The WBG ESMAP Gender and Energy Program will be available to provide assistance to the SRMI facility, as part of its mandate to strengthen women's role in the energy sector as consumers, employees and entrepreneurs. The Facility will encourage the recruitment of a balanced female-to-male employment ratio within the PMU team.

Obtaining stakeholders buy-in will be key in ensuring that interventions to promote women's employment and enhance their livelihood are effectively implemented with measurable outcomes. Countries adherence to international agreements and national normative frameworks on gender equality,

will be a good starting point to strengthen government's support. At the project and program level, the following approach will be followed:

a. Data collection and initial assessments:

- Build upon existing country gender diagnostics and action plans;
- Document the baseline situation and identify gaps through collection of sex-disaggregated data;
- Conduct stakeholders' consultations involving female participants, to understand men and women differentiated needs; and
- Include an analysis of the differentiated impact of the SRMI project on men and women in initial assessments, through the involvement of gender and energy experts (Environmental and Social Impact Assessment, Resettlement Action Plans etc.).

b. Identifying and implementing relevant actions:

- Identify and implement relevant interventions to narrow gaps in terms of employment and energy access, based on the initial diagnosis and inclusive stakeholders' consultations; and
- These may include (i) actions to promote female employment in the renewables sector (including capacity building and programs for women to gain hands-on experience in the energy sector such as support for school-to-work transition), (ii) financing of local communities projects empowering women in the area of implementation of the RE projects (such as activities generating revenues), (iii) implementation of prevention measures to mitigate GBV in the area of implementation of the RE projects and (iv) facilitating of the energy access for FHH).

c. M&E and Knowledge management:

- Monitor results throughout the project cycle to assess impact on men and women and allow for real-time adjustments; and
- Expand the knowledge base by documenting lessons learnt from prior projects, knowledge production and exchange.

d. Strengthening female participation and voice during Stakeholders Consultations:

- Organize stakeholders' consultations throughout the project cycle. Ensure effective participation of women while conducting consultations to gain knowledge about their specific concerns during project preparation, and to monitor impacts and allow for real-time correction during implementation;
- Use local dialects and involve female facilitators/local women leaders, as well as women representatives from the utility company whenever possible;
Organize women-only discussion groups if appropriate; organize consultations and focus groups at a convenient time for women; and create partnerships with women's groups and nongovernmental organizations (NGOs) to assist in the process and ensure the projects' sustainability.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Each EE of loan/grant financing under the SRMI Facility will be required to conduct financial management and procurement according to the World Bank policies and procedures.

■ Procurement

Each project supported by the SRMI Facility will follow the World Bank's Procurement Regulations.¹⁵ World Bank procurement staff provide implementation support to the EEs throughout the procurement process. They work with governments to achieve the highest bidding and contract management standards to get the best development results. Each project is developed and supervised by a World Bank task team that includes a procurement specialist. Each project with the exception of the guarantees is also subject to the 'World Bank's Anti-Corruption Guidelines, dated October 15, 2006 and revised in January 2011 and as of July 1, 2016'.¹⁶

■ Financial Management

Each project with the exception of the guarantees supported by the Facility is required to maintain financial management arrangements (including planning and budgeting, accounting, internal control, funds flow, financial reporting, and auditing arrangements of the EE and other entities responsible for project implementation) acceptable to the World Bank. For each project, the task team includes an accredited Financial Management (FM) specialist who provides support during the entire project cycle (preparation, supervision and implementation) in accordance with the Bank's operational policies and procedures (to work with partner countries to design financial management and disbursement arrangements and to supervise and support the performance of Bank-financed operations). The team's FM specialist closely monitors the performance of the EE's FM arrangements and their continued capacity to provide reasonable assurance that financing proceeds are being used for the purposes intended.

Financial reporting will also be required for each project and (with the exception of the guarantees) is specified in each project's Disbursement and Financial Information Letter (DFIL). Unless otherwise agreed by the Bank, the EE must submit annual audited project financial statements six months after the close of the financial year and unaudited interim financial (IFR) reports periodically. Audits are carried out by auditors with independence and capacity acceptable to the Bank, under terms of reference acceptable to the World Bank.

G.4. Disclosure of funding proposal

- ☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.
- ☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

¹⁵ <https://www.worldbank.org/en/projects-operations/products-and-services/brief/procurement-new-framework#framework-with-the-exception-of-guarantee-projects,-to-which-these-regulations-do-not-apply>.

¹⁶ Guarantees are subject to a similar anti-corruption regime defined in the Anti-Corruption Guidelines for World Bank Guarantee and Carbon Finance Transactions (<https://policies.worldbank.org/sites/ppf3/PPFDocuments/Forms/DispPage.aspx?docid=1971>) and to similar restrictions on participation by debarred entities.

- ☐ full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- ☐ redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☒ Environmental and Social Sustainability Framework (ESSF)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☐ Annex x Other references

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

TELEPHONE: 3950100

FAX: 3956086

Toll free: 0800600773



REPUBLIC OF BOTSWANA

MINISTRY OF FINANCE AND ECONOMIC
DEVELOPMENT
PRIVATE BAG 008
GABORONE

REF: MFED 170/1/4A

Date: 22nd April 2020

Executive Director
Secretariat of the Green Climate Fund
175 Art Center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea

Dear Madam/Sir,

RE: Funding Proposal for the Green Climate Fund (GCF) by the World Bank Regarding the Solar Risk Mitigation Initiative (SRMI) Facility

We refer to the renewable energy project in Botswana as included in the funding proposal submitted by The World Bank for the Solar Risk Mitigation Initiative (SRMI) Facility.

The undersigned is the duly authorized representative of Ministry of Finance and Economic Development, the National Designated Authority of Botswana.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the renewable energy project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The Government of Botswana has no-objection to the renewable energy project as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Botswana's national priorities, strategies and plans; and
- (c) In accordance with the GCF's environmental and social safeguards, the renewable energy project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the renewable energy project as included in the funding proposal has been duly followed.

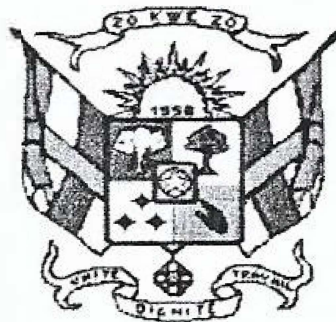
We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the renewable energy project.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours faithfully,

A handwritten signature in black ink, appearing to read 'Boniface G. Mphetlhe', written over a horizontal line.

Boniface G. Mphetlhe
for/Permanent Secretary



20 AVR 2020

ATTENTION : Le Fonds Vert Climat (FVC)

Re : Proposition de financement pour le FVC de la Banque mondiale concernant la facilité SRMI

Chère Madame, Monsieur,

Nous nous référons au projet d'énergie renouvelable en République Centrafricaine tel qu'il est inclus dans la proposition de financement.

Le soussigné est le représentant dûment autorisé de la Coordination Nationale Climat, l'autorité nationale désignée nationale pour le FVC en République Centrafricaine.

Conformément à la décision FVC B.08/10, dont nous reconnaissons avoir examiné le contenu, nous communiquons par la présente notre Non-Objection au projet d'énergie renouvelable tel qu'inclus dans la proposition du financement.

En communiquant notre non-objection, il est sous-entendu que :

- (a) Le gouvernement de la République Centrafricaine n'a pas d'objection au projet d'énergie renouvelable tel qu'il est inclus dans la proposition de financement ;
- (b) Le programme tel qu'il figure dans la proposition de financement est conforme aux priorités, stratégies et plan nationaux de la République Centrafricaine ;
- (c) Conformément aux sauvegardes environnementales et sociales du FVC, le projet d'énergie renouvelable tel qu'il figure dans la proposition de financement est conforme aux lois et réglementations nationales applicables

Nous confirmons également que notre processus national de vérification de la non-objection au projet d'énergie renouvelable tel qu'inclus dans la proposition de financement a été dûment suivi.

Nous confirmons également que notre non-objection s'applique à tous les projets ou activités à mettre en œuvre dans le cadre du projet d'énergie renouvelable.

Nous reconnaissons que cette lettre sera rendue publique sur le site internet du FVC.

Respectueusement,

Nom : Denis BEINA

Titre : Point Focal FVC

The World Bank

INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT
INTERNATIONAL DEVELOPMENT ASSOCIATION

1818 H Street N.W.
Washington, D.C. 20433
U.S.A.

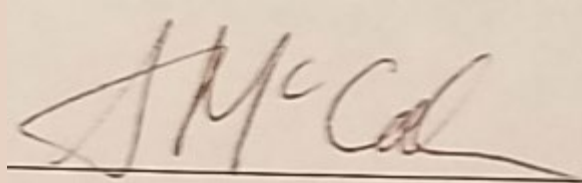
(202) 473-1010
Cable Address: INTBAFRAD
Cable Address: INDEVAS

August 5, 2020

To Whom It May Concern:

Certificate of Accuracy – Job No. 2021000388

I, Debyanne McCook, Senior Translator/Reviser in GCSTI, hereby certify on behalf of GCSTI that the attached translation from French to English of the document submitted under Job No. 2021000388 was revised by me and is, to the best of my knowledge and belief, a true and accurate translation of the original French document.

A handwritten signature in dark ink, appearing to read 'J McCook', is written over a horizontal line.

Debyanne McCook
Senior Translator/Reviser
English Translation Team (GCSTI)

Central African Republic
Ministry of Environment and Sustainable Development
Ministerial Directorate
Coordination Nationale Climat
No. 009/MEDD/DIRCAB

To: The Green Climate Fund ("GCF")

April 20, 2020

Re: Funding proposal for the GCF by the World Bank regarding the SRMI Facility

Dear Madam, Sir,

We refer to the renewable energy project in the Central African Republic as included in the funding proposal.

The undersigned is the duly authorized representative of the *Coordination Nationale Climat* [National Climate Coordination Office], the GCF National Designated Authority of the Central African Republic.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the renewable energy project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Central African Republic has no-objection to the renewable energy project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with the Central African Republic's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the renewable energy project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the renewable energy project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the renewable energy project.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

/s/

Name: Denis Beina

Title: GCF Focal Point

République Démocratique du Congo
Ministère de l'Environnement et Développement
Durable



Secrétariat Général à l'Environnement et Développement
Durable

Direction de Développement Durable
Coordination Nationale du Mécanisme de Développement
Propre et du Fonds Vert pour le Climat

L'Autorité Nationale Désignée

To: The Green Climate Fund ("GCF")

Kinshasa, 04/22/2020

Re: Funding proposal for the GCF by the World Bank entitled "DRC
Accelerating Electrification".

Dear Mr. Yannick, Glemarec

We refer to the Funding Proposals by the World Bank as accredited entity entitled "DRC Accelerating Electrification" submitted to us on 04/21/2020. The undersigned is the duly authorized representative of National Coordination of Clean Development Mechanism and Green Climate Funds, the National Designated Authority of the Democratic Republic of Congo (DRC).

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the Project as included in the Funding Proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the DRC has no-objection to the Project as included in the Funding Proposal.
- (b) The Project as included in the Funding Proposal is in conformity with DRC's national priorities, strategies and plans; and
- (c) In accordance with the GCF's environmental and social safeguards, the Project as included in the Funding Proposal is in conformity with relevant national laws and regulations.

Avenue Papa Ileo (Ex-des Cliniques) n°15 Kinshasa/Gombe
B.P.: 12348 KIN I – E-mail : hansandredjamba@gmail.com
Site Web: www.fvc-rdc.com

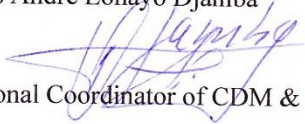
We also confirm that our national process for ascertaining no-objection to the Project as included in the Funding Proposal has been duly followed.

We also confirm that our no-objection applies to all activities to be implemented within the scope of the Project.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Hans Andre Lohayo Djamba



National Coordinator of CDM & GCF

Cc: Excellency Minister of Sustainable Development and Environment, General Secretary of the Ministry of SD& Environment, Unit of Energy Project Management

*Avenue Papa Ileo (Ex-des Cliniques) n°15 Kinshasa/Gombe
B.P.: 12348 KIN I – E-mail: hansandredjamba@gmail.com
Site Web: www.fvc-rdc.com*



REPUBLIC OF KENYA

THE NATIONAL TREASURY AND PLANNING

Telegraphic Address: 22921
FAX NO. 310833
Telephone: 2252299

THE NATIONAL TREASURY
P O BOX 30007 – 00100
NAIROBI

When Replying Please Quote

Ref: TNT/CONF/36/021/C/TY (45)

Date: July 9, 2020

Mr. Yannick Glemarec
Executive Director
Green Climate Fund
G-Tower, 24-4 Songdo-dong
Yeonsu-gu
Incheon City, Republic of Korea

Dear

Yannick

Re: Funding proposal for the GCF by The World Bank regarding ‘Sustainable Renewables Risk Mitigation Initiative (SRMI).’

We refer to the project “Sustainable Renewables Risk Mitigation Initiative (SRMI)” as included in the funding proposal submitted by the World Bank to us on June 30, 2020.

The undersigned is the duly authorized representative of the National Treasury, the National Designated Authority of Kenya.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The Government of Kenya has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with Kenya’s national priorities, strategies and plans;
- (c) In accordance with the GCF’s environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed. Further, we confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours

Sincerely



JULIUS M. MUIA, PhD, CBS
PRINCIPAL SECRETARY/NATIONAL TREASURY

Copy to: **Hon. (Amb.) Ukur Yatani Kanacho, EGH**
Cabinet Secretary
The National Treasury and Planning
NAIROBI

Hon. Charles Keter, EGH
Cabinet Secretary
Ministry of Energy
NAIROBI

Dr. Eng. Joseph K. Njoroge, CBS
Principal Secretary
Ministry of Energy
NAIROBI

Agence de l'Environnement et du
Développement Durable
(AEDD)



N° **0497**/MEADD-AEDD-DGA.

Bamako, le ... **10 4 JUIN 2020**

*The Director General of the Environment and
Sustainable Development Agency*
To

Mr Yannick Glemarec
Director Executive of Green Climate Funds
G-Tower, 24-4 Songodo dong Yeonsu gu
Incheon City Republic of Korea

Re: Funding proposal for the GCF by World Bank regarding the Programme « Sustainable Renewables Mitigation Initiative (SRMI) »

Dear Madam, Sir,

We refer to the Programme « Sustainable Renewables Mitigation Initiative (SRMI) » in Mali as included in the funding proposal submitted by the World Bank to us on the 28th of Mai 2020.

The undersigned is the duly authorized representative of Environment and Sustainable Development Agency, the National Designated Authority of Mali.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Mali has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with Mali's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

The Director general


Bouréma CAMARA
Chevalier de l'Ordre national





REPUBLIC OF NAMIBIA

MINISTRY OF ENVIRONMENT, FORESTRY AND TOURISM

Private Bag 13306
Windhoek
Namibia

Tel. no. 061 – 2842701
Fax. No. 061 – 240339
Corner of Robert Mugabe &
Dr. Keneth Kaunda Street

Enquiries: Petrus Muteyauli

Email: petrus.muteyauli@met.gov.na
pmuteyauli@yahoo.co.uk

Date: 20 April 2020

The Secretariat
Green Climate Fund
G-Tower, 24-4 Songdo-dong, Yeonsu-gu Incheon
Republic of Korea

RE: FUNDING PROPOSAL FOR THE GCF BY THE WORLD BANK REGARDING THE SRMI FACILITY

Dear Sir,

We refer to the renewable energy project in Namibia as included in the funding concept note submitted by The World Bank for the SRMI Facility.

The undersigned is the duly authorized representative of the Ministry of Environment, Forestry and Tourism, the National Designated Authority/focal point of Namibia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the renewable energy project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Namibia has no-objection to the renewable energy project as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Namibia's national priorities, strategies and plans;

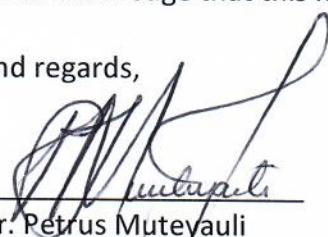
- (c) In accordance with the GCF's environmental and social safeguards, the renewable energy project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the renewable energy project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the renewable energy project.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Mr. Petrus Muteyauli

**DEPUTY DIRECTOR AND NATIONAL DESIGNATED AUTHORITY /
FOCAL POINT FOR THE GCF**

MINISTRY OF ENVIRONMENT AND TOURISM
2020 -04- 2 0
Private Bag 13306 WINDHOEK / NAMIBIA



№07-17 - 04359

To: The Green Climate Fund ("GCF")

Tashkent, 25/04/2020.

Re: Funding proposal for the GCF by The World Bank regarding the Sustainable Risk Mitigation Initiative (SRMI) Facility

Dear Madam, Sir,

We refer to the *Electricity Transmission Modernization and Market Development Project (ETMMD)* in Uzbekistan as included in the funding proposal submitted by the World Bank for the SRMI Facility which aims at unlocking RE deployment by mobilizing sustainable private investments by addressing key risks including integration risks. The ETMMD project supports digitalization of the electricity transmission sector and strengthening and modernization of the Uzbek power grid essential to enable efficient integration and balancing of higher targets of renewable energy resources aimed for development in Uzbekistan and included in the Government's electricity sector development program.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the *Electricity Transmission Modernization and Market Development Project* as included in the funding proposal. We also confirm that the project is part of the electricity sector development priorities and will be implemented in conformity with relevant national laws and regulations and environmental and social safeguards policies including those applicable to the project by the World Bank.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Name: Abidov Badriddin

Title: Deputy Minister, Ministry of Investment and Foreign Trade,

Republic of Uzbekistan, GCF focal point for Uzbekistan

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Sustainable Renewables Risk Mitigation Initiative (SRMI) Facility
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	World Bank
Environmental and social safeguards (ESS) category	Category A
Location – specific location(s) of project or target country or location(s) of programme	The target countries under the SRMI Facility are Botswana, Central African Republic (CAR), Democratic Republic of Congo (DRC), Kenya, Namibia, Mali, Pakistan, Uzbekistan and Vietnam.
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Sunday, June 21, 2020
Language(s) of disclosure	English, French, Uzbek, and Vietnamese
Explanation on language	English is one of the official languages of Botswana, Kenya, Namibia and Pakistan. French is one of the official languages of CAR, DRC, Mali. Uzbek is the official language of Uzbekistan. Vietnamese is the official language of Vietnam.
Link to disclosure	English: https://esmap.org/sites/default/files/esmap-files/A6_SRMI_ESSF_English_vF.pdf French: https://esmap.org/sites/default/files/esmap-files/A6_SRMI_ESSF_French_vF.pdf Uzbek: https://esmap.org/sites/default/files/esmap-files/A6_SRMI_ESSF_Uzbek_vF.pdf Vietnamese: https://esmap.org/sites/default/files/esmap-files/A6_SRMI_ESSF_Vietnamese_vF.pdf
Other link(s)	https://esmap.org/reportsrmi-facilityenvironmentalsocialsustainabilityframewo
Remarks	An ESIA template consistent with the requirements for a Category A project is contained in the “Environmental and Social Sustainability Framework”.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Sunday, June 21, 2020

Language(s) of disclosure	English, French, Uzbek, and Vietnamese
Explanation on language	English is one of the official languages of Botswana, Kenya, Namibia and Pakistan. French is one of the official languages of CAR, DRC, Mali. Uzbek is the official language of Uzbekistan. Vietnamese is the official language of Vietnam.
Link to disclosure	English: https://esmap.org/sites/default/files/esmap-files/A6_SRMI_ESSF_English_vF.pdf French: https://esmap.org/sites/default/files/esmap-files/A6_SRMI_ESSF_French_vF.pdf Uzbek: https://esmap.org/sites/default/files/esmap-files/A6_SRMI_ESSF_Uzbek_vF.pdf Vietnamese: https://esmap.org/sites/default/files/esmap-files/A6_SRMI_ESSF_Vietnamese_vF.pdf
Other link(s)	https://esmap.org/reportsrmi-facilityenvironmentalsocialsustainabilityframewo
Remarks	An ESMP template consistent with the requirements for a Category A project is contained in the “Environmental and Social Sustainability Framework”.
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	N/A
Place	N/A

Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	TBC
Date of GCF's Board meeting	Thursday, 22 October 2020

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP163

Proposal name:	Sustainable Renewables Risk Mitigation Initiative (SRMI) Facility
Accredited entity:	World Bank
Country/(ies):	Botswana, Central African Republic (CAR), Democratic Republic of the Congo (DRC), Kenya, Mali, Namibia, and Uzbekistan
Project/programme size:	Large

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
SRMI proposes a streamlined process with a one-stop shop approach from technical assistance to financing the critical public investments and the risk mitigation coverage needed to enable private investments at scale. This integrated approach can make a difference, particularly when countries are prioritizing the scarce resources available for an emergency response to COVID-19, focusing on healthcare.	The SRMI Facility and the projects for investment are at pre-concept stage; proper assurances, including implementation arrangements need to be in place to ensure the Facility will deliver the impact as indicated in the proposal. It would be important that the accredited entity strictly comply with the agreed set of parameters in the term sheet throughout the process.
GCF funding is critical to address the current renewable energy market failures identified and to ensure that countries continue their low carbon development pathway.	
The seven countries are representative of different types of markets/energy sectors, with various levels of electrification rates, low to high cost of generation, small to large grids and so on. The success and lessons learned from these seven countries will enable replication of the SRMI approach across countries and facilitate unlocking sustainable renewable energy markets for private investments.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVII, titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Achieving global goals of climate change mitigation and universal access to electricity will require rapid growth in the share of variable renewable energy (VRE) used in global energy generation. According to the Sustainable Development Scenario of the International Energy Agency (*World Energy Outlook*, 2018), capacity to generate 950 gigawatts (GW) using solar photovoltaics (PV) and 580 GW from wind power need to be installed by 2025 in developing countries. This represents an investment of USD 150 billion capital expenditure needed annually for wind and solar deployment. To reach this objective, large amounts of private finance will have to be unlocked to complement the limited public funding available.

4. The falling technology costs make investment in renewable energy (RE) commercially viable in more and more countries. Nevertheless, global investment remains below the targets needed to meet the Paris Agreement objectives and its potential. A range of constraints can obstruct private sector financing into RE projects, many of which are closely related to the national environment, including (i) the limited generation and transmission planning capacity of utilities to organize and contract future generation, which has resulted in additions to power generation capacity on an ad hoc and rarely optimum basis from a systems perspective; (ii) lack of adequate public-private partnership or independent power producer (IPP) regulatory framework that can ensure the development of IPP-owned generation; (iii) weak competitive procurement capacity to select the independent power producers that has often led to power purchase agreements (PPAs) with more expensive tariffs than if reached under structured competitive bidding; (iv) limited financial viability of off-takers, which resulted in poor creditworthiness to leverage private sector investments; and (v) grid integration challenges to support uptake of more VRE because of grid technical limitations and lack of flexibility in the grids.

5. The proposed SRMI aims to address these challenges by creating an enabling environment that is more conducive to private investment while maximizing socioeconomic benefits for the countries in the context of COVID-19, which has increased challenges and fragility. The programme proposes various de-risking instruments for each of the identified constraints with a one-stop shop approach from technical assistance to financing the critical public investments and the coverage needed for risk mitigation to enable private investment at scale. SRMI offers climate financing for three core aspects:

- (a) Technical assistance to help countries develop evidence-based VRE targets, implement a sustainable RE programme, and maintain robust procurement processes with transaction advisors;
- (b) Critical public investments to enable integration of VRE, finance solar/wind park infrastructure, and increase access to electricity; and
- (c) Risk mitigation instruments to cover residual risks perceived by private investors.

6. The SRMI Facility will start with seven countries, namely Botswana, CAR, DRC, Kenya, Mali, Namibia, and , Uzbekistan. The seven countries were selected based on the willingness and interest of their respective governments to deploy RE and combat the challenges they face in the implementation of their nationally determined contribution (NDC) and low-emissions development pathways. They are representative of different types of markets/energy sectors, with various levels of electrification rates, low to high cost of generation, small to large grids and so on. The success and lessons learned from these seven countries will enable the SRMI approach to be replicated across countries and facilitate unlocking sustainable RE markets for private investments.

7. Table 1 provides details of the proposed financing plan for the SRMI Facility. The plan is financed through GCF financing with co-financing from the International Bank for Reconstruction and Development/International Development Association (IBRD/IDA) in the

form of loan, and IDA and the Energy Sector Management Assistance Programme (ESMAP) in the form of grants.

Table 1: Estimated financing plan of Sustainable Renewables Risk Mitigation Initiative

Sources (USD million)		Uses of GCF proceeds	
GCF grant	24	Technical assistance and project management cost in seven countries	
GCF grant	30 (to CAR)	Finance to public infrastructure	
GCF concessional loan	176	(i) Common infrastructure for solar/wind parks: 48 (ii) Grid upgrade/resilience: 138 (iii) Mini grids: 20	
GCF reimbursable grant	30	Providing risk mitigation instrument (reimbursable grant) for CAR	
GCF guarantee	20	Providing guarantee for renewable energy-connected private projects as necessary in Botswana and Namibia	
World Bank ESMAP grant	18		
IDA grant	65		
IBRD concessional loans	450		
IDA concessional loans	750		
Total	1,563.5	Total (GCF)	280

Abbreviations: CAR = Central African Republic, ESMAP = Energy Sector Management Assistance Programme, IBRD = International Bank for Reconstruction and Development, IDA = International Development Association.

8. The total GCF funding contribution of USD 280 million is co-financed with USD 1.28 billion concessional finance from the accredited entity (AE). On top of AE co-financing, the AE further projects that SRMI would mobilize an estimated USD 3.3 billion of private capital to implement the RE projects downstream.

9. The majority of GCF concessional financing (USD 128 million out of 206 million) will be devoted to the grid upgrades because these upfront investments are preconditions for VRE integration and unlocking private sector finance for the RE plants that could be made commercially viable. Such upfront investments are usually borne by public parties and this constitutes an incremental cost at system level when investing in renewable energy, leading to a tough political and economic decision when a government has to prioritise. The concessional finance provided by GCF is key and catalytic particularly under the current COVID-19 crisis when countries are challenged and fragile and would prioritize public expenditure on the health and welfare of their population.

10. In addition to the concessional financing, the AE proposes to put in place a technical assistance grant of USD 22.5 million, out of which GCF has been requested to provide USD 20 million, the balance being co-financed by the AE. The grant component will be used to build institutional capacity of countries for energy planning, put in place the legal framework for IPP procurement, and preparation of preparatory and feasibility studies and environmental and social risk mitigation instruments, for the development of solar PV, solar PV with battery storage, CSP, and onshore wind parks in support of investment de-risking and investor selection process.

11. The programme is expected to contribute to the shift to low-emission sustainable development pathways as summarized by the AE in the funding proposal and as follows:
 - (a) Directly support the development of an estimated 2.4 GW of RE, (i) 1.7 GW of solar PV; (ii) 0.4 GW of concentrated solar plants (CSP); and (iii) 0.3 GW of onshore wind; Of this, 2 GW are expected to be built during the lifetime of the projects and 0.4 GW are expected to be built after the end of the projects. It will also enable the integration of another 1.9 GW of RE. The integration of other projects is enabled due to the VRE integration analysis and grid investments;
 - (b) Directly reduce greenhouse gas (GHG) emissions by an estimated 97.82 million tonnes of carbon dioxide equivalent (MtCO₂eq), made up of a reduction of 88.8 MtCO₂eq for the 2 GW RE built under the projects over the 25 years of their operation, and 9 MtCO₂eq for 0.4 GW of RE power plants over the 25 years of their operation that are expected to be prepared under the SRMI Facility projects but not built during their lifetime;
 - (c) Unlock an estimated USD 3.3 billion of private investment, including USD 3 billion for the 2 GW built during the lifetime of the projects and USD 0.3 billion for the 0.4 GW built after the end of the projects;
 - (d) Provide access to clean low-emission electricity to an estimated 4.2 million people through the electrification investments;
 - (e) Strengthen institutional and regulatory systems for planning and development; and
 - (f) Reduce locking-in long-lived climate-vulnerable infrastructure by increasing resilience of grid infrastructure.

2.1.1. Implementation arrangements

12. The Facility will be managed by the World Bank. In respect of each project in the seven countries, the World Bank will carry out due diligence to ensure that the individual projects meet the agreed eligibility criteria (section B.3 of the funding proposal), which are fully compliant with GCF investment criteria. The World Bank will provide implementation support to the projects, including necessary governance, oversight and quality assurance in accordance with its policies.

13. The Facility will be category A/I-1 following the GCF classification, and the relevant environmental and social safeguards (ESS) disclosure at the facility level has been completed in accordance with the GCF Information Disclosure Policy. There is also an agreed approach between GCF and the World Bank on the disclosure and communication process for approval of each individual project.

2.2 Component-by-component analysis.

Component 1: Technical assistance (total cost: USD 22.5 million; GCF cost: USD 21.5 million)

14. Component 1 (technical assistance) focuses on the country/macro-level risks and aims at supporting countries to transform their strategy to low-emission development, allowing them to harness RE resources at scale.

15. The first step to a sound deployment is to set a sustainable long-term target for renewables based on concrete analysis for demand, available resources and grid flexibility and stability. These targets will be the basis for a realistic but also ambitious plan to deploy RE capacity in the respective countries. Subcomponent 1.1 is therefore designed to support the seven countries to revise their generation plans and to identify the critical investment needs for

large-scale VRE integration while maintaining grid reliability and resilience. The targets will then be integrated into revisions of the NDC to ensure that they are implementable.

16. As an important aspect of the technical assistance, the SRMI will also identify all the legal changes that need to be implemented to attract private investments, and how the domestic lending sector can be leveraged to finance the debt part of privately owned power plants to reduce the risk of foreign exchange to the utility. Subcomponent 1.2, therefore, will focus on the legal, financial and procurement aspects, capacity-building and transaction advisory support to implement competitive and transparent tendering of privately financed RE projects.

17. Complementary to setting out the legal framework for IPPs and the competitive bidding scheme that is conducive to private sector investment, subcomponent 1.3 will cover the cost of preparing feasibility studies and environmental and social risk mitigation instruments in support of investment de-risking and the investor selection process. Preparatory and feasibility studies may include geospatial analyses of the land around the substation, topography and geotechnical analyses, site-specific interconnection studies, solar radiation/wind measurement analyses, all of which will be made available for bidders.

Component 2: Public investments (total cost: USD 1,466 million; GCF cost: USD 206 million)

18. This public investment component focuses on the grid- and infrastructure-level risk.

19. It will concentrate on public investments for:

- (a) Solar and/or wind park common/shared infrastructure, ranging from the evacuation line to basic elements such as fence, roads and street lighting. The Facility will use the solar/wind park competitive bidding concept to address the grid curtailment risks for deployment of renewables. The government will identify project sites, conducts land clearance, and construct sharing infrastructure for the solar/wind park. The government will also conduct the environmental and social assessments as well as the geotechnical studies. Once the solar/wind parks are ready for auction, the bidding procedure begins and competitively selected IPPs are responsible for financing, construction, and operation of the RE projects. This approach can contribute to reaching a fully cost-effective electricity tariff and can effectively mitigate the risk of curtailment and land ownership issues;
- (b) VRE integration grid upgrades (including publicly owned battery storage) and enhancing grid resilience. Deploying renewables requires significant expansion and modernization of electricity grids. The programme will support grid reinforcements through (i) addition or replacement of lines and transformers for grid extension and capacity enhancements; (ii) provision of equipment for smoothing the voltage plan, such as capacitor banks, and other reactive power compensators together with flexible alternating current transmission systems; and (iii) equipment for faster and more efficient grid operation, such as monitoring systems, demand and production forecasting systems, and automatic generation control systems for generation units and grid operations. Much attention will also be paid to increasing the grid's resilience to the impacts of climate change, which is expected to increase the incidence and severity of extreme weather conditions, putting the structural integrity of ageing electric infrastructure under greater strain. The majority of GCF finance will be devoted to this activity because upfront investments are usually borne by public parties and constitute an incremental system level requirement that is not covered by IPPs and thus becomes challenges for countries.
- (c) Electrification to increase resilience of populations. This subcomponent aims to invest in solar home systems, mini grids and grid connection, providing access to 840,000 households across different countries. This subcomponent is critical to enable public

investments in clean energy mini grids and connection to the grid, which will be enhanced by the other interventions under the programme.

Component 3: Risk mitigation (total cost: USD 50 million; GCF cost: USD 50 million)

20. This component focuses on the residual risks for the private sector (i.e. excluding risk related to grid integration). It is intended to provide support to mobilize the private and commercial capital for investment in renewables (i) through reimbursable grants in DRC; and (ii) through guarantees in Namibia and Botswana.

2.2.1. Reimbursable grant (Democratic Republic of Congo)

21. The RE sector in the DRC is characterized by its weak governance and regulatory framework associated with a restrictive licensing/permissions process, which affects its ability to crowd in private sector finance. The lack of a proven track record with a successful revenue model is a structural challenge with mini grids project in a nascent market. Although the existence of latent demand for electricity in underserved communities is well known, the expected level of demand and associated ability to pay in the early years will likely have a significant level of uncertainty. Market analysis shows that the first years of activity are critical because the behaviour of productive users is still difficult to anticipate, despite a good preparation underpinned by a sound business plan and deployment of demand simulation measures. Therefore, large mini grids are commonly perceived as an investment risk, given the slow uptake in the early stages.

22. The USD 30 million of GCF reimbursable grant aims to mitigate this demand risk during the initial years of operation. It is expected that this risk mitigation instrument will be made available as part of the tender for competitive bidding to select large-scale mini grid developers. Bidders will be requested to submit a business plan covering estimated demand as well as ability to pay and financial flows, to demonstrate how the expected payments from targeted clients will allow the special purpose vehicle to reimburse the lenders and equity providers. The business plans of the pre-selected bidders will be audited and agreed in advance with the AE to ensure it is robust. If the verified level of revenue from customers is below the pre-agreed minimum revenue level, the reimbursable grant (from the special purpose vehicle) will be triggered. The use of a reimbursable grant will not only crowd in the private sector, but it will also reduce the risk premium, ultimately ensuring a more affordable tariff in DRC.

2.2.2. Guarantee instrument (Namibia and Botswana)

23. In the context of Botswana and Namibia, limited financial viability of the utilities (payment and termination risk) may limit private sector interest in RE and therefore result in higher PPA prices. GCF guarantees will be used to strengthen the PPAs for project developers to improve private sector participation through mitigating two types of risks: the first relating to regular payments under the RE PPA contract, and the second relating to long-term recovery of capital investment. The guarantee merits for the former will ensure that the RE private investor receives the regular payments owed by the public entity or government under the RE PPA contract on a timely basis, whereas the latter will cover a one-off payment by the government following an early termination of the RE PPA. Further design of this activity will be finalized at the time of approval of subprojects and will be based on the AE's due diligence and further market sounding.

24. This component provides fit-for-purpose risk mitigation instruments based on needs in the individual countries and is expected to unlock the huge potential of the private sector in the targeted countries.

Project management (total cost: USD 25 million; GCF cost: USD 2.5 million)

25. This component will concentrate on the management of the projects and coordination and implementation of their activities, in compliance with the contractual obligations of the executing entities (EEs) and programme implementing entities, 'as included or referred to in the legal agreements with the AE. The requested GCF finance responds to the needs of the programme and is in compliance with GCF policy on fees for accredited entities.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: High

26. As a mitigation project, the proposal reports reduction of GHG emissions in the seven countries by 88.9 MtCO₂eq as a result of the 2 GW of RE built during the project implementation and an additional 9.0 MtCO₂eq resulting from the 0.4 GW RE investment that will be prepared but not built during the project implementation. These impact figures exclude the indirect GHG emission reduction resulting from another 1.9 GW of RE enabled by integration into core grid infrastructure and the VRE integration analysis funded by the GCF investment. The AE will use the International Financial Institutions dataset of grid factors for GHG accounting for emissions from grid-connected RE projects. The AE agreed to provide the methodology for off-grid sub-projects prior to them being financed by the programme.

27. Adaptation co-benefits resulting from the Facility include access to clean electricity provided to an additional 4.2 million people, including in areas where the population is under severe threat from climate change.

28. The programme's impact potential in the long term is notable. Impacts resulting from direct and indirect investments under the Facility are further broken down in the proposal, therefore there is minimal concern about potential double counting. However, it is uncertain at this moment if any of the GHG emission reductions achieved by the programme will be converted into offset credits causing concerns over double dipping of climate finance. In the absence of universally acceptable methodology on attribution, it was agreed with the AE that in case that any executing entity seeks to convert any GHG emission reductions achieved by the Facility it has to seek consent of GCF, whilst the secretariat will continue to explore plausible ways to deal with this matter for future transactions.

3.2 Paradigm shift potential

Scale: High

29. The key pillar of the programme's paradigm shift potential lies in the comprehensive approach taken to mitigate the risks identified in the renewables development sector encountered in developing countries. It is intended to utilize the limited public resources to unlock the maximum private financing for a capital expenditure-intensive sector.

30. At the strategic level, the SRMI aims at supporting countries to transform their strategy to low-emission development allowing them to harness their domestic resources at scale. The SRMI will assist countries to set long-term RE targets, based on concrete analysis of demand, grid flexibility and resilience, and available domestic renewable resources. The targets are then incorporated into the NDC revisions to guide the countries' priority in subsequent phases.

31. In terms of the grid- and infrastructure-level risk, SRMI, through its public investment component, will put in place the necessary common infrastructure for solar/wind parks which contributes to achieving a fully cost-effective electricity tariff and can effectively mitigate the risk of curtailment and land ownership issues. The SRMI focuses strongly on the grid upgrade and modernization based on countries' least-cost transmission plans to support the integration

of VRE and ensure a better quality of electricity service. This is critical to allow a higher penetration of RE and to mitigate the grid curtailment risk.

32. Residual risks are being addressed by the provision of GCF risk mitigation instruments under the Facility. Different instruments are carefully assessed and designed to fit the specific country contexts. In the case of DRC, where the RE development is nascent, the reimbursable grant is an attractive option for private mini grid developers to mitigate the risk of low demand and slow uptake in the initial years of operation. In Botswana and Namibia, the guarantee instrument is designed to assist in backing up the utilities' ability to pay for PPAs.

33. The Secretariat considers the risk mitigation model offered by SRMI to be comprehensive and innovative, and having potential for replication. The seven targeted countries represent a large sample of developing countries, covering around 500 million people, each having different constraints and contexts with regard to the development of RE. The results and lessons learned from this Facility will enable replication of the approach at scale and may easily be adapted to countries with similarities. Indeed, SRMI is currently being implemented/considered for another 30 countries in sub-Saharan Africa and Asia, which will benefit from the results and lessons learned from this first phase. The proposal has huge potential to transform the RE sector in the targeted countries; therefore, the programme paradigm shift potential is rated as "high".

34. The mitigation impact of the project has been reviewed as part of the interdivisional review. The project has used the IFI TWG grid emission factors per country for grid connected electricity generation sub-projects. It is also the understanding of the Secretariat that despite the fact that the methodology has not been clearly specified, the project is following the general framework of CDM methodology ACM 0002, which is acceptable. The project has not provided a methodology for estimation of GHG emission reductions from off-grid power generation. The AE has agreed to provide to the Secretariat that methodology prior to investing into any off-grid projects.

35. The project has also provided the data sources and references used in the estimates of mitigation impact.

36. However, it is important to note that there is uncertainty in relation to the generation of impact as it will depend significantly to the use of parallel financing for the construction of the generation facilities.

3.3 Sustainable development potential

Scale: High

37. At the country level, implementation of the SRMI will avoid countries becoming locked in to long-term high-emission development pathways. Deploying domestic renewables will reduce the need for fuel imports and the potential economic impact of fuel cost fluctuations. It will increase the energy security of the countries, diversifying their energy mix and improving their energy independence while reducing price volatility. Clean grid connected RE parks, RE mini-grids and household services it is expected to bring in considerable environmental and health benefits thanks to reduced air and water pollution.

38. The Facility will also support the socioeconomic benefits of the RE projects from the perspectives of local communities and local industries. Sites with good solar/wind resources and large plots of land are usually located in areas where the standard of living is below the national average. The tailor-made development programmes offered by the Facility will ensure that local communities benefit positively from creation of jobs and development of skills, as well as increased resilience and enhanced livelihoods.

39. Regarding gender benefits, the programme recognizes the current gap in the RE industry especially the significant employment gap in large-scale solar and wind infrastructure

projects which generally benefit men more than women. A proactive policy will be put in place to promote female employment throughout the project implementation and beyond.

3.4 Needs of the recipient

Scale: High

40. Although the seven countries have different economic development levels, they are commonly undergoing dramatic tensions on their budgets due to the COVID-19 crisis, so they are likely to opt for a fast recovery strategy that would promote fossil fuel generation unless there is strong parallel support for RE development. Therefore, this is a critical time to support harnessing renewables through the best use of limited public finance. Private investments are a means to deploy renewables, which are generally more capital intense. To ensure that the trade-off between public financing (which is limited and could be used better) and private financing (which may be more expensive) is fair for the society, SRMI aims to bring down the cost of privately owned generation through a comprehensive and integrated risk mitigation framework that is able to leverage climate finance.

3.5 Country ownership

Scale: High

41. The AE reports that the seven countries targeted under the Facility in the initial phase were based on the willingness and interest of their respective governments to embrace RE deployment and combat the challenges they face in the implementation of their NDCs and low-emission development pathways. All seven countries have ambitious GHG emission reduction objectives as part of their NDCs and their current energy policy in place. The programme aims to help the countries achieve their NDCs. Moreover, the SRMI has the potential to shape the countries' NDCs in the coming years through the technical assistance offered by the programme. The SRMI is supporting the seven countries to revise their generation plans and to identify the critical investment needed to integrate VRE into their grids while improving grid reliability and resilience. Such revisions will then be integrated into revisions of the NDC that will guide the countries' action in mitigation and adaptation in the following phase.

42. In each country, the World Bank has extensive engagement with relevant stakeholders, and the funding proposal package contains a solid stakeholder engagement plan to ensure it continues engaging stakeholders throughout the project cycle.

3.6 Efficiency and effectiveness

Scale: Medium to high

43. The GCF proposed contribution of USD 280 million will be complemented by an expected USD 1.28 billion of public finance from the World Bank and its administered ESMAP, which in turn is expected to leverage USD 2.9 billion of private finance during the project implementation, and another USD 300 million beyond the project completion. The financial structure is a mix of grant for technical assistance, concessional loan for critical public infrastructure (except for CAR, where because of its financial and economic situation, grants are given for public investments), and risk mitigation instruments for residual risk for private RE developers (a reimbursable grant for DRC and guarantees for Botswana and Namibia). Each instrument is designed to address specific types of risks in order to achieve the programme's objective to support countries in their sustainable low-emission pathways. The expected GHG emission reductions to be achieved during the project implementation are 88.8 MtCO₂eq, with an extra 9.0 MtCO₂eq expected beyond project completion. This represents the cost of abatement to be USD 3/tCO₂eq for GCF, which is a low abatement cost.

44. The AE also presented country-specific economic and financial analysis. The economic and financial viability vary from country to country the financial analysis conducted for

activities under Components 2.1 and 2.2 resulted in financial internal rates of return (IRR) of 13 per cent (Botswana), 18 per cent (Kenya), 27 per cent (Mali), 12 per cent (Namibia) and 14 per cent (Uzbekistan). However, it is worth pointing out that the financial analysis shows that the GCF concessional financing increases the financial IRR on average by around 2 per cent and, more importantly, makes possible projects that these countries could not otherwise implement.

45. The Secretariat considers the programme's efficiency and effectiveness to be "high".

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

46. The SRMI Facility aims to support countries in developing sustainable solar and wind programmes that will attract private investments and reduce reliance on public finances, while maximizing the socio-economic benefits triggered by the projects deployed. It will finance seven projects in seven countries, and each project will have three components: (i) technical assistance grants to provide comprehensive and planning and advisory support; (ii) public investments in VRE infrastructure, in the form of solar and wind parks, wind and solar power plants, and grid integration upgrades, including storage infrastructure, and electrification projects; and (iii) risk mitigation instruments, which would provide risk mitigation instruments to private investors for RE/storage solutions. The Facility will be hosted at the World Bank, as the AE. Each project will individually go through the World Bank project cycle process, including safeguards review/appraisal and approval process. A project management unit will be established in each country.

47. **Environmental and social (E&S) risk category.** The Facility is proposed as category A. The Secretariat confirms this categorization, and it is within the AE's accreditation level. The investment projects expected to be funded under the programme include large solar PV and CSP parks/plants, as well as large offshore and inshore wind power plants/parks. The Facility is also expected to support solar home systems, mini-grids and grid connection. The larger solar and wind projects would most likely involve the acquisition of several hectares of land and would have involuntary resettlement issues involving formal and informal land occupants/settlers. Wind turbines may also: create visual and noise nuisance to the community residents in their vicinity; negatively impact birds; and, if built near the shore/offshore, negatively affect the marine ecosystem, economic activities and livelihoods, as well as cause coastal erosion due to alteration of bathymetry and wave patterns. Solar PV farms and solar PV with battery storage may also involve the handling of hazardous waste materials. Associated facilities and auxiliary infrastructure, such as transmission lines, substations and access roads, will have corresponding environmental and social impacts.

48. **Safeguards instrument and disclosure.** The Facility will cover seven countries, which would have their own safeguards systems. An indicative scope of the country projects has been provided in the funding proposal, but the specific projects to be funded would have to be identified and developed only during the programme's implementation. The AE has prepared a document titled Environmental and Social Sustainability Framework (ESSF). This document describes how the AE would conduct environmental and social (E&S) due diligence of each country project under its Environmental and Social Framework (ESF). It includes indicative outlines and frameworks of the environmental and social impact assessment, Indigenous Peoples Planning Framework, Resettlement Policy Framework and various other instruments for individual projects.

49. The World Bank's ESF requires that projects comply with its Environmental and Social Standards (ESS) (ESS1–ESS8 and ESS10), which fully covers the GCF environmental and social safeguards standards as well as the stakeholder engagement and grievance redress mechanism

requirements. The following describes how the programme would comply with these standards and requirements at the project level.

50. **ESS 1: Assessment and management of environmental and social risks and impacts.** The AE has highlighted that the projects under the programme are expected to bring substantial direct environmental and social benefits, such as with regard to GHG emissions, air and water pollution, and use of water resources, employment and economic development. However, if not properly managed, the development of the infrastructure for the solar/wind parks, as well as the construction and operation of the RE power plants, may result in adverse environmental and social impacts. Each project to be funded by the Facility will undergo risk categorization, E&S assessment and mitigation planning commensurate to its size and complexity, in accordance with ESS1. The E&S assessments and mitigation/management planning will cover risks and impacts relating to ESS2–ESS8 and ESS10. The assessments and management of E&S risks and impacts for each project will have to take into consideration the country-specific context in terms of legal and institutional frameworks, capacities, and the country and local environmental and social conditions. The exact roles and responsibilities will be defined on a per project basis after discussions with each EE and, in particular, as will be stated in the environmental and social commitment plan developed by the borrower/EE. This plan will specify which entity will be responsible for the various E&S commitments, as well as any necessary capacity-building activities on E&S aspects.

51. **ESS 2: Labour and working conditions.** The projects to be funded by the Facility are expected to involve the hiring of workers during construction and during the operation of the infrastructure. Under ESS1 and ESS2, the E&S assessments for these projects will cover issues relating to labour and working conditions, including possible violations of workers' rights, child labour exploitation, discrimination and other unfair labour practices, occupational health and safety, and workplace gender-based violence (GBV). Aside from the environmental and social management plan (ESMP), each project that would involve the hiring of workers will be required to prepare a separate occupational health and safety plan as per the World Bank Group's environmental, health and safety guidelines and a labour management procedure, which will also provide for the establishment of a dedicated grievance redress mechanism for project workers. All relevant measures in the ESMP, occupational health and safety plan, and labour management procedure will be reflected in the bidding and contractual documents of project contractors.

52. **ESS 3: Resource efficiency and pollution prevention and management.** The proposed Facility will contribute positively to the objectives and requirements of ESS3 in terms of RE sources and GHG reduction benefits. Even so, energy efficiency technologies and processes, energy conservation investments, best practices of RE, and GHG reduction benefits will be further considered in the design of individual projects. Moreover, the E&S assessment and mitigation plans will cover the downstream risks and impacts related to resource use and pollution, such as: potential water scarcity (especially in arid regions, and also considering seasonal water availability and any cumulative effects of multiple demands); noise and air emissions; soil waste generation; management of disposal of waste materials and hazardous wastes; and their effects on the local communities during construction and during the operations of the infrastructure facility. However, most of these potential impacts are expected to be low to moderate in intensity or significance, and to be reversible and localized in nature.

53. **ESS 4: Community health, safety and security.** ESS4 requirements will be taken into account in the design and implementation of individual projects. The E&S assessments and mitigation planning will cover the impacts of labour influx into the project sites communities, which may include: potential spread of infectious diseases; increase in criminality, alcoholism and GBV; traffic and construction site accidents; and conflict between community and migrant workers, and other security-related issues. Moreover, the E&S assessments and mitigation/management planning will also consider the effects of nuisance and pollution from

construction activities on the residents' health and welfare. All measures under the responsibility of the contractors, such as management of construction traffic, ensuring safety of residents at construction sites, control of noise and dust, generation of wastes, health screening of workers, and workers' conduct and good relations with communities, will be reflected in the bidding and contract documents.

54. **ESS 5: Land acquisition, restrictions on land-use and involuntary resettlement.**

Land requirements for solar and wind parks are substantial. Any potential land acquisition or land-use restrictions will be assessed and managed in a manner consistent with ESS5 requirements, in order to avoid, minimize, mitigate and compensate at replacement cost land acquisition based on due diligence and plans prepared in accordance with ESS5. An indicative outline of a resettlement policy framework has been included in the programme's ESSF. The EEs in the borrower country will be responsible for preparing, in a manner and substance satisfactory to the World Bank, the project-specific resettlement policy frameworks and/or resettlement action plans in line with its ESS5 requirements.

55. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.**

Solar and wind projects, if not properly designed and sited, may result in loss, degradation or fragmentation of natural habitats, and other adverse impacts on biodiversity and living natural resources. Care will be taken to ensure that the impacts of such projects on natural habitats, and on terrestrial and aquatic ecosystems and species are well assessed and managed. All projects will include measures to ensure full compliance with ESS6.

56. **GCF indigenous peoples policy and ESS 7: Indigenous peoples/sub-Saharan African historically underserved traditional local communities.**

The presence of indigenous peoples in the project areas or their collective attachment to such areas will be screened and confirmed by the World Bank in consultation with the national authorities. Once such presence is confirmed, consistency with ESS7 objectives and requirements will be integrated into the design of the project and E&S documents, including the development of an indigenous peoples' plan or other types of plans depending on the country and project context, and free, prior and informed consent (FPIC) in the circumstances specified in ESS7. An Indigenous Peoples Planning Framework is provided as part of the ESSF.

57. **ESS 8: Cultural heritage.** The EE of the borrower country will screen the project area and assess potential impacts on cultural heritage, both tangible and intangible. The site selection for each project will be mindful of cultural heritage, and the application of ESS8 will be further assessed under each project. Chance find procedures will be developed and adopted based on existing country laws and regulations, and on international good practice, and will be included as part of the ESMP.

58. **Stakeholder engagement and disclosure.** The AE has provided a stakeholder engagement approach and preliminary stakeholder engagements in the targeted seven countries where the SRMI Facility will be implemented. The various tasks on detailed stakeholder identification, analysis, information dissemination, consultation and engagement will be carried out later during individual project design and implementation. Stakeholders will include project site communities or individuals affected by the project, and their formal and informal representatives, national or local government authorities, politicians, religious or community organizations and civil society groups with special interests, academic communities, and businesses. The identification of project-affected parties (individuals or groups) will also include those who, because of their circumstances, may be disadvantaged or vulnerable; that is, those who may be more likely to be adversely affected by the project impacts and/or more limited than others in their ability to take advantage of the project benefits. A stakeholder engagement plan will be developed for each project to ensure transparency and meaningful consultation with the project-affected and interested parties. Stakeholder engagement and consultations will be conducted throughout the project cycle. These will include discussions of project design and impacts as well as multi-stakeholder discussions on these issues during the

preparation phase. The stakeholder engagement plan, along with other E&S instruments, will be subject to public consultation and disclosure as per the requirements of ESS10, and it will be treated as a living document to be regularly updated as needed during project implementation.

59. **Grievance redress mechanism.** The ESSF describes the requirements for establishing project-specific grievance redress mechanisms under the responsibility of the EE of the borrower country. In addition, the World Bank Grievance Redress Service may also be available for use by any aggrieved party. Communities and individuals who believe that they are adversely affected by a World Bank-supported project may submit complaints by phone or online.¹ Moreover, project-affected communities and individuals may also submit their complaints to the World Bank's Inspection Panel.

4.2 Gender policy

60. The AE has provided a gender assessment and gender action plan, and therefore complies with the requirements of the GCF Gender Policy.

61. The AE has provided information on the levels of enabling environment that exist in all seven countries to promote women's empowerment and gender equality. An initial review of each country's economic and social development reveals the different stages each of the countries are at but also demonstrates, regardless of the differences, that all share and face gaps in addressing prevailing gender inequalities in general and in the energy sector in particular. Some countries are more unequal than others and provide limited socioeconomic opportunities for women and their empowerment.

62. At the national level, most of the countries are signatories of, for example, the Convention on the Elimination of All Forms of Discrimination Against Women, and the Beijing Declaration and Platform for Action. In addition, most have also adopted national strategies and mechanisms to promote gender equality. While national-level commitments exist in many of the countries, customary law and discriminatory norms prevail, curtailing the advances that women and girls can make in the social, economic and political spheres.

63. In the energy sector, the assessment indicates that women suffer most from insufficient access to electricity and clean-cooking facilities, as they are mostly responsible for household chores. Moreover, women are found to have limited employment opportunities in the formal sector, including in the energy industry, due to limited education, qualifications and skills. Women's limited political participation and representation results in lower levels of female access and participation in energy-related decision-making. In sub-Saharan Africa, the electrification rate stands at less than 48 percent, leaving nearly 600 million people without access to electricity, and further negatively impacting female economic empowerment, decision-making, education and health outcomes. The assessment indicates that, every year, 4.3 million people – mainly women and children – die as a result of indoor air pollution. Overall, the level of access to electricity is skewed against women. However, there are marked differences among the target countries in terms of overall access to electricity. For example, Kenya is the only country in East Africa on track to meet the Sustainable Development Goal of universal electricity access by 2030, while Uzbekistan is the second-most energy-intensive country worldwide and has achieved near universal electricity access, as has Viet Nam (to be covered in the next phase). On the other hand, access to electricity stands at around 8 percent in the Central African Republic, and 15 percent in the Democratic Republic of the Congo, with many of the population and especially women relying on fuelwood and charcoal for energy. Worldwide, only about 32 percent of people employed in the RE industry are women, with possible increases in the number of available jobs in the sector from 10.3 million in 2017 to 29 million in 2050. However,

¹ World Bank. 2021. *Grievance Redress Service*. Available at <<http://www.worldbank.org/en/projects-operations/products-and-services/grievance-redress-service>>.

the increased employment opportunities in RE might be constrained by existing cultural norms that limit women's choice and access to non-traditional fields. Even when employment opportunities are created, the assessment indicates that these jobs are created in traditionally feminine sectors such as hospitality or administrative support.

64. Furthermore, the assessment identifies female-headed households (FHHs) as being significantly more likely to be economically poorer than male-headed households. FHHs are characterized by lower levels of education, lower salaries and lower levels of access or control of resources. They are more dependent on unreliable sources of income for subsistence, such as remittances, maintenance, and destitute allowances. Because poverty levels are highest among FHHs, especially in rural areas, these households also have lower access to energy services, resulting in low electricity connection rates. Gender gaps in access to assets and financial literacy make it more difficult for women to access credit. For example, in Kenya, FHHs also have lower access to electricity, often because of high connection/subscription fees to the grid.

65. Although more data are needed in order to assess the prevalence of GBV, studies indicate that women and girls often experience high levels of sexual harassment and intimate partner violence in target countries. Legal loopholes persist, especially with regard to GBV, and some legal provisions are directly discriminatory against women. The main drivers of GBV include social tolerance to violence against women, poverty, and the lack of a protective legal framework.

66. The gender action plan provided complies with the requirements of the GCF Gender Policy, and includes activities, indicators, targets, timeline, human resources, and budget. As indicated in the assessment, further and country-specific analysis will be required at the project level, as will collection of sex disaggregated data including those on GBV. The gender analysis will be conducted for the required countries at the initial stages, ensuring the participation of communities, civil society organizations and women's organizations. While the assessment will assist in reaffirming activities to be undertaken, with targets to be set per activity, it will also conduct country-level policy dialogue at the Facility level to close policy gaps as these relate to the sector and as will be identified through the assessments. The activities included will contribute in the 7 countries in the areas of increased employment in the energy sector by: breaking down the barriers to sex-based segregation of occupations; training programmes to build women's skills; and creating income-generating opportunities for women. Thereby, the activities will help to improve women's access to electricity and combat GBV, while at the same time making investments in providing evidence and best practices for improved gender equality and women's empowerment in the countries and in the energy sector. The AE will provide technical and financial support to the EE to design and implement the required gender-related activities, while the recruitment of social and gender specialists in each of the seven countries will be the responsibility of the project management unit.

4.3 Risks

4.3.1. Overall programme assessment (medium-high risk)

67. The project aims at tackling the lack of sustainable and bankable pipeline for RE projects in developing countries to achieve low-emission development pathways. The project consists of three main components: (i) grants for capacity-building, generation plans, transaction advisory, and solar and wind park preparatory studies; (ii) loans and grants for public investments in solar and wind park infrastructure, grid resilience, grid VRE integration and electrification; and (iii) reimbursable grants and guarantees for risk mitigation instruments for mini grids and grid-connected RE projects.

68. The total project cost is USD 1.56 billion, of which USD 280 million is GCF finance. GCF financing consists of USD 84 million in grants (of which USD 30 million is reimbursable), USD

176 million in loans and USD 20 million in guarantees. Co-financing is provided by several World Bank entities. IBRD will provide USD 460 million in senior loans, IDA will provide USD 750 million loan and USD 60 million grant. To achieve the impact as presented in the funding proposal under A.5. an additional USD 3 billion in parallel financing is needed. However, this additional financing is beyond the scope of the project and not arranged by the AE. This leads to increased uncertainty about the impact materializing as envisaged.

4.3.2. Accredited entity/executing entity capability to execute the current programme (medium risk)

69. The World Bank is the AE for the proposed programme. Given the structure of the programme, the broad eligibility criteria, and limited visibility at this stage regarding the end projects, the success of the programme depends to a great extent on the capability of the AE. The AE has a strong institutional capacity and experience in executing similar programmes.

70. For the purposes of the loans and grants, the respective countries represented through Ministries or a project implementing entity (PIE) will act as the Executing Entities (EEs). For the purposes of the guarantees, the private sector RE project companies will be the EEs. Thus, some of the potential EEs under the program are not yet identified and their capacity will be assessed by the AE during the appraisal stage of each project. In assessing the capacities of the EE the AE will apply its internal policies to ensure that the EE has the technical, financial management, procurement and ESS capacities for implementation.

71. In addition to the EEs, the project is dependent on private sector entities for developing the estimated 2 GW RE projects in the solar parks and supporting the project's grid-strengthening activities. The project's impact is contingent on the execution and operational capacities of the private sector entities.

4.3.3. Programme-specific execution risks (high risk)

72. Impact risk: direct impact as presented in the funding proposal (A.5. Expected mitigation impact) is estimated based on new RE capacity of 2 GW operating over 25 years. These RE power plants will be located in the RE parks and will benefit from the grid-strengthening activities under the project, but the financing and construction of RE projects is beyond the scope of this funding proposal. A total of USD 3 billion in parallel financing is needed to achieve the intended direct impact. Lack of appetite for financing for the RE power plants may reduce the impact. Comfort can be derived from the AE's experience in working with the countries covered under the project.

73. The construction of the 2 GW RE capacity will be contingent on the success of the activities undertaken under Components 1 and 2 to generate and implement necessary plans and reforms followed by construction activities. Lack of institutional capacity at the country level, as well as political and security risks, could impact the ability to complete this work, thus impeding the ability of the programme to generate viable RE plants.

74. The AE will monitor the project impact for 12 years of the implementation period. Accordingly, the final target for emission reductions is stated in the log frame as 28 MtCO₂eq as against the impact potential of 89 MtCO₂eq as stated in A5. This 28 MtCO₂eq is in turn based on the 2 GW capacity operating for 7 years during the 12-year implementation period. The impact quantum will be reduced if the financing and development of the RE projects is delayed.

75. Sovereign credit risk: GCF will provide up to USD 176 million in sovereign loans under the programme, in addition via the risk mitigation mechanism will have indirect exposure to sovereign credit risk. The project countries have different levels of credit ratings ranging from investment grade to substantial risk. GCF loan tenor may exceed tenor of the senior loans provided by the AE. While it will be the lender of record for GCF loans, the AE has stated that

GCF loans will not enjoy the AE's preferred credit status in those countries, thus exposing GCF to higher risk.

76. Carbon credits: while the FP estimates an impact of 89 million tCO₂eqT, the impact is exposed to high risk of dilution. The term sheet proposed by AE provides that if the EEs seek to convert GHG emission reductions into offset credits, they are required to negotiate and agree with GCF on volume of impact directly attributable to GCF which may not be converted into offset credit. The attribution methodology is not defined. Further, the term sheet does not put any restriction on off-setting impact not attributable to GCF, but which would be included in the 89 million tCO₂eq of estimated emission reduction. In addition, it is also possible that the impact will be generated at the private sector RE generating projects level and not at EE level; the term sheet doesn't put any restriction on the private sector RE generators to off-set the climate impact. Also, the GHG emissions converted into offset credits will be deducted from the impact reported to the Fund. It is advised that the AE clearly provides in the funding proposal the impact that will be attributable to GCF and requires the EE as well as the private sector RE generating projects to retire the carbon credits generated under the program.

4.3.4. Project viability and concessionality

77. The AE proposes providing a high concessional loan with equal terms for all seven countries in the programme. The recipient countries are at different levels of RE development. There are also meaningful differences in terms of credit ratings between the countries, for example Mali has Caa1 rating (Moody's) while Botswana has an investment grade rating A2 by Moody's. Given the current limited visibility of the financed activity and the structural differences between the seven countries, it is advised that the AE considers differentiation in financing terms among the countries and apply these terms in line with the minimum concessionality principle.

4.3.5. Compliance risk (high risk)

78. When asked whether it had conducted any assessment of risks to the project, the AE (World Bank) referred to its AMA requirements and said that it would apply those against the project. Accordingly, the AE has acknowledged that it has not conducted any risk assessment specific to this project and its activities. Without such project-specific assessment, GCF is unable to evaluate and provide an accurate and informed Compliance Risk rating for this project. In light of specific information directed toward this project, the Compliance risk rating is rated as "HIGH".

4.3.6. GCF portfolio concentration risk (low risk)

79. In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is not material.

4.3.7. Recommendation

80. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment		Rationale
Overall programme	Medium-high	The project has benefit of an experienced AE – the World bank who is also co-financing along with GCF under the program.
Accredited entity/executing entity capability	Medium	
Project-specific execution	High	

GCF portfolio concentration	Low	However, the climate impact estimated under the program is exposed to high risk because of i. need of additional funding of USD 3 bn for the RE generating projects to cause the impact but the funding is beyond the scope of the project and ii. term sheet not restricting the private sector RE generators from off-setting the GHG emission and introducing an attribution approach (not defined) for EEs to attribute some of the emission reductions to GCF while permitting the EEs to off-set the balance emission reductions.
Compliance	High	

4.4 Fiduciary

81. As the AE, the World Bank will be responsible for the management and administration of GCF resources and will carry out such in accordance with its policies, procedures and practices, and the relevant provisions of the funded activity agreement (FAA) and accreditation master agreement (AMA). The World Bank will apply its own fiduciary principles and standards relating to any integrity checks, anti-corruption, countering the financing of terrorism, fraud, financial sanctions, embargoes and anti-money laundering.

82. During implementation of the Facility, the World Bank will be responsible for providing the necessary governance, oversight and quality assurance in accordance with its policies, procedures and any specific requirements in the AMA and FAA.

83. The World Bank team will manage and be responsible for the Facility's overall objectives and ensure that the projects selected meet all the set eligibility criteria. The World Bank will inform the GCF Secretariat about projects selected in accordance with the eligibility criteria as part of the periodic reporting.

84. Implementation of the Facility-supported projects (project implementation through completion and evaluation) follows the World Bank's applicable policies and procedures. For each project, a project management unit will be established with a team expected to include at least a team lead, technical expert, procurement specialist, financial management specialist, environmental specialist and social/gender specialist, and a lawyer. Each project will have its own monitoring and evaluation and reporting process, and the EE will be accountable for its implementation.

85. Each EE of loan/grant financing under the SRMI Facility will be required to conduct financial management and procurement according to the World Bank policies and procedures. Financial reporting will also be required for each project and (with the exception of the guarantees) is specified in each project's Disbursement and Financial Information Letter. Unless otherwise agreed by the Bank, the EE must submit annual audited project financial statements six months after the close of the financial year and unaudited interim financial reports periodically. Audits are carried out by auditors with independence and capacity acceptable to the Bank, under terms of reference acceptable to the World Bank.

4.5 Results monitoring and reporting

4.5.1 Results monitoring and reporting

86. The project is a mitigation project focusing on the development of the infrastructure and support of investments in RE from public and private sources in seven countries in Africa and

Asia. The project provides a set of de-risking instruments including development of wind and solar parks, upgrading grid infrastructure and financial guarantee instruments that are expected to attract more investments in the sector. Project interventions are expected to reduce GHG emissions by 88.9 MtCO₂eq over 25 years as a result of the supply of carbon-free electricity to national grids.

87. The AE has developed a detailed GHG emission estimation methodology, which is attached in Annexes 22 of the project proposal. However, the actual emission reduction calculations are provided not in Annex 22(b) as required in the programming manual, but as Annex 3. The assumptions in the methodological approach is generally transparent, conservative and aligned with the grid emission factors and methodologies recommended by the International Financial Institutions Technical Working Group on harmonization of GHG accounting. As the impact of the project will continue to be generated beyond the end of the 12-year implementation period, the AE will arrange for impact reporting beyond that through performance reports. However, it is important that the AE specify which is the exact methodology that they are using, e.g. CDM ACM002, and how the additionality is established.

88. As mentioned in earlier sections of this assessment, it is uncertain at the moment whether executing entities will seek to convert part of the GHG emission reduction achieved by the SRMI facility into carbon credits to be sold at international market. In the absence of an agreed attribution methodology and GCF policy, it was agreed that in the event that an executing entity seeks to convert, prior consent by GCF is needed.

89. The logical framework is aligned with the GCF results management framework (RMF) and performance measurement frameworks (PMF). The appropriate indicators have been selected and targets have been set up accordingly.

90. Project performance indicators have been designed to clearly capture the results of the project and tests the linkages in the Theory of Change, as well as allow to measure whether the project is on track to deliver results.

91. As this is a multi-country project covering seven countries, Annex 17 has been developed and the breakdown of impact per country has been provided. However, the lifetime emission estimates are different from those in the logframe.

92. The project activities and sub-activities have also been clearly articulated. It is noteworthy, however, that a large share of the impact will be generated by activities that are enabled by the project, but not directly financed by it. Measures and additional conditions to guarantee the realization of impact are currently being discussed with the AE.

93. Annex 11, M&E plan has also been completed but monitoring activities that are linked to project indicators are not clearly budgeted for.

4.5.2. Theory of change

94. The theory of change has been provided and there are clear linkages between the various elements of the theory of change and the recognition of certain assumptions with respect to the project. The barriers and how the project helps overcome those barriers are clearly articulated.

4.5.3. Logical framework

95. The Fund-level impacts and outcomes sections (section E.3 and E.4, respectively) include baselines, mid-term and final targets and are compliant with the GCF results management framework/performance measurement framework. Additionally, the assumptions on the different indicators are clearly provided.

96. Section E.5 has also been completed and contains relevant indicators to monitor project performance and results.
97. Section E.6 may benefit from further elaboration of some of the activities.
98. Section E.7 is clearly elaborated and describes the monitoring and evaluation activities to be undertaken by the funded activity.

4.5.4. Implementation timetable

99. The current version of Annex 5 – implementation timetable, is not fully compliant with GCF requirements due to the lack of sufficient data prior to FP approval. However, the AE has agreed to elaborate on the deliverables and key milestones for each activity at the stage when these are available and provide them as part of the APR.

4.6 Legal assessment

100. The accreditation master agreement was signed with the AE on 13 November 2017 (the “AMA”), and it became effective on 1 February 2019.
101. The AE has provided a legal opinion/certificate confirming that it has obtained all internal approvals and that it has the capacity and authority to implement the programme.
102. The proposed programme will be implemented in seven countries: Botswana, the Central African Republic, the Democratic Republic of the Congo, Kenya, Mali, Namibia and Uzbekistan.
103. GCF has signed a bilateral agreement on privileges and immunities with Namibia.
104. GCF is not provided with privileges and immunities in the six other countries. This means that, among other things, GCF is not protected against litigation or expropriation in such countries, the risks of which need to be further assessed.
 - (a) With respect to Botswana, the discussion on the privileges and immunities agreement with the Government of Botswana has not yet started.
 - (b) With respect to the Central African Republic, the discussion on the privileges and immunities agreement with the Government of the Central African Republic has not yet started.
 - (c) With respect to the Democratic Republic of the Congo, the Secretariat submitted a draft privileges and immunities agreement to the Government of the Democratic Republic of the Congo on 9 October 2018 and a bilingual version on 1 March 2019. The agreement is currently under review by the Government.
 - (d) With respect to Kenya, the Secretariat submitted a first draft privileges and immunities agreement to the Government of Kenya on 30 September 2015. Following several rounds of negotiations, the Government responded that the revised draft privileges and immunities agreement required further review. The agreement is currently under review by the Government.
 - (e) With respect to Mali, the Secretariat submitted a first draft privileges and immunities agreement to the Government of Mali on 7 December 2015 and a bilingual revised version on 21 December 2017. A meeting was held at the GCF Global NDA Conference on 9 October 2018. The agreement is currently under review by the Government.
 - (f) With respect to Uzbekistan, the Secretariat submitted a draft privileges and immunities agreement to the Government of Uzbekistan on 10 December 2018. The agreement is currently under review by the Government.

105. The Heads of the Independent Redress Mechanism and the Independent Integrity Unit have both expressed the view that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF be made only after GCF has obtained satisfactory protection against litigation and expropriation in the countries, or has been provided with appropriate privileges and immunities.

106. In order to mitigate risk, it is recommended that any approval by the Board be made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the Secretariat within 180 days from the date of Board approval, or the date on which the AE has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and
- (b) Completion of the legal due diligence to the satisfaction of the Secretariat.

Independent Technical Advisory Panel's assessment of FP163

Proposal name:	Sustainable Renewables Risk Mitigation Initiative (SRMI) Facility
Accredited entity:	World Bank
Country/(ies):	Botswana, Central African Republic (CAR), Democratic Republic of the Congo (DRC), Kenya, Mali, Namibia, and Uzbekistan
Project/programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential

Scale: High

1. In 2015, 196 Parties to the United Nations Framework Convention for Climate Change (UNFCCC) came together under the Paris Agreement to transform their development trajectories so that the world can proceed on a path to sustainable development, which aims at limiting warming to between 1.5°C to 2°C above pre-industrial levels. One of the components of the strategies to achieve this goal is to reduce greenhouse gas (GHG) emissions. The Agreement also included establishing clear pathways for financial flows to facilitate low GHG emissions, and climate-resilient development. Nationally determined contributions (NDCs) are prepared and submitted to the UNFCCC by all the signatory countries. The strategies that each country will utilize to reduce national emissions and adapt to the impacts of climate change are stipulated in their NDC. The developing countries included in this proposed World Bank (WB) intervention have submitted their NDCs to the UNFCCC.

2. The NDC of each country is expected to clearly spell out domestic measures that countries will take to achieve the objective of the proposed contributions. The contributions of the 196 Parties are expected to provide metrics that can be progressively checked to determine whether the long-term goals of the Paris Agreement are achieved and greenhouse gas (GHG) emissions are limited. This is expected to be followed by rapid reductions thereafter, to achieve a balance between anthropogenic emissions by sources and removals by sinks of GHGs in the second half of this century. The Paris Agreement also recognize that financial barriers can limit developing and least developing countries (DCs and LDCs) from achieving these targets. These countries were therefore expected to present their contribution in two forms: those that are conditional to having international financial support; and those that they can implement themselves (not conditional to having international funding). All seven countries that are covered by the proposed interventions of the sustainable renewables risk mitigation initiative (SRMI) have submitted their NDC to the UNFCCC. The proposed countries include: Botswana, Central African Republic (CAR), Democratic Republic of Congo (DRC), Kenya, Mali, Namibia and Uzbekistan. The SRMI Facility aims to be scaled up by adding countries, namely Indonesia and Vietnam (Phase 2), that would have projects following the agreed upon eligibility criteria, as per the term sheet and the present funding proposal.

3. The seven countries submitted an NDC to the UNFCCC that covered conditional and unconditional measures, while Botswana's NDC states that only unconditional measures will be implemented to meet its stated mitigation objective. While unconditional interventions contained in a country's NDC will be affected by the availability of public and private

investment, it can also be argued that conditional measures, which will heavily depend on the availability of international financial support, may also require some level of public funding that will be needed as counterpart funding. According to implicit information derived from the funding proposal, most of the nations that will be covered by SRMI were already gearing up for the NDC projects through allocation of public funding prior to the shutdown of economies caused by COVID-19. The losses suffered by these nations as a result of the pandemic are likely to have limited the availability of funds for NDC intervention thus making it almost impossible to maintain the low carbon pathway that would have resulted from implementation of NDCs. Renewable energy systems (RES) include: solar and wind – together referred to as variable renewable energy (VRE); biomass and waste-to-energy; geothermal; marine; and small hydro, many of which are zero carbon or low carbon energy generation facilities. These RES facilities are suitable to deliver a low carbon future and can enable the project countries to contribute their quota for global GHG mitigation plans to achieve the long-term goals of the Paris Agreement. Prior to the COVID-19 pandemic, many countries, including the developing countries, submitted NDCs which described their actions to contribute to global GHG mitigation. These NDC documents clearly stipulated actions that will be implemented by the countries with funding from national public and private sources, and actions that will be implemented within the country but will be conditional on having, international support in place to fund low carbon projects.

4. Another pertinent observation made in the funding proposal is the fact that a significant proportion of the energy sectors of these seven countries is currently supplied by coal, with plans for capacity expansion in the future. According to information presented in the funding proposal, total installed coal capacity in the seven countries, is expected to increase from 0.5 GW to 7 GW between now and 2030. As per their current plans, by 2030, 8.5 GW of solar and wind generation will be built out of a total of 60 GW installed capacity – representing 14 per cent of total mix. When this share is estimated using GWh energy production, the solar and wind share would drop to between 5–6 per cent. The funding proposal also states that according to the International Energy Agency and estimations by the World Bank, around 20 GW and 45 GW of renewable energy (RE) (including large hydropower), will need to be deployed in the seven countries by 2025 and 2030 respectively, as part of their electricity mix according to their NDCs and to achieve universal access. This will call for an investment of around USD 15 billion for new solar and wind generation alone by 2025.

5. While the implementation of RES has been recognized as an important strategy for achieving the targets set in each of the seven country's NDCs, real barriers exist, impeding implementation of low carbon systems. The limited installed capacity of cost-competitive solar and wind powered energy in many developing countries, including the project countries, is the result of financial, regulatory, structural and technical constraints. The financial constraints include the scarcity of public funds to finance continuous debottlenecking of power transmission and distribution infrastructure, which are usually nationally owned. For example, the existing infrastructure usually requires continuous maintenance and technical strengthening to enable the integration of variable renewable energy (VRE) into power grids. Without such interventions it is difficult to develop grid connected VRE systems. The above constraints together represent serious obstacles to the flow of private sector investors to the development of large RES systems in the project countries and may be compounded by the COVID-19 crisis.

6. Currently these countries are prioritizing the immediate emergency response to manage the current COVID-19 crisis and public investments are being channelled to address the impacts of the crisis. The funding proposal concludes that the need for economic recovery post COVID-19 will become crucial in shaping the long-term pathways for emissions and determining whether the NDC targets can be achieved. If governments do not roll out low carbon development strategies and policies in response to the coming economic crisis, emissions could rebound and even overshoot the previously projected levels by 2030, despite lower economic

growth in the same period. Stated in another way, although the seven countries have ambitious GHG emissions reduction objectives as part of their NDCs, their current energy policies, the planned deployment of coal and the COVID-19 crisis may inhibit these NDC achievements.

7. To tackle exacerbation by the pandemic of existing risks, the World Bank (the accredited entity/AE for this submission) developed the SRMI facility and has applied for GCF funding to support the initiative in Botswana, Central African Republic (CAR), Democratic Republic of Congo (DRC), Kenya, Mali, Namibia, and Uzbekistan. The underlying concept of the proposed programme is described above and the proposal was reviewed by the independent Technical Advisory Panel. According to the information provided in the funding proposal and related annexes, the project has been developed around four major components, summarized as follows:

Component 1: Technical assistance

8. The independent Technical Advisory Panel (TAP) review of the funding proposal and its annexes indicated that this component will focus on the following:

- (a) Comprehensive and integrated planning support to develop least-cost generation plans combined with VRE integration analysis and dispatch and capacity-building for planning;
- (b) Capacity-building for the legal, financial and procurement aspects and transaction advisory support to implement competitive and transparent tendering of privately-financed RE projects; and
- (c) Technical and environmental and social studies to prepare solar and wind parks for future tenders.

Component 2: Public investments

9. This component will concentrate on public investment for:

- (a) Solar and/or wind park common/shared infrastructure;
- (b) VRE integration grid upgrades (including publicly owned battery storage) and enhancing grid resilience to climate change; and
- (c) Electrification to increase the resilience of populations.

Component 3: Risk mitigation instruments

10. Activities at this component level will concentrate on providing risk mitigation instruments to private investors for grid-connected and off-grid renewable energy/storage projects. The risk mitigation instruments that will be employed under this component will focus on the residual risks for private investors aiming to ensure sustainable leveraging of private investments in grid-connected and off-grid RE projects in the seven countries.

Component 4: Project Management

11. This component will concentrate on the management of the projects and coordination and implementation of their activities, in compliance with the executing entities' (EEs) and programme implementing entities' (PIEs) contractual obligations included or referred to in the legal agreements entered into with the accredited entities (AEs).

12. The subcomponent activities to be carried out within the four components described above, and the funds earmarked for them, in each of the seven countries, are summarized in Table 1.



Table 1: Project components and activities (*indicative financing in million USD*)¹

Components	Total Financing Instrument	Botswana	CAR	DRC	Kenya	Mali	Namibia	Uzbekistan
Component 1: Technical Assistance								
Sub-Component 1.1 Generation plans with VRE integration and Capacity Building (technical)	Grants	0.5		1	1	1	1	0.5
Sub-Component 1.2 Transaction Advisory and Capacity Building (procurement, legal)		1.5	1	1	1	1	1	1.5
Sub-Component 1.3 RE Park Infrastructure studies		1.5	0.5	1	1	2.5	1.5	1.5
Component 1 indicative budget per country (all)	22.5	3.5	1.5	3	3	4.5	3.5	3.5
Component 1 indicative budget (GCF)	20	3.5		3	3	4	3	3.5
Component 2: Public Investments								
Sub-Component 2.1 Public investments in solar and wind park infrastructure	Loans (with the exception of CAR which is grant)	30	20			25	21	50
Sub-Component 2.2 Public investments for VRE integration grid upgrades and grid resilience		68	10		75	46	40	250
Sub-Component 2.3 Public investments for electrification (PPP mini-grids, grid extension)		49	60	500	125	62	35	
Component 2 indicative budget per country (all)	1466	147	90	500	200	133	96	300
Component 2 indicative budget (GCF)	206	30	30		45	38	20	43
Component 3: Risk Mitigation Instruments								
Sub-Component 3.1 Risk mitigation instrument for grid-connected RE	Reimbursable Grants			30				
Sub-Component 3.2 Risk mitigation instrument for mini-grids	Guarantees	10					10	
Component 3 indicative budget per country (all)	50	10		30			10	
Component 3 indicative budget (GCF)	50	10		30			10	
Component 4: Project Management (Grants)								
Component 4 indicative budget per country (all)	24.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
Component 4 indicative budget (GCF)	4	0.5	0	1	0.5	1	0.5	0.5
Total indicative budget per country	1563	164	95	536.5	206.5	141	113	307

Abbreviations: CAR = Central African Republic, DRC = Democratic Republic of Congo, PPP = public-private partnership, RE = renewable energy, VRE = variable renewable energy

13. The total financing requested (GCF financing plus co-financing) for this project is USD 1,563 million. This is expected to be contributed as follows:

(a) **GCF:** USD 280 million made up of:

- (i) Senior Loan: USD 176 million with 40 years tenor, 10 years grace period, and 0.25 per cent annual interest rate;
- (ii) Guarantee: USD 20 million with a tenor of 25 years;
- (iii) Reimbursable grants:² USD 30 million with a tenor of 15 years; and
- (iv) Grant: USD 54 million.

(b) **Co-financing:** USD 1,283 million sourced as follows:

- (i) The World Bank – Energy Sector Management Assistance Programme (ESMAP) Grant: USD 18 million;
- (ii) The World Bank – International Development Association (IDA) Grant: USD 65 million;

¹ All the financing amounts presented in Table 1 are estimates based on the current scope of the projects. These are indicative and subject to approval by the World Bank Board and agreements with the EEs.

² The reimbursable grant will be used as a risk mitigation instrument.

- (iii) The World Bank – International Bank for Reconstruction and Development (IBRD) Senior Loan: USD 450 million with a tenor of 20 years and an annual interest rate of 3.4 per cent; and
 - (iv) The World Bank – International Development Association (IDA) Senior Loan: USD 750 million with a tenor of 30–40 years, a grace period of 5–10 years and an annual interest rate of 0.75 per cent.
14. The expected potential outcomes that will be delivered by this project as estimated and presented in the funding proposal and the annexes submitted for this review have been evaluated by the independent TAP as conservative estimates. These are summarized below:³
- (a) It has been estimated that about 2.1 GW of VRE installed capacity will be delivered directly by the project in the seven countries that will be covered by the SRMI. These are expected to be built during the lifetime of the SRMI programme;
 - (b) It has also been estimated and presented in the funding proposal that the SRMI will result in the preparation of an implementation plan for 400 MW of VRE, which are expected to be built after the end of the SRMI programme;
 - (c) The SRMI Facility will also enable the integration of another 1.9 GW of RE. The integration of other projects will be enabled through the VRE integration analysis that will be conducted under subcomponent 1.1 and subcomponent 2.2 under which core grid infrastructure will be financed;
 - (d) The SRMI Facility, through the projects described above, will unlock USD 3.3 billion of private investments, including USD 3 billion for the 2.1 GW that will be built during the lifetime of the projects and USD 300 million for the 0.4 GW that will be built after the end of the projects. These private investments are expected to be leveraged by the activities of subcomponent 1.2 that will deliver transaction advisory and bankable project support. It is expected that this will be provided through the general SRMI support of technical assistance to governments under Component 1. In addition to this, the critical infrastructure under Component 2 and guarantees and risk mitigation instruments under Component 3 are expected to leverage private sector investment in VRE in seven countries; and
 - (e) The independent TAP reviewed the methodologies used in the calculation of GHG emission reductions expected to be delivered by these SRMI components. The methodological approach in all the calculations presented in the submission were judged to be sound and the tools used are expected to deliver reliable and conservative estimates of GHG emission reductions. The total mitigation potential of the SRMI measures when implemented has been reliably estimated to be about 97.93 million metric tCO₂eq made up of:
 - (i) 88.98 million metric tCO₂eq reduction for the 2.1 GW RE that will be built under the projects; and
 - (ii) 8.95 million metric tCO₂eq for the 0.4 GW of RE projects that will be prepared under the projects but not built during their lifetime.⁴
15. In addition to the project mitigation benefits reported above, the project will provide access to clean low-emission electricity to 4.2 million people through the electrification investments that will be engendered by the SRMI interventions.

³ See details for each of the seven countries in Table 3 of the funding proposal.

⁴ GHG emissions calculations were made following the World Bank standards: the official combined margin grid emission factor per country was used.

16. Other benefits that are expected to be delivered by the SRMI in the target countries will include:
 - (a) Strengthening institutional and regulatory systems for planning and development; and
 - (b) Reduction of lock-in long-lived, climate-vulnerable infrastructure by increasing resilience of grid infrastructure.
17. The independent TAP concludes that this project intervention will deliver a high climate impact potential.

1.2 Paradigm shift potential

Scale: High

1.2.1. Comprehensiveness

18. In the absence of this SRMI intervention, the following baseline conditions will continue to hold in all the countries covered:
 - (a) The inability of the existing grid to take electricity from VRE will continue, thus the development of VRE systems will be very constrained. The implication of this is that most of the solar and wind power systems that many of the seven countries relied on in their NDCs to deliver deep GHG emissions reduction may not materialize;
 - (b) A important outcome of the situation described in (a) above is that in many of the seven countries covered by the SRMI, especially those with baseline coal power plants and good access to coal, use of coal-based systems to deliver necessary power generation will be ramped up with strong negative consequences for the ability of countries to meet the target for low carbon emissions included in their NDCs. If they cannot meet these targets, the intended goal of these countries to contribute to limiting global warming to between 1.5°C and 2°C above pre-industrial levels will be thwarted;
 - (c) In the absence of the SRMI investment, public funding will be scarce and consequently investments in power capacity additions and the preparation of national grids for VRE will not take place; and
 - (d) The in-country capacity to carry out renewable power transactions, plan capacity expansion, regulate and create an enabling market environment for power systems' capacity expansion will remain undeveloped, particularly as regards uptake of RE.
19. The COVID-19 pandemic has severely affected the world economy and exacerbated the situations of the seven project countries. The availability of public funding has been further constrained, making the SRMI intervention even more necessary to support the countries in achieving their intended NDCs and making enhanced energy access a reality.
20. During the assessment the independent TAP discussed project risks with the AE. The AE explained why these risks will not constrain the comprehensiveness of the SRMI interventions as summarized below:
 - (a) The independent TAP noted that a number of international (bilateral and multilateral) agencies, including the World Bank, are already working in each of the seven nations on issues similar to the proposed SRMI interventions (strengthening of existing grid to take VRE capacities by supporting the public sector, identifying and implementing solar and wind projects, etc.). The AE pointed out that the risk of double counting will be eliminated through the establishment of a multi-stakeholder group – as an informal high-level platform – to discuss SRMI core principles and hence share each other's vision and programme. This will ensure that overlaps in the programme are avoided but synergies are enhanced. The AE also pointed out that the individual projects to be

funded under the SRMI programme will be selected according to GCF project metrics and criteria thereby ensuring that there is no duplication;

- (b) A second type of risk raised during the independent TAP discussions with the AE is lack of the necessary public and private investment for VRE in each of these countries as a result of the impact of COVID-19. The pandemic will limit public investments for grid infrastructure as public financing is being diverted to meet the needs of critical health and jobs investment during this crisis. In addition, utilities are facing challenges with payment issues from consumers, for example, reduced demand, leading to an increased risk perception on the part of private sector investors. In turn this will increase the premium on the cost of capital and increase the power purchase agreement (PPA) price. The World Bank maintains this risk will be mitigated as summarized in (c) below;
- (c) The SRMI has been designed to support first the governments of the seven countries to leverage private investments in optimized conditions, while maximizing socioeconomic benefits. Private investments are a means to deploy VRE – that are capital expenditure intensive – without impacting public finances. The main constraint with having private investments is that in general, the cost of capital is higher than if it were fully publicly financed. However, to ensure that the trade-off between public financing (that is limited and could be used better) and private financing (that may be more expensive) is fair for the society. SRMI aims to do everything possible to reduce the cost of privately owned generation through an integrated risk mitigation framework leveraging climate finance by reducing the risk premium embedded in the tariff and the cost of financing. The World Bank proposed to minimize public investment to those critical assets that will be publicly owned, such as the grid, while ensuring that any new project is inscribed in a medium-term evidence-based plan and independent power producers (IPPs) are selected in a competitive and robust, as well as transparent, manner with risk mitigation instruments available to reduce further the residual risks (political, utility payments, etc.); and
- (d) The independent TAP identified a further residual risk in the possibility of not having an adequate number of IPPs to bring in private investment for VRE in some of the countries. The success of the SRMI will hinge on: effective public investment that will be utilized to debottleneck existing power grids to facilitate the uptake of renewable power investment from private investors; availability of guarantees as tools for risk mitigation to the private investors who will fund VRE power plants; capacity-building and technical support for public sector stakeholders in the power sector of each of the nine countries for planning, regulation and sound policies that can drive activities of the sector. If the SRMI cannot attract good and financially capable IPPs in some of the seven countries, then the goals of the intervention, namely adding tangible VRE capacities to the national grids, will not be realized. In its response to the issue of IPP investments, the AE pointed out that the main goal of the SRMI is to unlock affordable power purchase agreements. In order to deliver comprehensive outcomes, competition among IPPs must be engendered. This will call for competitive bidding, open to all IPPs, that will attract low-cost capital and reduced costs from equipment providers. This could only happen if the IPP engagement process is open and competitive. The World Bank expects an open and competitive bidding process, attracting IPPs from various locations, including those that are already on ground. To ensure that the IPP acquisition process is followed, the World Bank informed the independent TAP that it will refine the analysis presented in the Feasibility Study (annex A2), which presented the status in each of the seven countries regarding deployment of renewable energy projects by the private sector. In its reaction to the independent TAP on this residual risk, the AE promised to refine this analysis at the project stage to inform the planned activities.

21. The independent TAP believes that the structure of the SRMI as described in the funding proposal and annexes, is sufficiently comprehensive to deliver significant solar and wind power capacity additions during and after the project period. This will be facilitated by the availability of concessional loans that will provide public investment to adapt the grids of these countries to solar and wind power at scale, compared to the status quo ante. This will also remove some of the existing barriers to private sector investment in renewable power development. The provision of guarantees for private sector investments in renewable power capacity will also mitigate barriers and facilitate the flow of funding to develop renewable power projects in these countries. The independent TAP also concurs with the AE that if the process of acquisition by IPPs in each country is open and competitive, then they will be able to participate in the SRMI without any hindrance. To ensure that this is achieved, the AE must refine the analysis regarding deployment of renewable power projects by the private sector in each of the seven countries.

1.2.2. Innovation

22. Innovation is partially delivered through changes to the status quo that result in value creation, reform and renovation of failing systems (inability to deliver sound VRE projects), and ensuring sustainability of the process. The SRMI programme can deliver such results. The development of renewable energy in general and VRE in particular is hampered by various barriers in the seven countries. The proposed SRMI intervention is expected to mitigate these barriers and deliver a low carbon power sector in the selected countries. The strategy to achieve this was presented convincingly in the funding proposal and annexes. Although the technologies (solar, wind, storage batteries, etc.) that are likely to be used by the SRMI in all the countries may not entirely reflect innovation, the SRMI process itself can be considered as innovative as its development will increase the penetration of renewable power systems in project countries.

1.2.3. Creation of an enabling environment/sustainability of outcomes

23. The SRMI process as described above is highly likely to deliver sound outcomes. Activities that will sustain and deliver sound outcomes can be summarized as follows:

- (a) Technical assistance to the public sector of the seven countries to enable them to:
 - (i) Support the public institutions in each country to carry out comprehensive and integrated planning to expand the capacity of the energy supply using least-cost generation tools, which will take into consideration, among other things, the integration of VRE into the grid and dispatch analysis. The technical assistance is also expected to build the capacity of key personnel of the relevant public institutions to carry out these planning exercises once the SRMI intervention ends;
 - (ii) The technical assistance will also build the capacity of relevant professionals in public sector agencies in the legal, financial and procurement areas, by providing transaction advisory support to implement competitive and transparent tendering of privately-financed RE projects. Local personnel will therefore be capable of managing these activities once the SRMI expires; and
 - (iii) The technical support will also deliver technical, environmental, and social studies for solar and wind parks for future tenders, with robust local capacity-building;
- (b) Public Investments through very concessional loans from the SRMI process, which will deliver those facilities, which would have been met with public financing, if private investment in renewable power and especially VRE systems are to become a reality. Key activities that will be covered by this investment will include:

- (i) Solar and/or wind park common/shared infrastructure;
 - (ii) VRE integration grid upgrades (including publicly-owned battery storage);
 - (iii) Enhancing grid resilience to climate change;
 - (iv) Electrification to increase population resilience; and
 - (v) An enabling environment and sustainability of SRMI outcomes will be created by the above activities as follows: a common shared infrastructure will be built and will serve the solar and wind power plants delivered through the SRMI intervention; the RE capacity developed will continue beyond the programme lifetime; upgrading and adding battery storage capacities to the national grids will facilitate integration of variable solar and wind renewable power plants; resilience of grids to climate change will be enhanced beyond the SRMI time period; and ongoing plans to install coal power plants in some countries may be scaled down thereby delivering tangible and measurable GHG emission reductions;
- (c) The Project Management component of the SRMI is also expected to deliver an enabling environment as local personnel will benefit from the knowledge and skills of international experts. The Project Management component has been conceptualized to manage all programme interventions as well as the coordination and implementation of all activities; and
- (d) The SRMI also provides risk mitigation instruments to private investors for grid-connected and off-grid renewable energy/storage projects. The risk mitigation instruments will be focused on reducing and, where possible, eliminating the residual risks for private investors inherent in the present power system. This will enable the SRMI process to crowd in private sector investment and establish a robust tradition of private sector investments in renewable power plant infrastructure in the electricity sector of each of the seven countries.
24. The ability to sustain the sound outcomes of this SRMI programme will only be engendered if the project itself is designed to deliver an enabling environment to do so. In reviewing the funding proposal and related annexes, the independent TAP specifically studied the documentation to identify the enabling attributes of the project. The independent TAP is of the opinion that the activities of the SRMI as presented in this submission are capable of delivering an enabling environment and sustainability of the expected outcomes, even beyond the project lifetime.

1.2.4. Potential for scaling up and replication

25. A key objective of the SRMI is to scale up the global performance of renewable energy power projects in developing countries. According to information available in annex A2 of the submissions reviewed by the independent TAP in 2018, renewable energy plants represented 76 per cent of the net power capacity additions globally, while the share of new fossil fuel power plants was 24 per cent.⁵ While countries like Denmark and Iceland achieved over 40 per cent of RE in their power generation mix by 2018, most countries in Africa and other developing countries were well below 5 per cent.⁶ Inadequate regulatory frameworks, limited planning, inadequate procurement planning and implementation capacity, limited financial viability of off-takers and VRE grid integration challenges were among the key barriers that have been identified as limiting the deployment of RE power infrastructure in many developing countries. The need to mitigate these barriers prompted the World Bank, in partnership with the Agence

⁵ UNEP, 2019, Global Trends in Renewable Energy Investment 2019.

⁶ African Energy, 2018, African Energy Atlas.

Française de Développement (AFD), the International Renewable Energy Agency (IRENA) and the International Solar Alliance (ISA), to develop a new risk mitigation concept that proposes a comprehensive and integrated approach to renewable energy deployment – the SRMI. The SRMI was launched at the Conference of the Parties (COP) 24 (December 2018) by the World Bank and its partners listed above as an integrated approach to RE deployment that will leverage private sector investments, bilateral and multilateral development finance, and concessional climate finance in developing countries.

26. In terms of the potential for scaling up and replication, should the launch of this initiative be successful, the approach will prove to be convincing, thus encouraging scale-up and replication in other countries during and after the programme. There are still a few questions on this initiative but these can only be answered after the programme commences. The independent TAP is therefore of the opinion that implementation of the initiative will lead to scale-up of the deployment of REs and VREs. However, the extent cannot be clearly estimated until the performance of the SRMI demonstrates whether the level and extent of the guarantees included in the initiative will deliver the critical number of IPPs to each country. We believe the answer to this question will depend on the baseline power sector markets in each country. However, the countries where IPP presence has not been outstanding can be monitored as part of this current intervention, thus providing answers that will improve the replication and scalability of the SRMI programme in other countries and regions.

1.2.5. Potential for knowledge-sharing and learning

27. The design and implementation plan of the SRMI include various strategies for knowledge-sharing and learning. Some are described below to illustrate the potential for knowledge-sharing and learning built into the SRMI process:

- (a) The use of financing and transaction advisors for the first round of the RE programme is expected to build the skills needed by in-country personnel, who will work with advisors under a "learning by doing" process to gain much needed hands on experience. This is expected to create cadres of trained local experts within the relevant institutional and regulatory frameworks, who will have the skills to replicate the projects, and fully integrate the changes in their energy systems as SRMI is implemented into countries' long-term strategies;
- (b) The funding proposal section on capacity-building and technology development/transfer states categorically that the strategic analysis included in a component of the SRMI was designed to identify the roles and responsibilities of key stakeholders and to assess the potential of local industries in relation to their position in the value chain. This analysis is expected to define a concrete action plan which will develop the skills of key stakeholders and improve their chances of benefitting from their own RE projects. Furthermore, the plan will benefit from the support of transaction advisors, financed under the SRMI, who will support countries through a "learning by doing" approach for pre-identified stakeholders; and
- (c) The funding proposal also states that capacity-building will be mainstreamed throughout the RE projects in order to strengthen the countries' frameworks and develop the skills needed (including through "learning by doing" with the support of transaction advisors). This approach will enable the seven countries to replicate the projects efficiently.

28. The proposed project will deliver comprehensive and innovative attributes that the independent TAP believes will contribute strongly to a high paradigm shift potential. It will also deliver an enabling environment that will make the tangible outcomes of the intervention sustainable, while also enhancing replicability of the proof of concept to relevant stakeholders. The development of the knowledge management and training platform under the "learning by

doing" approach, will enhance delivery of robust knowledge-sharing and learning attributes that are essential for the scalability and replication of the project. In the light of these observations, the independent TAP concludes that the paradigm shift potential is high.

1.3 Sustainable development potential

Scale: High

1.3.1. Alignment with the Sustainable Development Goals

29. The independent TAP reviewed the funding proposal's claims that successful delivery of the project would be aligned with specific Sustainable Development Goals (SDGs). The independent TAP agrees with the notion that if the project is successfully implemented, it will directly contribute to the achievement of the following SDGs:

- (a) SDG 7 (Ensure access to affordable, reliable, sustainable and modern energy for all): the sustainable development benefits that the project will deliver under SDG 7 include facilitating access to cleaner and lower-carbon electricity to grid connected and off-grid customers in the seven countries. It has been estimated that the SRMI programme will provide access to clean, low-emission electricity to 4.2 million people through the investments that will be engendered by the SRMI interventions; and
- (b) SDG 13 (Take urgent action to combat climate change and its impacts): the sustainable development benefits that the project will deliver under SDG 13 include implementing NDC actions at scale and reducing GHG emissions owing to the increased ability of grids to integrate RE infrastructure provided by implementation of the SRMI programme. In addition, it is expected that coal power capacity additions to the grid will be displaced and electricity access to rural populations will be improved, reducing their dependence on higher carbon supply options.

1.3.2. Economic co-benefits

30. The systems planning aspects of SRMI will enhance the planning of RE infrastructure facilitating investment by private IPP investors and providing guarantees that mitigate their perceived risks. In the countries with substantial coal power generation and coal resources or traditional access to coal resources, SRMI will harness domestic solar and wind resources. This will reduce the need for fuel imports and the potential economic impact of fuel cost fluctuations thereby delivering economic co-benefits to the implementing country. Over time, as a result of the diversification of the energy mix, the country's economy will be protected from the negative impacts of fossil fuel fluctuations. The funding proposal also states that the SRMI process will develop a tariff that is long-term (about 25 years), thus mitigating price volatility, which usually affects net fossil fuel imports and delivering further economic co-benefits.

31. COVID-19 has resulted in the diversion of a sizeable proportion of public funds available in most countries. With available public finance targeted at critical COVID-19 challenges in the project countries, the implementation of SRMI will reduce the economic losses that will arise due to non-implementation of the NDC targets.

32. By leveraging private investments at scale, the SRMI will spare limited public sector finance for the countries, which could be directed to other sectors in need such as healthcare and post-COVID-19 strategies for other needy sectors such as education, infrastructure, welfare and so on. These are co-benefits that will be delivered by a successful implementation of the SRMI programme.

1.3.3. Social and health co-benefits

33. At national levels, the significant penetration of RE power infrastructures will deliver social and health co-benefits. By displacing fossil fuel power generation in almost all the countries, emissions of obnoxious gases and particulates will be reduced. This will deliver tangible health and social co-benefits for people living near the displaced fossil fuelled power plants. The independent TAP concluded that a successfully implemented SRMI programme will deliver tangible social and health co-benefits in each of the seven target countries.

1.3.4. Environmental co-benefits

34. The independent TAP agrees with the view presented in the funding proposal that the project is capable of delivering environmental co-benefits through the increased penetration of RE power infrastructures in the target countries. The RE systems will displace high carbon power generation systems that have potential for expansion, thus yielding the biggest environmental co-benefits. Key environmental co-benefits will include reduction in the release of obnoxious gases and particulates with significant GHG emissions. In the opinion of the independent TAP, the success of the SRMI will deliver tangible penetration of RE power infrastructure that will translate to the above environmental co-benefits.

1.3.5. Gender-related co-benefits

35. The gender co-benefits of the proposed SRMI were well articulated in the funding proposal and in annex A8 (Gender Assessment and Action Plan). It was stated that the SRMI intervention has been planned to enhance female economic participation in the energy sector and to create income-generating opportunities for women in the project countries, in line with the World Bank Group Gender Strategy (2016–2021). Annex A8 details the strategic approach that will be adopted by the SRMI Facility to effectively address identified gender challenges and ensure the investment projects contribute to promoting female development, for instance through employment in the energy sector, while enhancing the livelihoods of women and men in affected communities. The two key ways in which gender actions will be implemented were discussed in annex A8 as follows:

- (a) At the facility level – a systematic approach will be followed, which will include:
 - (i) Support for the projects with a focus on data collection, generating knowledge and providing training and capacity-building activities;
 - (ii) Utilization of the Gender and Energy Program of the World Bank Energy Sector Management Assistance Program (ESMAP), at every stage of the project cycle, to provide technical assistance to the project team and the client country, especially on gender mainstreaming;
 - (iii) At the facility level, recruitment of a balanced female-to-male employment ratio will be encouraged within the Project Management Unit team to the extent possible; and
 - (iv) Key stakeholders' buy-in will be encouraged to ensure that the promotion of women's employment and enhanced livelihoods are effectively provided with measurable outcomes; and
- (b) At the project-specific level – the following approach would be followed:
 - (i) Data collection to document the baseline situation and identify gaps: this will be implemented through the collection of baseline data to set targets and measure progress during implementation, with a focus on countries where existing baseline sex-disaggregated data are weak;

- (ii) Identifying and implementing relevant actions: based on the initial diagnosis and inclusive stakeholder consultations, targeted activities will be identified to address key gender gaps;
- (iii) Monitoring gender issues: the World Bank Group strengthened monitoring system related to gender, which focuses on results and outcomes, will be implemented at the project level; and
- (iv) Strengthening female participation and voice during stakeholder consultations: consultations and public meetings with stakeholders will be conducted throughout the project cycle in a participatory manner and with a special focus on engaging women.

36. In view of the above-mentioned contributions to various elements of sustainable development, the independent TAP concludes that the project will generate significant sustainable development potential and is capable of delivering all the benefits discussed in Section 1.3 above. Therefore, the independent TAP concludes that the sustainable development potential of this proposed project is high.

1.4 Needs of the recipient

Scale: High

1.4.1. Needs of the country

37. Each of the seven countries is susceptible to climate change. As such, they have subscribed to taking actions in their countries to contribute to the central aim of the Paris Agreement limiting global temperature rise to below 2°C. A key sector where the greatest contribution can be made to this global goal in-country is the energy sector, which leads all other sectors in terms of GHG emissions. As stated in many of the important climate change and national planning documents (e.g. NDCs) of the project countries, implementation of renewable energy power generation will yield tangible results that will enhance their energy access and energy security and enable them to contribute to the global goal of limiting temperature increase.

38. Even without the impact of the COVID-19 pandemic on the global economy, in most developing countries (including the seven SRMI countries), barriers exist making the deployment of RE power infrastructure very difficult, despite the fact that the specific cost of generating power (USD/kWe) from these clean resources has become competitive with the costs of traditional fossil fuel alternatives. The emergence of COVID-19 has made it even more difficult to deploy RE systems in these countries as their governments are prioritizing the use of public financing to tackle critical health and employment challenges. The public funding needed to unlock private investments is therefore no longer available.

39. Furthermore, utilities in these countries are facing tremendous challenges as a result of the ongoing pandemic. Missed payments and lower demand for power are both affecting their revenues and therefore their financial ratings. Private investors, who are expected to bring in funds for the deployment of RE, are discouraged by poor financial ratings of utilities, which are then perceived as riskier as the premium on the cost of capital may increase, leading to a rise in the price of electricity. Clearly, in the absence of the SRMI intervention, these barriers will become entrenched and RE systems will not be built at the level envisaged in the countries' NDCs.

40. The SRMI has been designed to address these barriers. It is expected to: provide funds to meet the funding responsibilities of the seven governments to provide public financing for strengthening common facilities in the electricity sector industry; provide funds for stakeholder engagement, which will develop their relevant skill sets for planning energy systems and preparing and implementing robust bid evaluations to select IPPs; strengthen existing

institutional infrastructure in the electricity sector (e.g. regulatory and policy framework); provide guarantees to IPPs to reduce their perception of residual risks, among others. These actions will assist SMRI participants to achieve their primary goals of reducing GHG emissions (particularly the power sector) and improving electricity access and sector diversification.

1.4.2. Economic and financial needs of the country

41. The economic standing of the participating countries varies and the level to which they will fulfil economic and financial needs will depend on their baseline economic levels. According to the information contained in the funding proposal and some of the annexes, their economic standing can be summarized as follows:

- (a) GDP per capita below USD 1,000: CAR, DRC and Mali;
- (b) GDP per capita between USD 1,000 and 2,500: Kenya and Uzbekistan; and
- (c) GDP per capita > USD 5,000: Botswana and Namibia.

GDP growth in each of the seven countries is on average quite strong with the exception of Namibia which was hit with droughts in recent years.

42. The economic impact of COVID-19 on the economy of each of the seven nations can be described as significant, therefore the concessional debt that will flow into the countries through the SRMI process, is likely to have a positive impact on the performance of each economy especially in the area of funding RE needs of the countries, with some tangible multiplication effects. The kind of very concessional debt that will flow into the economy of these nations would not have happened in the absence of the SRMI process.

43. The concessional loan from GCF is expected to crowd in financing from World Bank sources. In addition, grant funding from GCF and World Bank sources will fund various components of the SRMI. Strengthening of the financial market will be essential to change the baseline situation in which RE projects have not received funding. As stated in the funding proposal, government financing to strengthen power grids is constrained by the need to channel funds towards mitigating the health impacts of COVID-19. The continuing scarcity of public funding will also constrain post-COVID-19 recovery. Public funding needed to catalyse a low carbon future will therefore be unavailable in the absence of the SRMI.

1.4.3. Institutional needs

44. Institutional needs will be served in each country using grant funding to implement some of the SRMI activities. Specifically, the baseline of relevant public institutions in the power sector of these countries can be described as follows: very low capacity to carry out generation expansion capacity planning; very low capability to carry out technical analysis for determining how VRE can be integrated into the grids of their country; low capacity to organize competitive bidding for the selection of IPPs, which requires experience in transaction advisory experiences; and low capacity to plan the development of RE Park infrastructure. These weaknesses will be resolved in participating countries by the SRMI programme which will provide a transaction advisory team in each country to lead local national teams in carrying out these duties. As stated in the funding proposal and in many of the annexes reviewed by the independent TAP, the work of these transaction advisory teams will involve building the capacity of the local teams through a "learning by doing" approach. This will deliver some level of local capacity before the end of the SRMI programme, which may then redress the institutional weaknesses in the baseline.

45. The various institutional needs, including those that will benefit the country as a whole and those that will benefit directly or indirectly key stakeholders of the project, will be positively affected by the success of the project, as discussed above. In addition, successful implementation and replication after the end of the intervention, will be enhanced as the

enabling environment for planning, implementing and managing RE programmes will be built into the country systems. This indicates that the needs of the recipient country, its people and institutions, will be adequately met by the project. As a result, the independent TAP concludes that the proposed project has a high potential to meet the needs of the country and its people.

1.5 Country ownership

Scale: Medium to high

1.5.1. Evidence of country ownership through participation of relevant stakeholders

46. Country ownership is strongly related to national stakeholder engagement. Annex A7 of the funding proposal provides information on the approach to stakeholder engagement adopted by the AE in development of the SRMI concept. According to information available in annex A7,⁷ the AE will use a two-tier approach to stakeholder engagement: at facility level and at the project level. The first (facility) tier is expected to engage key stakeholders in each of the seven countries in a comprehensive discussion of the objectives, the components and the important elements of the SRMI programme.

47. As stated in Annex A7, as part of the facility-level stakeholder engagement, preliminary stakeholder engagements were held in the seven countries with discussions during these engagements related to the deployment of renewables and in particular presentations/engagements on the core objectives and steps of SRMI, and details on the technical assistance that will be provided to countries that are now enabling the development of those new projects. This facility-level stakeholder engagement will be continued as soon as GCF funding is approved. The facility-level stakeholder engagement will be finalized after the funding proposal is approved, as part of each country's project design and implementation as presented in each project SEP, and will cover tasks, including stakeholder identification, analysis, information dissemination, consultation and engagement. The independent TAP noted the strength of prior engagement of the World Bank in these seven nations, and considers that familiarity of the AE with the institutional structure in each of the seven country is adequate to enhance the participation of relevant stakeholders in the SRMI programme.

48. The second (project) tier stakeholder interactions will be carried out after each project activity has commenced. A full stakeholder analysis and identification exercise will be conducted for each project. According to the SRMI Stakeholders' Engagement Approach (SEA) (annex A7), the analysis for stakeholder identification will be informed by the level of potential risks and impacts affecting them and will be carried out for all project components. The findings of this exercise will be documented in the project Stakeholders' Engagement Plan (SEP). The independent TAP is of the opinion that the outcomes of the second tier stakeholder engagement should be regularly monitored and reported throughout the SRMI programme duration.

49. Evidence of country ownership is also represented by the availability of a no-objection letter (NOL) from each participating country. All seven countries, namely Botswana, CAR, DRC, Kenya, Mali, Namibia and Uzbekistan, submitted NOLs as part of the SRMI programme submission.

1.5.2. Alignment of the project with relevant national strategies

50. The SRMI programme aims to deliver a much higher level of RE capacity than that shown in the baseline. This plan aligns with each country's NDC, relevant national plans, and/or enabling policies and institutional frameworks. To illustrate this alignment, we present a summary of key commitments in the national documents of each country:

⁷ SRMI Stakeholders Engagement Approach, June 2020.

- (a) *Botswana* – The country intends to achieve an overall emissions reduction of 15 per cent by 2030, taking 2010 as the base year. The emission reduction target was estimated based on the baseline inventory for the three GHGs being CO₂, methane (CH₄) and nitrous oxide (N₂O). The reductions will be realized from the energy sources which are categorized as stationary and mobile. The country is still working on developing initiatives for emission reductions in terms of a long-term low carbon strategy.
- (b) *Central African Republic (CAR)* – CAR aspires to reduce its emissions by 5 per cent and 25 per cent for the 2030 and 2050 horizons respectively, in comparison to its reference business as usual emissions and to increase its sequestration potential.
- (c) *Democratic Republic of Congo (DRC)* – DRC has committed to reduce its GHG emissions by 17 per cent by 2030, focusing specifically on three sectors – energy, agriculture and land use, land use change, and forestry (LULUCF) – which collectively produce nearly 98 per cent of its GHG emissions.
- (d) *Kenya* – Climate Fund to be a financing mechanism for priority climate change actions and interventions approved by the high-level National Climate Change Council chaired by the President. In addressing climate change issues, public entities are required to undertake public awareness and consultations, and ensure a focus on gender, in line with the Constitution and the Climate Change Act (2016).
- (e) *Mali* – The Ministry of Environment is responsible for adoption of NDC targets. There is no stakeholder involvement or multisector commission for the development of NDC targets. Currently, the country has no climate policies in place to mitigate climate change. However, Mali has requested technology transfer to provide the means of implementation.
- (f) *Namibia* – Namibia aims at a reduction of about 89 per cent of its GHG emissions at the 2030 time horizon compared to the business as usual scenario.
- (g) *Uzbekistan* – In the long-term perspective up to 2030, Uzbekistan intends to strengthen measures and actions to combat climate change, in order to achieve the mitigation objectives by decreasing specific emissions of greenhouse gases per unit of GDP by 10 per cent by 2030 from the 2010 level. Achievement of the INDC long-term objective envisages support from international organizations and financial institutions, ensuring access to advanced energy saving and environmentally sound technologies, and resources for climate financing.

51. It is the opinion of the independent TAP that the SRMI as proposed is in alignment with the climate change mitigation objectives of these seven countries, given that GHG emissions are targeted for reduction and the fact that the leading contribution to national GHG emissions inventories in most of these countries, is represented by their energy sectors.

1.5.3. Executing capacity of the executing entity

52. As stated in the funding proposal, the executing entity (EE) has been carefully chosen to combine the public entity responsible for managing external fund inflows in the country (e.g. the Ministry of Finance) and the public entity responsible for overall energy matters in the country (e.g. the Ministry of Energy). In five of the countries, the national utility was also included in the function of EE together with the supervising Ministry of Energy. It was stated in the funding proposal that for the purposes of the loans and grants, the EEs would be the borrowers/recipients of the loans/grants (e.g. for Botswana, the Ministry of Finance, acting through the Ministry of Energy and the state utility, the Botswana Power Corporation. For the purposes of the guarantees contemplated under Component 3.2, the EEs would be the private sector RE project companies.

53. The Ministries of Finance (in CAR, the Ministry of Economy, Planning and Cooperation; in Kenya, the National Treasury) are responsible for the management of funds from international institutions on behalf of the Government and ensuring that they are correctly disbursed and managed. The ministries in the seven countries have lengthy experience and are capable of serving in the capacity of EE. The Ministries of Energy in the participating countries have statutory responsibility for all activities in the national energy sector, including the power sector. They are thus fit for the role in the SRMI programme. In CAR, DRC, Kenya and Uzbekistan the national power utility has not been included in the EE arrangement presented in the funding proposal but the Ministries of Energy are likely to seek their participation. The utility companies have power development experience but in many of the countries, limited RE power experience. This is not likely to hamper the success of the SRMI programme as the transaction advisors will train country stakeholders through "learning by doing", which will provide them with skills not only to participate in the projects, but continue to apply their skills in replication projects, post SRMI.

54. Based on the analysis presented in 1.5.1, 1.5.2 and 1.5.3 above, the independent TAP considers that the country ownership of this SRMI project submission can be ranked as high.

1.6 Efficiency and effectiveness

Scale: Medium to high

1.6.1. Financial structure

55. The proposed project is requesting grant and debt financing from GCF and other co-financing from World Bank sources to the amount of USD 1,563 million. The GCF contribution is a total of USD 280 million made up of: grant – USD 54 million; reimbursable grant – USD 30 million; guarantee – USD 20 million; and senior loan – USD 176 million. The complementary World Bank co-financing will total USD 1,283 million that will be sourced as follows: grant from the World Bank ESMAP – USD 18 million; grant from the World Bank IDA – USD 65 million; and senior loan from the World Bank IBRD – USD 450 million and another senior loan from the World Bank IDA – USD 750 million. The appropriation and terms of these funds were discussed in an earlier section of this report (see table 1 and section 1.1, para. 13 on pages 6 and 7).

56. The breakdown of the funding contribution from these sources, the types of fund and the total that will be provided for the project, is presented in table 2.

Table 2: Breakdown of project funding

S/N	Source	Type of fund	Amount (USD million)
1.	GCF Financing		
a.	GCF	Grant	54
b.	GCF	Reimbursable grant	30
c.	GCF	Guarantee	20
d.	GCF	Senior loan	176
Subtotal GCF			280
2.	Co-financing		
a.	World Bank - ESMAP	Grant	18
b.	World Bank - IDA	Grant	65
c.	World Bank - IBRD	Senior loan	450
d.	World Bank - IDA	Senior loan	750
Subtotal co-financing			1,283
Project total funding			1,563

Abbreviations: ESMAP = Energy Sector Management Assistance Program, IDA = International Development Association, IBRD = International Bank for Reconstruction and Development.

57. The total fund requested from GCF is about 17.9 per cent of total fund required for implementation of the programme. As a percentage of total SRMI programme funding for the seven countries, GCF requested contributions can be summarized as follows:

- (a) Grant funding is 3.45 per cent;
- (b) Reimbursable grant funding is 1.9 per cent;
- (c) Guarantee is 1.3 per cent; and
- (d) The senior loan is 11.3 per cent.

58. The grant funding from GCF has been earmarked as follows:

- (a) About 37 per cent of the GCF grant funding will be spent to support each of the activities of Component 1. Activities under this component will cover preparation of the generation plan with VRE integration and capacity-building in the nine countries.
- (b) About 55.6 per cent of GCF grant will be spent on the activities of Component 2 only in CAR. Activities of this component will focus on public investment in solar and wind park common infrastructure, VRE integration upgrades and grid resilience. These activities in the remaining six countries will be financed by loans.
- (c) The balance of 7.4 per cent of GCF grant has been earmarked for project management costs.

59. Reimbursable grant, guarantees and senior loan requested from GCF will be utilized as follows:

- (a) The entire USD 176 million senior loan from GCF is planned to be utilized for funding activities of Component 2 in five countries namely: Botswana, Kenya, Mali, Namibia and Uzbekistan;
- (b) Activities of Component 2 in CAR (the sixth country) will be financed by grant from GCF;
- (c) The reimbursable grant of USD 30 million has been entirely earmarked to provide a risk mitigation instrument for grid connected RE in DRC; and
- (d) The USD 20 million guarantee funding from GCF will be spent in equal proportion to provide risk mitigation instruments for mini-grids in Botswana and Namibia.

1.6.2. Co-financing, leveraging and mobilizing long-term investment

60. The USD 280 million requested from GCF will be complemented by about USD 1,283 million from the World Bank. The GCF fund as well as the complementary World Bank fund are expected to be public investments in the electricity sectors of the countries, which will enable leverage of private investment for the countries to build RE power plants. Successful implementation of these public investments is expected to leverage around USD 3 billion in private investments into the seven countries. This will yield an indicative leveraging between public investments (World Bank and GCF financing) and private investments of about 1 to 2.4. This can be considered an effective and efficient use of funds for the SRMI programme. As stated in the funding proposal, the leveraged private investment will only occur if the public investment is successful as laid out in the project components. Therefore, the leveraged USD 3 billion estimate is not considered part of the co-financing for this programme.

61. The independent TAP is of the opinion that this financial leveraging ratio indicates sound cost-effectiveness of the project when compared with benchmarks of similar projects recently funded by GCF.

1.6.3. Cost efficiency

62. The project is expected to deliver about 16.1 MtCO₂eq of GHG emission reductions over a 20-year period. The efficiency metrics that will be delivered by this project intervention were estimated as follows:

- (a) Total project financing = USD 1,563 million
- (b) Requested GCF amount = USD 280 million
- (c) Expected lifetime GHG emissions = 88,978,000 tCO₂eq
- (d) Estimated cost per tCO₂eq (a/c) = USD 17.57/tCO₂eq
- (e) Estimated GCF cost per tCO₂eq (b/c) = USD 3.15/tCO₂eq

63. The independent TAP agrees with the conclusion in the funding proposal that the estimated GCF mitigation cost of USD 3.1/tCO₂eq, as presented above, when added to the project's additional benefits (better electricity access, resilience of power infrastructure, etc.) represents excellent value for money. This conclusion will hold when the GHG emissions reduction cost is compared to estimations for other similar energy projects already approved and funded by GCF.

1.6.4. Economic and financial rate of return of the investment

64. The independent TAP reviewed the economic and financial analysis carried out for the programme implementation in each of the seven countries submitted by the AE to the GCF Secretariat (see annex A3 – SRMI Economic and Financial – 30 November 2020) and came to the following conclusions:

- (a) Results for all the countries analysed showed that GCF financing increased the financial internal rate of return (IRR) by an average of about 2 per cent. The concessional GCF financing is therefore critical to achieve a feasible SRMI programme in these countries; and
- (b) The financial analysis conducted for five of the countries, namely Botswana, Kenya, Mali, Namibia, and Uzbekistan for activities under Components 1 and 2, yielded financial IRR of 13 per cent, 18 per cent, 27 per cent, 12 per cent and 14 per cent respectively. The concessionality of the GCF loan can therefore be said to be instrumental to achieving an acceptable IRR for investment in those countries.

65. The independent TAP therefore concludes that the efficiency and effectiveness of implementation of this proposed project will be moderately high.

II. Overall remarks from the independent Technical Advisory Panel

66. The independent TAP recommends that the Board approve the SRMI.

67. The independent TAP further recommends that, should the funding proposal be approved, the AE shall agree to the inclusion of the following reporting requirements in the funded activity agreement to be signed between GCF and the AE:

- (a) Provide under each APR a programme monitoring report at the Facility level, also covering Project-level activities information, as and when available, at the minimum covering the following:
 - (i) report detailing the success and challenges of the gender mainstreaming process presented in the Gender Action Plan;

-
- (ii) report of the lessons learned, and successes and challenges, of stakeholder engagement at Project level as well as reflecting process and measures undertaken or to be undertaken in the future ensure that pitfalls are corrected in a timely way; and
 - (iii) the GHG emission reductions achieved using the monitoring protocol of the emission reduction estimation method adopted, and covering, at the minimum:
 - (1) Project parameters monitored by name, value and confidence interval;
 - (2) Emission reduction calculation methodologies utilized;
 - (3) The procedure used to develop the database; and
 - (4) The quality assurance/quality control methods used in preparing and reporting the data and the emission reduction calculations.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP 163)

Proposal name:	Sustainable Renewables Risk Mitigation Initiative (SRMI) Facility
Accredited entity:	The World Bank
Country/(ies):	Botswana, CAR, DRC, Kenya, Mali, Namibia and Uzbekistan
Project/programme size:	Large

Impact potential

Thank you for the assessment. The SRMI Facility aims to be scaled up by adding new countries. We want to take this opportunity to thank the GCF Secretariat that worked with us very hard on making this program impactful, implementable and replicable. In the context of COVID-19, we strongly believe that this Facility is critical in supporting countries in their energy transition.

Paradigm shift potential

Thank you for the assessment. As agreed with iTAP, the World Bank team will adapt its support to each country based on the status of solar/wind independent power producer (IPP) deployment (i.e., if it had existing bilateral deals, feed-in-tariff or already existing tenders). At the project stage, a more thorough analysis of the status will be performed, and the support of the transaction advisors under SRMI will be critical in promoting a sustainable and bankable transaction.

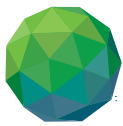
We agree that the need for risk mitigation instruments/credit enhancements (under Sub-Component 3.2) will have to be evaluated at a project level with the support of the transaction advisors and when developing the tenders. We fully agree with iTAP that the need for such support depends on the baseline power sector markets in each country. It is for that reason that SRMI aims to be an integrated approach that tackles all key risks from a country, sector and project perspective (as a comprehensive and articulated risk mitigation framework). Not only by implementing SRMI's methodology, countries will be able to unlock private investments but they are also ensuring that they are on the right track to improve their sector bankability – via the right plans reducing the average generation costs, and via upgrades on the grid to reduce losses.

Sustainable development potential

Thank you for the assessment.

Needs of the recipient

Thank you for the assessment.



Country ownership
Thank you for the assessment.
Efficiency and effectiveness
Thank you for the assessment.
Overall remarks from the independent Technical Advisory Panel: The WB team appreciates ITAP's review and assessment outcomes and notes with thanks the positive recommendation from iTAP to the GCF Board, in particular that the Board approve the SRMI.



SRMI FACILITY GENDER ASSESSMENT AND ACTION PLAN

November 2020

SRMI Sustainable Renewables
Risk Mitigation Initiative

Disclaimer: The gender profiles were prepared, using publicly available sources. The World Bank does not guarantee the accuracy or completeness of the information provided in the present document. To the extent that it reflects opinions, they are the opinions of the individual authors and do not necessarily reflect the views of the World Bank, its staff or Directors. Nothing in this summary should be taken as legal advice.

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1. INTRODUCTION

The Sustainable Renewables Risk Mitigation Initiative (SRMI) Facility aims to support developing countries in advancing sustainable solar and wind programs and attracting private investments. The proposed Facility's objective is to enable seven countries to unlock GWs of solar and wind generation in the medium-term with financial support from the Green Climate Fund (GCF). Targeted countries are Botswana, Central African Republic (CAR), Democratic Republic of Congo (DRC), Kenya, Mali, Namibia and Uzbekistan. The SRMI Facility will deploy its support through three components, namely (i) Technical Assistance, (ii) Public Investments and (iii) Risk Mitigation Instruments.

The present Gender Assessment and Action Plan provides an overview of issues affecting men and women related to the Facility's implementation in the seven countries. It details the strategic approach that will be adopted by the SRMI Facility to effectively address these challenges and ensure the investment projects contribute to promoting female development, for instance through employment in the energy sector, while enhancing women and men's livelihoods in affected communities. Planned interventions under the present Gender Action Plan (GAP) are aligned with the GCF third policy objective to "contribute to reducing the gender gap of climate change-exacerbated social, economic, environmental vulnerabilities and exclusions through GCF climate investments that mainstream gender equality issues."

The Facility's activities will also directly support the World Bank Group 2016-2023 Gender Strategy and its four pillars of action¹. Projects will include, to the extent possible and subject to funding availability, interventions to facilitate women's access to energy services which has the potential to enhance maternal outcomes and safety (Pillar I); promote proactive policies to increase female employment along the energy value chain (Pillar II); create income-generating opportunities through access to grants or access to micro-finance for women living in the areas of implementation of the Renewable Energy (RE) projects (Pillar III) promote women's engagement in stakeholders' consultations; and implement rigorous mechanisms to prevent and respond to gender-based violence (GBV) in project areas, thus enhancing women's voice and agency (Pillar IV).

¹ The WBG Gender Strategy has four objectives: (i) Improving Human Endowments, (ii) Removing Constraints for More and Better jobs, (iii) Removing Barriers to Women's Ownership and Control of Assets and (iv) Enhancing Women's Voice and Agency and Engaging Men and Boys.

2. COUNTRY BACKGROUND

Despite varying economic and social development levels, the seven countries targeted by the SRMI Facility face similar gender gaps and energy challenges. Relations between men and women and levels of inequality vary considerably across areas: the 2020 World Economic Forum's Gender Gap Index ranks Namibia 12th out of 153 countries, while Mali and the DRC are respectively ranked 139th and 149th. Nonetheless, all countries suffer from social norms that confine women to unpaid household and care work resulting in lower education, occupational sex-segregation and limited political participation.

In low to middle income countries where access to energy is limited, women suffer most from insufficient access to electricity and clean cooking facilities as they are mostly responsible for household chores¹. In Sub-Saharan Africa electrification rate stands at less than 48 percent, leaving nearly 600 million people without access to electricity². This issue negatively impacts female economic empowerment, agency, education and health outcomes. For instance, every year, 4.3 million people – mainly women and children – die as a result of indoor air pollution³.

Women have limited employment opportunities in the formal sector, including in the energy industry, due to limited education, qualifications and skills. They are therefore more likely to occupy lower-paid, lower-quality jobs in family or home-based businesses, such as farming and handicrafts. Men predominate in Science, Technology, Engineering and Mathematics (STEM) fields and are disproportionately active in the construction and information technology industries. For instance, in Kenya, 11 percent of employees in STEM fields are female, against 21 percent for men. This education gap is reflected in energy infrastructure projects where female employment is lower due to a lack of technical construction and professional engineering skills⁴. Worldwide, the RE industry employs only about 32 percent of women, while it is estimated that the number of available jobs in the sector will increase from 10.3 million in 2017 to 29 million in 2050⁵. Female access to new employment opportunities created by RE projects is further constrained by cultural norms that limit women's choice and access to non-traditional fields. Employment opportunities created by women in large-scale power projects are therefore usually in traditionally feminine sectors such as restauration and hospitality or administrative support⁶.

Women's limited political agency and representation in political institutions translates into lower female access and participation in energy-related decision-making. With the exception of Namibia where the share of seats held by women in national parliament reaches 46 percent, none of the targeted countries has more than 30 percent of female parliamentarians. In DRC, CAR, Mali and Botswana, the rate of female in Parliament does not

¹ Resurrección and B.Boyland,M. 2017. Gender Equality in renewable energy in the Lower Mekong: assessment and opportunities – USAID clean power Asia.

² World Bank Data. 2018.

³ The United Nations Development Programme (UNDP). 2016. Gender, Climate Change and Food Security.

⁴ United States Agency for International Development (USAID). 2016. Engendering Utilities: Improving Gender Diversity in Power Sector Utilities.

⁵ International Renewable Energy Agency (IRENA).2019. Renewable Energy: a gender perspective.

⁶ The World Bank Group. Energy Sector Management Assistance Program (ESMAP). 2018. Getting to Gender Equality in Energy Infrastructure.

exceed 10 percent. Women's representation and participation in the energy sector, including the RE sector, is therefore limited because women are not thought of as key stakeholders.

Although more data is needed to assess the prevalence of gender-based violence, studies indicate that women and girls routinely experience high levels of sexual harassment and intimate partner violence in target countries. This phenomenon is reinforced by social norms normalizing violence against women. For instance, and although Namibia is recognized for its progress towards equality between men and women, 33 percent of ever-married women aged 15-49 have experienced intimate partner violence, and 28 percent of women aged 15-49 approve the practice of husbands beating their wives to discipline them (Namibia Demographic Health Survey, 2013). The influx of migrant workers created during the realization of large-scale RE projects may increase GBV risks and should receive specific attention throughout the projects-cycle.

At the national level, most countries are signatories of international and regional gender equality agreements (e.g. the Convention on the Elimination of All Forms of Discrimination against Women or CEDAW, the Beijing Declaration and Platform for action). Most countries have also adopted national strategies and mechanisms to address inequality between men and women. During project implementation, national and international legal frameworks may therefore be used to strengthen stakeholders' commitments to promote women's economic empowerment and agency, as it is often aligned with governmental priorities.

2.1 Botswana Profile

Botswana is a small, landlocked country located in the heart of Southern Africa. With a density population of less than 4 people per sq. km and a population estimated at 2.3 million, Botswana are highly dispersed across the country. Three major ethnic groups coexist in the country, namely the Tswana (80 percent), Kalanga (11 percent), and the Sarwa (3 percent). More than 70 percent of the country's territory is covered by desert lands, making agricultural conditions difficult. Botswana is rich in natural resources with abundant supply of diamonds, silver, copper, nickel etc. The country's economy is heavily reliant on diamond, making it highly vulnerable to the market's price volatility.

Botswana is a stable, peaceful country that has experienced one of the world's fastest economic growth in the past 50 years. It is regularly quoted as an example of good governance and responsible resource management with significant socio-economic gains. Botswana is classified as an upper middle-income country: between 2003 and 2010 poverty went down from 30.6 percent to 19.4 percent. The country has also experienced high rates of urbanization in recent years: in 1990, only 40 percent of Botswana lived in urban areas while this was the case for more than 69 percent of the population in 2018.

Despite high national investments in the delivery of social services and infrastructure, human development outcomes remain low by global standards. The country continues to grapple with unequal distribution of wealth, high levels of poverty and unemployment, as well as HIV. Botswana's national Gini coefficient is estimated at 55.3 (2015) making it one of the most unequal societies in the world. The country's extremely low population density is a key barrier to effective service delivery. Only 60 percent of national households in Botswana have access to electricity, with significant disparities in access between urban and rural households and across the income

distribution. In rural areas, only 42 percent of households have access to proper sanitation and 22 percent to electricity.

With the exception of the African Charter on Elections and Governance (ACDEG) and the Maputo protocol, Botswana is signatory of key international and regional agreements related to women's rights. The Government of Botswana has ratified the International Labour Organization (ILO) Convention on equal pay for work of equal value, the Convention on the Elimination of all Forms of Discrimination against Women, the Protocol to the African Charter on Human and People's Rights on the Rights of Women in Africa (1989), adopted the Beijing Platform of Action in 1995, signed the Southern African Development Community Declaration on Gender and Development (1997) as well as its addendum on the Prevention and Eradication of Violence Against Women and Children.

At the national level, Botswana has adopted a comprehensive legal framework to protect women's rights and promote gender equality. The Constitution guarantees equality for all persons irrespective of gender. The country's main reforms on women's rights are chronologically listed below:

- The **1993 amendment to the Tribal Land Act** allows any citizen, including women, to apply for any Botswana land;
- The **1996 Mines and Quarries Act** removes restrictions placed on women working underground in the mines;
- The **1996 Employment Act** guarantees equal remuneration for work of equal value;
- The **1998 Penal Code Act** stepped up penalties for sexual offences and criminalized the intentional spread of HIV;
- The **2001 Marriage Act** prohibits child marriage;
- The **2003 amendment of the Citizenship Act** allows women married to foreign nationals to pass citizenship on to their children;
- The **2004 amendment to the Penal Code** introduces a minimum sentence of ten years for rape;
- The **2004 Abolition of Marital Power Act** provides equal power for men and women within common law marriages excluding customary and religious marriages;
- The **2008 Domestic Violence Act** provides protection to survivors of domestic violence. However, it does not acknowledge marital rape.

Several national plans were adopted by the government in recent years to advance women's rights, while gender issues have been embedded across sectoral policies. Following the ratification of CEDAW in 1981, the government established a Women's Affairs Unit. In 1995, Botswana developed its first Policy on Women and Development, followed by a Development Plan (2010-2016) targeting women and marginalized groups who engage in small-scale agriculture. The National Gender-Based Violence Strategy 2015-2020 aims to support design and implementation of integrated actions to eliminate gender-based violence.

Like many other Sub-Saharan nations, Botswana continues to operate under a dual legal system, Common Law and Customary Law, with the effect of preserving some discriminations against women in practices. For instance, the Abolition of Marital Power Act exempts customary laws. However, the judicial system has recognized the principle of equality between men and women within the household in its decisions and has protected women's right to own and inherit property. This indicate progressive trends in the society that have yet to be codified.

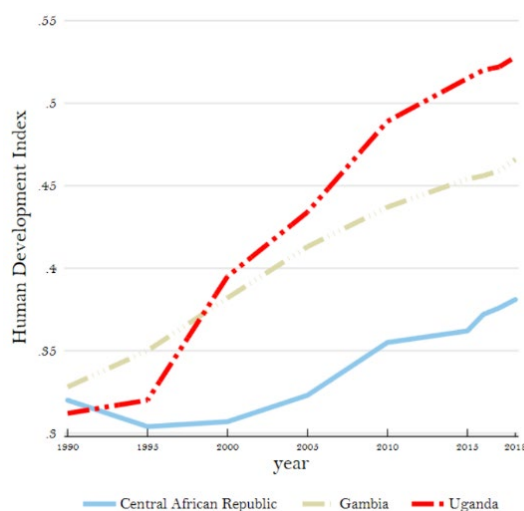
2.2 Central African Republic Profile

The Central African Republic is a landlocked country with important natural resources and an estimated population of 4.9 million. The country is rich in natural resources including diamond, gold, copper, forestry and agricultural land. CAR has one of the lowest density population in the region, with 8.3 people per square kilometer, making delivery of basic public services difficult. The country's economy relies mostly on the agriculture and livestock sectors that employ nearly 80 percent of the active population. The majority of the Central African population (60 percent) lives in rural areas.

Since 2012, the country is facing the most complex and important conflict in its history due to its intensity, duration and territorial scope. According to UN estimates, the conflict caused more than 6000 death. The Government of CAR has authority over less than 40 percent of the territory while the rest of the country is under the control of various armed groups. The political crisis turned into a humanitarian crisis, causing population displacement, the disruption of economic activity and the destruction of social infrastructure (health, education). At the national level, more than one quarter of households have been displaced. This conflict led to a 58 percent decline in crop production, increased food prices, and an elevated risk of malnutrition. The democratic and peaceful election of a new President in 2016 raised hope for stabilization, soon dashed following the resurgence of armed clashes in 2017. A new Peace Agreement was concluded between the authorities and 14 armed groups in February 2019.

CAR is one of the poorest countries in the world and was ranked second to last out of 188 countries on the United Nations 2019 Human Development Index (HDI). Repeated conflicts in recent years worsened living conditions for the population, in a country that was already plagued by high-levels of poverty, inequality, unemployment and low access to basic services. The majority of the population lives below the international poverty line and is food insecure, especially in rural areas. Life expectancy is among the lowest in the world (52 years), low even by Sub-Saharan standards. Government and institutional capacity and presence across the territory is scant; the population heavily relies on the UN peacekeeping mission MINUSCO for security and on international actors for the provision of health and education services.

Figure 1: HDI trends for Central African Republic, Uganda and Gambia, 1990-2018



Source: 2019 UNDP Human Development Report

The collapse of state institutions caused by the insecurity and poor governance has resulted in extremely limited supply of basic services and infrastructure, even in the capital city of Bangui. Only about 8 percent of the population has access to electricity which is far below the average for Sub-Saharan Africa. Outside of Bangui, access to electricity is practically non-existent (under 2 percent). Access to clean water and sanitation is a challenge across the country; 70 percent of households are without access to drinking water and 80 percent live without latrines. In areas outside of Government authority, armed groups compete for control over land and resources, hampering the provision of social services and humanitarian support to the population.

Although the Central African Republic has ratified most international and regional human rights instruments and adopted some laws and policies in favor of gender equality, legal texts remain largely unapplied and unknown to the general population. CAR ratified the African Charter on Human and Peoples in 1986, the Convention on the Elimination of Discrimination against Women (CEDAW) in 1991, the Convention on the Rights of the Child in 1992, signed the Beijing Declaration and Platform for action in 1995, signed the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in 2008 (without ratification), and committed to the implementation of UN Security Council Resolution in 2014 through the adoption of a National Action Plan.

Despite some progress in recent years, the national legal framework remains insufficient to protect women's rights and is weakened by lack of respect for the rule of law. The country's major milestones in the field of gender equality are listed below:

- **Ordinance No. 66/16 of 1966** prohibits the practice of female genital mutilation (excision);
- **Ordinance No. 66/26 of 1966** promotes the advancement of girls and their continuation in the education system until the age of 21 years;
- **Act No. 97/013 of 1997 on the Family Code** requires the consent of both spouses for a marriage to be lawful, bans early marriage and provides that divorce may be granted at the request of either spouse. The Code

guarantees women's capacity to inherit. However, many discriminatory provisions persist. Article 254 and 255 state that the husband is the head of the family and chooses the marital home. Polygamy is authorized and a prosecutor may authorize individuals under the age of 18 to marry;

- **Act No. 06.030 of 2006 establishing the rights of people living with HIV infection/AIDS** upholds the principle of non-discrimination against people living with HIV infection/AIDS;
- **Act No. 06/032 of 2006 protecting women against violence** in the Central African Republic prohibits rape (excluding marital rape) and provides a definition of gender-based violence. However, there is no minimum sentence for rape;
- **Act No. 06.005 of 2006 on reproductive health** protects women's access to health services so that they can have a healthy pregnancy and delivery. Abortion remains prohibited with some exceptions;
- **Act No. 09.004 of 2009 establishing the Labour Code** guarantees that all workers, regardless of origin and gender, receive an equal wage for equal work, qualifications and output;
- **Act No. 06.005 of 2016 on parity** requires 35 percent of elected and nominated women in decision-making bodies; and
- **In 2016, a new Constitution was promulgated.** It prohibits discrimination based on gender (Article 6), requires that political parties respect the principle of representation of gender (Article 131) and upholds the principle of equality between men and women (Article 148).

At the policy level, the country has adopted several national strategies to promote women's rights in 2005¹, 2007² and 2011³ but has failed to translate its commitments within sector-specific policies. Furthermore, customary law remains widely implemented resulting in systematic discrimination against women and girls.

2.3 Democratic Republic of Congo Profile

DRC is the largest country in Sub-Saharan Africa, with an estimated population of 84 million in 2018, the third largest in the continent after Nigeria and Ethiopia. DRC's population is diverse: while the majority belongs to the Bantu ethnicity, more than 200 ethnic groups have been identified. The country's natural resources are among the richest and the most diverse in the world (copper, cobalt coltan, gold, diamonds, oil, etc.). DRC accounts for about 16 percent of the world's total production of diamonds and is the largest copper producer in the continent. The Congolese population is young and mostly rural; 33 percent of the population is under 15 years old while 70 percent of the population lives in rural areas. Central DRC where both Kasai provinces are located (and where the SRMI Facility project will focus) is the center of the diamond trade and has the largest accumulation of industrial diamonds in the world.

Despite its natural resource wealth, DRC is one of the poorest countries in the world and is not on track to achieve any of the United Nations' Sustainable Development Goals (SDGs) by 2030. The country's Human Development Index value for 2018 is 0.459 which puts it in the low human development category. DRC is ranked 179 out of 189 countries and territories by the 2019 Human Development Report. About two-thirds of the

¹ Government of DRC. 2005. Politique Nationale de Promotion de l'Egalité et de l'Equité.

² Government of DRC. 2007. Plan d'Action de la Politique Nationale de Promotion de l'Egalité et de l'Equité.

³ Government of DRC. 2011. Stratégie Sectorielle Egalité de Genre et Réduction de la Pauvreté.

population lives below the poverty line. The proportion of poor people has only declined slightly in past decades, with disparities across regions. For instance, in both provinces of Kasai, poverty incidence increased between 2005 and 2012¹. DRC's population is growing at a very fast pace, resulting in an increased absolute number of poor people in recent years. Gender gaps are observed in the distribution of poverty: widows and female-headed households are more likely to be poor due to lower access to assets and resources.

Between 1996 to 2003, DRC endured two wars that have generated massive flows of internally displaced people and refugees and killed millions of people. In the eastern region and Kasai province, the country continues to suffer from conflict and instability fueled by ethnic tension, disputes over land ownership and resources as well as foreign interventions. According to the Internal Displacement Monitoring Center, nearly 1.7 million new displacements were recorded in 2019. Félix Tshisekedi was elected president in January 2019, following an electoral process that marked the first peaceful transition of power since the country's independence in 1960.

The country's socio-economic development is hindered by poor infrastructure coverage, affecting the Congolese's population access to electricity, improved water resources and road transport. Only half of DRC's territory is accessible by road or rail transport and the country's electricity access rate is far below the regional average. Only 15 percent of the population has access to electricity and an estimated 90 percent rely on fuelwood and charcoal for energy. Lack of infrastructure particularly affect women who are responsible for water and fuel collection and are more likely than men to die from respiratory diseases and accidents caused by the low quality of cooking appliances.

In recent years, DRC has introduced a number of reforms that have removed some legal barriers to gender equality. However, discriminatory provisions remain in the Family Code as well as significant gaps with regards to gender-based violence, economic and political participation. For instance, to this date, the Congolese legal framework does not contain a specific law on domestic violence. Furthermore, a 2015 revision of the electoral law removed the requirement for a 30 percent quota of female candidates during national elections, which is considered a setback for women's political participation.

DRC has ratified most international and regional human rights instruments. They include the SADC Memorandum of Understanding on Gender and Development in 2008; the Convention on the Elimination of All Forms of Discrimination Against Women (1986); the Maputo Protocol (The Protocol to the African Charter on Human and People's Rights on the Rights of Women in Africa) in 2009; the UN Convention on the Rights of the Child (1990); the Universal Declaration of Human Rights; and the UN Security Council Resolution on Women, Peace and Security (2010).

However, at the national level, implementation of international commitments remains weak. Customary law continues to be widely implemented across the country, especially in rural areas, even though its discriminatory provisions contradict positive law. Early marriage remains a reality for many girls, and women don't have equal access to inheritance due to the application of traditional practices. The main reforms towards gender equality adopted in the past decades are listed below:

¹ World Bank Group. 2018. Democratic Republic of Congo Systematic Country Diagnostic.

- **The 2002 Labor Code** prohibits sexual or psychological harassment in the workplace (Article 73);
- **The Constitution of the Democratic Republic of Congo** promulgated in 2006, establishes that all human beings are equal under the law (Article 11), specifically bans discrimination against women (Article 14) and prohibits sexual violence (Article 15);
- **The Sexual violence laws of 2006** expands the definition of rape and defines the rights of survivors;
- **The 2015 Parity Act** guarantees the right of women to entrepreneurship and promotes access to savings, credit, and new technologies without discrimination on the basis of gender;
- **In 2016, the Labor Code** was amended to allow women to work at night in the mining, manufacturing and construction industries. The revised Code also establishes the principle of non-discrimination in employment.

A New Family Code was promulgated in July 2016, introducing several improvements with respect to women's rights: early marriage was banned, and equal rights recognized for husbands and wives in the household. Wives can now work without asking for their husband's authorization and their right to ownership and inheritance is recognized. The new legislation on nationality also enables women to pass their Congolese nationality on to their children. At the policy level, DRC developed a National Gender Strategy for the period 2017-2021 to promote gender equality and women's empowerment with five strategic axes¹. The Ministry of Gender, Family and Children is responsible for overseeing the Strategy's implementation, and a gender focal point has been appointed in each line ministry.

2.4 Kenya Profile

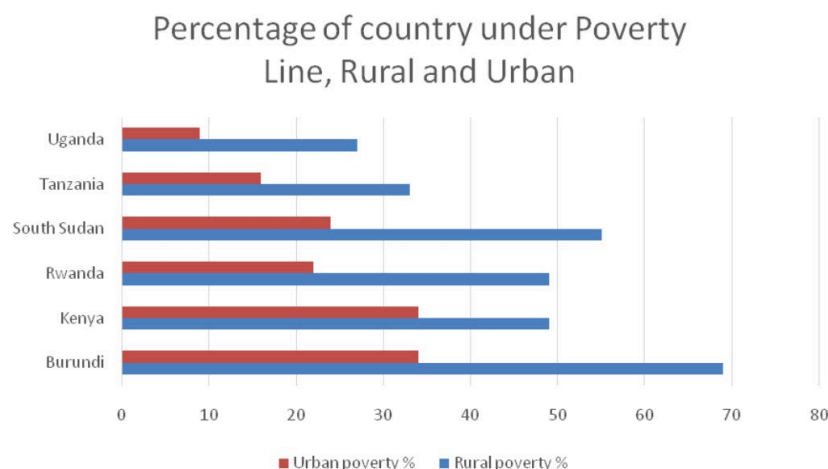
With a fertility rate estimated at 3.5 births per woman and a population of 51 million people, Kenya is among the world's 33 countries projected to see their populations increase five-fold or more by 2100². Despite rapid rate of urbanization, nearly 73 percent of the Kenyan population continues to live in rural areas. Only 20 percent of the country's land is arable, with the rest of the territory being classified as arid or semi-arid. Kenya achieved lower-middle-income status in 2012. The country's economy is driven by the agriculture and tourism sector.

Although Kenya has experienced sustained economic growth, social development and political gains over the past decade, high levels of poverty and inequality persist. In 2015, 36 percent of the population lived below the national poverty line and 56 percent of the urban population lived in slums. With a Gini coefficient of 40.8, Kenya remain an unequal society with wide regional disparities between arable and arid regions. In the latter areas where 20 percent of the population lives, provision of basic social services remain a challenge due to low population density, infrastructure deficits and insecurity.

¹ These are : (i) consolidation of gender equality and women's empowerment (ii) strengthening the role and place of women and girls in the economy and employment; (iii) increased access of women and girls to decision-making spheres; (iv) fight against forms of sexual violence; and (v) strengthening the institutional mechanism for implementation of the National Gender Policy.

² The United Nations Department of Economic and Social Affairs (DESA).2017.

Figure 2: Percentage of country under Poverty Line, Rural and Urban (percentage)



Source: Population Reference Bureau and EACREE report, 2018

With an agriculture sector accounting for more than a fourth of the country's gross domestic product, Kenya is highly vulnerable to climate change. In recent years, the frequency of droughts has increased and exacerbated the vulnerability of the population living in arid and semi-arid areas. In the remaining 30 percent of the territory, climate change has intensified pressure on existing natural resources, especially forest and soil. The tourism industry is also likely to be impacted by the anticipated loss of bio-diversity. The country will experience increased flooding, particularly affecting urban areas through the destruction of infrastructure and homes and the development of water borne diseases.

Kenya has achieved rapid progress in electrification and is the only country in East Africa on track for meeting the Sustainable Development Goal of universal electricity access by 2030. In less than a decade, Kenya more than tripled electricity access in the country. In 2017, about 64 percent of the population had access to electricity compared to 19.2 percent in 2010. However, 17 million people remain without access to electricity, especially in rural areas, owing to high electricity fees and geographical inaccessibility. Traditional use of biomass for cooking (wood fuel, dung, charcoal, and agricultural waste) continues to predominate across the country with negative impact on health incomes. 34 million people do not have access to clean cooking in Kenya¹.

Figure 3: Access to electricity in Eastern African Countries

Country	Total population (Millions)	Access to electricity (%) of population)	Access to electricity, rural (% of rural population)	Access to electricity, urban (% of urban population)
Burundi	11.1	11.0	3.4	61.6
Kenya	51.3	75.0	71.6	83.9

¹ East African Centre for Renewable Energy and Energy Efficiency (EACREEE). 2018. Situation Analysis of Gender and Sustainable Energy in the East African Community.

Rwanda	12.3	34.7	23.4	89.0
South Sudan	10.9	28.2	23.6	46.8
Tanzania	56.3	35.5	18.8	68.3
Uganda	42.7	42.6	38.0	57.5

Source: World Bank, World Development Indicators, 2018

Kenya has ratified most international, regional Treaties and Agreements related to gender equality and enacted a progressive Constitution in 2010 that protects women's rights. The Government of Kenya has ratified the Convention on the Elimination of all forms of Discrimination against Women (CEDAW), the Protocol to the African Charter on Human and People's Rights on the Rights of Women in Africa (Maputo Protocol) and has joined the 1995 Beijing Platform for Action.

The new Constitution adopted by Kenya in 2010 recognizes women's rights as human rights, while removing several discriminatory provisions towards women. Article 27 guarantees equality and freedom from discrimination. It states that every person is equal before the law and has the right to equal protection and equal benefit of the law. The Constitution promotes equality in marriage, employment, access to education and guarantees women's rights to inheritance and own land. Articles 97 and 98 introduce quota in the National Assembly and the Senate. The Constitution also calls for a 30 percent government procurement preference for youth, women and persons with disabilities and requires both public and private entities to comply with gender equality principles. Furthermore, all discriminatory customary practices are prohibited.

At the legislative level, Kenya has adopted a progressive normative framework to promote women's rights (excluding the Marriage Act of 2014):

- **The Married Women's Property Act of 1882** provides married women equal rights to own property. It applies to all systems of marriage recognized under Kenyan law;
- **The Sexual Offense Act of 2006** provides a definition of sexual offences excluding marital rape and protects all persons from unlawful sexual acts;
- **The Anti-FGM Act of 2011** prohibits Female Genital Mutilation and imposes tough penalties on perpetrators;
- **The National Gender and Equality Commission Act of 2011** promotes gender equality and freedom from discrimination in accordance with Article 27 of the Constitution. This applies to both public and private sectors;
- **The Election Act of 2011** promotes women's participation in political and public life;
- **The Land Act of 2012** grants all citizens the right to own land and promotes gender equality in land distribution;
- **The Matrimonial Property Act of 2013** guarantees equality between spouses in matrimonial property;
- **The Marriage Act of 2014** allows the customary practice of polygamy; and

- **The Protection Against Domestic Violence Bill of 2015** outlines specific types of domestic violence including physical abuse, forced and child marriage, female genital mutilation (FGM), sexual violence, economic and inheritance abuse, emotional abuse, harassment, etc. it also provides for protection measures for victims.

At the policy level, Kenya has regularly enacted National Strategies and sectoral policies to promote gender equality. For instance, the Government of Kenya adopted a National Policy on Gender and Development in 2000 and a National Policy on Prevention and Response to gender-based violence in 2014. The country's national strategy Kenya Vision 2030 (2013-2017) includes a social pillar dedicated to ending discrimination of women in economic and social sectors and empowering them to actively participate in economic, social, and political activities. In the energy sector, the Ministry of Energy not only produced an audit of its energy policies in 2007 but also developed a gender policy in 2019.

However, the adoption of the Marriage Act in 2014 that allows polygamy indicates that discriminatory customary practices are still widely tolerated in the Kenyan society. Customary law systems are based on patriarchal traditions in which the principal decision-making power is allocated to men, and men inherit and control land and property.

Figure 4: Broad Principles of Customary Laws in Kenya

- Married women do not inherit from their parents.
- Unmarried women inherit less from their parents than their brothers do.
- Women with sons may retain their husbands' property, but only to hold it in trust for the sons.
- Women with no children or with daughters are not likely to inherit from their husbands, and the estate is given to male relatives as if she were unmarried.
- Divorced or separated women are expected to leave the matrimonial home and return to their parents with only personal items.
- Unmarried daughters can use land in the paternal household, but only for certain crops.
- Married women have less control over significant family property than men do.
- Usually property that a wife acquires before and during a marriage is controlled and essentially owned by the husband.
- In some communities, the hut that a widow shared with her husband is destroyed after his death, and she is built a new (and often inferior) hut after she is cleansed or inherited.

Source: Human Rights Watch, 2003 and World Bank report, 2007

2.5 Mali Profile

Mali is a large landlocked country with a population estimated at 19 million in 2018. With two-third of the country's land area being arid or semi-arid, economic activity is confined to the Niger River area mainly in the agriculture sector. Mali has been experiencing high demographic growth in the past decade: in 2017, the country's

average fertility rate was still at 5.9 children per woman. The population is unevenly distributed between rural and urban areas. More than 70 percent of Malians live in rural areas while the urban population is concentrated in the regional capitals and in the capital city of Bamako (4 million inhabitants)¹.

Mali is among the poorest countries in the world, with important regional disparities in wealth distribution. The country is ranked 184 out of 189 nations in the 2019 Human Development Index. The country's main drivers of poverty include increasing food insecurity due to demographic pressure, lack of education and employment opportunities as well as political instability : 49 percent of the population lives below the extreme poverty line. Poverty is concentrated in the rural areas of southern Mali (90 percent of poor people) where the population density is the highest.

In 2012, a coup d'état in Bamako along with attacks from extremist groups in the northern and central part of the country provoked prolonged insecurity and political instability. Despite a peace agreement signed in 2015, insecurity persists and has weakened the authorities' ability to provide basic services to the population, especially in areas that remain outside of its control. This political and security crisis has increased the population's vulnerability to poverty, especially affecting the 218 534 internally displaced persons (IDPs)² from the conflict.

Mali's economy is highly vulnerable to climate change, with the agriculture, health and energy sector classified most at risk in the country's National Adaptation Programme of Action. Almost three-quarters of the Malian labor force is engaged in agriculture. Desertification, high variability in rainfall, silting of the Niger river and rapid deforestation around large urban centers are among Mali's main environmental challenges. This climate stress can lead to increased food insecurity and poverty, particularly in rural areas where Malians are mostly engaged in subsistence agriculture.

Access to energy in the country is low and unreliable, with wide disparities between rural and urban areas. Despite significant progress over the last decades, the country's electrification rate does not exceed 49 percent (less than 1 percent in 1995)³. Urban households are more than twice as likely as rural households to have electricity (85 percent versus 38 percent). This disparity is partly due to the high cost of electricity in remote areas that are not connected to the main grid or isolated centres, where tariffs range between USD 0.25/kWh and USD 0.51/kWh⁴. In rural households, low availability of energy supply means that most households rely on biomass and kerosene lamps for cooking and lighting. This increases women's burden as they are traditionally responsible for fuel collection and cooking.

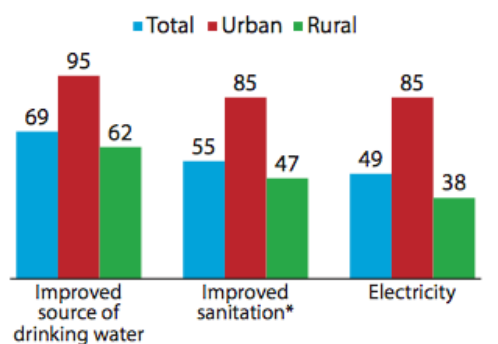
Figure 5: Water, Sanitation and Electricity in Mali by Residence

¹ Government of Mali. 2018. Mali Demographic and Health Survey or EDSM-VI.

² United Nations High Commissioner for Refugees (UNHCR) Data.2020.

³ Government of Mali. 2018. Mali Demographic and Health Survey or EDSM-VI.

⁴ International Renewable Energy Agency (IRENA).2019. Renewable Energy: a gender perspective.



*Including any improved sanitation facility, whether shared or not by other households

Source: EDSM-VI and UNICEF report, 2018

Despite having ratified most international treaties and convention, Mali's national legal framework on gender equality remains weak. Several gaps persist notably with respect to gender-based violence, early marriage as well as ownership and property rights. At the international level, Mali has ratified (i) the Convention on the Elimination of all forms of Discrimination against Women (CEDAW) in 1986, (ii) the Convention on Children's rights in 1990, (iii) the Maputo Protocol to the African charter on Human and Peoples' Rights in 2004 and (iv) the 2008 Protocol on the Statute of the African Court of Justice and Human Rights. In 2018, the said court found the Government of Mali in violation with the Maputo protocol, for condoning child marriage and discriminating against women in the Malian Family Code¹.

At the national level, the normative framework is poorly harmonized with international and regional instruments and Mali has yet to enact key legislation. Legal loopholes persist especially with regards to gender-based violence, and some legal provisions are directly discriminatory against women. Furthermore, customary law is routinely applied at the expense of positive law and to women's disadvantage, especially in the field of inheritance on rural land tenure.

In 2009, the Government of Mali attempted to reform the 1962 Family Code, to improve equality between men and women. The revised version of the code included provision to forbid child marriage, improve women's inheritance and property rights and increase women's agency in the household. However, the adoption of the Code was blocked by wide-scale protests led by conservative groups. The new 'Code du Mariage et de la Tutelle' adopted in 2011 did not include rights previously granted in the draft version. Despite this significant setback, the country has made some progress in the promotion of gender equality:

- **Article 2 of the Constitution** enshrines the principle of equality between men and women;
- The 2006 **agricultural orientation law** protects women's property rights;
- In 2010, Mali adopted a **National Policy for the Permanent Eradication of Female Genital Mutilation (FGM)**;

¹ Stockholm International Peace Research Institute (SIPRI). 2019. A Study of Insecurity and Gender in Mali.

- Since 2011, Mali has a **National Policy for Gender Equality** based on six strategic directions¹;
- Since **2015**, the **Law**² requires a 30 percent quota for female appointments to national institutions and on electoral list; and
- A **2017 Law**³ allocates 15 percent of arable governmental land for the use of women and young people.

2.6 Namibia Profile

With an estimated population of about 2,4 million (2018) and a surface area of 825,615 km², Namibia is one of the least densely populated countries worldwide. Most inhabitants are located in the north, where six of the most populous regions of Namibia can be found. Rural to urban migration is a growing trend across the country : in 2018 nearly half of the population lived in urban areas (49.9 percent) from 42.8 percent in 2011. Namibia is rich with natural resources and political stability as well as sound economic management has allowed it to reach the upper-middle-income development level.

While Namibia has achieved significant reduction of poverty since its independence in 1990, it remains one of the most unequal countries in the world. Between 1993/94 and 2015/16, the proportion of Namibians living below the national poverty line went from 69.3 percent to 17.4 percent. Poverty remains a challenge, exacerbated by extreme socio-economic inequalities inherited from the apartheid period, as well as high unemployment rate (34 percent). Female-headed households (FHH), ethnic minorities and subsistence farmers are particularly vulnerable to poverty.

Namibia's vulnerability to climate change is extremely high. 92 percent of the land area is defined as arid or semi-arid. The country is expected to experience increased occurrence of drought, desertification, land and water resource degradation and loss of biodiversity. Agriculture, fisheries and water have been identified as the most vulnerable sectors to climate change, in a country where subsistence farming is widespread. In the past 5 years, Namibia has experienced the worst drought in a century, with devastating impact on the farming community. As a result, more than 500,000 people suffered from increased food insecurity and water shortages.

Only half of the Namibian population has access to electricity, with wide disparities between urban and rural areas. In 2013, 72 percent of urban households had access to energy services, compared to 21 percent of rural households⁴. The country's large land mass and low population density make it harder for the government of Namibia to provide basic services, including access to electricity, to its rural communities. An estimated half of the Namibian population uses traditional sources of energy such as firewood for cooking, with negative impact on health outcomes. Women and girls are particularly affected as they are traditionally responsible for the collection of water and firewood.

¹ Those are: (1) equal rights for women and men, (2) development of human capital (for women and men), (3) the integration of women into productive channels, (4) equal participation of women and men in the spheres of decision, (5) establishing the values and egalitarian behavior in Malian society, and (6) the inclusion of man-woman equality as a guiding principle of good governance.

² Loi n°2015-052/PRM du 18 décembre 2015.

³ Loi N°2017- 001/ du 11 avril 2017 portant sur le foncier agricole.

⁴ Government of Namibia.2016. Namibia Inter-Censal Demographic Survey.

Namibia has a comprehensive legal framework to protect equality between men and women, and is regularly adopting national policies and strategies to further the gender equality agenda. The Namibian government is signatory of the main international and regional agreements and conventions. The country has ratified the Convention on the Elimination of All forms Discrimination against Women (CEDAW) in 1992, the Protocol to the African Charter on Human and People's Rights on the Rights of women in Africa in 2004 and is a signatory of the Beijing Declaration and Platform for Action (1995) and the SADC Protocol on Gender and Development (2008). At the national level, Namibia has one of the continent's most progressive legal framework on gender equality:

- The **Namibian constitution** establishes the principle of equality before the law without discrimination based on sex (Article 10), and authorises affirmative action to encourage women to participate in the *“political, social, economic and cultural life of the nation”* (Article 23);
- The **Married Persons Equality Act 1 of 1996** abolishes the marital powers of the husbands, and puts wives and husbands on equal footing within civil marriage;
- The **Affirmative Action (Employment) Act 29 of 1998** establishes an Employment Equity Commission and authorizes affirmative action measures to promote equal opportunity in employment for women;
- The **Combating of Rape Act 8 adopted in 2000** criminalizes marital rape;
- The **Communal Land Reform Act of 2002** protects the rights of rural widows to remain on land allocated to their deceased husbands. The widow's right to remain on the land is not affected by re-marriage;
- The **Combating of Domestic Violence Act 4 of 2003** provides a broad definition of domestic violence including physical, sexual, psychological, economic abuse and intimidation. This act makes domestic violence a specific crime;
- The **Children's Status Act of 2006** provides for equal guardianships and custody for unmarried parents;
- The **Labour Act 11 of 2007** prohibits sexual harassment in the workplace and forbids direct or indirect discrimination in any employment decision based on sex, marital status, family responsibilities, or pregnancy;
- The **National Policy on Sexual, Reproductive and Child Health adopted in 2013** includes specific measures to provide equitable, accessible and affordable health and nutrition information and services to the population;
- The **Child Care and Protection Act 3 of 2015** requires a minimum age of 18 for marriage.

At the policy level, Namibia has regularly enforced strategies and policies to promote gender equality. The National Land Policy adopted in 1998 promotes women's rights to maintain their lands after their husband's death. The country's first National Gender Policy was adopted in 1997 and its revised version in 2010 for the period 2010-2020¹. The 2010 Education Sector Policy on the Prevention and Management of Learner Pregnancy is a key policy that provides measures to encourage mothers and fathers to complete their education. In 2012, the country adopted a National Plan of Action on gender-based violence to coordinate and guide interventions, programmes and projects on violence against women in Namibia.

¹ The National Gender Policy 2010-2020 can be accessed here:

http://www.africanchildforum.org/clr/policypercent20perpercent20country/namibia/namibia_gender_2010-2020_en.pdf

2.7 Uzbekistan Profile

Uzbekistan is a Central Asian landlocked country sharing border with five Asian countries¹, with an estimated population of 32.1 million people in 2017. The country is rich in natural resources, notably copper, gold, oil, natural gas and uranium. Since the election of the president Mirzioiev in 2016, the government of Uzbekistan committed to a radical transition from a state-controlled to a market-oriented economy. Although the country is experiencing increased urbanization, nearly half of the population (49.4 percent) still lives in rural areas.

Uzbekistan is a lower-middle-income country whose poverty levels have declined significantly in the past decades, going from 27.5 percent in 2001 to 11.4 percent in 2018. Uzbekistan is ranked 108 out of 189 countries in the United Nations 2018 Human and Development report, and is included in the high human development category. However, large regional disparities remain, causing increased labor migration within and outside the country.

As one of the most water-dependent countries in the world, Uzbekistan is highly vulnerable to climate change. The country is located in one of Central Asia's driest regions and is prone to natural disasters such as floods and drought, whose occurrence may increase with climate change. The Aral Sea has been gradually shrinking, losing 90 percent of its volume in less than 60 years. More than 80 percent of the country's renewable water resources originate in neighboring countries.

Uzbekistan is the second most energy-intensive country worldwide and has achieved near universal electricity access. However, gaps in power supply are common due to inadequate maintenance of infrastructure, especially in rural areas, resulting in frequent interruptions and poor power quality. As women are mostly responsible for household chores, they are more affected by the unreliability of power supply and the inability to use time-saving energy appliances.

Uzbekistan is signatory of the main international conventions related to the promotion of women's rights: in 1995 the government ratified the United Nations Convention on the elimination of all Forms of discrimination against Women and joined the Beijing Declaration and Platform for Action. The government of Uzbekistan also ratified key International Labour Organization conventions (ILO) related to gender equality, and incorporated them in its national legislations.

At the national level, despite important progress in the most recent years, significant gaps remain in the legislation. The Constitution of the Republic of Uzbekistan guarantees equal rights without discrimination based on sex. This principle is reflected in national law; both the criminal code and the labor code contain provisions against discrimination on the basis of sex. Relevant milestones (laws, national policies, strategic documents) in advancing the normative framework for gender equality are chronologically listed below:

- **Creation of the Women's Committee (WCU), 1991:** the WCU was established to develop and implement state policy on women's rights. It has branches in every administrative territorial body;

¹ Afghanistan, Kazakhstan, the Kyrgyz Republic, Tajikistan and Turkmenistan.

- **National Action Plan for improving the status of women in Uzbekistan and improving their status in society, 1995:** this is a strategic document drafted in the context of the Beijing Platform for Action;
- **The Elections Act of the Oliy Majlis, 2004:** this act was key in increasing female political participation in the National Assembly. It requires that the number of women nominated by political parties for parliament should be no less than 30 percent of the total number of candidates;
- **First National Action Plan for the Implementation of the Concluding Observations of the CEDAW Committee, 2006;**
- **Second National Action Plan for the Implementation of the Concluding Observations of the CEDAW Committee, 2010;** and
- **National Development Strategy for 2017–2021** adopted by presidential decree in 2017: the document outlines 5 priority areas and includes objectives to “increase the social and political activity of women”.

In September 2019 and for the first time in the country’s history, the President of Uzbekistan signed a law on gender equality. The law on “Guarantees of Equal Rights and Opportunities for Women and Men” guarantees equal access to property and material resources and bans sexual harassment. This is an important milestone in a country that was one of the few nations in the world without legislation on gender-based violence.

3. KEY GENDER GAPS

The following section identifies key issues that may impact projects under the SRMI facility. Preliminary gender assessments for each of the seven countries under the SRMI facility highlighted the following gaps and their potential mitigation actions which will be explored during project design.

3.1 Botswana Gender Gap

Botswana is a patriarchal society ranked 73 out of 153 countries by the World Economic Forum 2020 Global Gender Gap. Men are traditionally heads of households, while women are mostly responsible for cooking food, collecting water and caring for the sick and elderly. Yet, unlike most countries in Sub-Saharan Africa, Botswana is at an advanced stage of demographic transition with a fertility rate estimated at 2.9 births per woman in 2017. This indicates progress in women's development driven by i) effective family planning, ii) better child survival, iii) increased age at first birth, iv) increased female education attainment and v) improved women's participation in the labor force¹. Despite these better outcomes, several gaps remain and will be explored below.

- **Education.** Together with Lesotho and Namibia, Botswana is one of the three Sub-Saharan countries to have already achieved parity in education². In the past decades, the government significantly increased public investment in education to ensure equal access for all, including girls and women. As a result, the enrolment of girls in education is higher than the enrolment of boys at all levels. 88.7 percent of women are literate compared to 86.7 percent of men. However, many girls continue to drop out of school because of early pregnancies or marriage (allowed under customary law). To mitigate the consequences of girls dropping out of school, Botswana has adopted policies to promote school re-entry for student mothers.

Figure 7: Botswana's Educational Attainment (percentage)

BOTSWANA'S EDUCATIONAL ATTAINMENT (percentage)		
	Women	Men
Enrolment in primary education	88.1	87.3
Enrolment in secondary education	63.8	55.8
Enrolment in tertiary education	29.2	20.5

Source: Based on Data from the World Economic Forum 2020 Global Gender Index

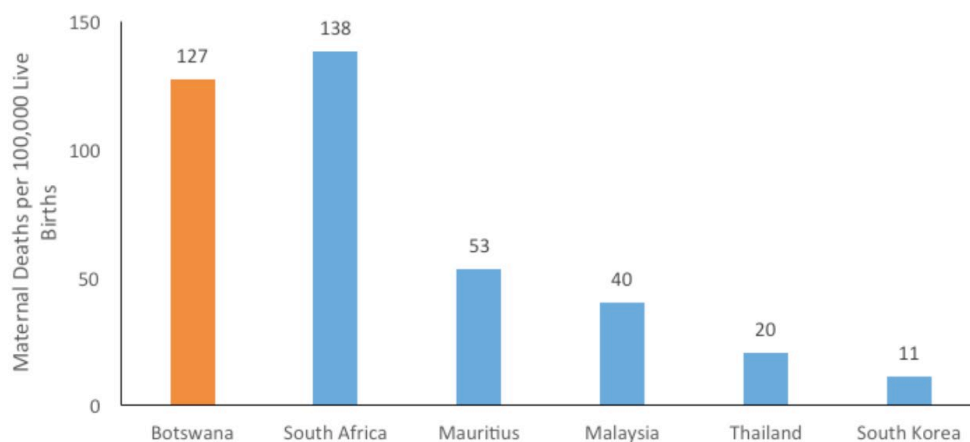
¹ The United Nations Population Fund (UNFPA). 2018. Opportunities and Policy Actions to Maximize the Demographic Dividend in Botswana.

² World Economic Forum (WEF). 2020. Global Gender Gap Index.

Educational sex-segregation remains a reality in the country, affecting women’s participation in STEM fields. For instance, female students only account for 21 percent of all students in engineering and technology subjects at the University of Botswana¹.

- **Health.** Heavy public investments have allowed Botswana to significantly improve health outcomes, but maternal mortality rates remain at high levels compared with other middle-income countries. Despite low population density, 84 percent of the population is within 5 km of a health facility and nearly 99 percent of birth in the country are attended by skilled birth personnel. However, the country continues to grapple with high mortality rates at 144 per 100,000 live births, putting Botswana off track in the achievement of the third Sustainable Development Goal (Good Health and Well Being).

Figure 8: Comparing Botswana’s Maternal Mortality Ratio (MMR) with those of selected countries (2015)



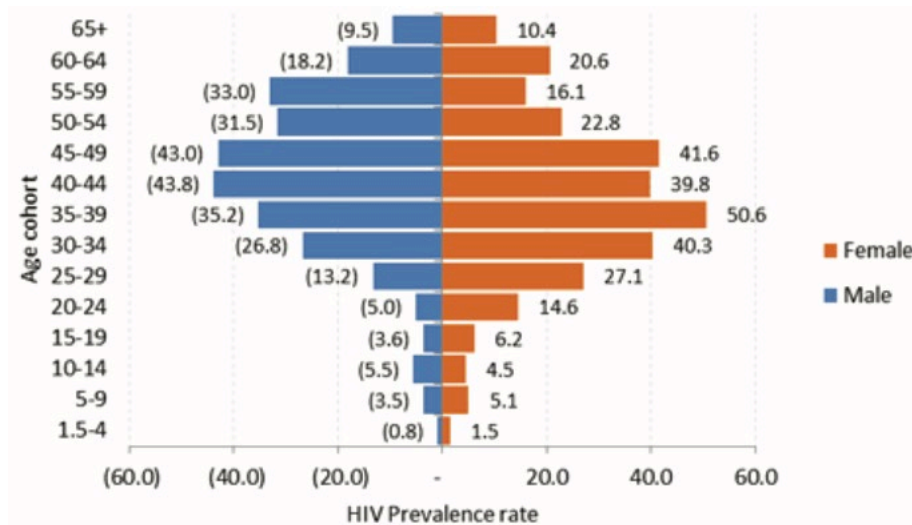
Source: Statistics Botswana, 2017, World Population Prospects, the 2015 Revision and UNFPA report 2017

Botswana has one of the highest prevalence of HIV in the world, with a major impact on women’s health outcomes. Almost twice as many women aged 15-24 (8.9 percent) than men (4.9 percent) are infected with HIV. Key factors of women and girls’ increased vulnerability to HIV include their inability to negotiate safe sex with their partners and their increased likelihood to engage in transactional sex². Women are not only more vulnerable to the HIV infection than men, they also shoulder the responsibility of caring for sick relatives.

¹ The United Nations Population Fund (UNFPA). 2018. Opportunities and Policy Actions to Maximise the Demographic Dividend in Botswana.

² United States Agency for International Development (USAID). 2016. Botswana PEPFAR Gender Analysis.

Figure 9: HIV prevalence in Botswana by age cohort and gender, 2013



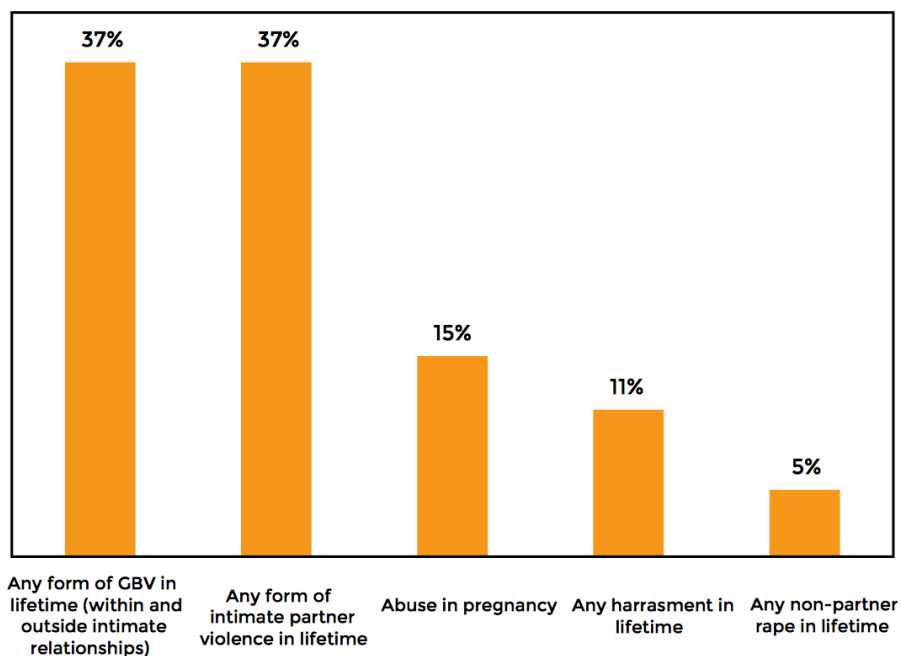
Data source: BAIS IV Survey, Statistics Botswana (2014)

- Gender-Based Violence.** GBV is a widespread phenomenon in Botswana, with 67 percent of women having experienced some form of GBV. As in many other countries, strong patriarchal norms and social tolerance towards violence against women discourages victims from reporting incidents due to shame and fear of reprisal. As a result, in the overwhelming proportion of cases, GBV is not reported to authorities. Only 1.2 percent of women have reported violence to the police. The most common type of gender-based violence observed in Botswana is intimate partner violence: domestic violence makes up 62 percent of GBV cases in the country¹. Key drivers behind domestic violence include alcohol abuse as well as women's ignorance of the law. Indeed, 54 percent of women are not aware of Botswana's protective legislative framework against domestic violence².

¹ United States Agency for International Development (USAID). 2014. MSH Fact Sheet Botswana GBV.

² Ibid.

Figure 10: Forms of violence experienced by women in a lifetime in Botswana



Source: Botswana Relationship Study, 2017 and UNFPA report 2019

- **Women's agency and political participation.** Botswana women remain under-represented in key decision-making institutions and their level of participation in politics has declined in the past year. During the 2019 general election, only 5 percent of parliamentary candidates were women, while they accounted for about 9 percent of candidates in 2014. No woman ran for the Presidency. Since the 2019 election, women make up 9.5 percent of members of Parliament and hold 15.8 percent of ministerial seats. These figures are far below the commitment to achieve at least a 30 percent quota for women in political and decision-making structures, made by the Government of Botswana when it signed the SADC Declaration on Gender and Development in 1997. Key barriers preventing women to participate in politics include patriarchal beliefs that women are not as capable as men in political affairs as well as legal gaps (there is no quota law). Despite being highly visible during grassroots campaigns in support of male candidates, women too often lack the opportunity to get involved in decision-making processes.
- **Female employment and access to decent work.** In Botswana, the labor market continues to reflect substantial barriers to women's economic participation. Men's labor force participation rate is 11.1 percentage points higher than female participation (80.2 percent versus 69.1 percent) and their unemployment rate is 7 percentage points lower (15.3 percent versus 22.3 percent¹). Occupational sex-segregation remains a key characteristic of the country's economy. Women are over-represented in the

¹ World Economic Forum (WEF). 2020. Global Gender Gap Index.

agricultural and informal sectors where most low paying, low-skill jobs can be found. They are 58 percent more likely than men to be unpaid family workers¹. When women are paid, they earn 29 percent less than men on average². While women are under-represented in the private sector across all skill levels, they predominate in the public sector: more than 70 percent of college-educated Botswana women work for the government. One of the main factors contributing to these disparities is the fact that women continue to play a reproductive role in the household. They remain mostly responsible for childbearing and domestic tasks and are the ones caring for sick relatives and the elderly, negatively impacting their presence and participation in the labor market. These barriers are reflected in women's access to entrepreneurship: only 8.9 percent of firms have female majority ownership³.

- **Female-Headed Household access to ownership and finance.** Female-headed households are significantly more likely to be poor than male-headed households. In Botswana, 58 percent of poor households are headed by females⁴; 33 percent of female-headed households live below the poverty line while this is the case for 27 percent of men-headed households⁵. A combination of factors contributes to this disparity: FHH are characterized by lower levels of education, salaries as well as lower access or control of resources. They are more dependent on unreliable sources of income for subsistence such as remittances, maintenance, and destitute allowance⁶. In rural areas, female-headed households have poor access to livestock and land which hinders their capacity to invest in sustainable livelihood. While the positive law protects women's right to ownership, the Tswana tradition prohibits women from inheriting property such as land, housing, and cattle. 75 percent of female-headed households do not own livestock⁷. Because levels of poverty are highest among FHH, especially in rural areas, they also have lower access to energy services resulting in low electricity connection rates⁸.

3.2 CAR Gender Gap

Gender inequality in the Central African Republic is one of the highest in the world and has increased over the past decades, owing to repeated conflicts and poverty. The country is ranked 159 out of 162 countries by the United Nations' 2019 Gender Inequality Index (GII). Gender gaps are observed at all levels with respect to women's participation and presence in political, economic, social and cultural life. The Central African society is patrilinear,

¹ The African Development Bank Group (AfDB). 2011. Profil Genre de la République Centrafricaine.

² The World Bank Group. 2015. CAR Systematic Country Diagnosis.

³ World Economic Forum (WEF). 2020. Global Gender Gap Index.

⁴ Ibid.

⁵ World Food Programme (WFP). 2019. Evaluation thématique sur les questions de genre dans les interventions du PAM en République centrafricaine 2014-2018.

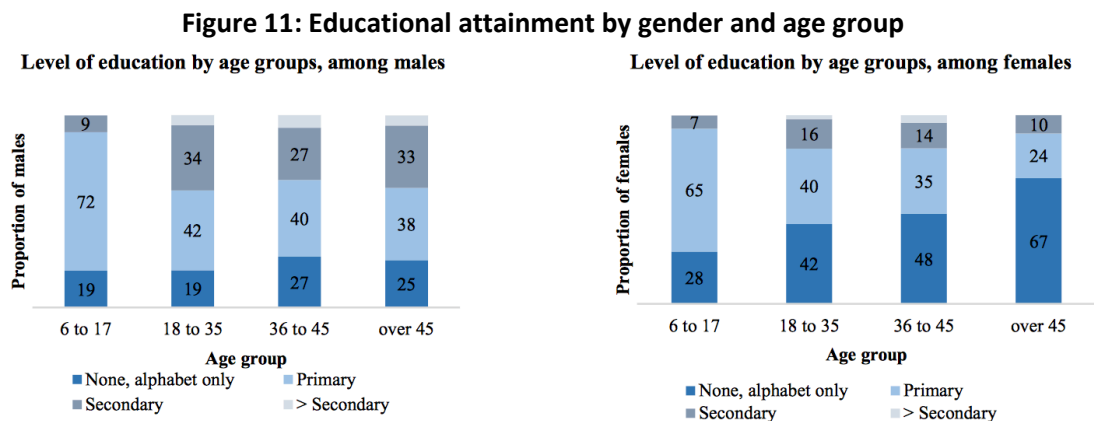
⁶ Lesetedi, N. 2018. A Theoretical Perspective on Women and Poverty in Botswana. Journal of International Women's studies.

⁷ Ibid.

⁸ Energia. Botswana Power Corporation. 2011. Gender Mainstreaming in the Botswana Power Corporation.

authority is essentially held and exercised by men. Women are traditionally confined to the role of housewives and mothers, and their social value heavily depends on their marital status¹.

- Education.** CAR’s educational gender gap is significant, the enrolment rate of girls in school is consistently lower than the one of boys. While the whole population has been affected by the destruction of school facilities and materials during the conflict, the educational system is characterized by significant gender disparities. Only 28.7 percent of women are literate, compared with 47.8 percent of men. More than half of girls do not complete primary school compared to 30 percent of boys. Most schools do not have water supply or latrines, particularly affecting girls’ attendance during menstruation periods. Only 26 percent of girls reach higher education and among them the rate of women who specialize in STEM fields is even lower. In 2011, only 8 percent of female medical students completed their degree at the University of Bangui and 2 percent of female sciences students². This disparity may be explained by the fact that technical subjects (carpentry, masonry, mechanics, electricity, refrigeration, etc.) are traditionally regarded as male fields³. Girl’s access to education is further constrained by the prevalent practice of early marriage and early childbearing. 29 percent of girls are married by age 15 and 68 percent by age 18⁴. Early marriage and high fertility rates are linked to low educational attainment, but also take a heavy toll on women and girls’ health outcomes due to higher maternal and infant mortality rates.



Source: ENMC, ICASEES 2017 and World Bank CAR Systematic Country Diagnosis, 2019

- Health.** In CAR women’s health outcomes are extremely low: with 829 deaths per 1000,000 live births, the maternal mortality rate is among the highest in the world. Conflicts have caused the destruction of health facilities and limited women’s mobility, thereby exacerbating structural deficiencies and impacting provision of gynecological and post-partum care. The maternal mortality rate increased by 29 percent and the risk of child marriage by 9 percent in the past 30 years⁵. Traditionally, the decision to use family planning methods

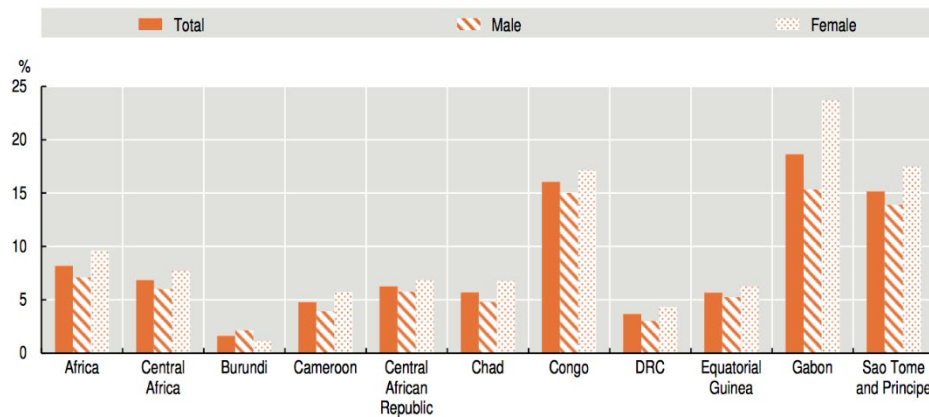
¹ The African Development Bank Group (AfDB). 2011. Profil Genre de la République Centrafricaine.
² Ibid.
³ Committee on the Elimination of Discrimination against Women (CEDAW). 2012. Consideration of reports submitted by States parties under Article 18 of the Convention on the Elimination of All Forms of Discrimination against Women.
⁴ United Nations Children’s Fund (UNICEF). 2016. Multi-country real time evaluation of UNICEF Gender-based violence in emergencies programme.
⁵ Government of CAR. 2019. Rapport National Volontaire de suivi de mise en œuvre des ODD.

is taken by the husband, at the expense of women's agency and right to choose what is best for their health and well-being. In 2017, women had on average 4.8 children. Women are also more affected than men by epidemics such as HIV/AIDS, with disastrous consequences on their social life and survival as they face stigmatization and exclusion. In 2018, female accounted for 56.2 percent of the population aged 15 living with HIV.

- **Gender-Based Violence.** Although data on the prevalence of GBV are lacking, studies report high-level of sexual and gender-based violence across the country. Human Rights Watch reported in 2017 widespread and systematic use of rape as a tactic of war during the conflict, particularly affecting women and girls and to a lesser extent men and boys. In addition to sexual violence, many women have been abducted by armed groups and forced to collect water, cook and wash clothes. Most women did not report the incidents, nor did they receive medical care or other support. Many suffered from social stigma within their community and households, following the events. GBV is exacerbated by high levels of social tolerance towards violence against women. According to the 2010 CAR Multiple Indicator Cluster Survey, 80 percent of men and women believe wife beating is justified. Furthermore, some harmful traditional practices are still widespread in some areas, such as female circumcision and widowhood rites (confiscation of property, levirate). Nearly 25 percent of women have suffered from Female Genital Mutilation.
- **Women's agency and political participation.** Women's political presence and participation in CAR's decision-making institutions is extremely low due to traditional norms and lack of education. Yet, from 2014 to 2016 and for the first time in the country's history, one woman served as interim President of the Central African Republic. Women hold 8.6 percent of seats in National Parliament and 18.2 of ministerial positions, well below the 35 percent quota required by the law. Women's higher illiteracy and lack of political instruction is a major obstacle to their participation in public life. Central African Women also lack time to dedicate to political affairs, as they are mostly responsible for household duties, as well as the care of children and sick people in their community. When women exercise their right to vote, they usually choose male over female candidates, due to internalized prejudice towards female politicians¹.
- **Female employment and access to decent work.** The Central African Republic's economy is characterized by high levels of occupational sex-segregation and employment gap. When they are employed, Central African women predominantly work in informal sectors, in inferior position and with lower income. Employed women are far more likely than men to work without pay at all, mostly as unpaid agricultural laborers in their own households. Women's labor force participation rate reaches 65 percent compared to 80 percent for men. More than 7 percent of women are unemployed, while this is the case for only 5.9 percent of men.

¹ Committee on the Elimination of Discrimination against Women (CEDAW). 2012. Consideration of reports submitted by States parties under Article 18 of the Convention on the Elimination of All Forms of Discrimination against Women.

Figure 12: Average unemployment, 2000-15 (percentage)



Source: OCDE report 2018

Female employment is concentrated in agriculture (75 percent), in sectors where productivity and profitability are lower. On average, female farmers have limited access to production factors and exploit smaller lands. Women are under-represented among salaried workers; only 2.4 percent of women are salaried workers compared with 8 percent of men. When women manage to enter the formal sector, their wages are lower. The average salary for women employed in the public administration is 357.66 FCFA/h compared to 602.63 FCFA/h for men¹. Working conditions are particularly precarious in the artisanal mining sector, characterized by lower wages and higher risks of sexual exploitation and gender-based violence.

- **Entrepreneurship and access to ownership and finance.** Although the law does not discriminate against women in inheritance and property rights, women's access to ownership and finance is constrained by traditional practices and patriarchal norms. Traditionally, women in male-headed households have no control over their personal earned income. In the agriculture sector, key gaps include land access and cattle ownership. Only 19 percent of female-headed household own heads of livestock compared to 27 percent of male-headed households². In a country where cattle ownership is seen as a form of saving, this disparity makes access to finance even more difficult for women. 63 percent of FHH have access to a plot of land, while this is the case for 69 percent of male-headed households³. Female-headed households are also less likely to own household and transport equipment such as a bike, car, radio, TV and mobile phone⁴. In rural areas, widows often experience economic and social discrimination. They often struggle to exercise their succession rights, due to customary practices and ignorance of the law. Gender gaps in access to assets and financial literacy make it more difficult for women to access credit. Female entrepreneurs in the Central African Republic are mostly active in the informal sector and are limited to

¹ Government of CAR. 2019. Rapport National Volontaire de suivi de mise en œuvre des ODD.

² Government of CAR. 2017. Enquête Nationale de Sécurité Alimentaire (ENSA) en RCA.

³ World Food Programme (WFP). 2019. Evaluation thématique sur les questions de genre dans les interventions du PAM en République centrafricaine 2014-2018.

⁴ The World Bank. 2012. Etude Diagnostique sur la Situation du Genre en République Centrafricaine.

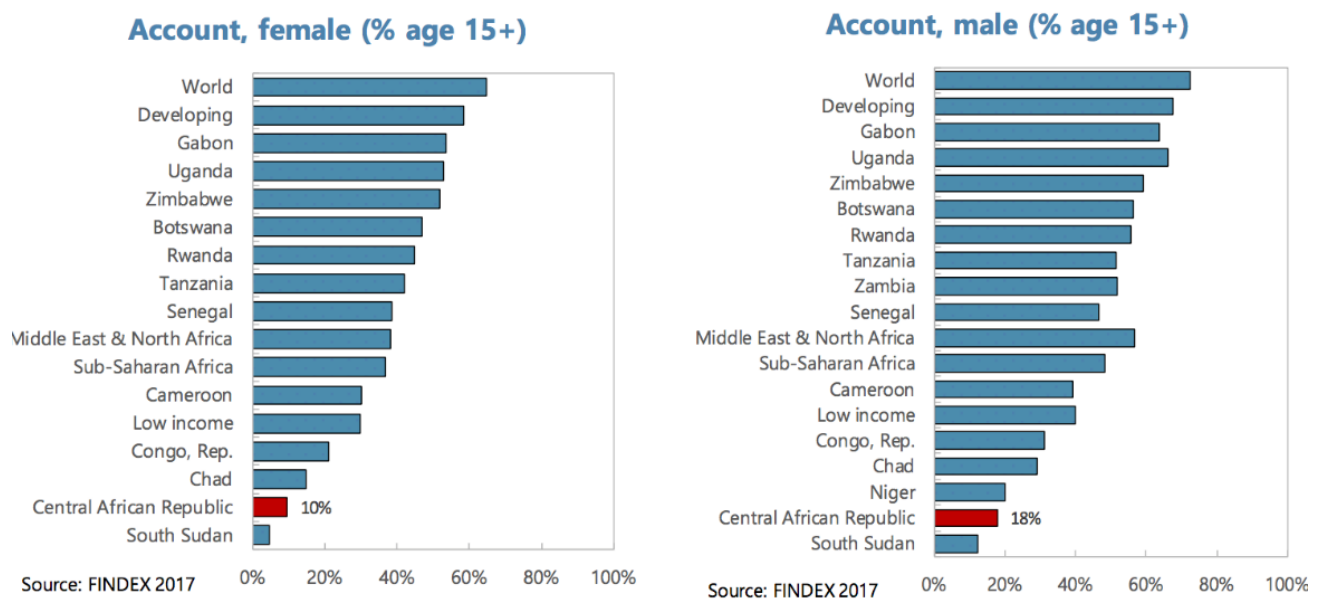
low-income subsistence activities. In many cases, they cannot provide collateral. Furthermore, female entrepreneurs are more likely to be risk-adverse than men and do not submit as many credit applications¹.

Figure 13: Credit applications in CAR, by sex

Year	Members		Credit applications	
	Men	Women	Men	Women
2008	26 131	28 816	1 947	704
2009	9 775	10 783	2 113	889

Source: *Crédit Mutuel de Centrafrique (CMCA), 2009 and CEDAW report, 2012*

Figure 14: Percentage of Women with access to Banking Services, 2017



Source: *FINDEX report, 2017 and International Monetary Fund, 2018*

3.3 DRC Gender Gap

DRC has some of the highest gender gaps in the world. Women are discriminated against with respect to education, health care, economic opportunities, access to ownership and political representation. The World Economic Forum 2020 Global Gender Gap Index ranked DRC 149 out of 153 countries, last in Sub-Saharan Africa.

¹ Committee on the Elimination of Discrimination against Women (CEDAW). 2012. Consideration of reports submitted by States parties under Article 18 of the Convention on the Elimination of All Forms of Discrimination against Women.

Despite a diversity of norms and traditional practices, the patriarchal model of family is predominant and has been imposed, including in matrilineal regions, since colonization. Religious and customary institutions are norms maker in the country and have been thought to play a key role in promoting gender inequalities¹. Their influence is exacerbated by a lack of governance and poor rule of law at the State level.

- **Education.** DRC has yet to close 40 percent of its educational gender gap and the enrolment rate of girls in school is consistently lower than the one of boys. Only 66 percent of women are literate, compared with 88 percent of men; 61 percent of boys are enrolled in secondary education while this is the case for only 38 percent of the girls. The gap keeps widening as the level of education increases: almost twice as many men (8.5 percent) as women (4.7 percent) are enrolled in tertiary education². Although the Constitution stipulates free mandatory education and promotes the eradication of illiteracy, primary education is not free of charge³. Families will often give preference to the education of boys, especially in rural areas, as they cannot afford to send all their children to school. In addition, girls often drop out of school due to early marriage and pregnancy or sexual and gender-based violence committed by teachers.
- **Health.** The country's healthcare infrastructure often lacks the capacity to offer the most basic services, particularly affecting women's maternal mortality in a country that has one of the highest fertility rates worldwide (6 children per woman on average in 2017). Many pregnancies are occurring too early due to the persistence of child marriage and poor access to sexual and reproductive health services, especially among indigenous women. Abortion is criminalized under the Penal Code and the prevalence of unsafe abortion remains high⁴. Many women do not have access to basic emergency obstetric and neonatal care. As a result, for every 100,000 live births, 693 women die from pregnancy related causes⁵. Malnutrition is another health challenge impacting women, particularly widows who have been found to have significant nutritional disadvantage⁶. Women are also more affected than men by epidemics such as HIV/AIDS and Ebola, with disastrous consequences on their social life and survival as they face stigmatization and exclusion. Health care facilities consistently lack properly trained staff, equipment, and supplies.

¹ Swedish International Development Cooperation Agency (SIDA). 2009. The DRC Country Gender Profile.

² World Economic Forum (WEF). 2020. Global Gender Gap Index.

³ Committee on the Elimination of Discrimination against Women (CEDAW). 2019. Concluding observations on the eighth periodic report of the Democratic Republic of the Congo.

⁴ Ibid.

⁵ The United Nations Development Programme (UNDP). 2019. Briefing note for the Democratic Republic of Congo on the 2019 Human Development Report.

⁶ Djuikom, Marie Albertine and Dominique van de Walle. 2018. Marital Shocks and Women's Welfare in Africa.

Figure 15: Percentage of health facilities per province with regular access to electricity

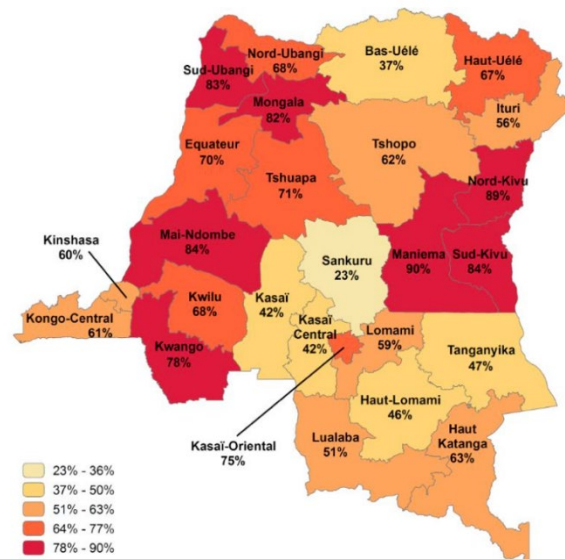
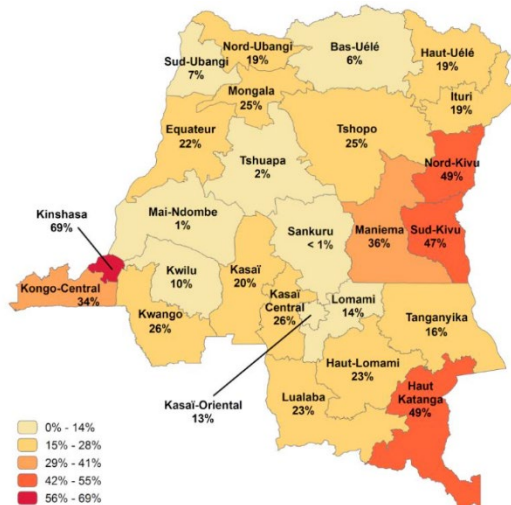


Figure 16: Percentage of health facilities per province with access to sanitary latrines for outpatients' services



Source: Evaluation des prestations des services de soins de santé - EPSS 2017-2018

- Gender-Based Violence.** High levels of Sexual and gender-based violence have been documented in DRC, with lasting impact on women's capacity to fully engage in social and economic life. GBV was used as a weapon of war during the conflicts and perpetrated by both militia groups and governmental forces. It has spread into a wider social phenomenon, threatening women's safety in schools, the workplace, their households and in the streets. According to the 2014 Demographic and Health Survey (DHS), 52 percent of women aged 15-49 have experienced physical violence while 27 percent have experienced sexual violence. Rates of intimate partner violence are even higher: 57 percent of married women have

experienced domestic violence. GBV rates vary across provinces; they are particularly high in Kasai Occidental and Kasai Oriental where the prevalence of physical and sexual violence is respectively of 30.5 percent and 57.4 percent¹. The main drivers of GBV include social tolerance to violence against women, poverty and the lack of a protective legal framework. DRC has the highest level of acceptance of intimate partner violence in the region: 74.8 percent of women and 59.5 percent of men aged 15-49 approve the practice of husbands beating their wives to discipline them². Sexual trafficking of women and girls is also widespread in DRC, particularly affecting the most vulnerable such as IDPs and mine workers. While Congolese women are often unaware of their rights, the Government of DRC lacks capacities and funds to enforce legal provisions against GBV. For instance, domestic violence cases are often judged by community and religious leaders who usually follow patriarchal norms, often at the expense of women's right to access justice.

- **Women's agency and political participation.** Women's presence and participation in decision-making institutions is weak at every level due to traditional norms, lack of education and limited economic opportunities. Women only account for 10 percent of members of Parliament and hold 17 percent of ministerial seats. In 2015, the electoral law was amended to remove a 30 percent quota for women's representation at the national level, thus weakening women's opportunities to access male-dominated power structures. Women's lack of visibility and influence in the political life reflects their lower decision-making power in the household. According to the 2014 DHS Survey, 53 percent of women do not participate in key household decisions. Although women are traditionally responsible for household chores and care for their family, many married women do not have control over their income.
- **Female employment and access to decent work.** Congolese women face unequal treatment with respect to labor force participation and have lower access to skilled higher-income jobs. They are mostly confined to subsistence agriculture and small business in the informal sector. Female labor force participation rate is 61.7 percent, compared with 66.5 percent for men. The gap is even wider with respect to access to skilled jobs; women only account for 24.4 percent of professional and technical workers (75.6 percent are men)³. Agriculture is the largest sector in terms of employment, with more than 70 percent of the population dependent on farming for their livelihoods. Women account for 70 percent of labor in agricultural production and make up 60 percent of workers in the informal sector. Only 2.8 percent of salaried workers are women. When they are paid, women earn lower salaries than men; on average USD 15 per month versus USD 20 per month for men⁴. Working conditions are particularly precarious in the artisanal mining sector, an industry that attracts many migrant women with poor education and lack of other employment opportunities. Female employment in the mining sector is characterized by lower wages and higher risks of sexual exploitation and gender-based violence⁵.

¹ The World Bank. 2018. Democratic Republic of Congo Systematic Country Diagnostic.

² Government of DRC. 2014. Demographic and Health Survey (DHS).

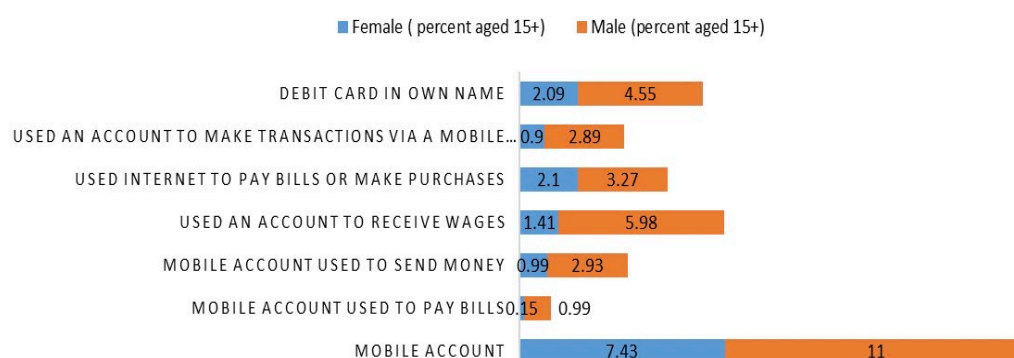
³ World Economic Forum (WEF). 2020. Global Gender Gap Index.

⁴ Embassy of Sweden. 2014. Profil du pays en matière d'égalité de genre.

⁵ Committee on the Elimination of Discrimination against Women (CEDAW). 2019. Concluding observations on the eighth periodic report of the Democratic Republic of the Congo.

- **Entrepreneurship and access to ownership and finance.** Women in DRC are discriminated against with respect to land tenure, property ownership, and face legal barriers and social stereotypes in entrepreneurship. Although women's right to inherit land is legally guaranteed, customary law continues to be applied across the country. In many cases, women are prohibited to own land by tradition and assets obtained within marriage are registered under the husband's name. Widows are particularly vulnerable to land predation after their husband's death, putting them at increased risk of poverty. In DRC, 61.5 percent of female-headed households are under the poverty line compared with 54.3 percent of male-headed households¹. Lack of access to land and financial resources limit the ability of female entrepreneurs to invest in new technologies and productive assets. More than 95 percent of the Congolese population is unbanked and there is no significant gap in account ownership. However, female financial inclusion is lower than men's and they face more challenges in accessing domestic credit due to lower financial literacy and limited access to professional networks. In 2013, only 3.6 percent of female-headed firms had a bank loan, compared to 10.2 of male-led companies².

Figure 17: Access to Financial Services in the Democratic Republic of Congo



Source: DRC Survey, START Center - March 2017

3.4 Kenya Gender Gap

Although Kenya has made great progress in closing gaps across all sectors, significant inequalities between men and women remain in education attainment, health outcomes, political empowerment and participation in the labor market. Kenya is ranked 109 out of 153 countries in the World Economic Forum 2020 Global Gender Gap Index.

- **Education.** Following the adoption of free primary education in Kenya in 2003, near parity has been achieved in enrolment at the primary level. 81.2 percent of girls are enrolled in primary school while this is the case for 78.3 percent of boys³. However, educational outcomes remain unequal because girls are more likely than boys to drop out of school. 78.2 percent of women are literate compared with 85 percent

¹ Japan International Cooperation Agency (JICA). 2017. Country Gender Profile Democratic Republic of The Congo Final Report.

² World Bank Group. 2014. Enterprise Survey.

³ World Economic Forum (WEF). 2020. Global Gender Gap Index.

of men. Significant disparities persist in secondary and university education. Only 45.8 percent of women reach secondary education (versus 49 percent of men); 9.7 percent enroll in tertiary education while this is the case for 13.2 percent of men. Key drivers behind the gap in educational outcomes include unequal household responsibilities, high cost of education and early marriage¹. Girls are often asked to help their mothers with household chores and care of relatives, with negative impact on their school attendance. Poorer households, especially in rural areas, tend to give preference to the education of boys at higher levels when costs increase. Teenage pregnancies and early marriage also contribute to lower female enrolment rates. 13.2 of girls between the age of 15 and 19 are married². In addition, insecure learning environments in some areas disproportionately affect girls, who are more vulnerable to sexual abuse. When they reach higher levels, women are less likely to pursue STEM studies. Only 11.1 percent of female students are enrolled in STEM fields, while this is the case for 20.8 percent of male students³.

- **Health.** Despite significant improvement in health outcomes in recent years, women continue to suffer from high maternal mortality rates especially among underserved communities. Maternal mortality is at 342 deaths per 100,000 live births and 61.8 percent of birth are attended by skilled personnel, with wide regional disparities. A study produced by UNFPA in 2014 indicates that 15 out of 47 counties accounted for 98.7 percent of the total maternal deaths in Kenya. Women and girls are also disproportionately affected by indoor pollution due to lack of access to clean cooking. Across the country, more than half (54 percent) of households use wood as their main source of cooking fuel, more than 84 percent of households in rural areas. Women and girls are the main consumers of electricity at the household level due to traditional gender roles; they collect at least 60 percent of all firewood supplies in Kenya⁴. Their chronic exposure to hazardous air pollutants (HAP) has been linked to increased prevalence of respiratory infections and injuries such as burn⁵.

Figure 18: Indoor air pollution death in the East African Community

	Total deaths from HAP	Deaths of 0-4 year olds	% of total	Deaths of males 18+	% of total	Deaths of females 18+	% of total
Burundi	9,375	5,189	55	2,268	24	1,918	20
Kenya	21,691	11,314	52	5,658	26	4,719	22
Rwanda	6,291	2,650	42	1,876	30	1,765	28
South Sudan	9,200	5,062	55	2,145	23	1,993	22
Tanzania	21,671	9,219	43	6,633	31	5,820	27
Uganda	20,954	10,039	48	5,682	27	5,233	25
Total	89,182	43,473	49	24,262	27	21,448	24

Source: World Health Organization, 2012 and EACREE 2018

¹ Malonza, R. Fedha, ML. 2015. An Assessment of Gender and Energy in Kenya: The Underlying Issue.

² World Economic Forum (WEF). 2020. Global Gender Gap Index.

³ Ibid.

⁴ United States Agency for International Development (USAID). 2016. Gender Analysis Report.

⁵ Government of Kenya, Ministry of Energy. 2019. Gender Policy in Energy.

- **Gender-Based Violence.** GBV is a widespread phenomenon in Kenya, commonly affecting women and girls. This is a major human rights and public health issue reinforced by social acceptance of violence against women and high levels of impunity. Under customary law, domestic violence can be justified as an acceptable way of reacting to women's refusal to obey her husband. The 2014 Kenya Demographic Health Survey (KDHS) indicates that 39 percent of ever-married women and 9 percent of men aged 15-49 report having experienced domestic physical or sexual violence. 45 percent of women between the age of 15 and 49 have experienced physical violence. Women living in internally displaced persons camps are even more vulnerable to GBV¹. Kenyan girls are particularly vulnerable to harmful traditional practices and early marriage. 21 percent of Kenyan girls have been subjected to FGM, mostly in the Somali, Samburu, Kisii, and Masaai communities. In the energy sector, several organization have introduced policies to prevent gender-based violence. For example, the Kenya Power and Light Company (KPLC) has introduced a Sexual Harassment Policy².

- **Women's agency and political participation.** Since the introduction of a gender quota in the 2010 Constitution, women's participation in political institutions has slowly improved. Following the 2013 general elections, female representation in Parliament increased from 22 to 86 representatives, while 96 women were elected as members of country assembly. However, women continue to be under-represented in political positions at all levels. Since the 2017 General Election, women account for 21.8 percent of members of Parliament and hold 25 percent of ministerial seats³. At the county level, women represent 34 percent of Members of the County Assembly (MCAs). Of that number, only 13 percent were elected, the remaining female members were appointed⁴.

- **Female employment and access to decent work.** Kenyan women continue to be under-represented in the labor market and are more likely to work in the informal sector in lower-quality, lower-paid jobs. Women's participation rate in the labor force reaches 64.1 percent (versus 69.6 percent for men), while 44.2 percent of women are employed part-time compared to 28.9 percent of men⁵. Gender gaps in outcomes are significant: men's estimated earned income is \$3,600 compared to \$2,500 for women. A 2019 study of 60 Kenyan companies shows wide gender disparities in access to higher-level positions in the private sector. Key barriers affecting women's career development include gender bias and stereotypes, lack of mentor, limited informal networks, sexual harassment as well as issues maintaining work-life balance⁶. However, Kenya does outperform global averages on female employment in companies per Figure 5 below.

¹ United States Agency for International Development (USAID). 2016. Gender Analysis Report.

² Clancy, Nthabiseng and Mohlakoana, 2020. Gender audits: An approach to engendering energy policy in Nepal, Kenya and Senegal.

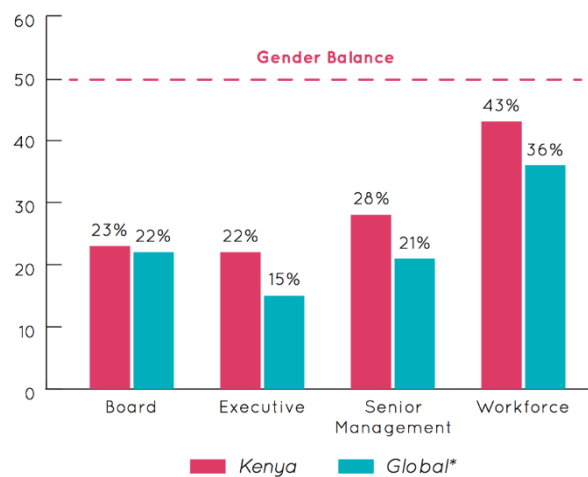
³ World Economic Forum (WEF). 2020. Global Gender Gap Index.

⁴ United States Agency for International Development (USAID). 2020. Kenya Gender Equality and Female Empowerment.

⁵ World Economic Forum (WEF). 2020. Global Gender Gap Index.

⁶ United States Agency for International Development (USAID). 2016. Engendering Utilities: Improving Gender Diversity in Power Sector Utilities.

Figure 19: Female employees by company level (in percentage)



*Equileap assesses over 3,500 companies globally.

Source: Equileap, 2019

Women's access to economic opportunities is hampered by gender roles that confine them to domestic tasks, especially in rural areas. As a result, women have less time than men to dedicate to economic activities. For instance, Kenyan women usually spend more than 5 hours a day near a fire, whereas men will spend less than an hour¹. The labor market is segregated by sex according to gender roles, resulting in women lower representation in the energy sector. Female employment is largely concentrated in the service (mostly in education) and agriculture industries. In the agricultural sector, most women are subsistence farmers providing unpaid labor. Only 1.8 percent of women work in engineering, manufacturing and construction while this is the case for 6.2 percent of men. The state-utility KPLC started embedding gender in its corporate policies in 2008, to achieve greater balance among its workforce. In 2015, less than 20 percent of its employees were women.

Figure 20: Percentage of women employees within selected utilities

	Macedonia	Georgia	Jordan		Kenya	Nigeria	
As a percent of employees	EVN Macedonia	Energo-Pro	EDCO	IDECO	KPLC	EKEDP	IBEDC
Women Employed %	19.6	10.8	6.3	5.6	19.4	20.2	19.0
Women Managers %	22.7	16.8	8.8	15.5	21.2	30.5	25.0
Women Engineers %	21.1	15.6	13.7	25.7	9.2	3.7	9.1
Total Employment	1,973	5,224	1,297	1,172	10,590	2,154	2,351

Source: USAID report, 2016

- **Entrepreneurship and access to ownership and finance.** Although women own 48 percent of businesses in Kenya, female entrepreneurs face more challenges than male entrepreneur in their professional

¹ Ibid.

development. Female-owned businesses are smaller in size and are most likely to be operating from home. These businesses include activities such as roadside stands selling vegetables or clothes and retail businesses. Due to lack of access to collaterals and limited land ownership, only 7 percent of female business owners have access to formal credit. Women are also less likely to have bank accounts than men, particularly due to the lack of bank branches in rural areas¹. In addition, female entrepreneurs face greater discrimination in access to electricity, experiencing delays in obtaining connections and the expectation that they will pay bribes to get them².

Female-headed households are particularly vulnerable to the lack of access to ownership and finance and experience higher poverty levels than men-headed households. They also have lower access to electricity, often because of high connection/subscription fees to the grid. According to Kenya's 2018 Multi-Tier Framework (MTF) survey, 50.4 percent of FHHs in rural areas have no access to electricity compared to 45.3 percent of male-headed households. In urban areas, 16.3 percent of FHHs have no electricity versus 9.3 percent of male-headed households³.

3.5 Mali Gender Gap

While Mali has made notable gains in gender equality in recent years, significant gaps remain in health outcomes, education, economic and political participation. Mali is among the top five most-improved countries in the 2020 World Economic Forum Global Gender Index, next to Ethiopia, Spain, Albania and Mexico. However, the WEF Index ranks Mali 139 out of 153 countries, as women and girls continue to suffer from harmful cultural norms and practices such as FGM and early marriage, educational gaps, unequal division of domestic labor and occupational sex-segregation.

- **Education and health.** Despite progress in overall enrolment levels and government efforts to decrease the disparity between males and females, Mali has yet to close more than 20 percent of its educational gender gap. Women's literacy rate in the country does not exceed 25 percent, while more than 46 percent of men are literate. At every level, women's enrolment is lower than men, with more than twice as many men than women reaching tertiary education.

Figure 21: Mali's Educational Attainment

MALI'S EDUCATIONAL ATTAINMENT (percent)		
	Women	Men
Enrolment in primary education	55.8	62.0
Enrolment in secondary education	26.8	33.0
Enrolment in tertiary education	3.2	7.7

¹ International Renewable Energy Agency (IRENA).2019. Renewable Energy: a gender perspective

² Ibid.

³ Government of Kenya. 2019. Gender Strategy in the Energy Sector.

Source: Based on data from the World Economic Forum 2020 Global Gender Index

Girl's access to education is constrained by the prevalent practice of early marriage and early childbearing, particularly in rural areas. Mali has one of the ten highest child marriage prevalence in Africa: more than 10 times as many women as men are married by age 18 (53 percent versus 4 percent) and 15 percent of girls are married before age 15¹. Early marriage and high fertility rates are correlated to low educational attainment, but also take a heavy toll on women and girls' health outcomes due to higher maternal and infant mortality rates. In 2018, women neonatal mortality reached 33 percent and post-neonatal mortality 22 percent.

- **Gender-Based Violence.** FGM is a widespread practice in Mali affecting 89 percent of women aged 15-49. Wide regional disparities have been found, varying from less than 2 percent in the areas of Gao and Kidal, to more than 95 percent in Koulikoro and Sikasso regions. Unlike neighborhood countries, Mali has not yet legislated on the prohibition of FGM. This practice is reinforced by cultural norms around female circumcision: nearly 70 percent of men and women aged 15-49 believe that FGM is a religious requirement². In addition to FGM, other forms of violence against women are widely accepted and practiced in the Malian society. For instance, more than 2 in 5 women aged 15-49 have experienced physical violence in their lifetime, and a majority of women (79 percent) and nearly half of men (47 percent) find justification in domestic violence³. Polygamous relationships have been associated with higher rates of intimate partner violence towards women than monogamous marriage⁴. Although polygamy has been slowly declining in the Malian society, 37 percent of women aged 15-49 and 19 percent of men were still in polygamous unions in 2018.
- **Women's political participation.** Malian women have extremely limited access to political institutions and are widely under-represented in decision-making structures. In 2018, women held 9.5 percent of women seats in the National Assembly, 34 percent of ministerial positions, less than 2 percent of mayoral posts and 1 governor position⁵. New legislative and municipal elections were originally scheduled in 2018, but were postponed to March/April 2020 due to political incidents and intensifying violence in the central and northern part of the country. At the time of writing, sex-segregated election results are not known. Women's low political participation indicates that the Malian society continues to predominantly perceive men as the main decision-makers, starting in the household. According to the 2018 Mali Demographic and Health Survey, 63 percent of women do not participate in key household decisions such as visiting their family or relatives, major households purchase or decisions about their own healthcare. Women's lack of agency in the household has spill-over effects in the labor market which is characterized by occupational sex-segregation, and employment gaps.

¹ Government of Mali. 2018. Mali Demographic and Health Survey or EDSM-VI.

² Ibid.

³ Ibid.

⁴ Cooperative for Assistance and Relief Everywhere (CARE). 2013. Men, Gender Equality and Gender Relations in Mali.

⁵ Stockholm International Peace Research Institute (SIPRI). 2019. A Study of Insecurity and Gender in Mali.

- **Female employment and access to decent work.** When Malian women are employed, they predominantly work in the informal sector with poor level of qualifications, in inferior positions and with lower, if any, income. Most of women's participation in the labour market (95 percent) is concentrated in agriculture, livestock farming, fishing, trade, processing activities, household activities as well as hotels, restaurants and catering services. Employed women are far more likely than men to work without pay at all, mostly as unpaid agricultural laborers in their own households : 77 percent of women farmers declared in a government survey that they have never received wages for their labor¹. When women are paid, whether in the formal or informal sector, their income is on average less than half the amount of what men receive. Women's participation in the labor market is further hindered by social norms making them responsible for domestic work, especially in rural areas. In addition to the time spent on farming, rural women dedicate on average 7 to 8 hours each day to domestic chores². This translates into a significant employment gap with only 63.4 percent of women participating in the labour force compared to 82.7 percent of men³. This employment gap is reflected in the energy sector, where women are under-represented and are not allocated the same career opportunities as men. In 2018, the state-owned utility Energie du Mali (EDM) only had 446 female employees out of 2234 staff, and only 15 percent of total trainings were allocated to women.
- **Entrepreneurship and access to ownership and finance.** Women play a major role in agricultural production but their access and ownership of resources remain extremely limited. Mali's customary law excludes women from land ownership and continues to be widely applied in rural areas, despite the adoption of the 2006 agricultural orientation law protecting women's property rights. Although all lands in Mali are State property by law, the allocation of lands is often managed by village chiefs and customary chiefs. Women's access to entrepreneurship is further hampered by lack of access to technology and credit, preventing women farmers from purchasing necessary inputs and services. In 2012, only 12 percent of total credit was allocated to women⁴. Women's limited access to entrepreneurship is compounded by legal barriers: according to article 38 of the Family Code women who wish to undertake a commercial business need their husband's authorization.

3.6 Namibia Gender Gap

The government of Namibia's efforts to promote equality between men and women has resulted in continued reduction of gender gaps in recent years. Namibia is today one of the most advanced countries worldwide with regards to gender equality. The country is ranked 12th out of 153 countries by the World Economic Forum Global Gender Index, second in Sub-Saharan Africa (behind Rwanda). Yet, significant disparities remain, especially in the fields of women economic participation and opportunities and gender-based violence. Key gender gaps that may impact the implementation of the RE project in Namibia are explored below:

¹ United States Agency for International Development (USAID). 2012. Mali Gender Assessment.

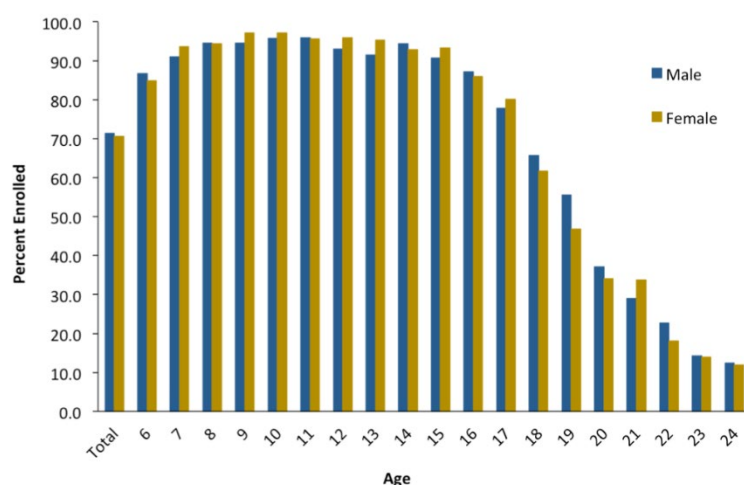
² Ibid.

³ World Economic Forum (WEF). 2020. Global Gender Gap Index.

⁴ United States Agency for International Development (USAID). 2012. Mali Gender Assessment.

- **Education and Health.** Alongside Botswana and Lesotho, Namibia is among three Sub-Saharan African countries to have already achieved parity in education. Namibian women have higher rates of enrolment in primary and secondary education, and are twice as many in tertiary education (30.3 percent versus 15.3 percent for men)¹. This is the result of a proactive policy from the Government of Namibia to make education accessible to all boys and girls, in both primary and secondary schools. Primary education in government schools became free in 2013, followed by secondary schools in 2016. Despite these significant progress, two challenges remain: i) persisting educational gap between rural and urban areas, and ii) occupational sex-segregation.

Figure 22: School enrolment of the school going population aged 6-24 years by sex



Source: Namibia Inter-Censal Demographic Survey (2016)

Gender gap in STEM remains high, with almost three times as many men as women enrolled in STEM fields (21 percent versus 7.71 percent for women)². For instance, the University of Namibia's 2015 annual report indicates that the faculty for engineering and IT has the lowest percentage of female enrolment rate (23 percent). The scarcity of female graduates in STEM fields translates into lower female participation in the energy sector.

Although the government of Namibia has improved health outcomes for both men and women in recent years, maternal mortality remains high with an estimated 385 deaths per 100,000 live birth³. Underlying causes include limited accessibility of health facilities, lack of skilled personnel and delays in seeking care. In 2013, 21 percent of the population lived more than 10 kilometers away from a health facility, making access to emergency obstetric care extremely difficult. On the other hand, the combination of increased knowledge about sexual and reproductive health, high educational attainment and increased female participation in the labour force has resulted in a significant drop in fertility rate. In 2017, Namibian women had on average 3.4 children compared to 6.1 children in 1991. Namibian women of all ages remain particularly vulnerable to HIV, which is a major public

¹ World Economic Forum (WEF). 2020. Global Gender Gap Index.

² Ibid.

³ Government of Namibia. 2013. Namibia Demographic and Health Survey (NDHS).

health issue in a country where the HIV prevalence rate is one of the highest in the world. In 2018 the rate of women aged 15 to 49 with HIV reached 14.1 percent compared to 9.3 percent among men of the same age.

- **Gender-Based Violence.** In Namibia, women and girls continue to experience high levels of sexual and GBV, exacerbated by substance abuse and harmful traditional practices. Harmful practices include genital stretching, testing of girls' sexual readiness by male relatives and widow cleansing,¹ especially in rural areas. The 2013 Namibian Demographic and Health Survey indicates that 32 percent of women aged 15-49 have experienced physical violence. Furthermore, more women than men feel insecure in the public space: 48 percent of women compared to 37 percent of men reported feeling unsafe walking in their neighborhoods. Although social tolerance to intimate partner violence has decreased over the years following legal reforms, 28 percent of women and 22 percent of men continued to justify domestic violence in 2013 (compared with 41 percent of men and 35 percent of women in 2006)².
- **Women's agency and political participation.** Namibia has one of the highest level of women's political participation in the continent. Since the November 2019 election, the rate of female members of Parliament reaches 46 percent, a 6 percent increase from last term. The election also resulted in increased female participation at ministerial levels: 38 percent of cabinet positions are held by women, a 28 percent increase, including in traditionally male-dominated ministries such as foreign relations, trade and justice. These progress indicate increasing acceptance of women's leadership in the political arena, although parity has not been achieved. A survey conducted in 2014/2015 by Afrobarometer revealed that 80 percent of men and women agreed with the statement that women should have equal opportunities to be elected to political office, while this was the case for only 67 percent of respondents in 2005/2006. However, women continue to report lower interest in politics and participation in politics than men : 63 percent versus 70 percent for men.
- **Female employment and access to decent work.** Although women are increasingly more present in the labor market, gaps remain with respect to earned income, vulnerable employment, and unemployment. Namibian women are indeed less likely to be employed: in 2018 unemployment rate was higher among female (34.3 percent) than among male (32.5 percent). Latest figures indicate that women's labour force participation rate reaches 58.7 percent, compared with 67.4 percent for men³. They are also more likely to earn a lower income and occupy informal employment: 61.2 percent of employed women are in the informal sector, compared to 54.1 percent of men⁴. In 2013, at least 60 percent of women earned less than their husbands and more than one-third of women and girls who engaged in agriculture work received no pay for it⁵. Rural women are particularly vulnerable to unemployment and vulnerable employment. In 2016, about 44 percent of the female rural labour force was unemployed, compared to

¹ Legal Assistance Centre. Delegation of the European Union to Namibia. 2017. Namibia Gender Analysis.

² Ibid.

³ World Economic Forum (WEF). 2020. Global Gender Gap Index.

⁴ Government of Namibia. 2013. Namibia Demographic and Health Survey (NDHS).

⁵ Ibid.

34 percent in urban areas¹. In rural regions, most active women are engaged in the agriculture, forestry and fishing sector as subsistence farmers or communal farm workers. They are therefore more exposed to the consequences of climate change and are likely to suffer disproportionate effects with the declining availability of water and other resources. For instance, in times of droughts, many women remain in the household while men seek work in urban areas. In 2016, 46 percent of Namibian households were headed by women (48 percent in rural areas and 41 percent in urban areas)². Research indicates that FHH are more likely to be affected by poverty than men-headed households. Occupational sex-segregation is highly present in the country, affecting women's participation in the energy sector as employees and entrepreneurs. Men predominate in the construction, transportation, wholesale and retail trade industries, while women are more likely to be found in the education, accommodation and food services sectors. For instance, in 2018, 8.9 percent of men in the labor market were employed as occupation plant and machine operators while this was the case for 0.4 percent of employed female³.

Figure 23: Employed persons by industry and sex (percentage)

Employed persons by industry and sex (percent)		
INDUSTRY	MALE	FEMALE
Manufacturing	7.8 percent	4.6 percent
Electricity, gas, steam and air condition	0.7 percent	0.2 percent
Construction	11.6 percent	0.9 percent
Accommodation and food service activities	5.3 percent	17.5 percent
Education	4 percent	9 percent
Human health and social work activities	1.5 percent	3.9 percent
Activities of households as employers	5.7 percent	14.2 percent

Source: Namibia Labor Survey 2018

- **Entrepreneurship and access to ownership and finance.** Although women's right to access and own land is protected by land reforms and laws, traditional customary law is still applied in some rural areas, affecting women's inheritance and property rights. Men and women enjoy relatively equal access to bank account ownership⁴. However, significant gaps remain when it comes to access to formal credit and entrepreneurship. Figures from the Agricultural Bank of Namibia (Agribank) indicates that in 2017, women accounted for 22 percent of all loans but only 10 percent of total loan amounts⁵. Poor financial skills, reliance on informal credit and lack of collateral are among the key barriers to accessing bank loans. In 2014/2015, only 26 percent of businesses had majority female ownership while 41 percent had female participation in ownership⁶.

¹ Legal Assistance Centre. Delegation of the European Union to Namibia. 2017. Namibia Gender Analysis.

² Government of Namibia. 2016. Namibia Inter-Censal Demographic Survey.

³ Government of Namibia. 2018. Namibia Labour Survey, 2018.

⁴ The World Bank Group. 2017. Global Findex.

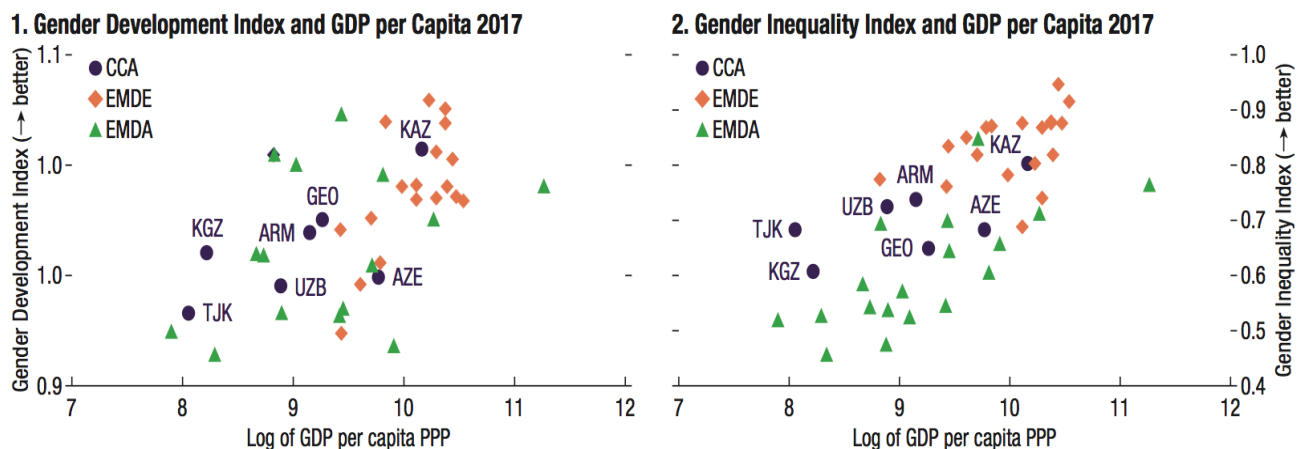
⁵ Legal Assistance Centre. Delegation of the European Union to Namibia. 2017. Namibia Gender Analysis.

⁶ The World Bank Group. 2014. World Bank Group, Enterprise Surveys, Namibia.

3.7 Uzbekistan Gender Gap

While Uzbekistan has significantly increased women's education and health outcomes in the past decades, women continue to have limited access to economic and political opportunities. Following Uzbekistan's independence in 1991, traditional values and patriarchal social norms have gained traction, affecting women's presence and participation in the political and economic arena¹. Uzbekistan is ranked 64 out of 162 by the 2018 UN Gender Inequality Index (GII), and 108 out of 182 in the Gender Development Index, lagging behind Kazakhstan, Azerbaijan, Armenia and Georgia. The GII measures gender inequalities in reproductive health, empowerment and economic status, while the Gender Development Index measures disparities between women and men in health, knowledge, and living standards.

Figure 26: Gender Index



Sources: United Nations, Human Development Report 2017; and IMF report 2019

Note: CCA = Caucasus and Central Asia; EMDA = emerging and developing Asia; EMDE = emerging and developing Europe.

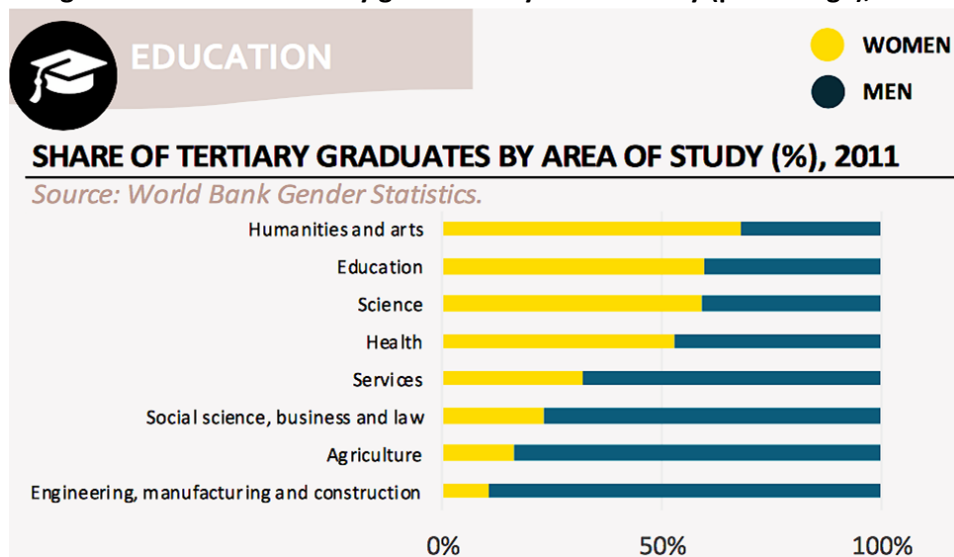
- Health and Education.** The Government of Uzbekistan has enacted policies to improve reproductive health indicators resulting in substantial decrease of maternal and infant mortality rates in past decades. The maternal mortality rate went from 41 per 100,000 live births in 2000 to 29 per 100,000 live births in 2017². Some barriers to accessing quality health care remain, especially in rural areas, owing to long distance to health facilities, and the high cost of care. Access to safe drinking water and sanitation is also unreliable in rural areas, particularly affecting women who are primarily responsible for water collection and household chores. Uzbekistan has achieved near parity in primary and secondary school enrolment. Nonetheless, two educational gap remain: (i) women's limited access to higher education and (ii)

¹ Rano Turaeva. 2017. Gender and changing women's roles in Uzbekistan: from Soviet workers to post-Soviet entrepreneur.

² World Bank Gender Data Portal.

educational sex-segregation. The Constitution of Uzbekistan guarantees equal access to education and near parity is observed in primary and secondary enrolment. In 2018, the female gross enrolment ratio in primary education was of 103.4 percent (105 percent for men) and 92.7 percent in secondary education (93.9 percent for men)¹. However, women's enrolment rates in higher education are significantly lower than men's. In 2018, only 8.2 percent of women were enrolled in tertiary education, compared with 11.8 percent of men². Key drivers behind women's lower tertiary education rates include social norms that encourage women to marry after completing secondary school and financial barriers such as the cost of tuition fees, accommodation and transportation to a higher educational institution. This particularly affects poorer households in rural areas who might prioritize the education of boys for economic reasons. In higher education, women and men's educational path is often determined by gender roles and stereotypes. Men are more likely to enroll in technical fields for instance in transport, communication, construction and agriculture, whereas women predominate in traditionally female fields of study such as education and healthcare. Educational sex-segregation directly affects women's economic opportunities because male-dominated fields generally lead to better-quality jobs with higher incomes. It also impedes women's presence and participation in the energy sector.

Figure 27 : Share of tertiary graduates by area of study (percentage), 2011



Source: World Bank Gender Statistics, 2015.

- **Women's political participation.** Since the adoption of a quota law in 2004 requiring at least 30 percent of female candidates in political party lists, the number of elected women in the National Assembly has consistently increased. However, women's political participation remains extremely low at the national and local level: women make up only 16 percent of the legislative chamber (9.4 percent in 2014) whereas few women hold leadership positions in local government. The local leaders, *hokim*, are traditionally male.

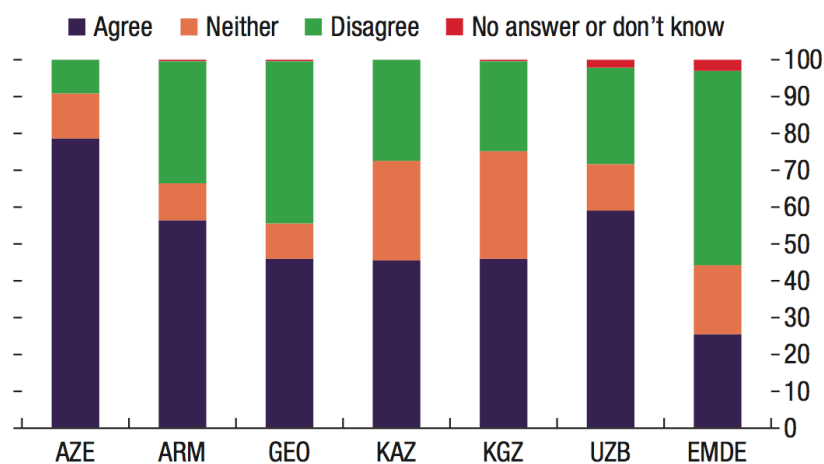
¹ UNESCO Database of Resources on Education.

² Ibid.

Women's presence and participation in the political arena is further hampered by gender stereotypes confining them to the household and reducing their free time. Women are estimated to spend three times more time than men in unpaid work (cooking, housekeeping etc.)¹.

- **Female employment and access to decent work.** Although Uzbekistan has a relatively high rate of female participation in the labour market, they remain under-represented, have limited access to formal employment and are concentrated in lower-income sectors. In 2019, 55.8 percent of women aged 15-64 participated in the labor force, compared with 81.7 percent of men². Women's share in formal employment (45.7 percent) is also lower than that of men (54.3 percent)³. One of the main drivers behind this gender imbalance is the prevalent social norm attributing responsibility for unpaid care work to women. According to a recent International Monetary Fund (IMF) survey, a majority of respondents (60 percent) agreed that men should be given preference over women in the job market when jobs are scarce. This indicates adherence to the patriarchal family model in which women's main responsibility lies with the household and children.

Figure 28: When jobs are scarce, men should have more right to a job than women (percent of respondents)



Source: IMF value Survey, 2019 Note: EMDE = emerging and developing Europe

Women are more likely to earn income through informal and family-based employment, for instance in farming or handicrafts, therefore increasing their vulnerability and dependency due to lower access to social benefits. Women's limited access to economic opportunities is exacerbated by occupational sex-segregation: female employment is concentrated in the public service sector (education, health care, social security) where salaries are inferior, while men predominate in technical fields such as construction, transport and communication. As a result, the gender wage gap between men and women is estimated between 60 percent and 53 percent⁴.

¹ Asian Development Bank. 2018. Uzbekistan Country Gender Assessment Update.

² World Bank Data Bank.

³ Asian Development Bank. 2018. Uzbekistan Gender Assessment.

⁴ Ibid.

Only 8 percent of workers in construction and 11.3 percent in transport and communications are women, while men only account for 18.4 percent of all employees in health care. This is reinforced by legal restrictions on women's entry to technical industries. Article 225 of the Labor code of Uzbekistan limits women's employment in several fields including transport, construction, natural resources extraction, manufacturing and industry.

Figure 29 : Sex-Disaggregation of the Employed, by Sector, 2015

Sector	Total (%)		Sector (%)	
	Women	Men	Women	Men
Total	100	100	45.7	54.3
Industry	10.8	14.5	38.5	61.5
Agriculture and forestry	27.6	27.7	45.5	54.5
Construction	1.7	16.2	8	92
Transport and communications	1.3	8.9	11.3	88.7
Trade, catering	13.3	9.7	53.7	46.3
Municipal services	3	4.3	36.7	63.3
Health, wellness, and social welfare	12.6	2.4	81.6	18.4
Education, culture, arts, science	19.9	6.5	72.1	27.9
Finance, banking, and insurance	0.4	0.5	41	59
Other	9.4	9.3	45.7	54.3

Source: Official statistics and Asian Development Bank Uzbekistan Gender Assessment, 2018

- **Female employment in the Energy industry.** Occupational sex-segregation translates into lower female presence and participation in the energy industry. Women are under-represented in this sector, especially in technical fields and in management positions, and are more likely to perform administrative tasks. Only 17 percent of all employees from the state-owned energy company Uzbekenergo are women. Women's under-representation in the energy sector can also be explained by limited opportunities for skills development and lack of awareness of employment opportunities in the sector.

Figure 30: Sex-Disaggregation of Uzbekenergo staff (percent)

Year	Total		Management		Technical staff		Maintenance	
	Women	Men	Women	Men	Women	Men	Women	Men
2014	17.2	82.8	23.1	76.9	44.8	55.2	14.7	85.3
2015	16.9	83.1	21.9	78.1	51.3	48.7	14.5	85.5
2016	17	83	20.2	79.8	46.5	53.5	15	85
2017 (Jan–June)			23.3	76.7	12.9	87.1		

Source: Asian Development Bank Uzbekistan Gender Assessment, 2018

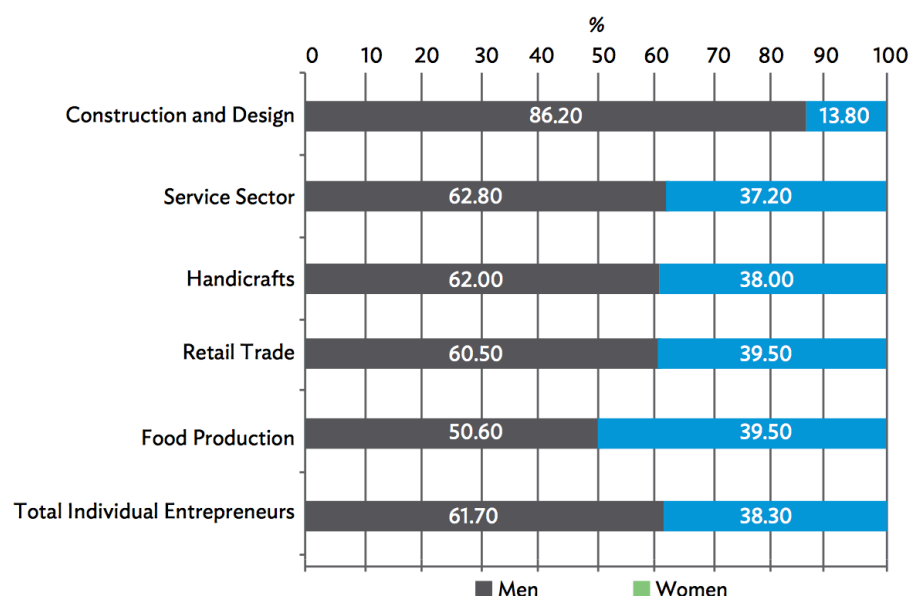
- **Entrepreneurship and access to finance.** While the private sector has been taking more space in the country's economy, the Government of Uzbekistan has enacted policies to promote women's entrepreneurship. Women represent around 38 percent of individual entrepreneurs in Uzbekistan, and are mostly working in handicrafts or food production in rural areas¹. Several constraints affect women's capacity to create and expand businesses: women are more likely to own small-scale businesses and have limited access to finance due to insufficient collaterals, as well as limited financial skills and knowledge about loan application processes. Women only own 22.3 percent of the total value of property officially registered² and only 33 percent of female entrepreneurs resort to bank loans for start-up capital³. Women's lack of access to entrepreneurship is compounded by psychological barriers affecting confidence and limiting risk-taking behaviors. Furthermore, occupational sex-segregation is reflected in entrepreneurship and women are under-represented in traditionally male sectors such as construction and design.

¹ Food and Agricultural Organization (FAO).2019. Gender, agriculture and rural development in Uzbekistan.

² Asian Development Bank. 2018. Uzbekistan Country Gender Assessment Update.

³ Business Forum of Uzbekistan.2014.

Figure 31: Male and Female Individual Entrepreneurs by Sector (percent)



Source: State Committee of the Republic of Uzbekistan on Statistics, 2012 and Asian Development Bank Uzbekistan Gender Assessment, 2018

- Gender-Based Violence.** Reliable data are still lacking to assess the prevalence of GBV in Uzbekistan, but attitudinal surveys show high levels of tolerance towards violence against women. One of the most common cause of domestic violence is the perception that women are challenging their submissive role in the household¹. The president signed a Law on the Protection of Women from Harassment and Violence in September 2019, yet services for victims continue to be scarce.

¹ Food and Agricultural Organization (FAO).2018. Uzbekistan Gender Assessment.

4. PROPOSED ENTRY POINTS

Building upon this preliminary assessment of key gender gaps in the seven targeted countries, four main areas of interventions have been identified, namely: i) Female employment and skill gaps in the energy sector, ii) Female employment and livelihoods in the areas of implementation of RE projects, iii) Gender-based violence and iv) women access to services as energy consumers. Potential mitigations actions for each area are highlighted below and will be explored during project design:

- a. **Female employment and skill gaps in the energy sector:** the main barriers to women participation in the energy sector, as employees and entrepreneurs, include social norms that confine women to unpaid household and care work, occupational sex-segregation, limited capacities and skills, as well as discriminatory work environments. In some countries, legislation may also prevent women from entering certain industries, affecting their employment in the energy sector. Without a proactive policy to promote female economic participation, men may disproportionately benefit from new employment opportunities across the RE value chain, particularly in construction, operation and maintenance. Women may already enjoy indirect employment opportunities in ancillary industries such as health, social services and catering. Increasing female employment in non-traditional fields can contribute to challenging social norms and stereotypes, ultimately enhancing women's voice and agency.

Entry points for addressing this issue include the promotion of women's participation in the realization of renewable energy projects, in energy generation, transmission and distribution, and the promotion of female-friendly infrastructure and services:

- Vocational trainings and capacity-building to increase women's technical skills (including programs for women to gain hands-on experience in the energy sector) and focus on reforms in the public utilities managing RE projects;
 - Partnerships with educational and technical training institutes to enhance female school-to-work transition in non-traditional fields;
 - During the procurement phase, technical support would be provided to national stakeholders to ensure a transparent and equitable process. Procuring entities would be encouraged to the extent possible to include non-price factors at relevant stage of the procurement process, to enhance female employment and the participation of women-owned businesses;
 - The adoption of equal opportunities practices at the corporate level would be encouraged through policy dialogue and capacity-building activities; and
 - Outreach initiatives to allow local players to be informed of the program's business opportunities. Special attention would be paid to the extent possible to the promotion of women-run businesses along the project development cycle.
- b. **Female employment and livelihoods in the areas of implementation of RE projects:** RE projects are frequently located in remote areas with good solar/wind resources and large plots of lands, but where living standards are below the country average. Local communities often face numerous challenges hindering their development. Women, if employed, are more exposed to lower-paid, lower-quality

employment in the informal sector (e.g. textile, crafts, food processing, small-scale farming and retail trading). They also have limited access to ownership of key inputs or assets. As a result, women may face more difficulties with regards to access to entrepreneurship and ownership of micro, small, and medium-sized enterprises.

The socio-economic needs of the local communities will be assessed, and actions empowering women will be prioritized to the extent possible along the electrification program. Those may include:

- Providing women and girls with sources of low-risk income and entrepreneurship opportunities while ensuring that community needs are accommodated. The activities to support would be selected after assessing the needs of the local communities and putting in place a scorecard to prioritize projects benefiting women. Actions may cover health services (such as mobile clinics), infrastructures to open-up a village or education and skills-building activities, for instance in traditional home-based activities such as handicrafts, or in agricultural activities. Particular attention would be paid to illiterate women when relevant. Support may be provided through grants, or as a requirement under local bids and/or part of private sector Corporate Social Responsibility (CSR) aspects;
 - Ongoing support would be offered to support female entrepreneurship notably through the creation of peer groups, in partnership with local NGOs.
- c. **Gender-based violence:** women routinely experience sexual harassment and intimate partner violence in target countries and the risk of violence against women can increase during the deployment of solar and wind projects, due to the presence of migrant workers on construction sites. Large-scale energy projects may also increase rates of gender-based violence in surrounding areas, according to studies. For instance, households with a rise in income due to employment in construction sites can see increased expenditures on alcohol, a known driver of GBV¹.

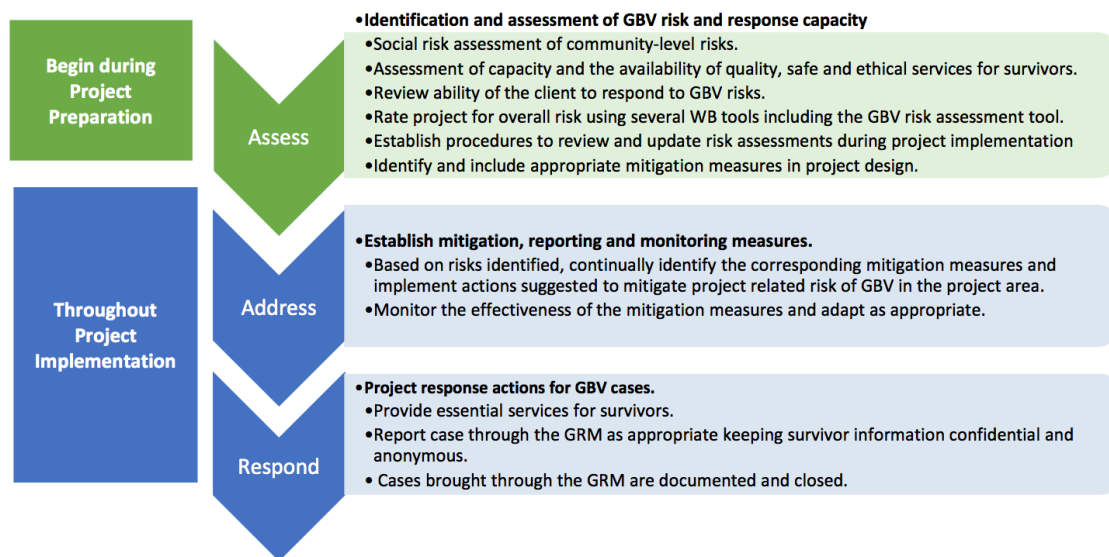
The risk of increased GBV, sex trafficking, and child abuse during the realization of large-scale infrastructure projects will be mitigated through adopting the following actions, required by the World Bank procedures and outlined in the WBG GBV Good Practice Note (2018):

- Assessment of GBV risks by the World Bank task team using the World Bank GBV risk assessment tools;
- Assessment of GBV risks by the client country during project preparation and particularly during community consultations. GBV considerations will be included in Safeguard documents (Environmental and Social Assessment ESA, Environmental and Social Management Plan);
- Development of a GBV Action Plan;
- Assessment of the project capacity to provide access to safe and ethical services for survivors;

¹ Kuriakose, A. and de Boer, F. 2015. Gender and Renewable Energy: A Literature Review.

- Establishment of GBV Risk and Mitigation Response Measures such as including specific requirements during procurement process based on ESA findings, and providing GBV trainings to contractors, consultants and clients;
- Monitoring and reporting of GBV activities;
- Responding to GBV incidents in cooperation with GBV service providers.

Figure 32: WBG mandatory requirements in Major Civil Works



Source: WBG GBV Good Practice Note (2018)

- d. **Women access to services as energy consumers:** Among project countries with lower electrification rates such as DRC and CAR, women suffer from time-poverty and a health-burden, as they shoulder the responsibility of food preparation, as well as fuel and water collection. In target countries, the main barrier to women accessing energy services from renewable energy may include high cost of connecting to the electrical grid, resulting in possible certain households being left behind. Women may resort to biomass fuels for cooking and heating purposes increasing their time poverty and limiting their ability to create income-generating opportunities. A study conducted by the Botswana Power Corporation found out that female-headed households were most affected by lack of access to modern energy, whereas these households had a high-demand for income-generating opportunities¹. Furthermore, women-headed businesses routinely face more constraints than men in accessing grid electricity, such as delays in obtaining electrical connections and the expectations that they will pay bribes for these services². Improving access to electricity services can therefore reduce the time and labor burden of women,

¹ Energia. Botswana Power Corporation. 2011. Gender Mainstreaming in the Botswana Power Corporation

² The United Nations Development Programme (UNDP). 2016. Gender, Climate Change and Food Security.

generate health and education benefits, increase female labor participation and incomes. Research shows that in rural areas, energy access is correlated with 59 percent higher wages¹.

The following interventions would be promoted to increase women's access to electricity:

- Conduct household surveys as a tool to assess affordability of services and willingness to pay, in order to better target financial support;
- Inclusion of women and men's differentiated needs in the design of electrification programs, including by promoting women's engagement during stakeholders' consultations;
- Targeted subsidies or credit schemes to the extent possible to provide electricity to some of the poorest and most under-developed regions, especially in countries where mini-grid projects will be deployed (DRC, CAR);
- Micro-financing schemes to the extent possible to help women-owned businesses overcome the initial cost of connection. Partnerships with local NGOs could be considered to manage loans.

¹ O'Dell, K., Peters, S., & Wharton, K. 2014. Women, Energy and Economic Empowerment; applying a Gender Lens to amplify the impact of energy access.

5. STRATEGIC APPROACH

5.1 Facility Level

The Facility will follow a systematic approach and provide support to the projects with a focus on data collection, generating knowledge and providing training and capacity-building activities. At every step of the project cycle, the Gender and Energy program of the World Bank Energy Sector Management Assistance Program (ESMAP) will be available to provide technical assistance to the project team and the client country. Kenya, Mali, Uzbekistan and DRC have already benefited from ESMAP Gender and energy regional programs that aim to strengthen women's role in the energy sector as consumers, employees and entrepreneurs.

The Facility will encourage the recruitment of a balanced female-to-male employment ratio within the Project Management Unit (PMU) team to the extent possible. The standard World Bank implementation process will indeed be put in place and include a PMU in each country, responsible for developing, implementing and monitoring each project. If needed, staff members including the Project Manager and the Environment and Social Protection Specialists may receive capacity-building trainings. This will ensure that identified gaps as well as sexual harassment and GBV risks are considered throughout the project cycle.

Obtaining stakeholders buy-in will be key in ensuring that interventions to promote women's employment and enhance their livelihood are effectively implemented with measurable outcomes. Institutional and behavioral change are a long-time process that will require sustained support of all relevant stakeholders (local councils, national energy policymakers, energy ministries and energy companies) throughout the program cycle. Countries adherence to international agreements and national legal frameworks on gender -equality, are a good starting point to strengthen government's support for gender-interventions. The following strategy will be followed to increase stakeholders' commitments, to the extent possible:

- Engage in policy-dialogue with client countries through dedicated meetings, to build awareness and sensitivity about the benefits of promoting women's economic empowerment and agency and remove legal and regulatory barriers to female employment in the energy sector;
- Closely align proposed activities with each country's national priorities and Gender Strategies, such as community development or female employment;
- Identify an institutional gender-focal point to serve as interlocutor during project implementation, if appropriate;
- Offer technical support to help utilities invest in workforce diversification, building on existing internal gender strategies when available. This support may take the form of workshops on gender-related concepts with senior management, or a review of training plans to better integrate female employees.

In the private sector, companies are increasingly adopting a focus on gender equality in CSR policies. Indeed, a growing body of research shows that mitigating risks for women and empowering them as consumers, employees and entrepreneurs can lead to increased firm productivity and better performance. During projects design, the private sector can be involved in financing interventions to i) create income-generating opportunities for women in affected areas, ii) implement mechanisms to prevent and respond to GBV incidents, iii) increase connection rates among female-headed households, for instance through capacity-building workshop on maintaining and developing energy services.

5.2 Project level

At the project level, the following approach would be followed:

- a. **Data collection to document the baseline situation and identify gaps.** Baseline data will be collected to set targets and measure progress during implementation, with a focus on countries where existing baseline sex-disaggregated data are weak. Research will build upon existing country gender diagnostics and action plan when available, to ensure national priorities are understood and considered. To support project design and implementation, stakeholders' consultations will involve female participants, to understand men and women differentiated needs. In addition, initial assessments will include an analysis of the differentiated impact of the SRMI project on men and women, through the involvement of gender and energy experts (Environmental and Social Impact Assessment, Resettlement Action Plans etc.).
- b. **Identifying and implementing relevant actions.** Based on the initial diagnosis and inclusive stakeholders' consultations, targeted activities will be identified to address key gender gaps. These may include (i) actions to promote female employment in the renewables sector (including capacity building and programs for women to gain hands-on experience in the energy sector such as support for school-to-work transition), (ii) financing local communities projects empowering women in the area of implementation of the RE projects (such as activities generating revenues), (iii) implementation of prevention measures to mitigate GBV in the area of implementation of the RE projects and (iv) facilitating energy access for FHH (including capacity-building workshops to maintain and develop energy services).
- c. **M&E and Knowledge management.** Since 2017, the World Bank Group strengthened its monitoring system with regards to gender, focusing on results and outcomes. A gender-tag was launched to identify projects who effectively promote equality between men and women and are aligned with the WBG gender strategy. The SRMI facility will adopt a gender perspective and aim to close key gaps in its sub-projects. Results will be monitored throughout the project and be used for real-time adjustments during implementation. To expand the knowledge base, projects will build on lessons learnt from prior projects as well as regional and country base knowledge production and exchange learning opportunities.
- d. **Strengthening female participation and voice during Stakeholders Consultations.** Consultations and public meetings with stakeholders will be conducted throughout the project cycle in a participatory manner and with a special focus on engaging women. This will help gaining knowledge about community

concerns during project preparation and help monitor impact and undertake real-time correction during implementation. Special attention will be paid to using local dialect when relevant and including women from minority ethnic groups and other socially vulnerable groups.

Building on prior experiences, consultations will be organized at a convenient time for women, taking into account their household and childcare responsibilities. Female facilitators and local female leaders will be identified to ensure that women's voice is heard during consultations. Women-only discussion groups may also be organized if appropriate. The participation of female representatives from utilities and local authorities will be encouraged. To ensure the project's sustainability, partnerships will be created to the extent possible with women's groups and nongovernmental organizations (NGOs).

6. GENDER ACTION PLAN

The Gender Action Plan proposed below is indicative at this stage and may be further amended and adapted according to knowledge gathered during the project inception phase, to reflect local context:

Activities	Indicators and Targets	Timeline	Responsibilities
Impact Statement: The program will follow a systematic approach and provide cross-support to sub-projects interventions to narrow gender gaps with a focus on policy dialogue, data collection, training and capacity-building activities. The expected impact includes enhanced female economic participation in the energy sector, increased access to electricity and the creation of income-generating opportunities for women in affected communities. Outcome Statement 1: The SRMI facility will focus on the following four key areas: i) sector-specific and country-level gender knowledge to support the design and implementation of projects is generated, ii) an enabling policy and legal environment for women’s economic participation is promoted through policy dialogue activities, iii) tailored project and program interventions to address identified gender gaps are designed and implemented iv) Knowledge and learning opportunities are enhanced.			
Output 1 <i>Sector specific and country-level gender knowledge is generated and shared to support the design and implementation of projects</i>			

1.1 Undertake sector gender assessments to identify key gender gaps 1.2. Organize multi-stakeholders workshops at country level to present key findings from the assessments 1.3. Conduct Gender need assessment of institutional partners (utilities) 1.4. Hold consultations with key stakeholders to gather knowledge on men and women differentiated needs with enhanced women's participation	1.1Number of gender assessment developed Target: 7 1.2.Number of multi-stakeholder workshops focused on gender equality Target: 7 1.3.Number of gender need assessments of institutional partners conducted Target: 3 1.4.Percentage of female participants in community consultations Target: 30 percent	At the time of project design	Borrowers with the support of the WB team
<i>Output 2 An enabling policy and legal environment for encouraging gender equality is promoted through policy dialogue activities</i>			
2.1 Organize dedicated meetings with client countries on women's economic empowerment and legal and regulatory barriers to female employment in the energy sector 2.2 Organize community and multi-stakeholder dialogue including with the Ministry of Women's affairs, private sector, NGOs, women's groups and Civil Society Organizations as applicable 2.3 Strengthen capacities of institutional stakeholders on the planning and design of gender-sensitive energy policies	2.1Number of meeting organized with client countries on women's economic empowerment in the energy sector Target: 3 2.2 Number of community and multi-stakeholder dialogue organized with the Ministry of Women's Affairs and private sector Target: 7 2.3Number of capacity-building workshop organized with institutional stakeholders Target: 2	By the Facility's end date	Borrowers with support of the WB team

2.4 Strengthen capacities of Project Management Unit staff on gender and energy issues.	2.4Number of capacity-building workshops organized with PMU staff Target: 3		
Output 3 Tailored project and program interventions are implemented to promote gender equality			
At the project and program level, the following approach would be adopted and tailored according to each sub-project context. Examples of interventions to e.g. promote female employment and enhance women's livelihood are listed below and classified by area of intervention. Final decisions will be made at project design: 3.1 Overarching the projects under this Facility will focus on meeting the Gender Tag per corporate target of the World Bank. Female employment and skill gaps in the energy sector: 3.2 Provide vocational trainings and capacity-building to increase women's technical skills (including programs for women to gain hands-on experience in the energy sector) 3.3 Build Partnerships with educational and technical training institutes to enhance female school-to-work transition in non-traditional fields 3.4 Launch outreach initiatives to allow local players to be informed of the program's business opportunities. Special attention would be paid, to the extent possible, to the promotion of women-run businesses along the project development cycle	3.1 Number of projects that fall under tag requirements that will meet the Gender Tag Target: 55 percent Female employment and skill gaps in the energy sector: 3.2Number of capacity-building programs to increase women's technical skills designed and supported in RE projects Target: 3 3.3Number of partnerships formed with training centers/schools Target: 3 3.4 Number of outreach initiatives to inform local players of the program's business opportunities Target: 3 Female employment and livelihoods in the areas of implementation of RE projects: 3.5 Number of programs fostering local socio-economic development (focused on women's empowerment) Target: 3	During the project cycle	Borrowers with support of the WB team

Female employment and livelihoods in the areas of implementation of RE projects: 3.5 Provide women and girls with sources of low-risk income and entrepreneurship opportunities while ensuring that community needs are accommodated 3.6 Organize meetings with female-led firms to share information on contracts/tenders opportunities brought by the RE projects Gender-based violence (GBV): 3.7 Assessment of GBV risks at project sites as per the World Bank Group GBV mandatory requirements 3.8 Development of GBV Action Plans as per the World Bank Group GBV mandatory requirements Women access to services as energy consumers: 3.9 Hold information sessions with local communities on connection procedures, costs, safety aspects and benefits of energy services (health, time-poverty) 3.10 Develop targeted subsidies or credit schemes to the extent possible to increase FHH connection to the grid 3.11Develop micro-financing schemes to the extent possible to help women-owned businesses overcome the initial cost of connection.	3.6 Number of information sessions held with female-led firms Target: country dependent Gender-based violence (GBV): 3.7 Number of assessments of GBV risks at project sites. Target: 7 3.8 Number of GBV Action Plan produced: Target: 7 Women access to services as energy consumers: 3.9 Percentage of female participants who attend information sessions Target: 30 percent 3.10Number of FHH households connected to the grid. Target: country dependent 3.11Number of women-owned businesses who benefit from micro-financing schemes to access electricity Target: country dependent		
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Output 4 Enhanced knowledge and learning opportunities			
4.1 Production of case studies showcasing best practices and findings 4.2 Organization of Forums focused on gender through the Facility with the participation of targeted country representatives 4.3 Undertake knowledge-sharing with local women's organizations and NGOs	4.1 Number of case studies produced by the facility Target: 2 4.2 Number of events with a focus on gender held through the facility Target: 3 4.3 Number of events held with local NGOs through the facility Target: 3	On a rolling basis	World Bank